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Nakata et al.

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(54) **DOCUMENT READING DEVICE AND
IMAGE FORMING APPARATUS**

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Ishikura**, Kanagawa (JP); **Takahiro
Iizuka**, Kanagawa (JP); **Hiroyuki
Tanaka**, Kanagawa (JP); **Shinya
Hasegawa**, Kanagawa (JP); **Kota
Tomioka**, Kanagawa (JP); **Yoichi
Yamakawa**, Kanagawa (JP); **Tomonori
Sato**, Kanagawa (JP); **Kazuyuki Koda**,
Kanagawa (JP); **Isamu Adachi**,
Kanagawa (JP); **Miho Morita**,
Kanagawa (JP); **Tomomi Ishida**,
Kanagawa (JP); **Shinnosuke Kondo**,
Kanagawa (JP); **Yuya Shiokawa**,
Kanagawa (JP); **Daisuke Ishihara**,
Kanagawa (JP); **Kohei Tachibana**,
Kanagawa (JP); **Shinichi Ohba**,
Kanagawa (JP); **Taisuke Endo**,
Kanagawa (JP)

(73) Assignee: **FUJIFILM Business Innovation
Corp.**, Tokyo (JP)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 402 days.

(21) Appl. No.: **17/715,062**

(22) Filed: **Apr. 7, 2022**

(65) **Prior Publication Data**

US 2023/0057320 A1 Feb. 23, 2023

(30) **Foreign Application Priority Data**

Aug. 23, 2021 (JP) 2021-135289

Aug. 23, 2021 (JP) 2021-135290

(Continued)

(51) **Int. Cl.**
H04N 1/08 (2006.01)
H04N 1/00 (2006.01)

(52) **U.S. Cl.**
CPC **H04N 1/0066** (2013.01); **H04N 1/00236**
(2013.01); **H04N 1/00522** (2013.01); **H04N**
2201/0084 (2013.01)

(58) **Field of Classification Search**
CPC H04N 1/0066; H04N 1/00236
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,614,991 A 3/1997 Moro et al.
5,764,385 A * 6/1998 Ohyama H04N 1/00591
399/373

(Continued)

FOREIGN PATENT DOCUMENTS

EP 0452931 10/1991
EP 1553462 7/2005

(Continued)

OTHER PUBLICATIONS

“Office Action of Europe Related Application No. 22169627.1”,
issued on Apr. 2, 2024, pp. 1-4.

(Continued)

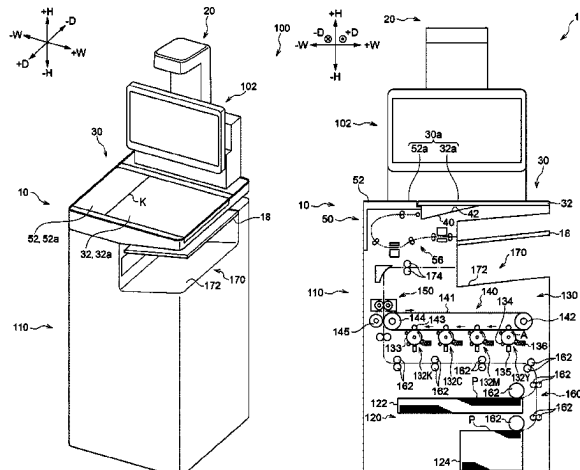
Primary Examiner — Ibrahim Siddo

(74) *Attorney, Agent, or Firm* — JCIPRNET

(57) **ABSTRACT**

A document reading device includes: an image capturing
member that captures an image of a document; a placement

(Continued)



portion that is disposed below the image capturing member and that has a boundary line between multiple surfaces at a position at which the boundary line is covered by the document when the document has a maximum size that the image capturing member is capable of imaging and the document is placed on the placement portion; and a document reading member that reads the document that is placed on the placement portion.

20 Claims, 78 Drawing Sheets

(30) Foreign Application Priority Data

Aug. 23, 2021	(JP)	2021-135424
Aug. 23, 2021	(JP)	2021-135483
Aug. 23, 2021	(JP)	2021-135487
Aug. 23, 2021	(JP)	2021-135489
Aug. 23, 2021	(JP)	2021-135490
Aug. 23, 2021	(JP)	2021-135491
Jan. 12, 2022	(JP)	2022-003293

(56) References Cited

U.S. PATENT DOCUMENTS

5,805,272	A *	9/1998	Nozawa	H04N 1/1043
5,940,128	A	8/1999	Morimura	355/25
6,011,635	A	1/2000	Bungo et al.	
6,371,476	B2	4/2002	Isogai et al.	
2002/0000689	A1	1/2002	Isogai et al.	
2003/0230841	A1	12/2003	Hiramoto et al.	
2005/0074266	A1	4/2005	Hiramoto et al.	
2007/0071492	A1	3/2007	Ito	
2007/0076268	A1	4/2007	Shojo et al.	
2007/0183810	A1	8/2007	Kamiya	
2013/0342878	A1	12/2013	Arson et al.	
2015/0139685	A1	5/2015	Hoshino et al.	
2016/0156247	A1	6/2016	Ishida et al.	
2019/0185279	A1	6/2019	Ando	
2020/0019354	A1	1/2020	Tsuchiya	
2021/0120137	A1	4/2021	Okamoto et al.	
2021/0218850	A1	7/2021	Ogawa	
2023/0053735	A1	2/2023	Shinoto et al.	
2023/0230841	A1	7/2023	Lin	

FOREIGN PATENT DOCUMENTS

EP	1562081	8/2005
EP	1793579	6/2007
EP	2911010	8/2015
EP	2962450	1/2016
JP	H09222762	8/1997
JP	H11278742	10/1999
JP	2003312870	11/2003

JP	2007214836	8/2007
JP	1358358	5/2009
JP	2010176104	8/2010
JP	2017175594	9/2017
JP	2020137163	8/2020
JP	2021068980	4/2021

OTHER PUBLICATIONS

“Office Action of Europe Related Application No. 22169652.9”, issued on Apr. 2, 2024, pp. 1-6.
 “Office Action of Europe Related Application No. 22169624.8”, issued on Apr. 2, 2024, pp. 1-6.
 “Office Action of Europe Related Application No. 22169641.2”, issued on Apr. 2, 2024, pp. 1-7.
 “Summaries with Thumbnails”, InnovationQ+, Jul. 16, 1996-Mar. 26, 2024, pp. 1-127.
 “Office Action of U.S. Related Application, U.S. Appl. No. 17/714,110”, issued on Sep. 13, 2024, p. 1-p. 25.
 “Office Action of U.S. Counterpart Application”, issued on Sep. 5, 2024, p. 1-p. 25.
 “Search Report of Europe Related Application, application No. 22169617.2”, issued on Oct. 5, 2022, p. 1-p. 9.
 “Search Report of Europe Related Application, application No. 22169619.8”, issued on Oct. 27, 2022, p. 1-p. 9.
 “Search Report of Europe Related Application, application No. 22169618.0”, issued on Oct. 21, 2022, p. 1-p. 11.
 “Search Report of Europe Related Application, application No. 22169627.1”, issued on Oct. 6, 2022, p. 1-p. 9.
 “Search Report of Europe Related Application, application No. 22169652.9”, issued on Oct. 5, 2022, p. 1-p. 10.
 “Search Report of Europe Related Application, application No. 22169633.9”, issued on Oct. 13, 2022, p. 1-p. 10.
 “Search Report of Europe Related Application, application No. 22169624.8”, issued on Oct. 6, 2022, p. 1-p. 9.
 “Search Report of Europe Counterpart Application”, issued on Oct. 13, 2022, p. 1-p. 13.
 Merriam-Webster Dictionary, “container”, retrieved on Dec. 5, 2024, Available at: <https://www.merriam-webster.com/dictionary/container>.
 “Summaries with Thumbnails”, InnovationQ Plus, retrieved on Dec. 11, 2024, pp. 1-127.
 Proquest, “Search Strategy from Dialog”, retrieved on Dec. 10, 2024, pp. 1-4.
 “Notice of allowance of U.S. Related Application, U.S. Appl. No. 17/714,152”, issued on Dec. 12, 2024, p. 1-p. 17.
 “Notice of allowance of U.S. Related Application, U.S. Appl. No. 17/714,110”, issued on Dec. 16, 2024, p. 1-p. 14.
 “Office Action of U.S. Related Application, U.S. Appl. No. 17/715,021”, issued on Nov. 1, 2024, p. 1-p. 24.
 “Office Action of U.S. Related Application, U.S. Appl. No. 17/715,025”, issued on Mar. 11, 2025, p. 1-p. 41.
 “Office Action of U.S. Related Application, U.S. Appl. No. 17/711,040”, issued on Mar. 14, 2025, p. 1-p. 52.
 “Office Action of U.S. Related Application, U.S. Appl. No. 17/714,123”, issued on Apr. 22, 2025, p. 1-p. 44.

* cited by examiner

FIG. 1-1

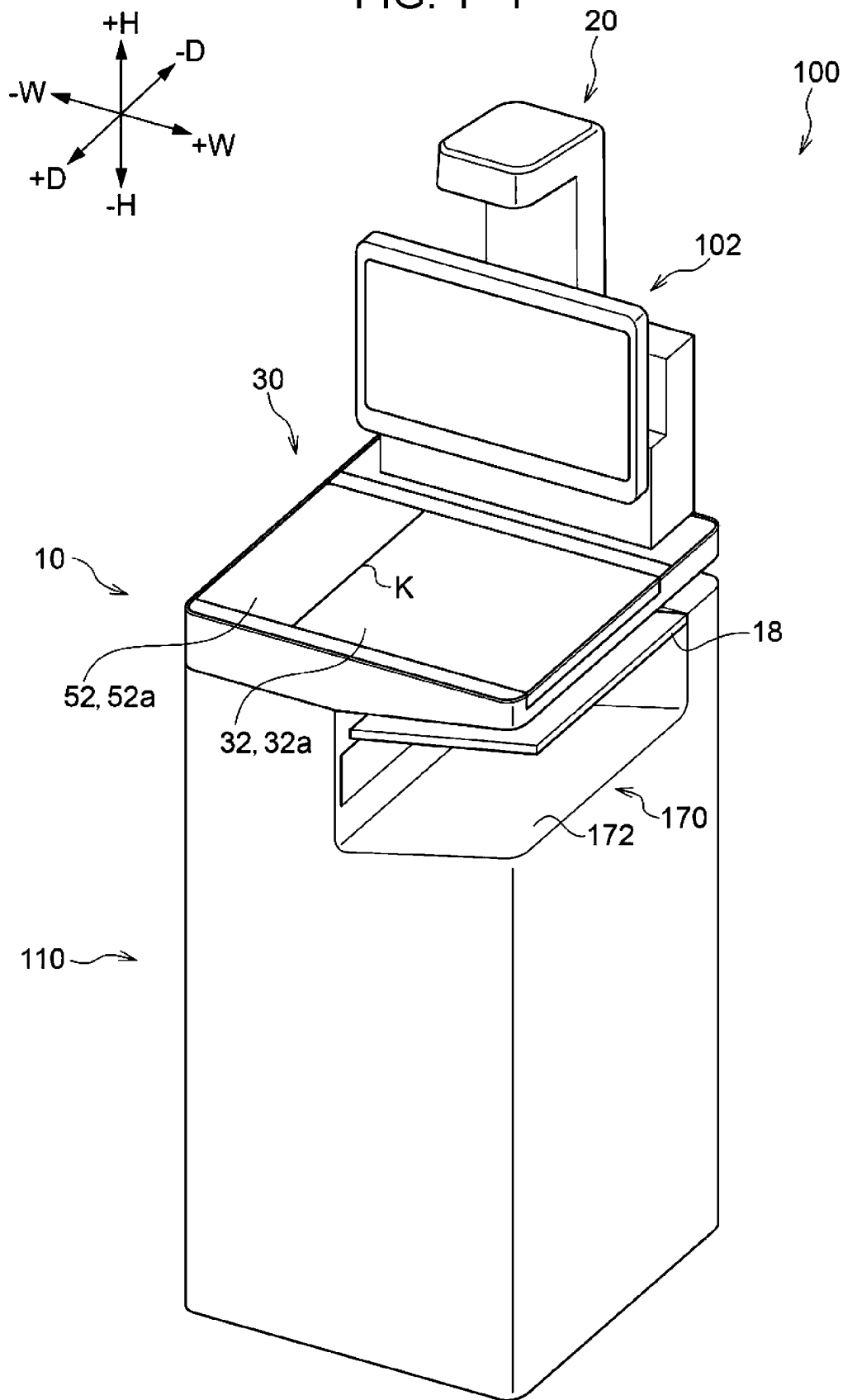


FIG. 1-2

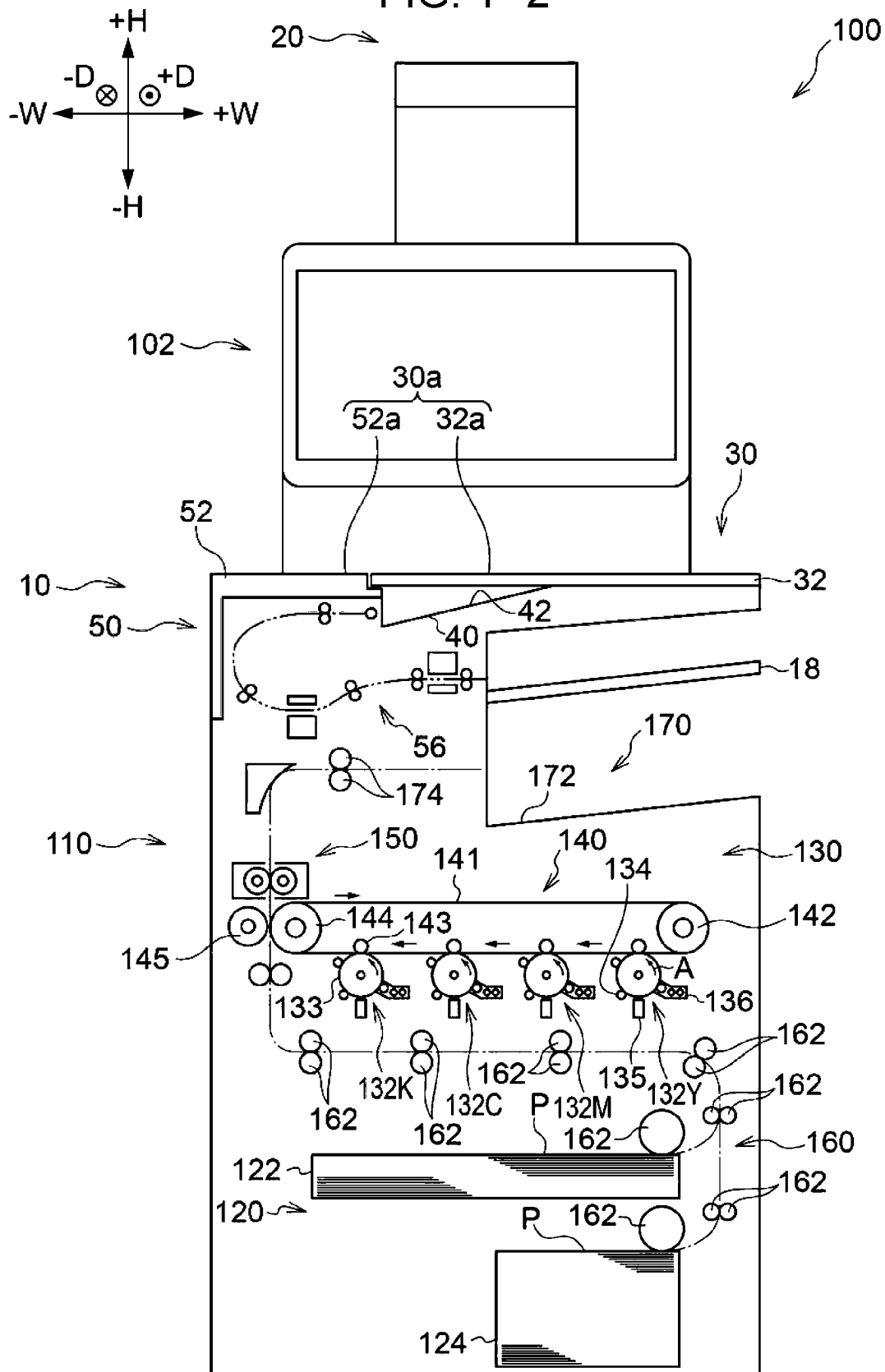


FIG. 1-3

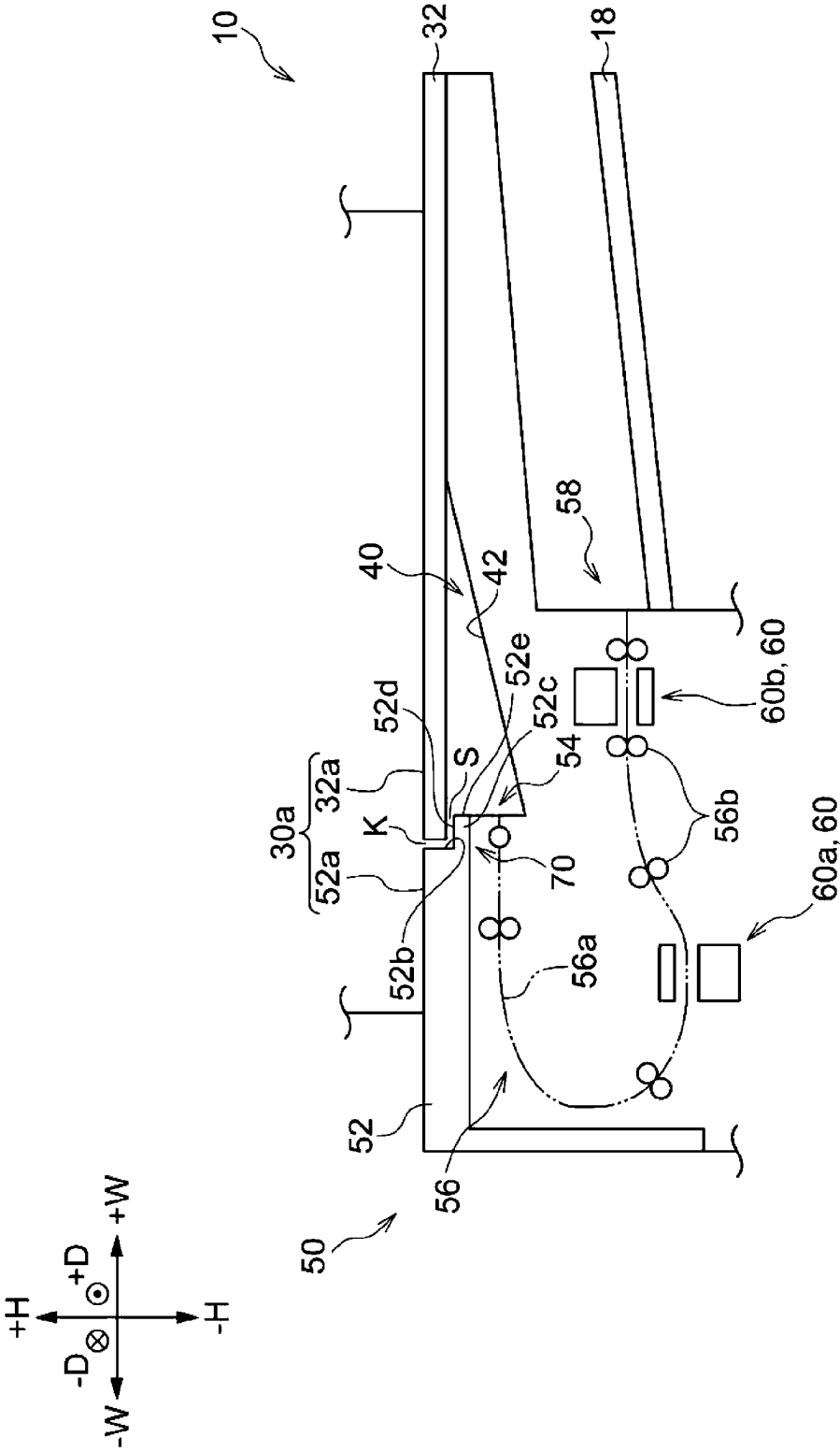


FIG. 1-4

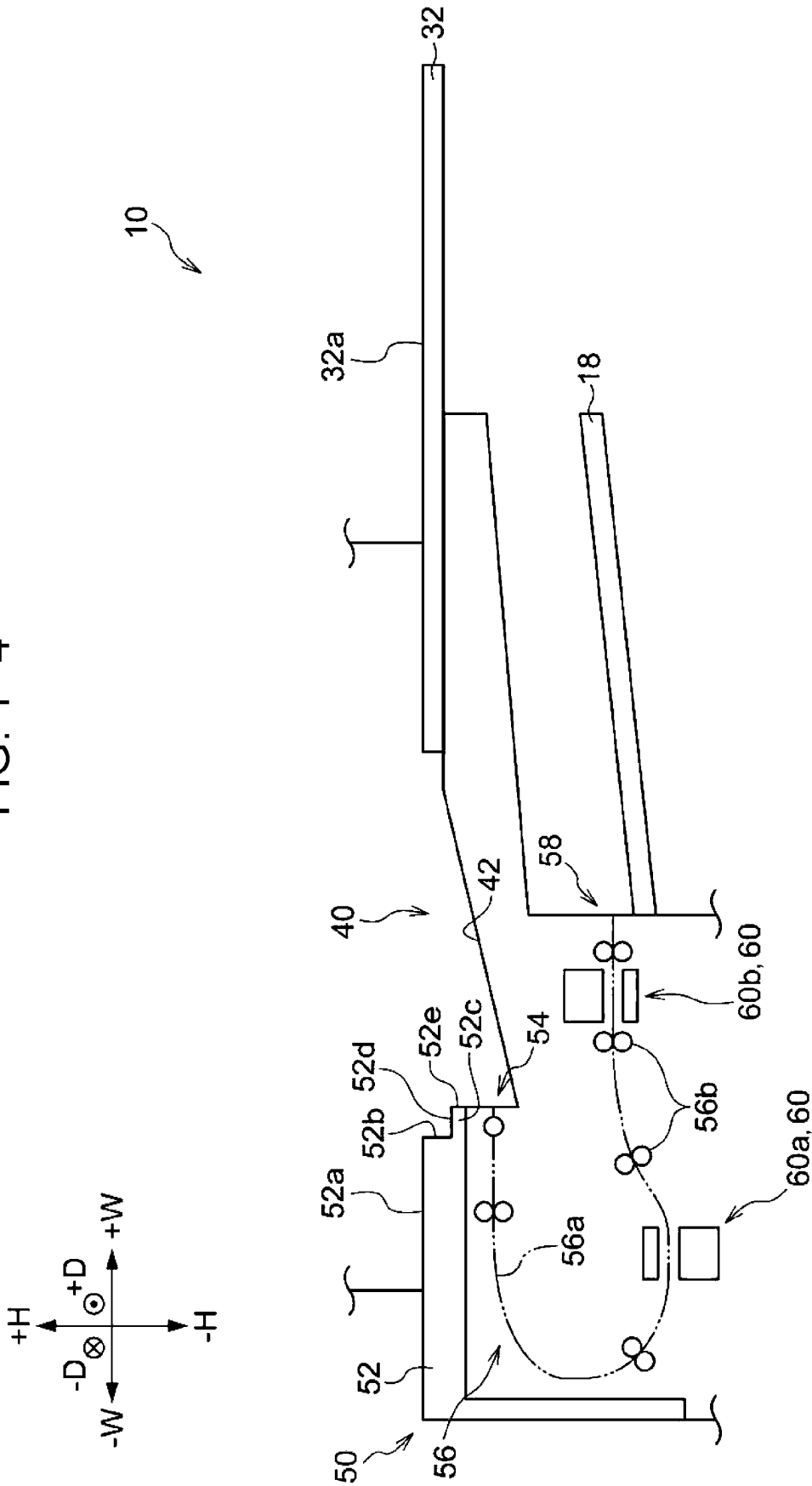


FIG. 1-5

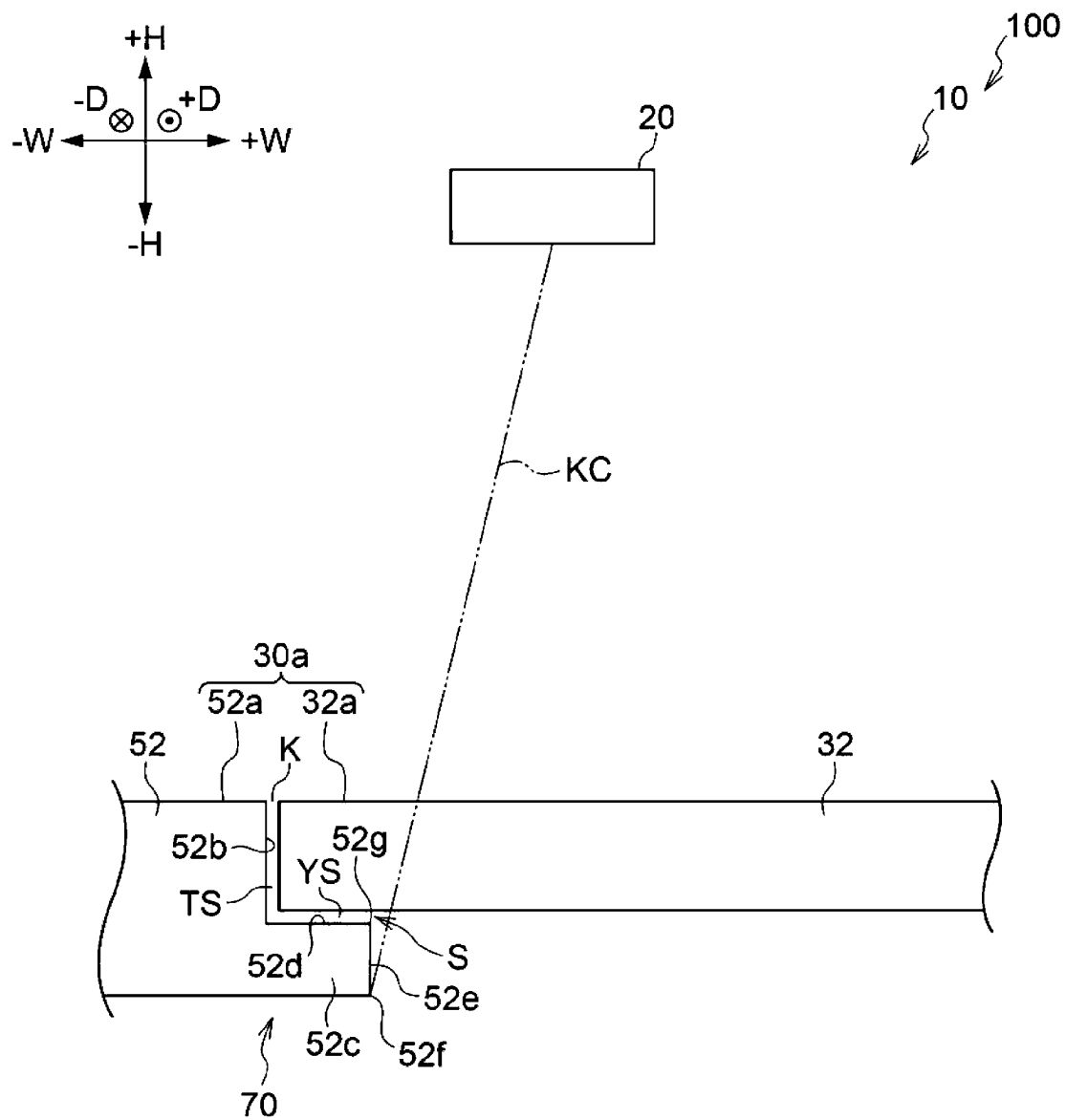


FIG. 1-6

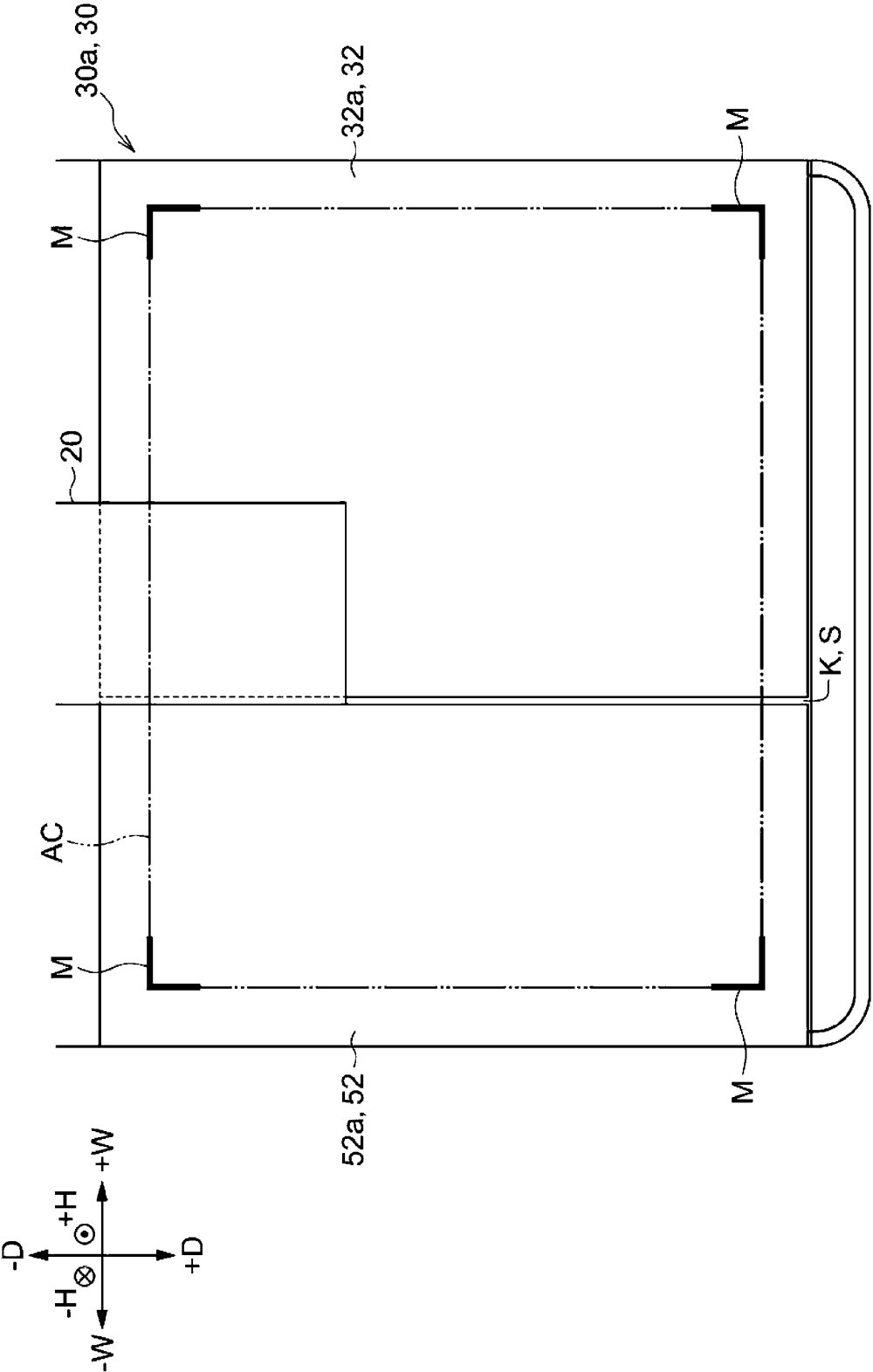


FIG. 1-7

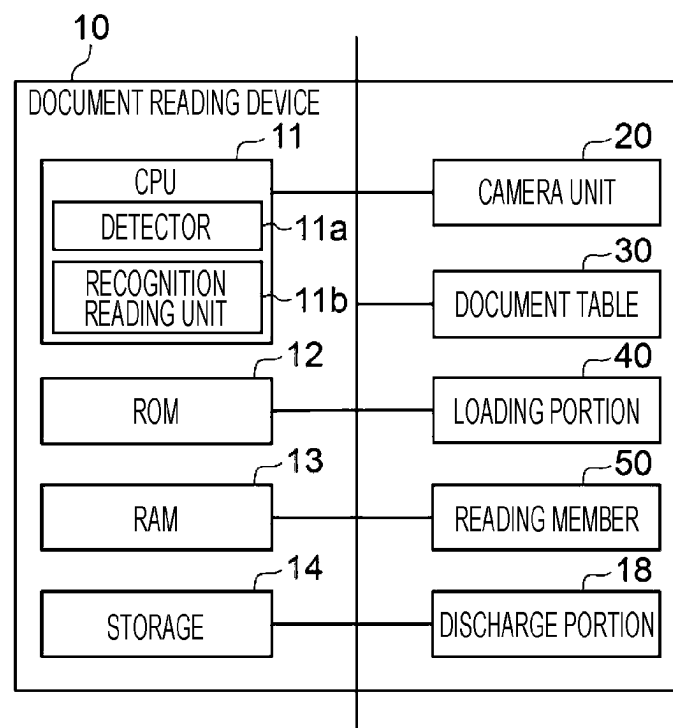


FIG. 1–8

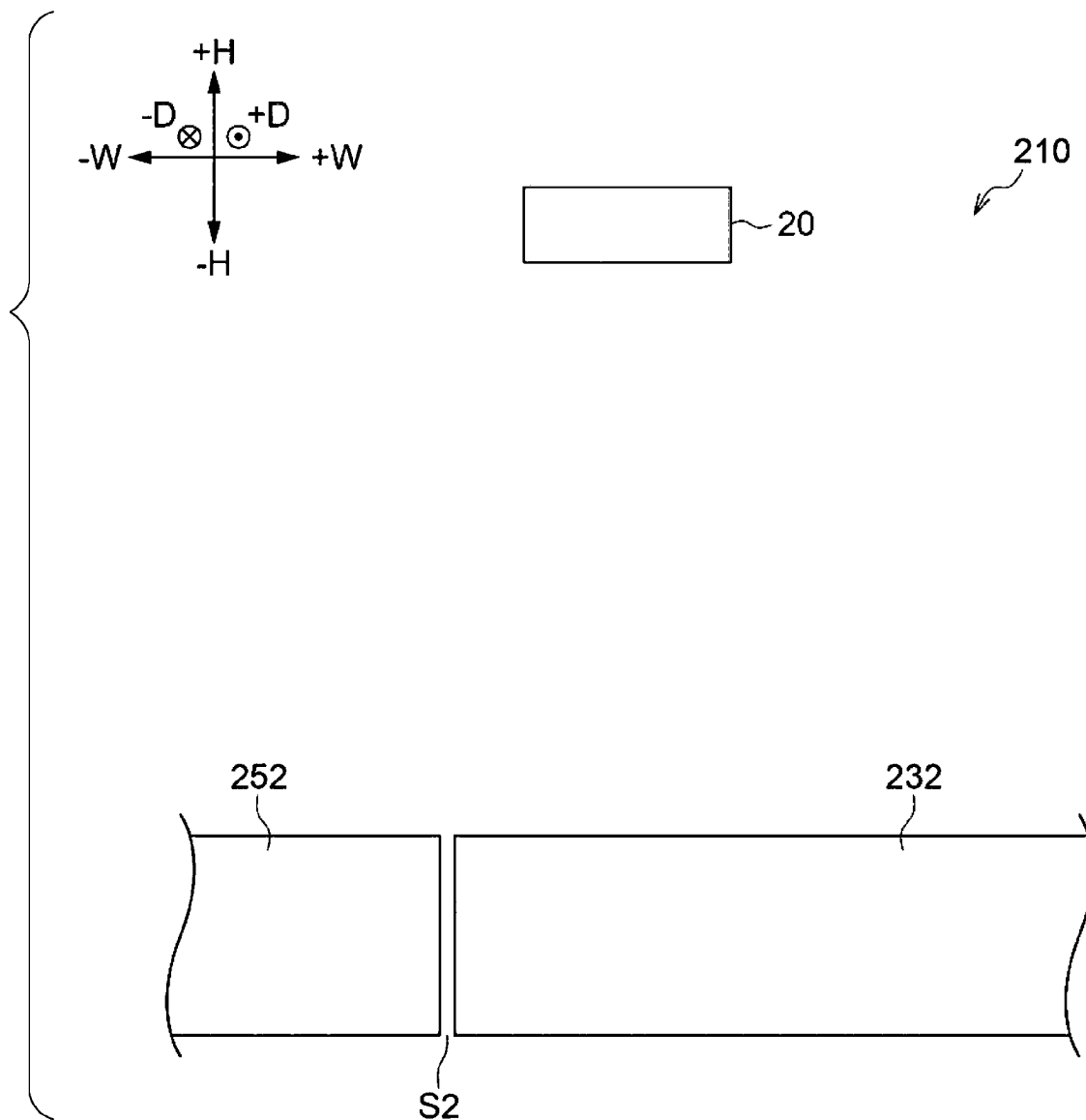


FIG. 1-9

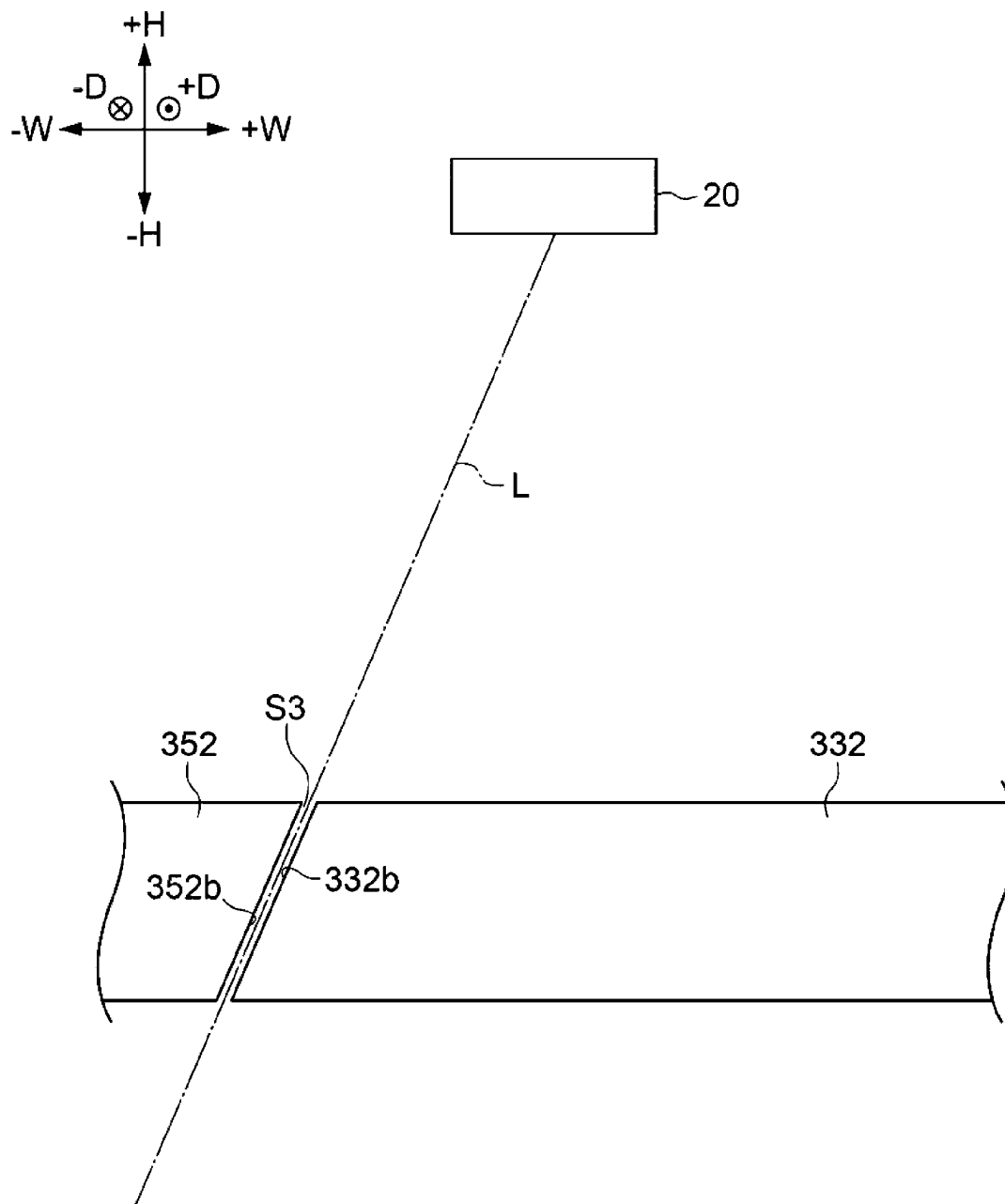


FIG. 1–10

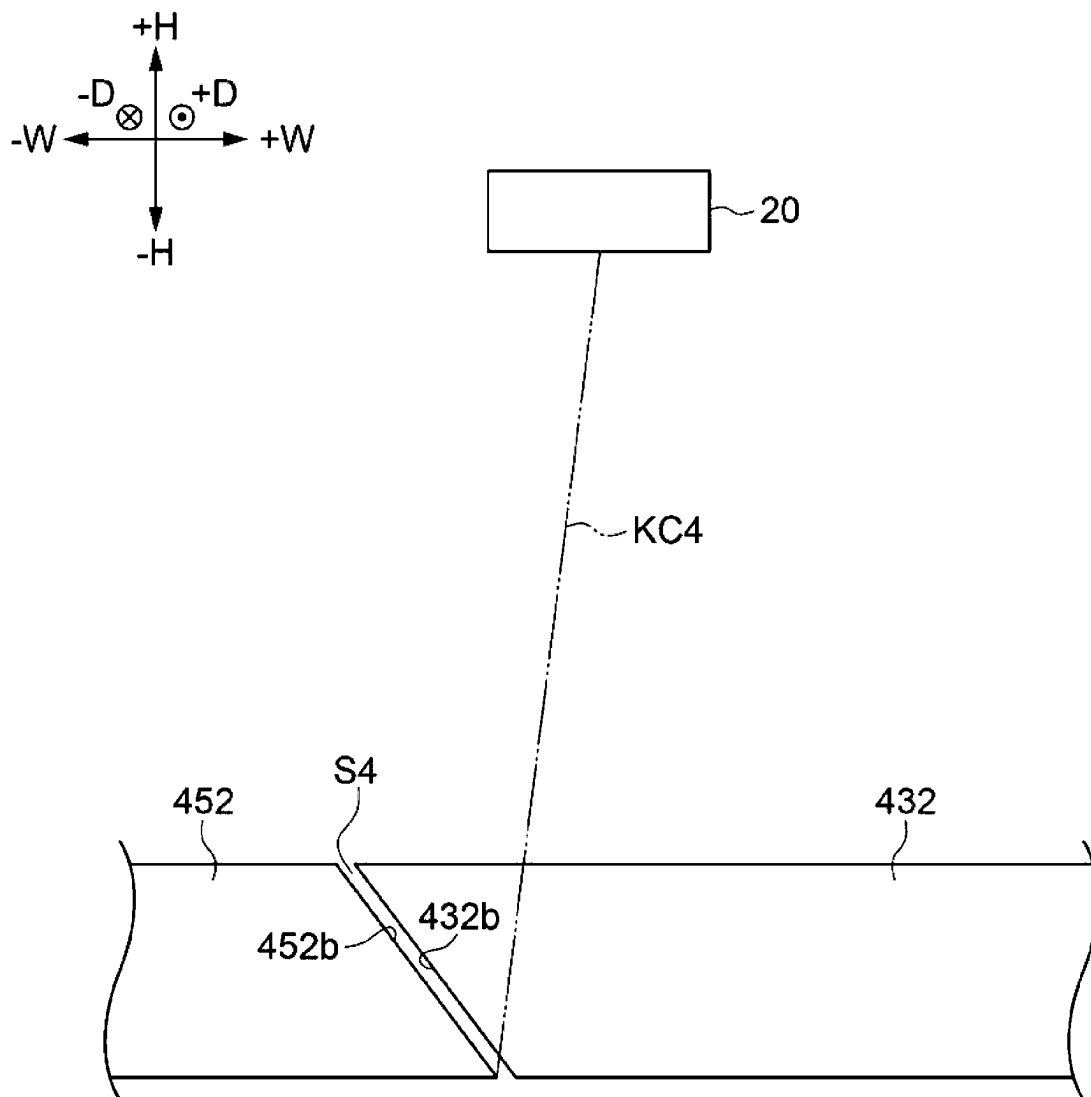


FIG. 1–11

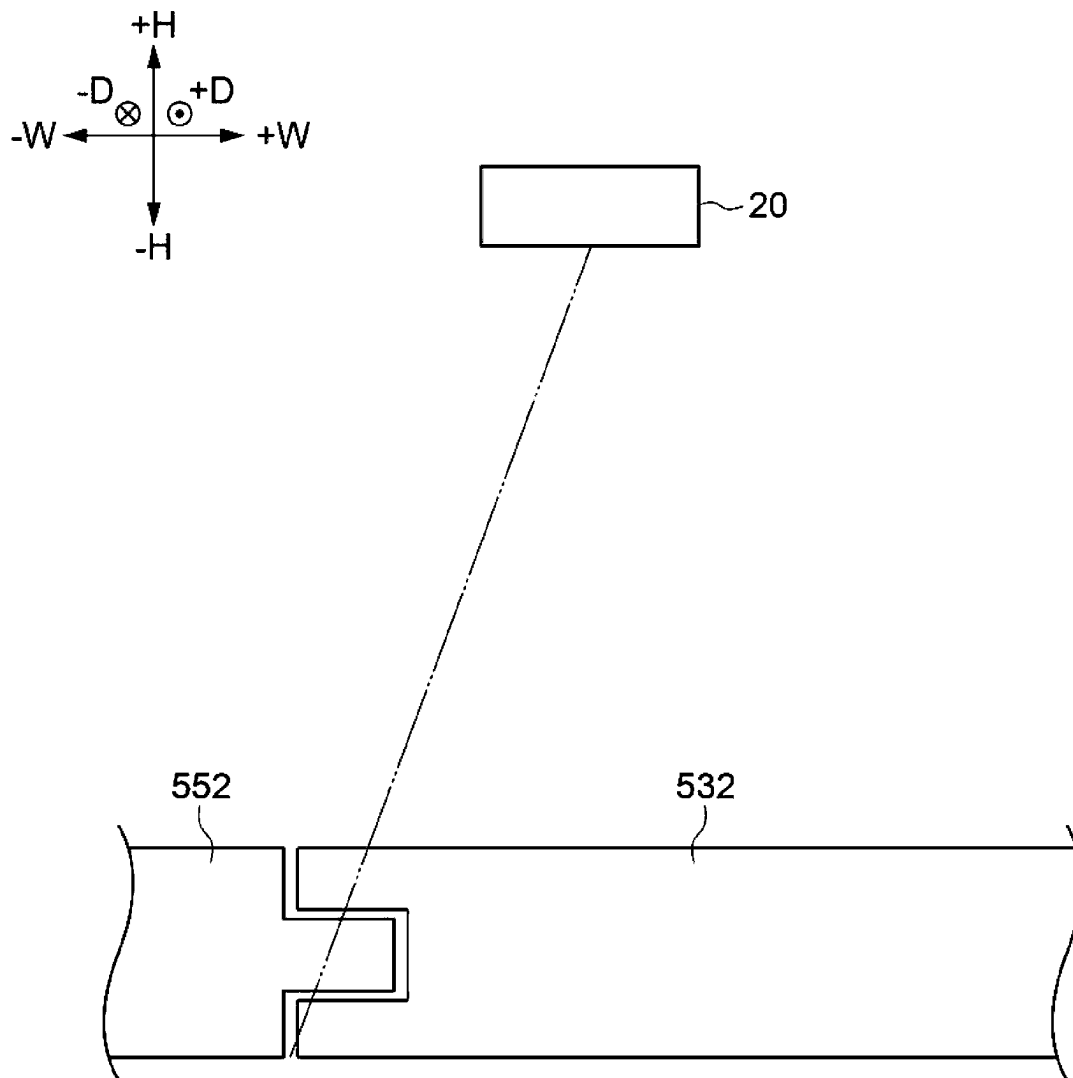


FIG. 1–12

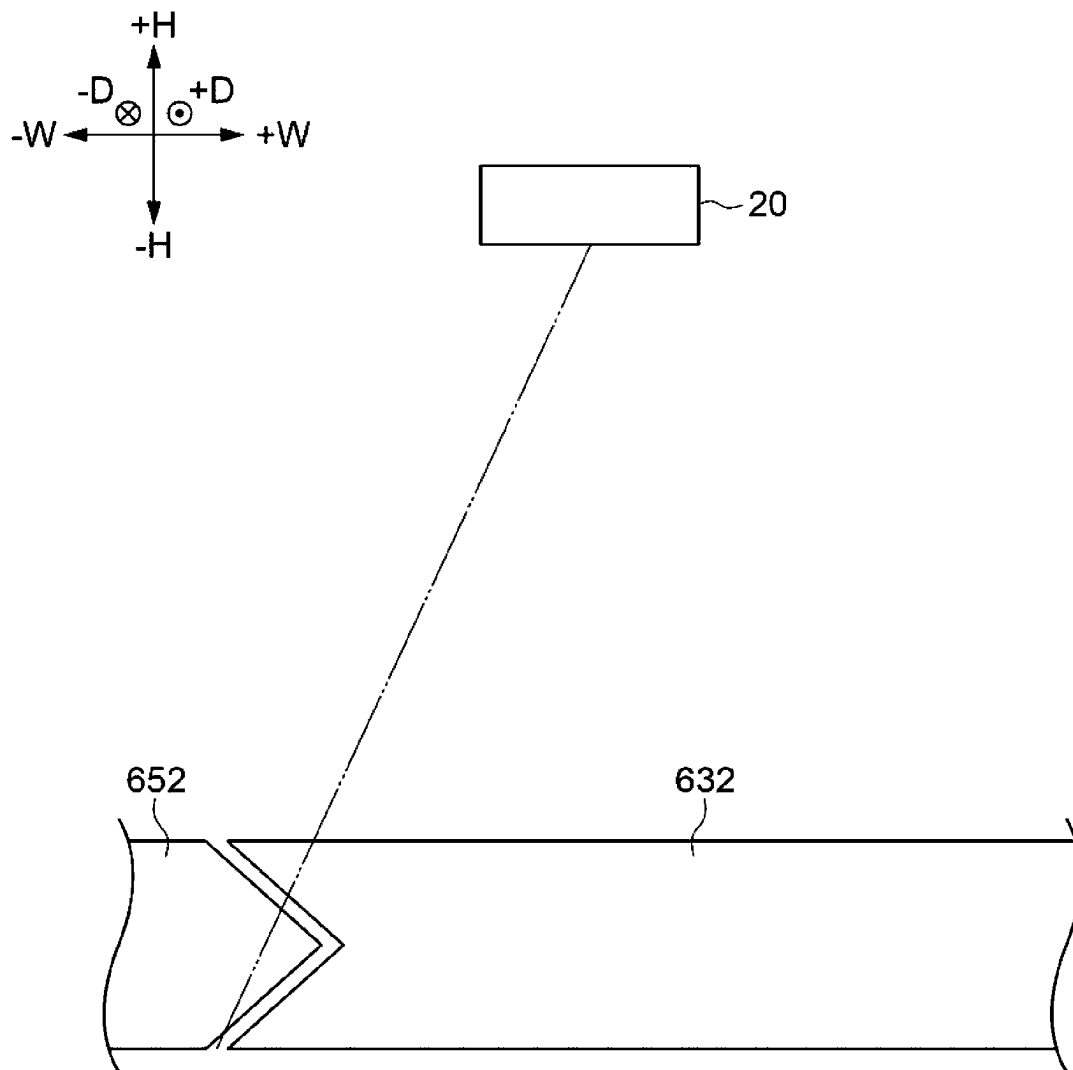


FIG. 2-1

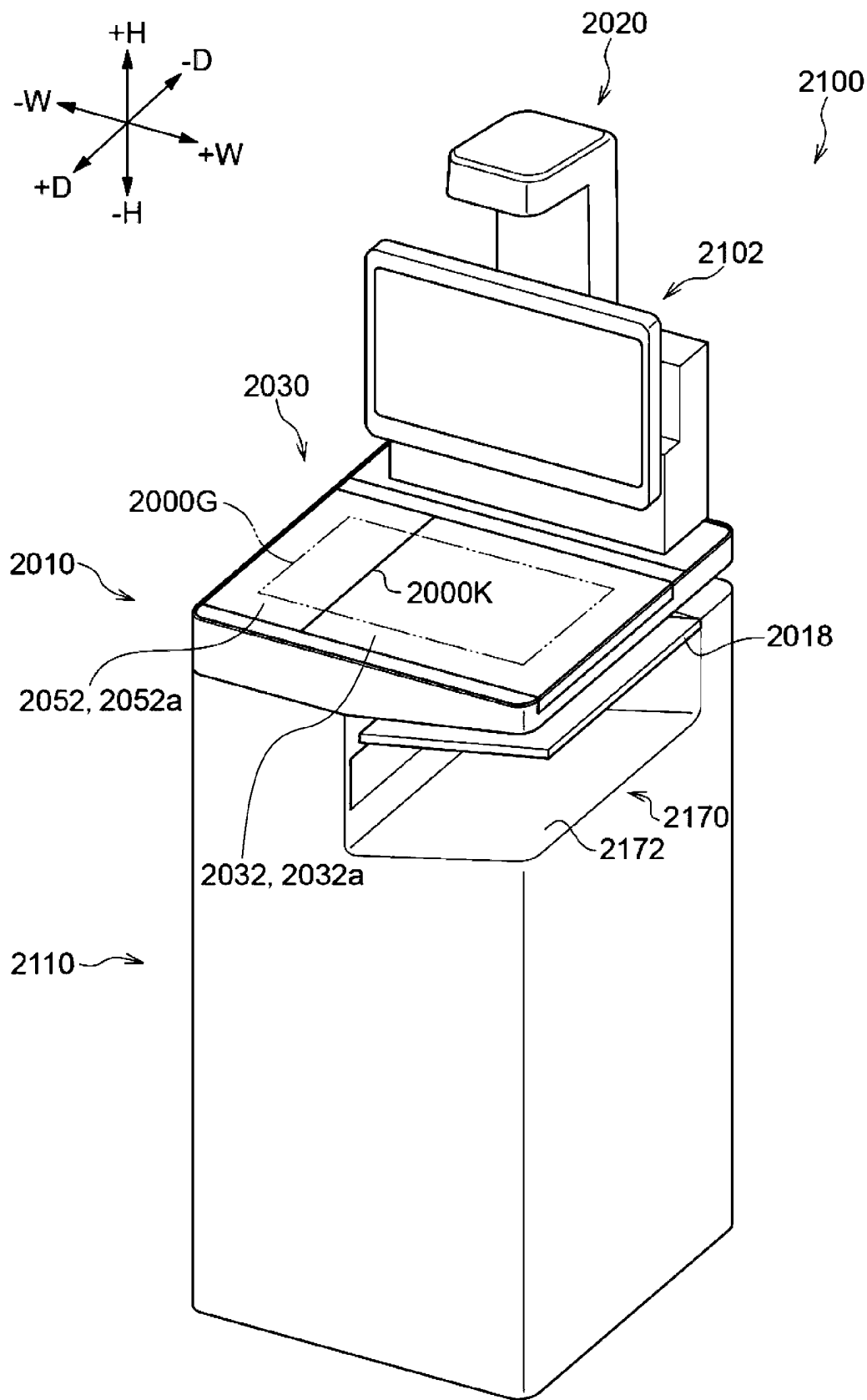


FIG. 2-2

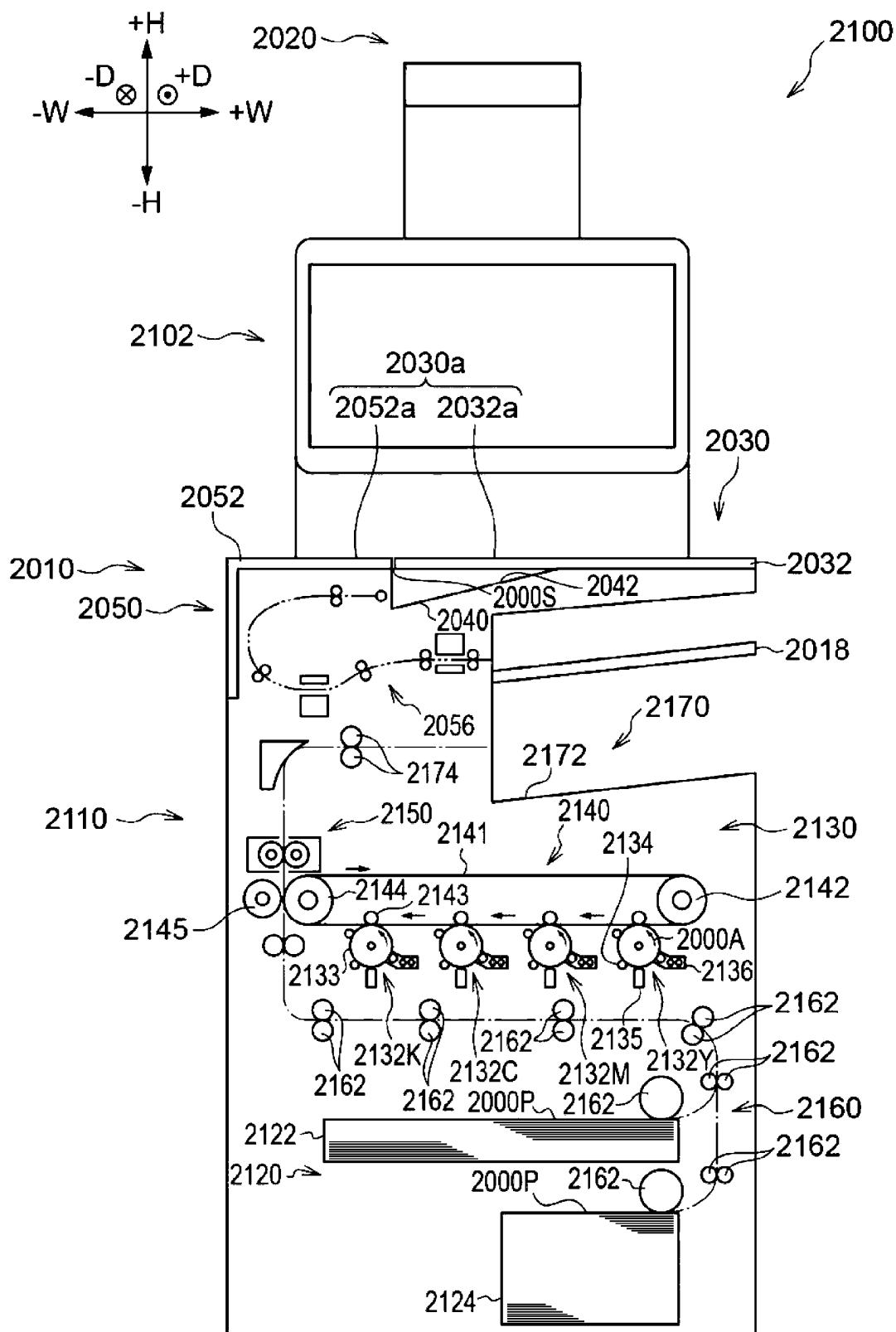


FIG. 2-3

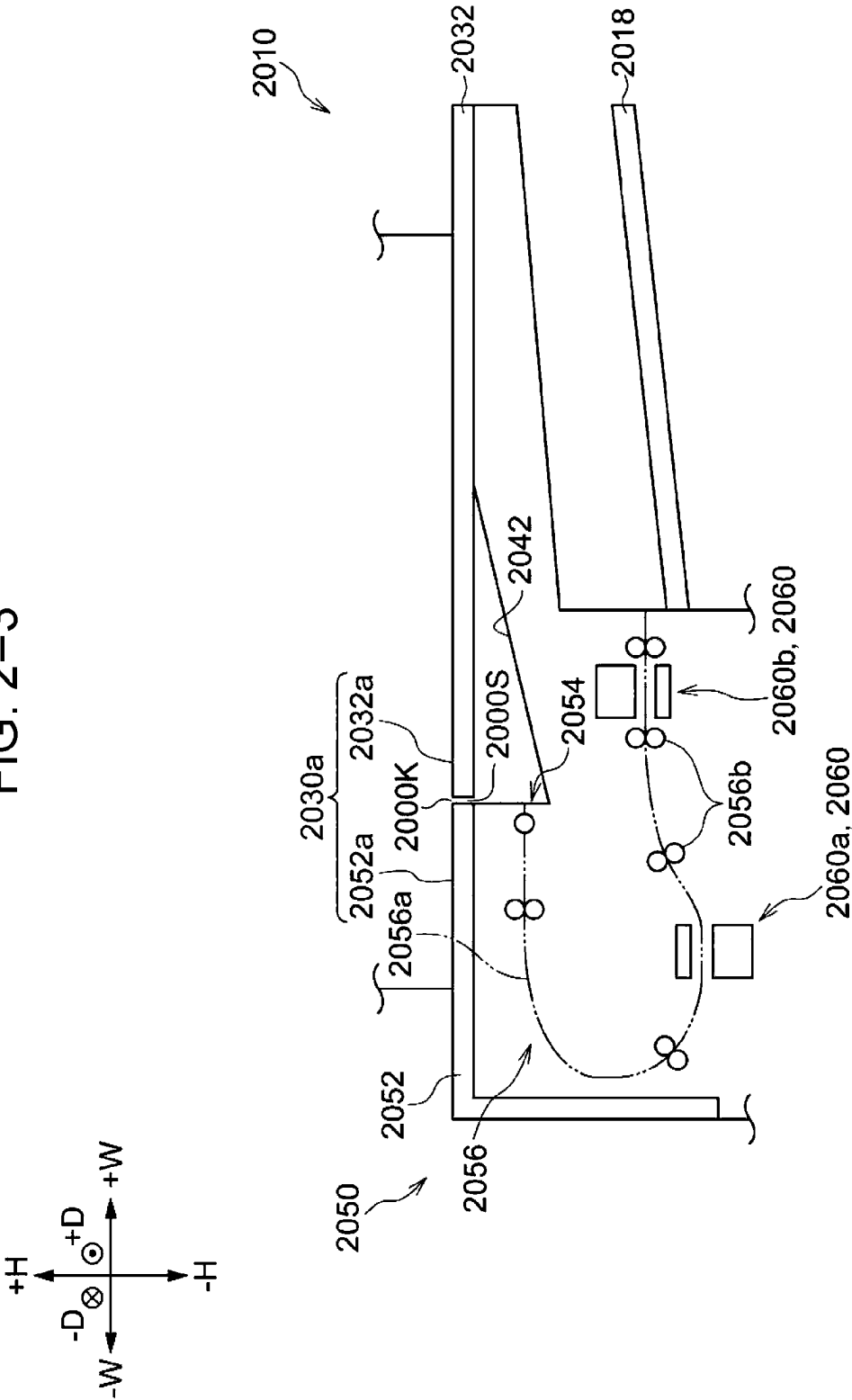


FIG. 2-4

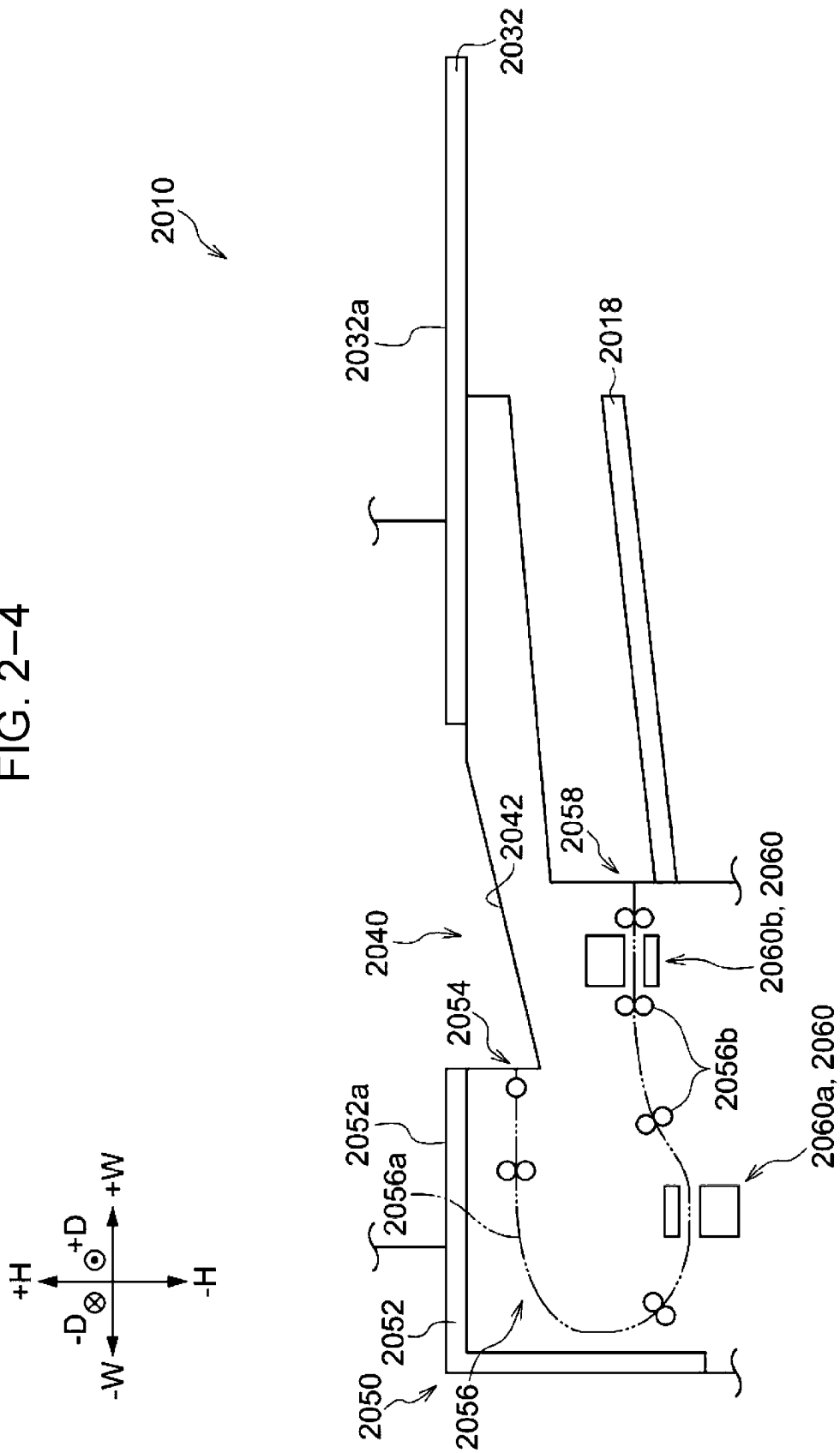


FIG. 2-5

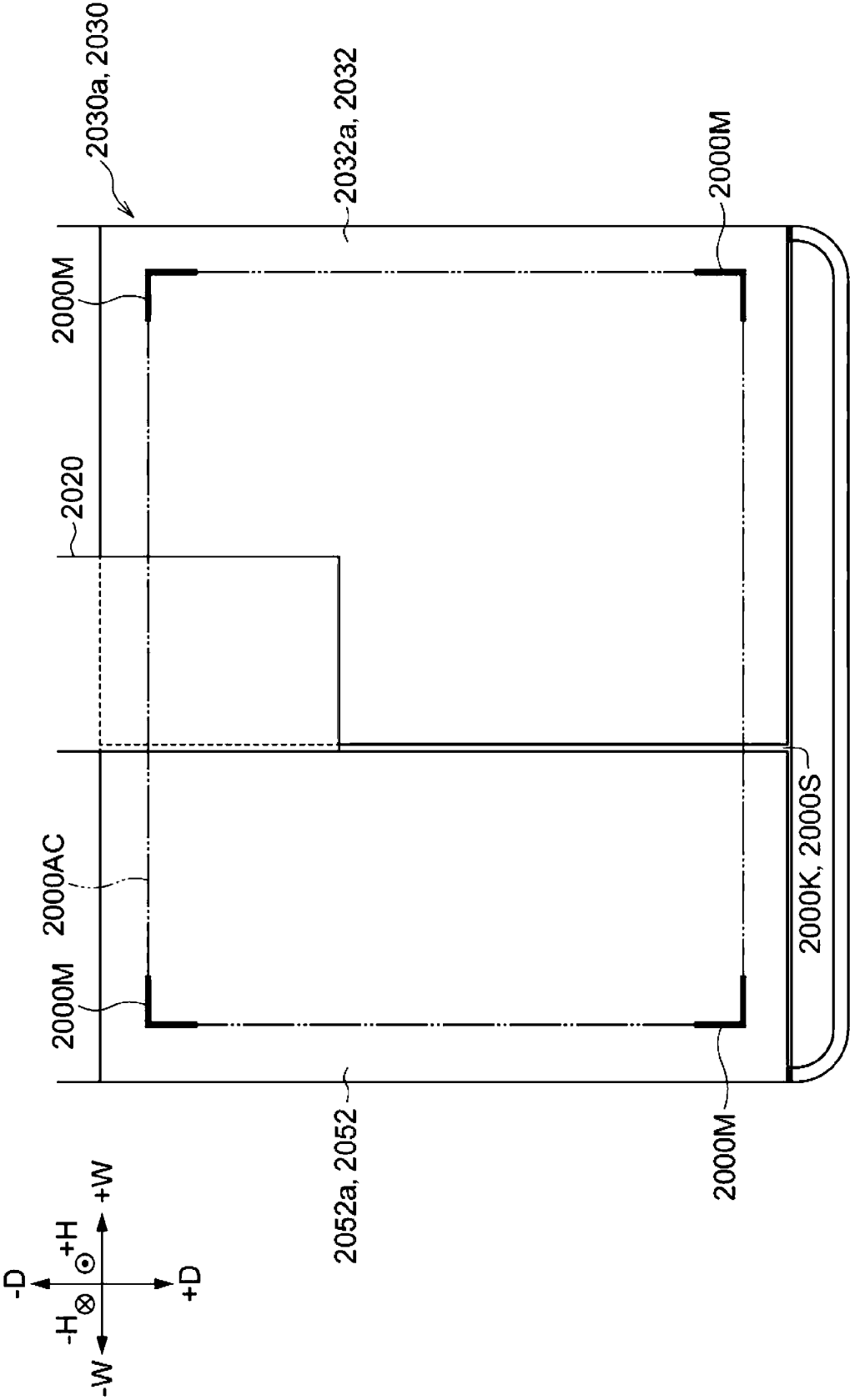


FIG. 2-6

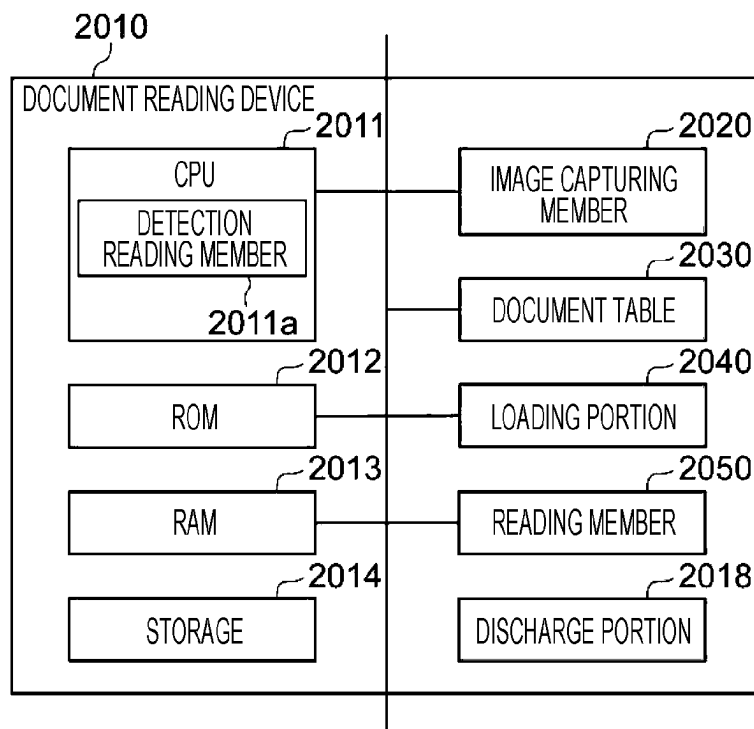


FIG. 3-1

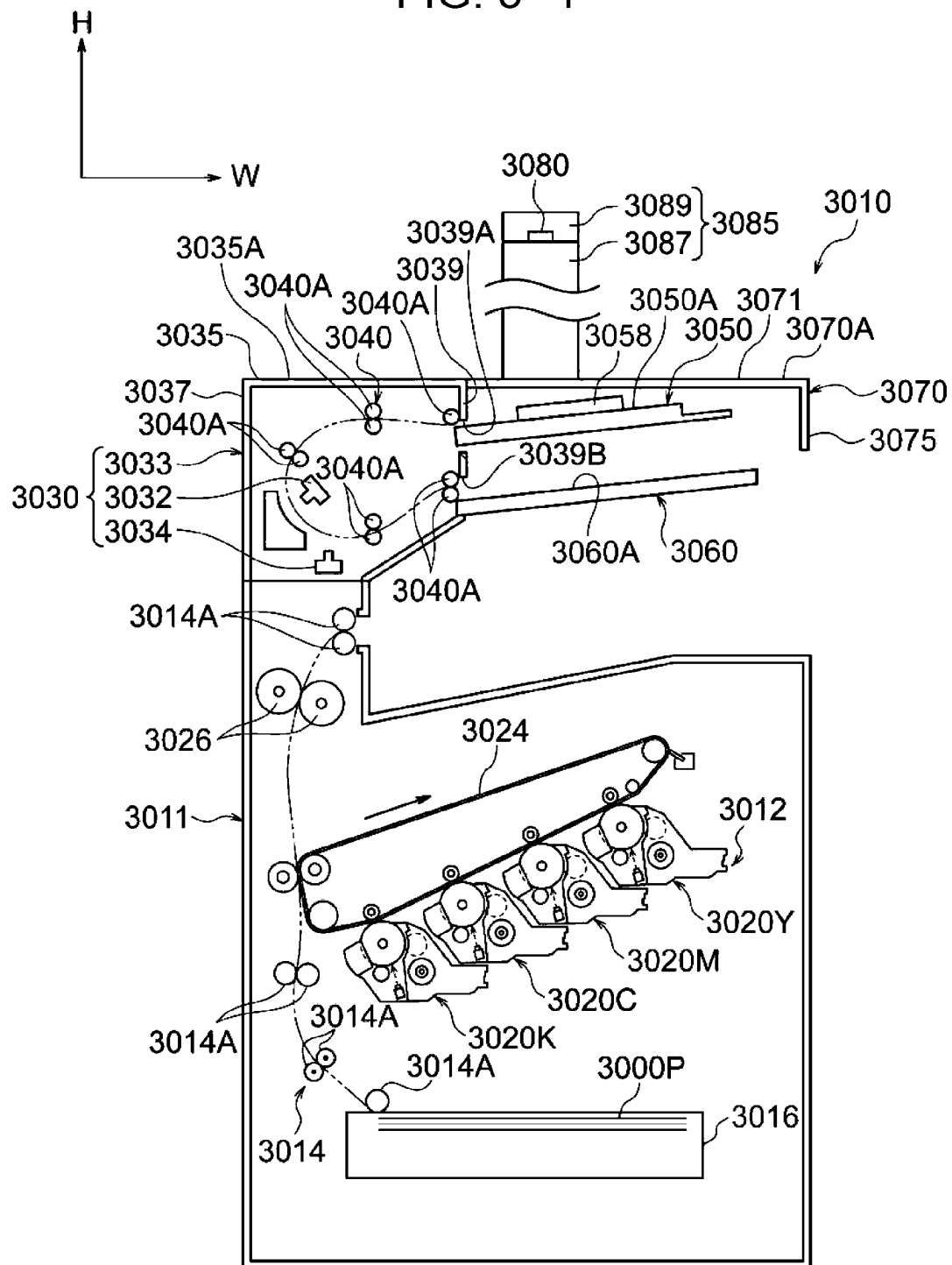


FIG. 3-2

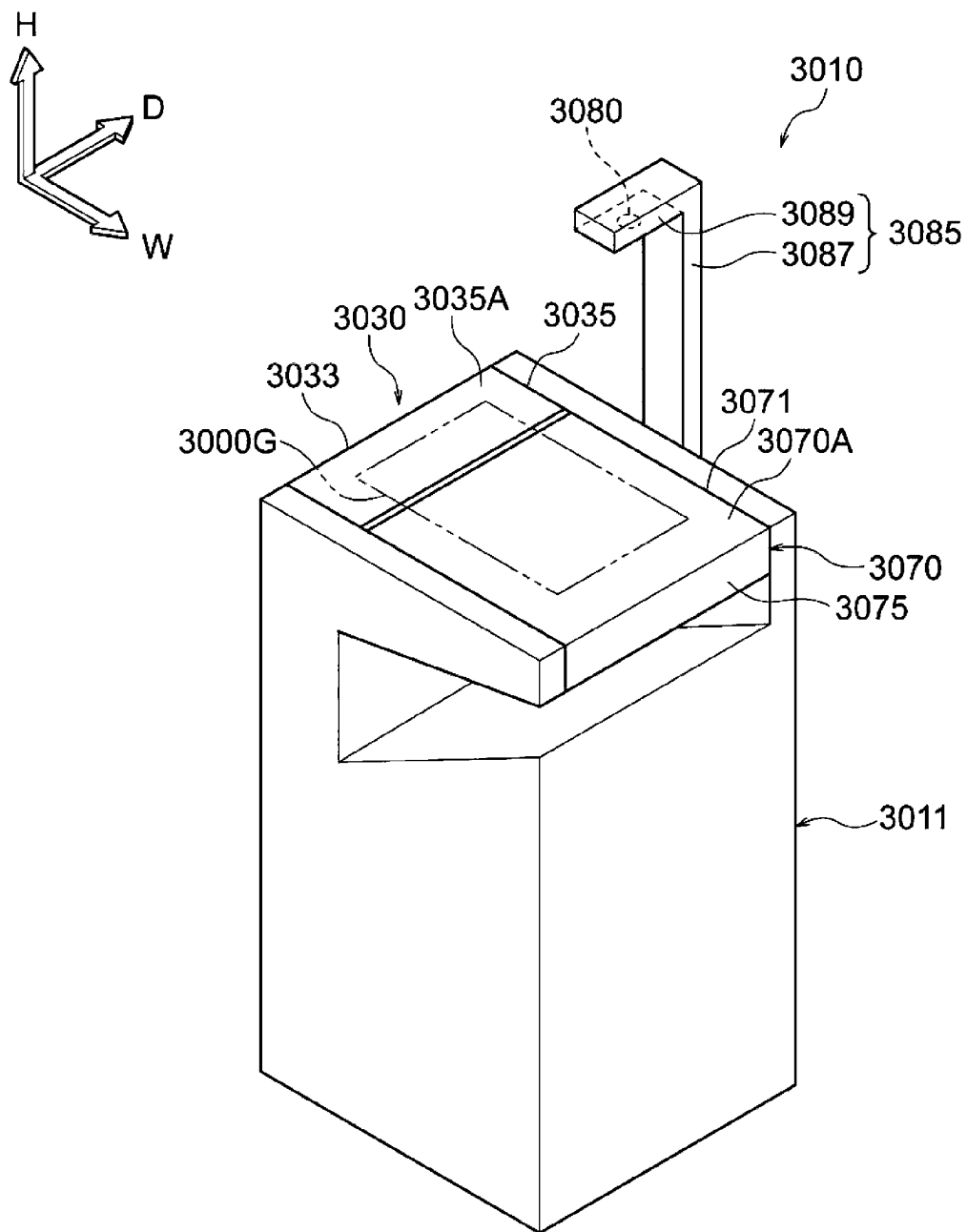


FIG. 3-3

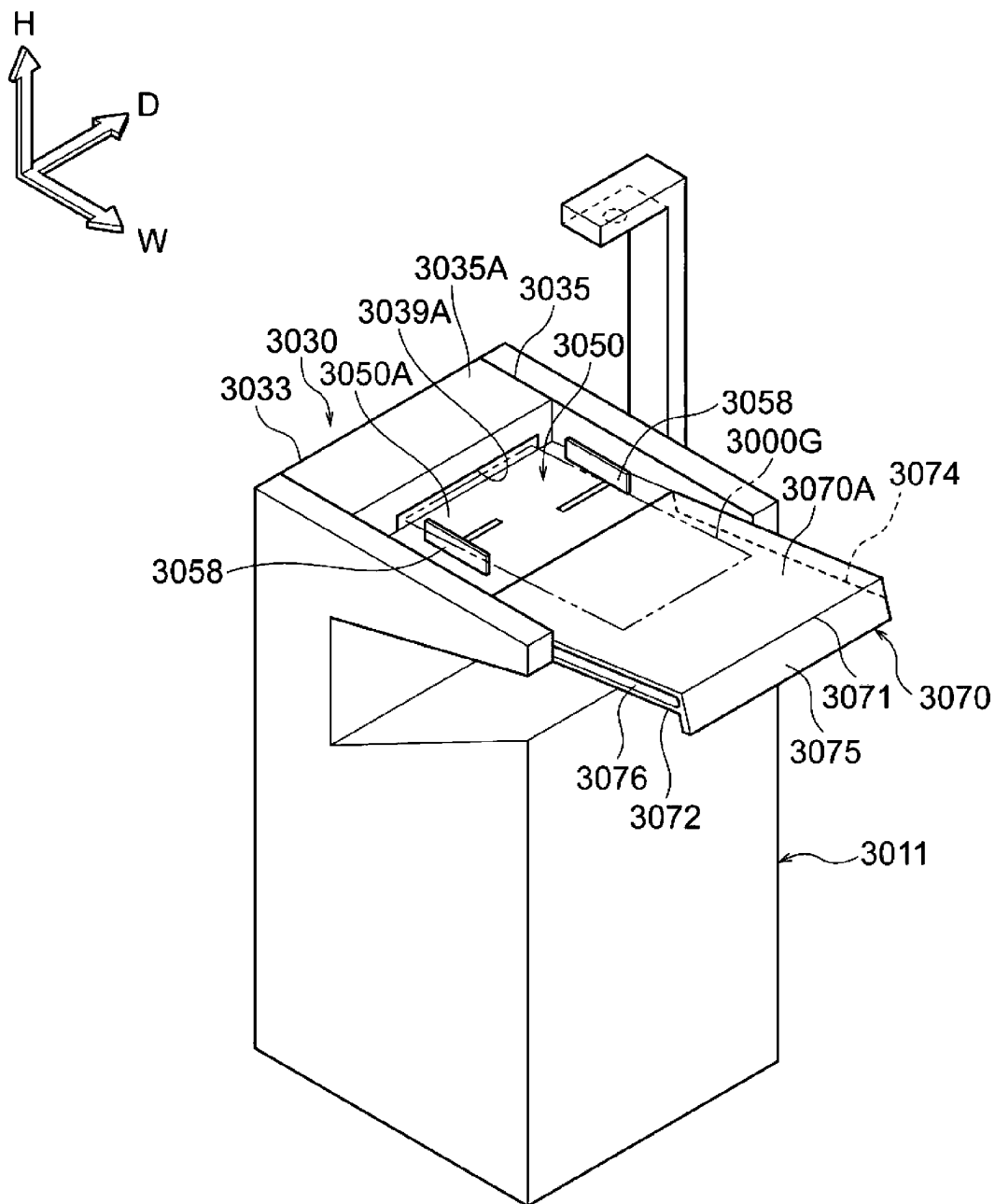


FIG. 3-4

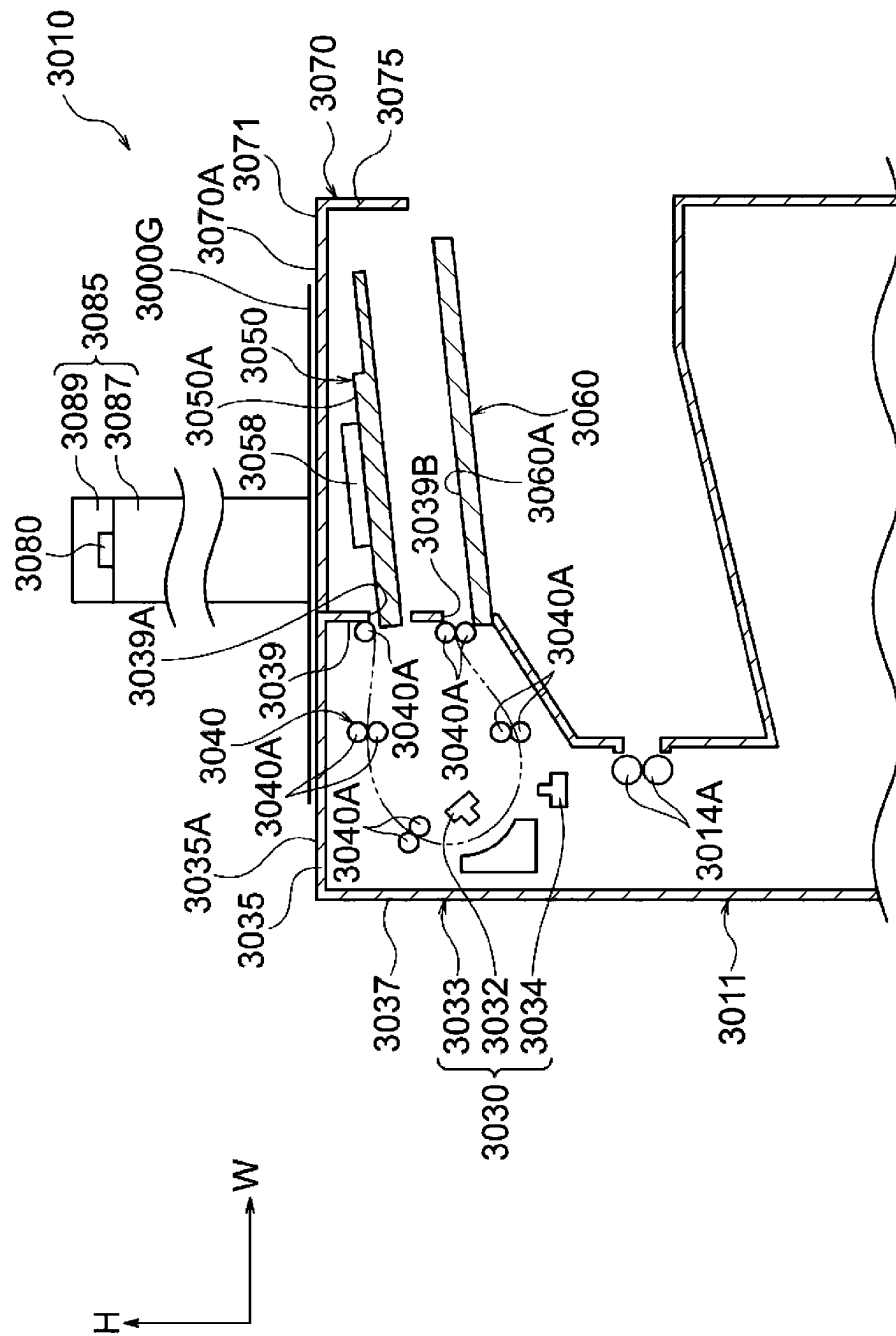


FIG. 3-5

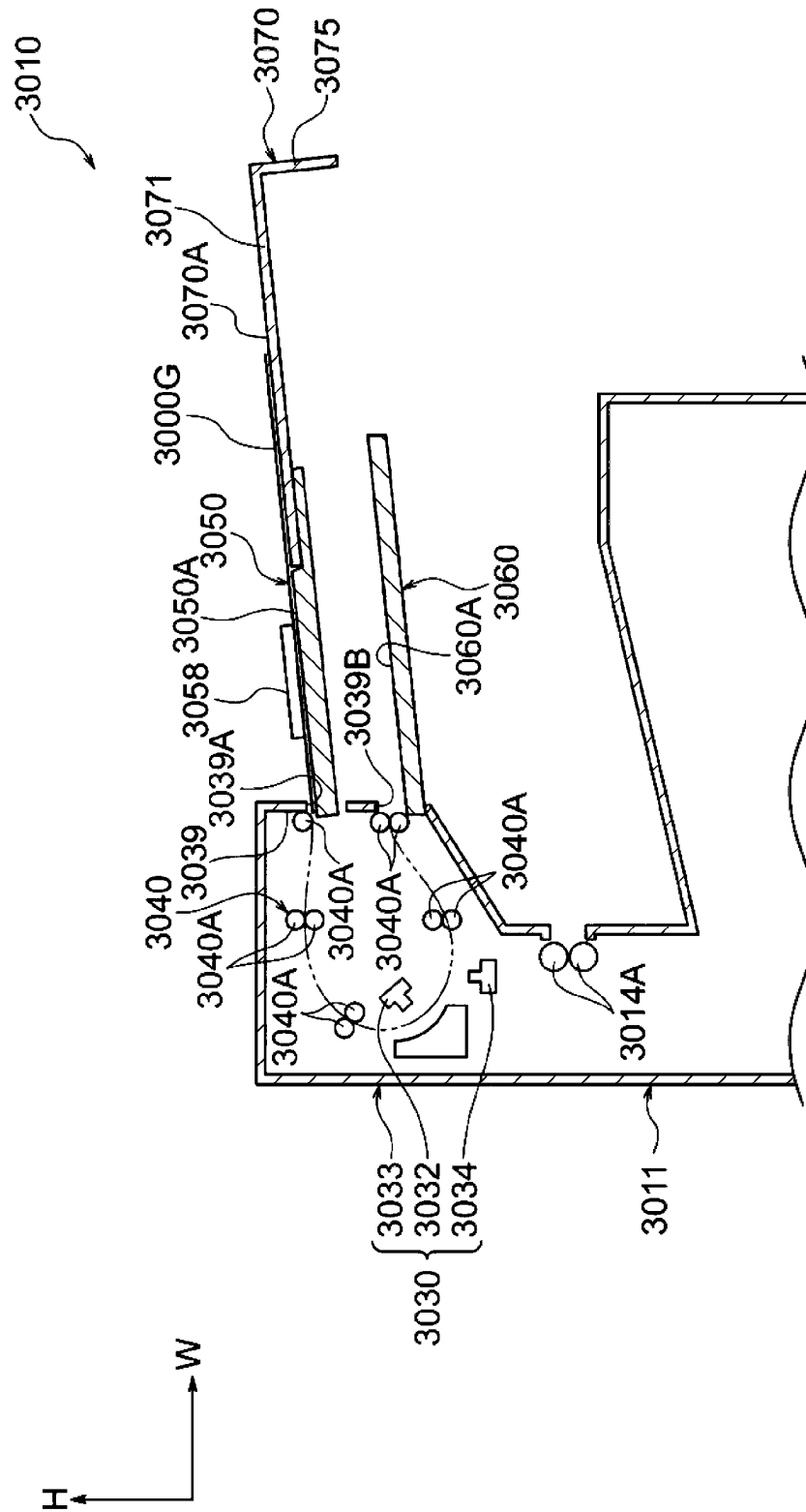


FIG. 3-6

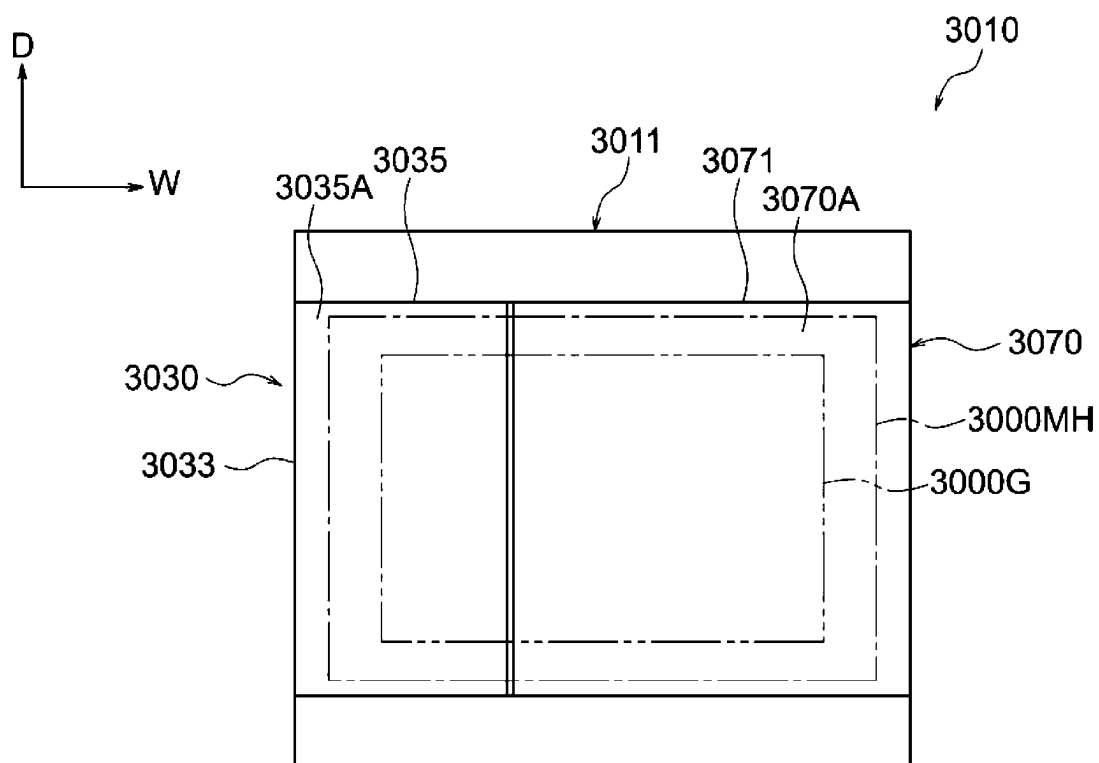


FIG 3-7

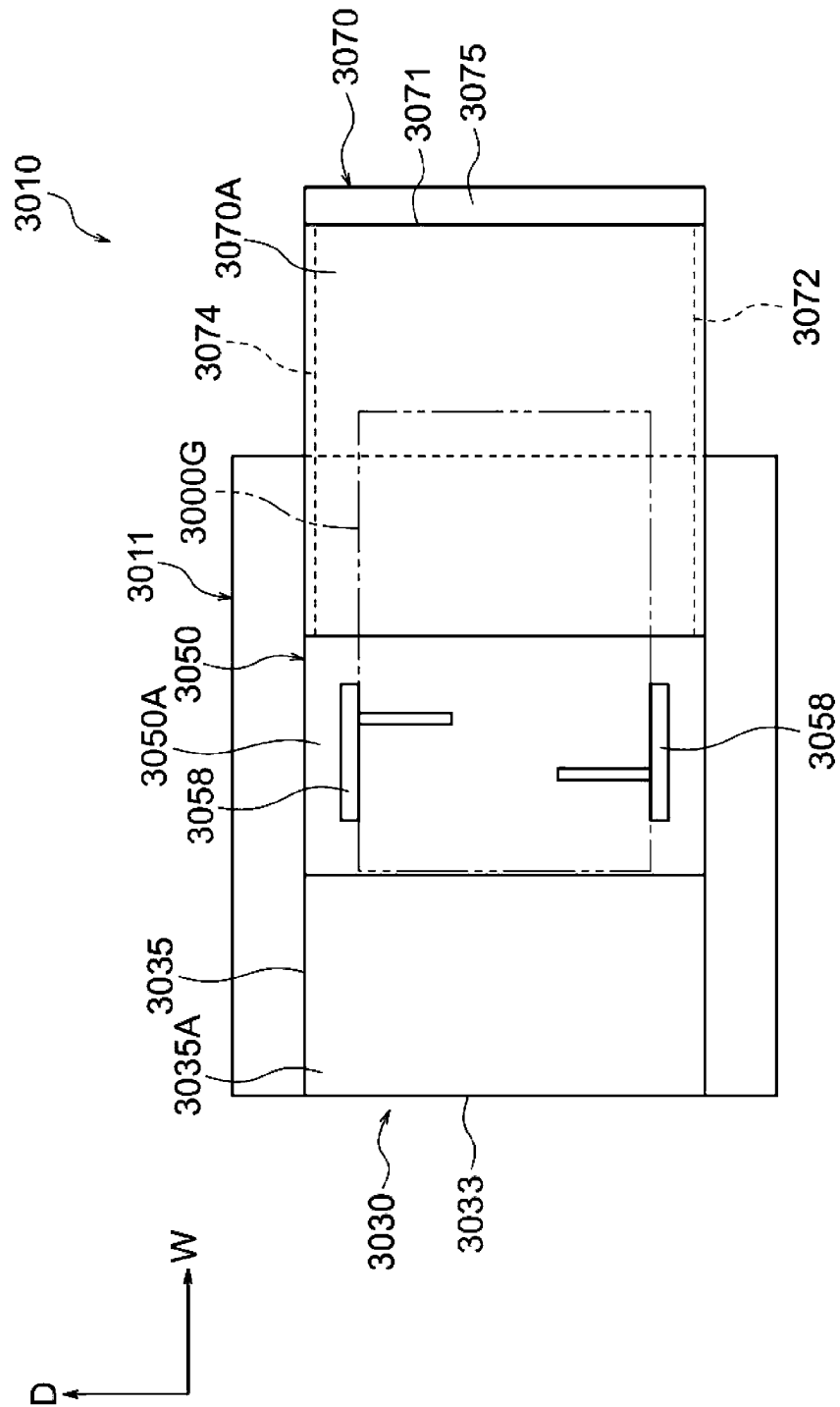


FIG. 3-8

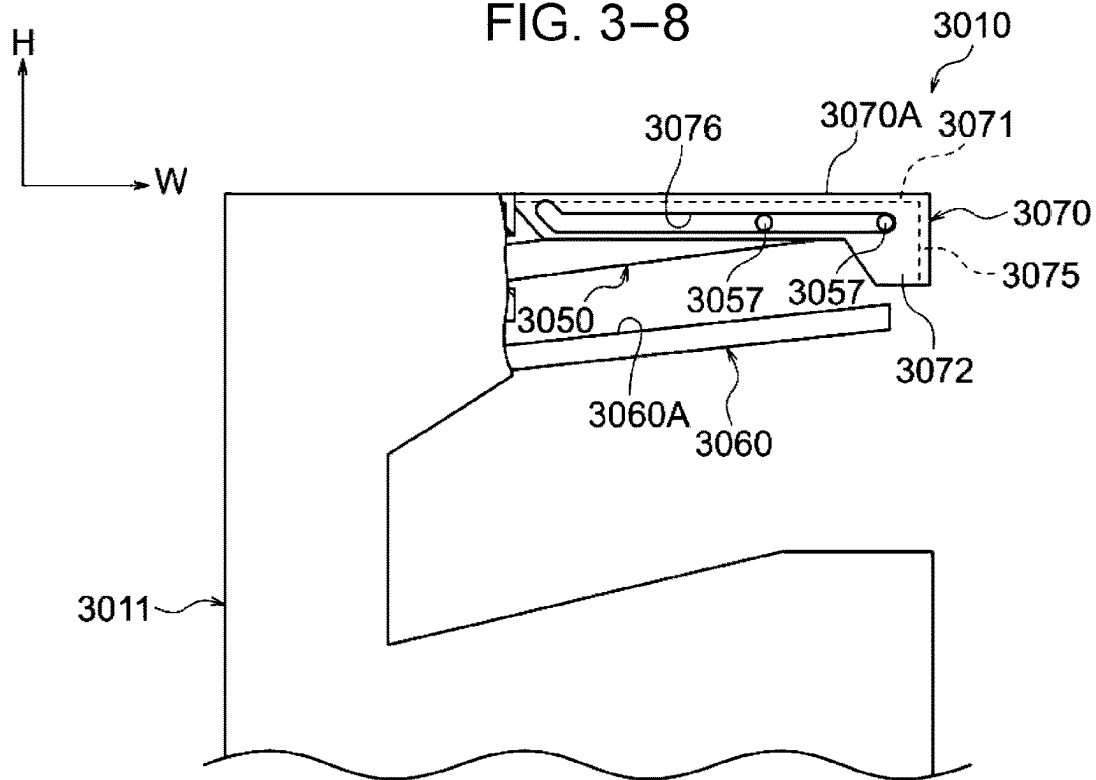


FIG. 3-9

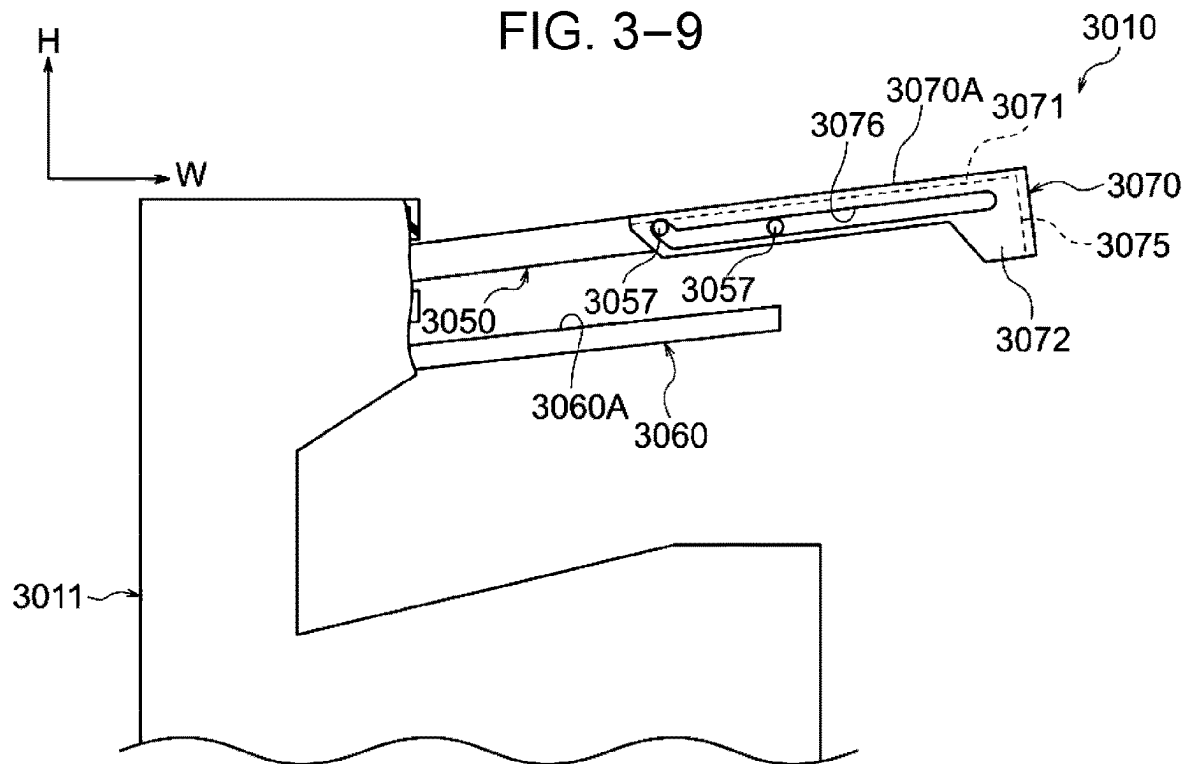


FIG. 4-1

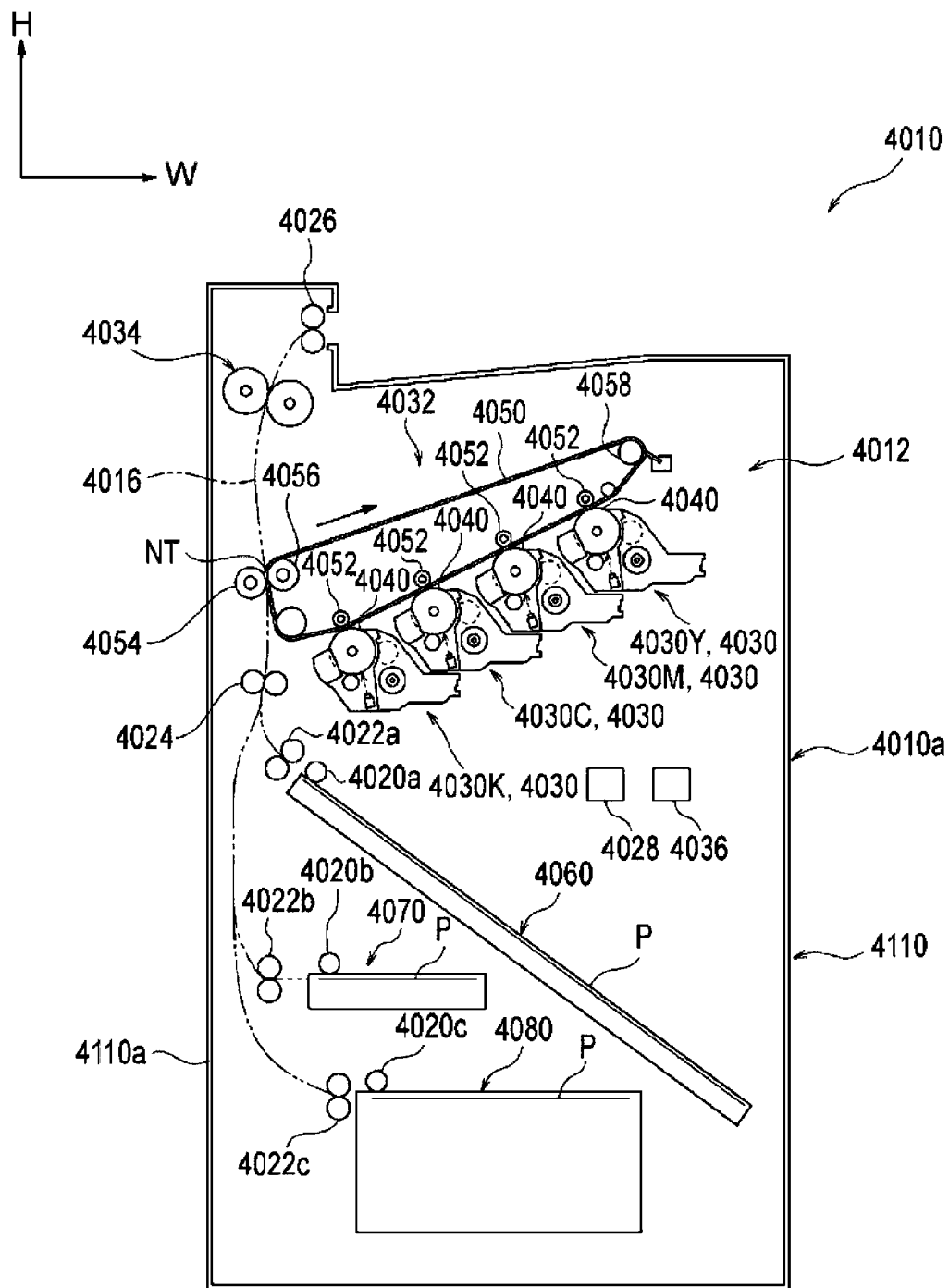


FIG. 4-2

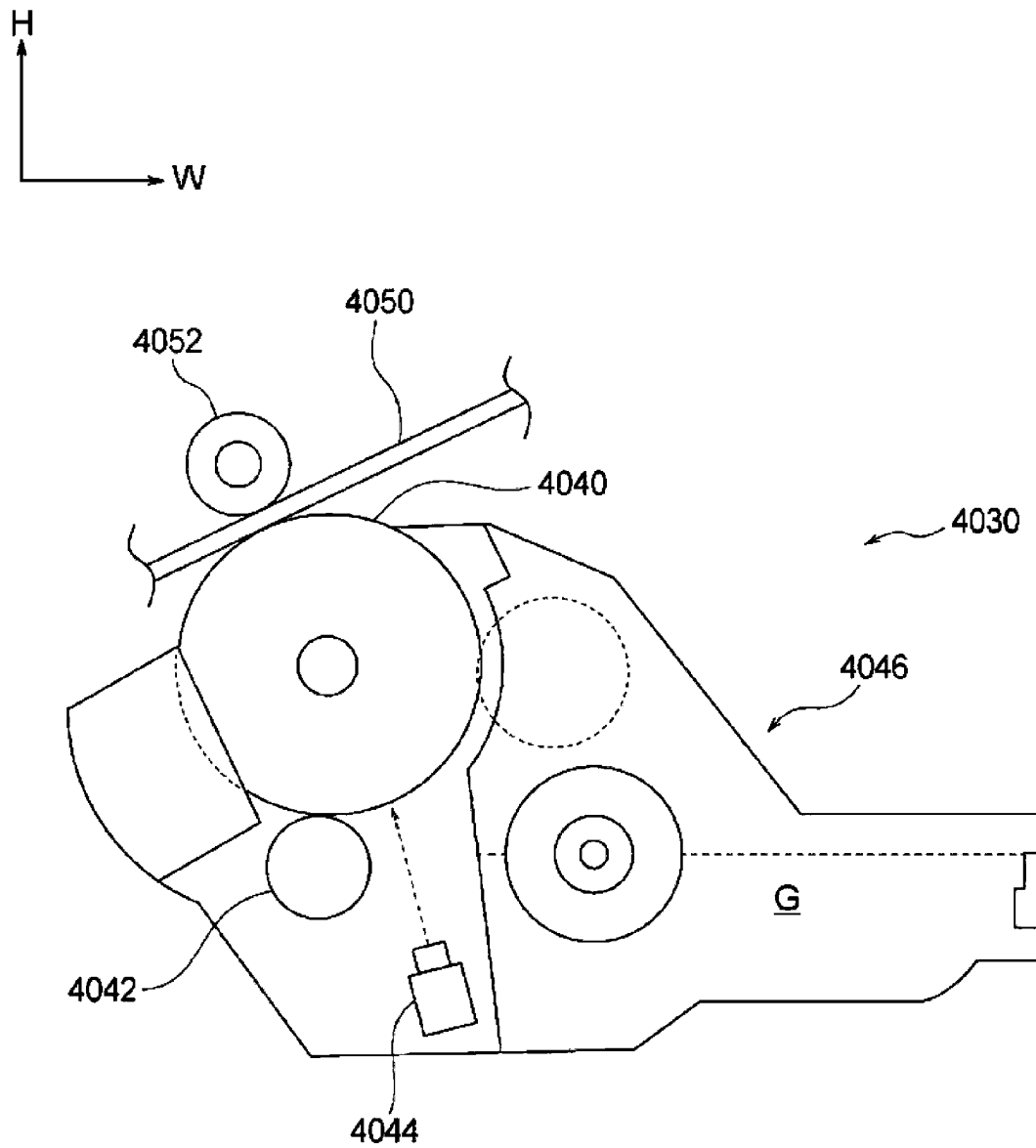


FIG. 4-3

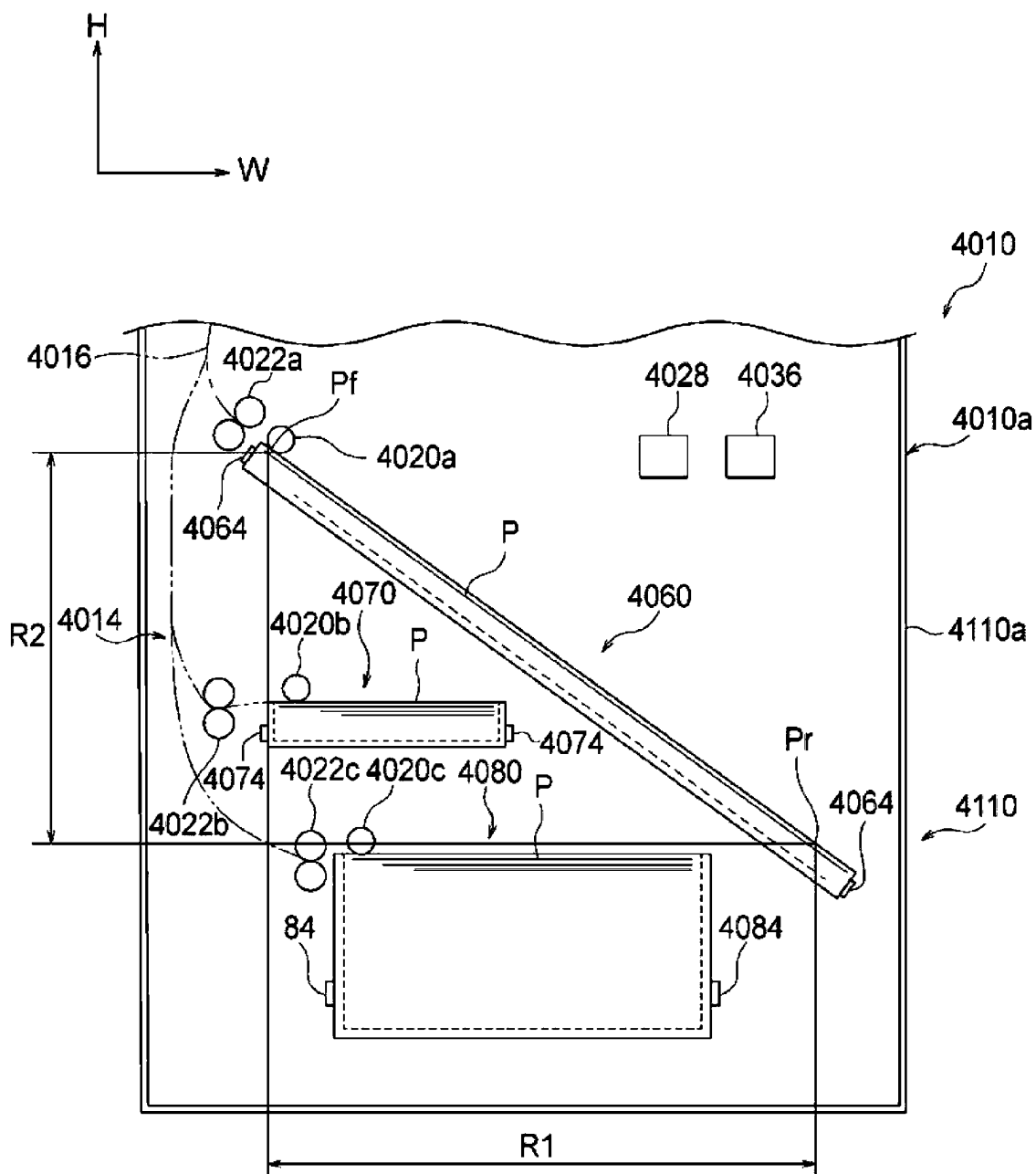


FIG. 4-4A

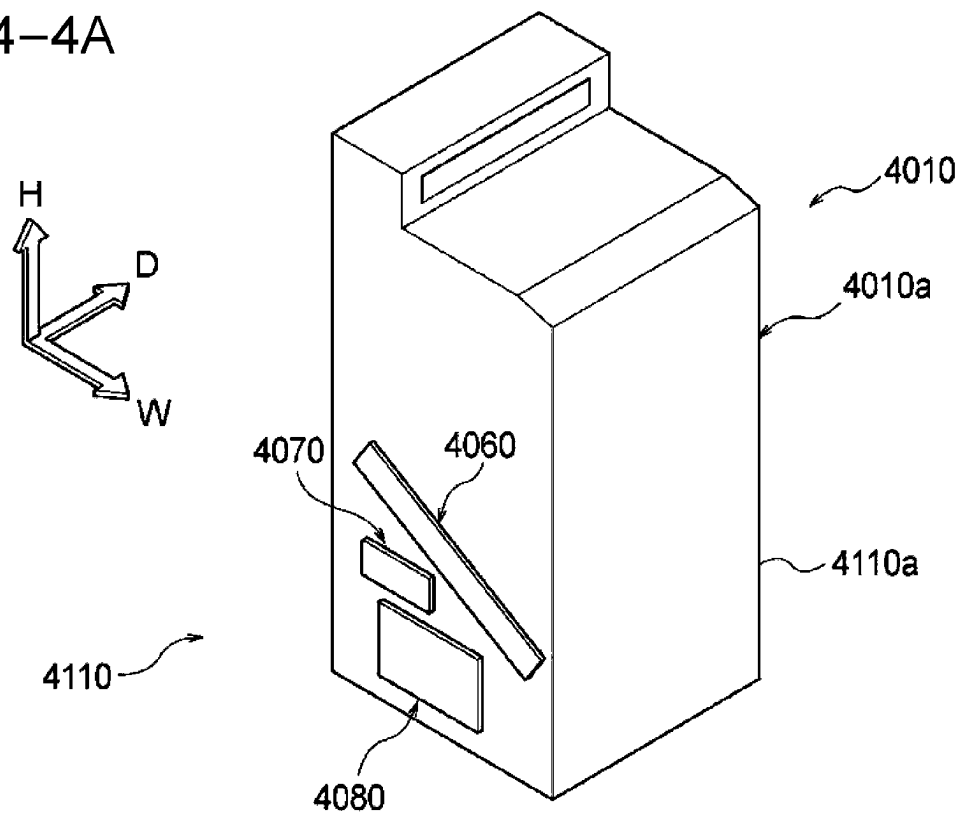


FIG. 4-4B

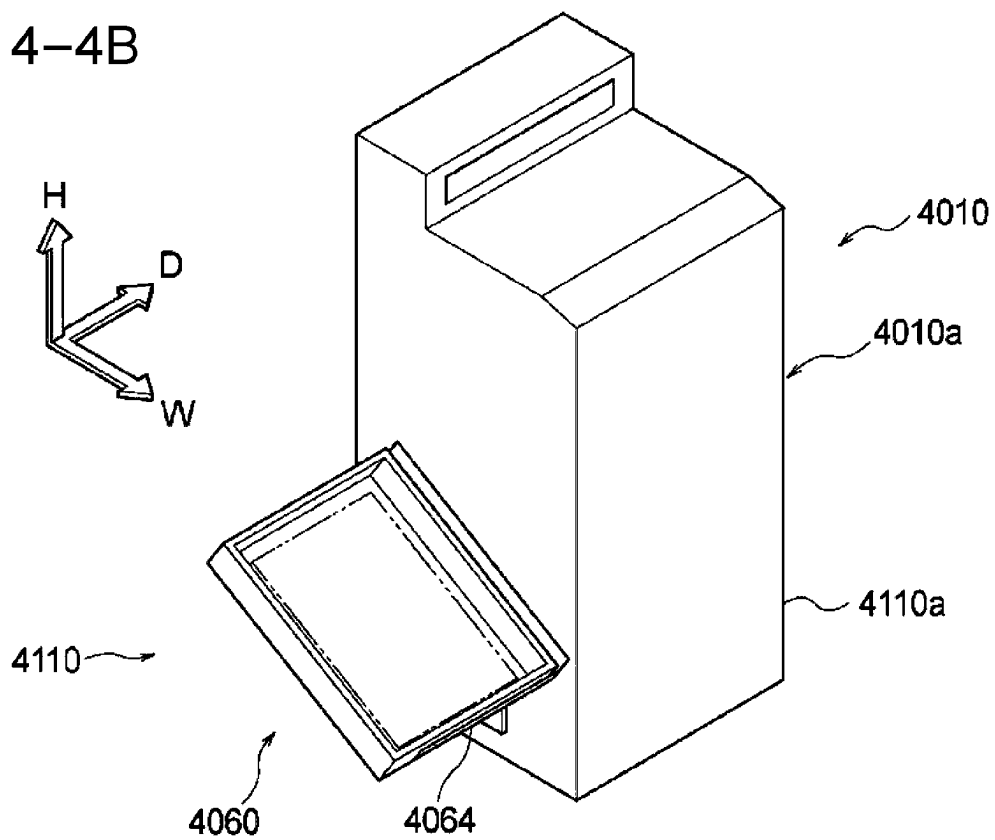


FIG. 4-5A

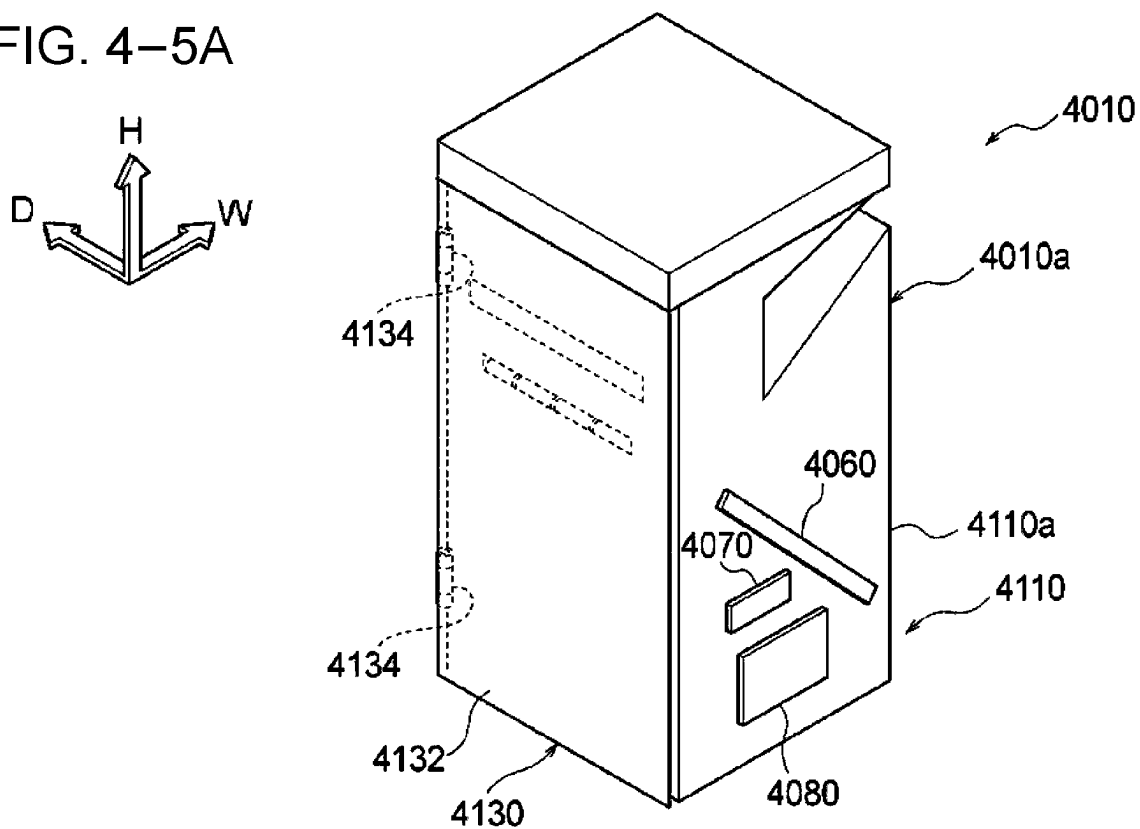


FIG. 4-5B

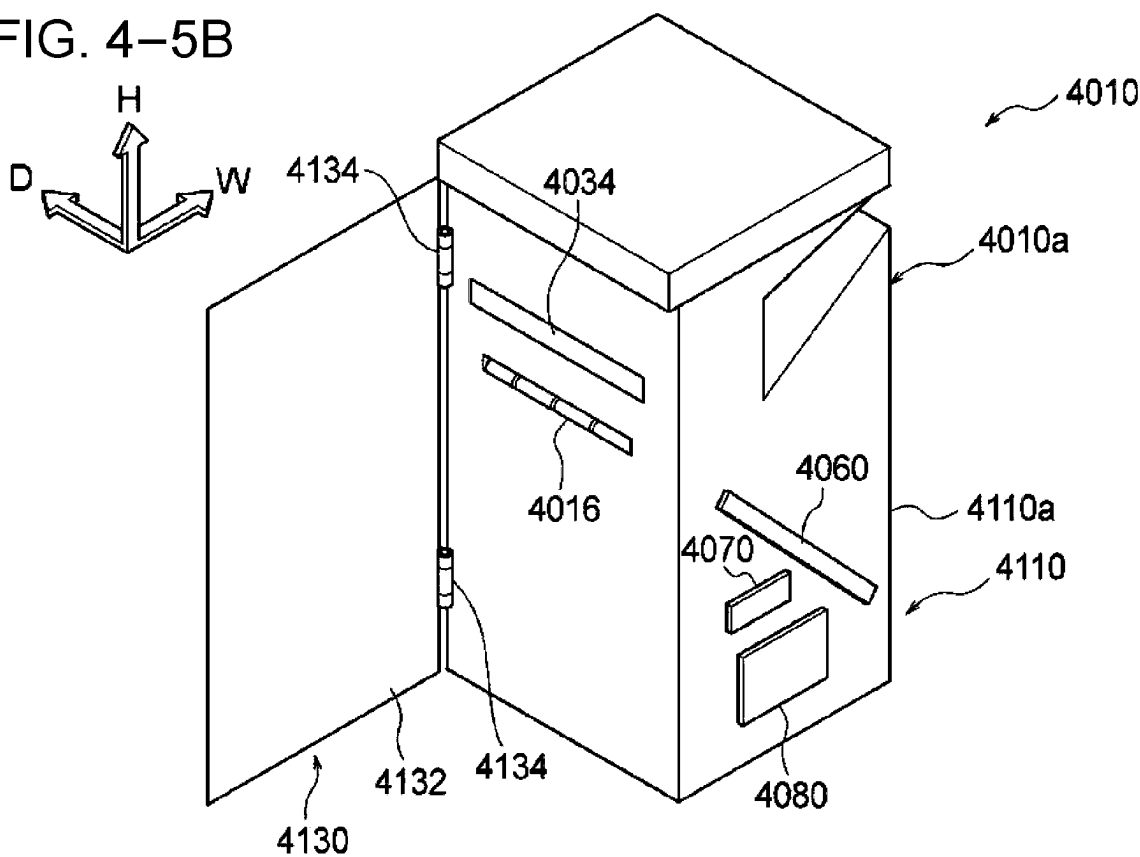


FIG. 4-6A

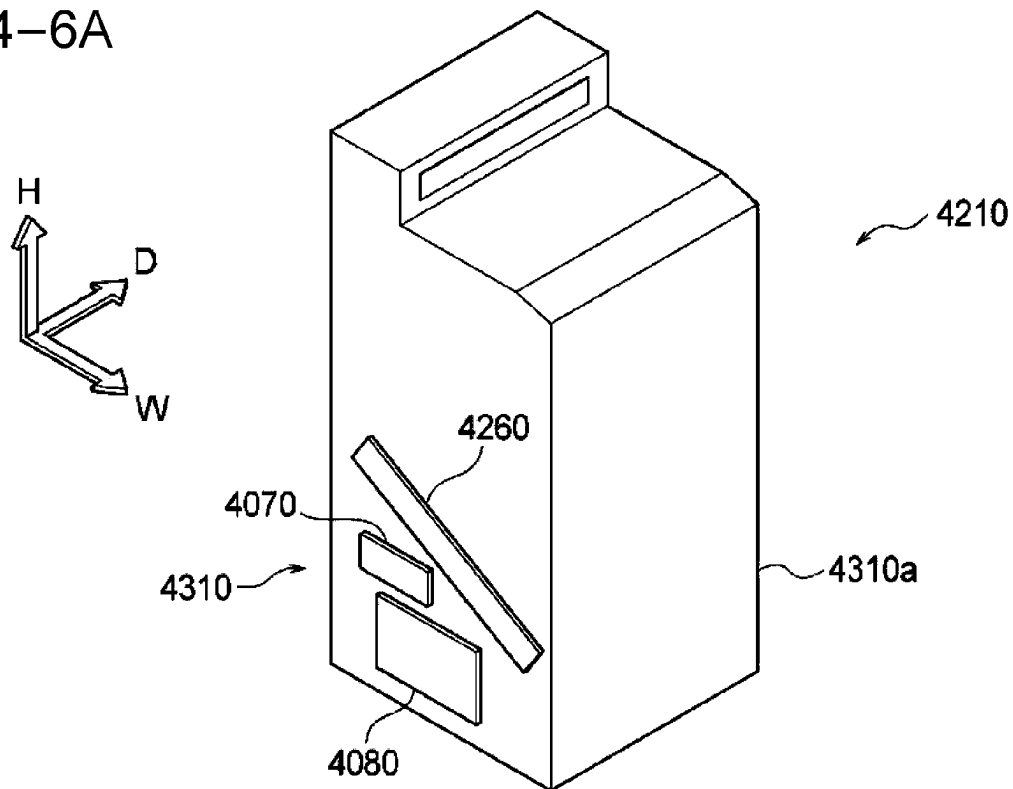


FIG. 4-6B

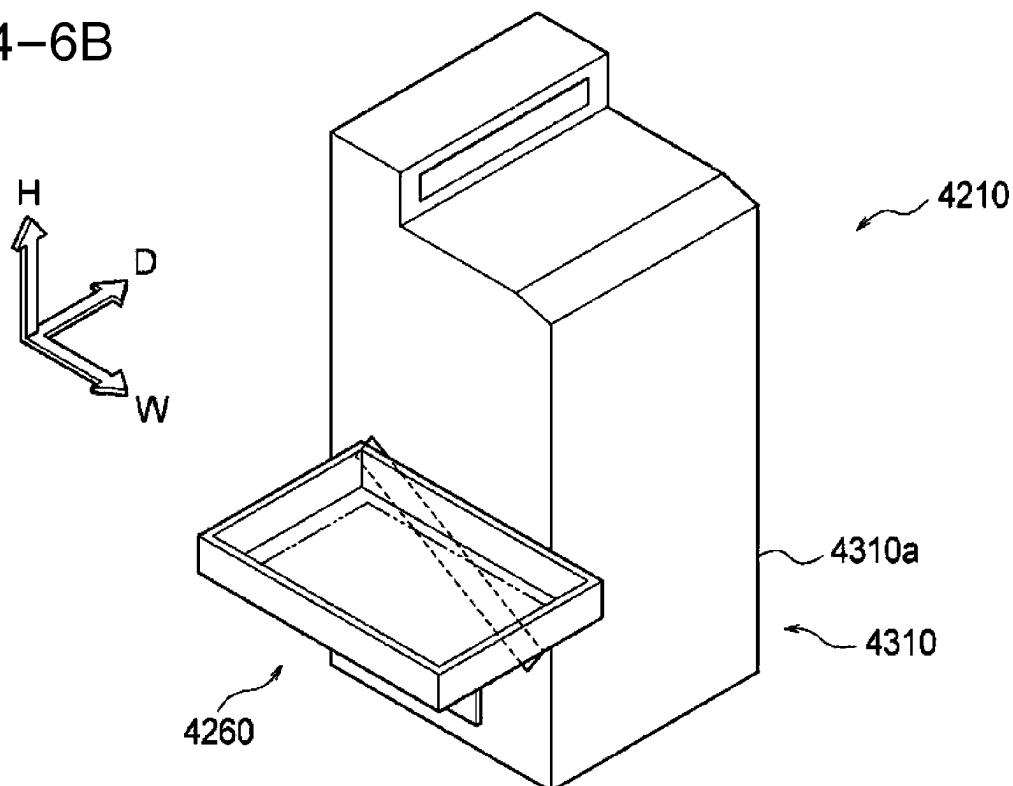


FIG. 4-7

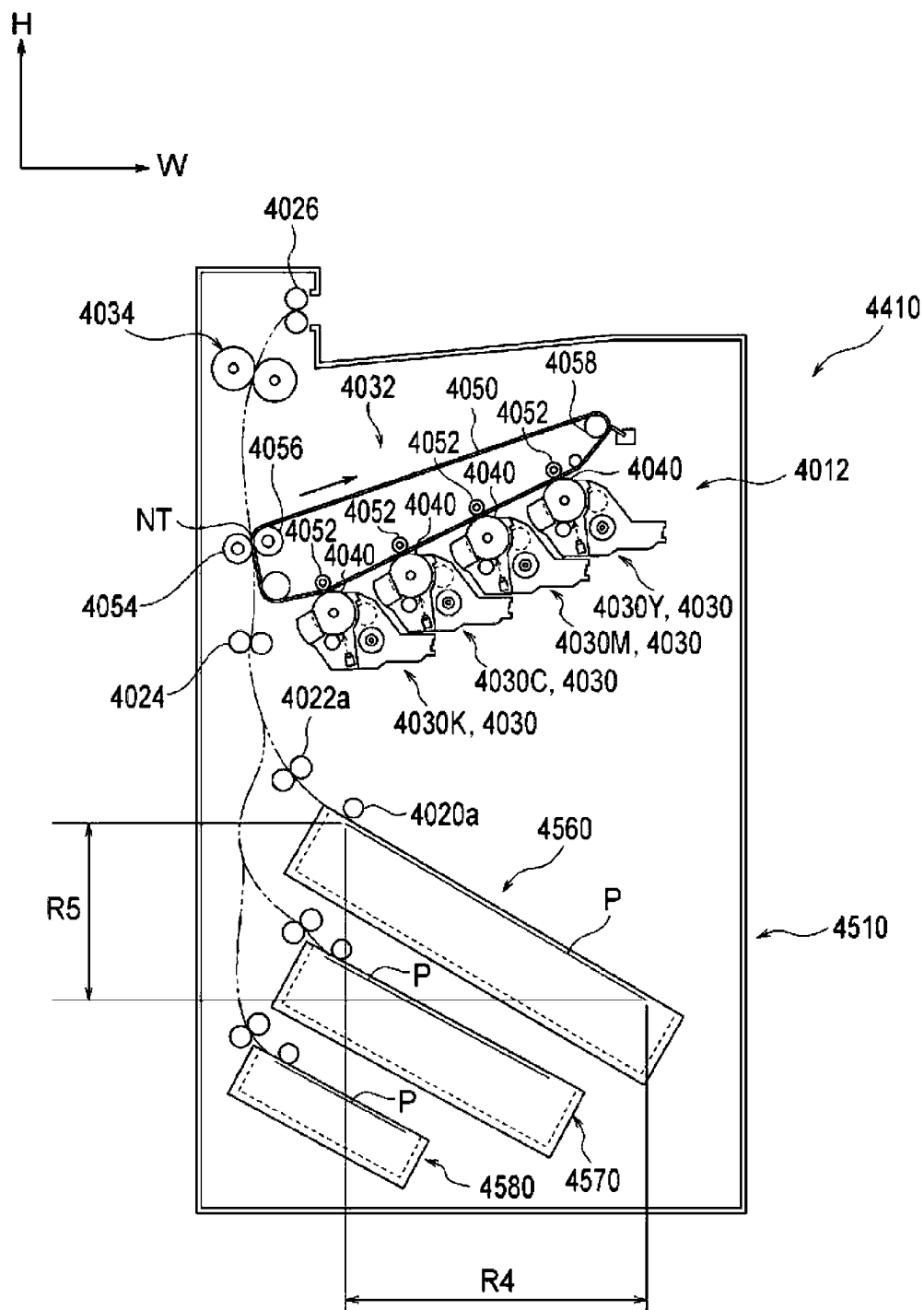


FIG. 4-8

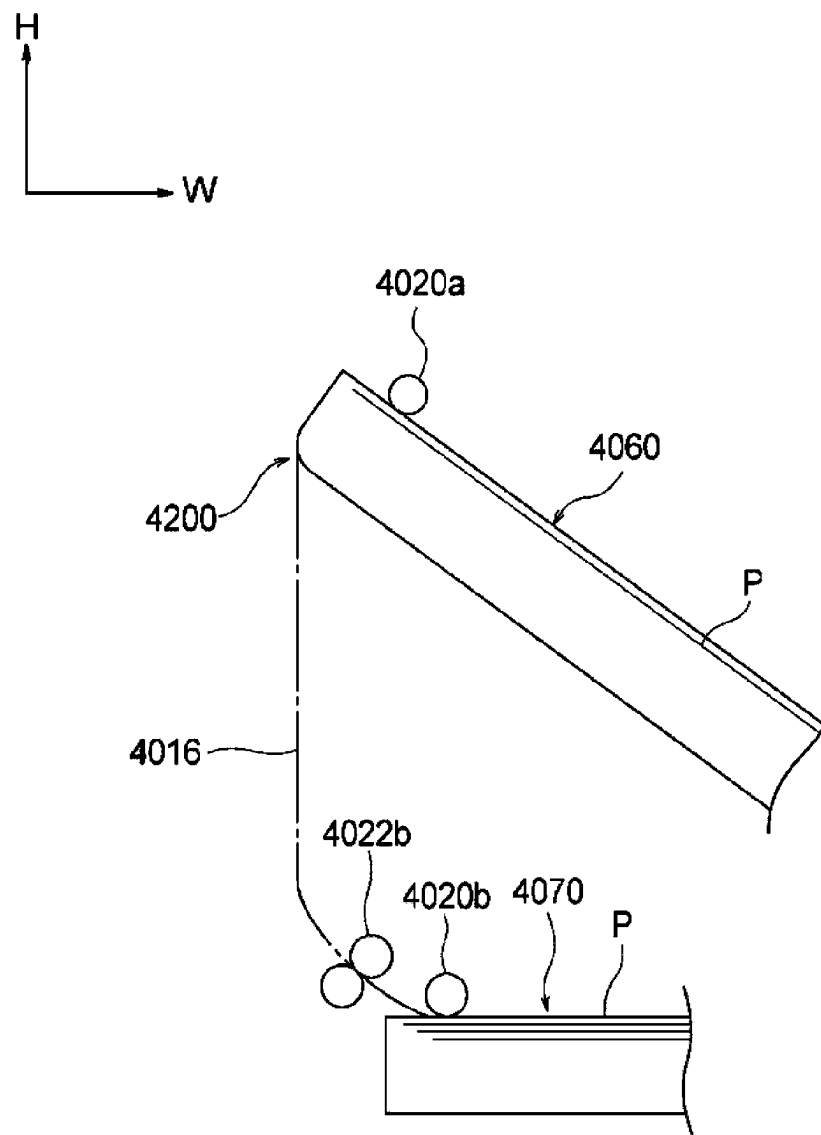


FIG. 5-1

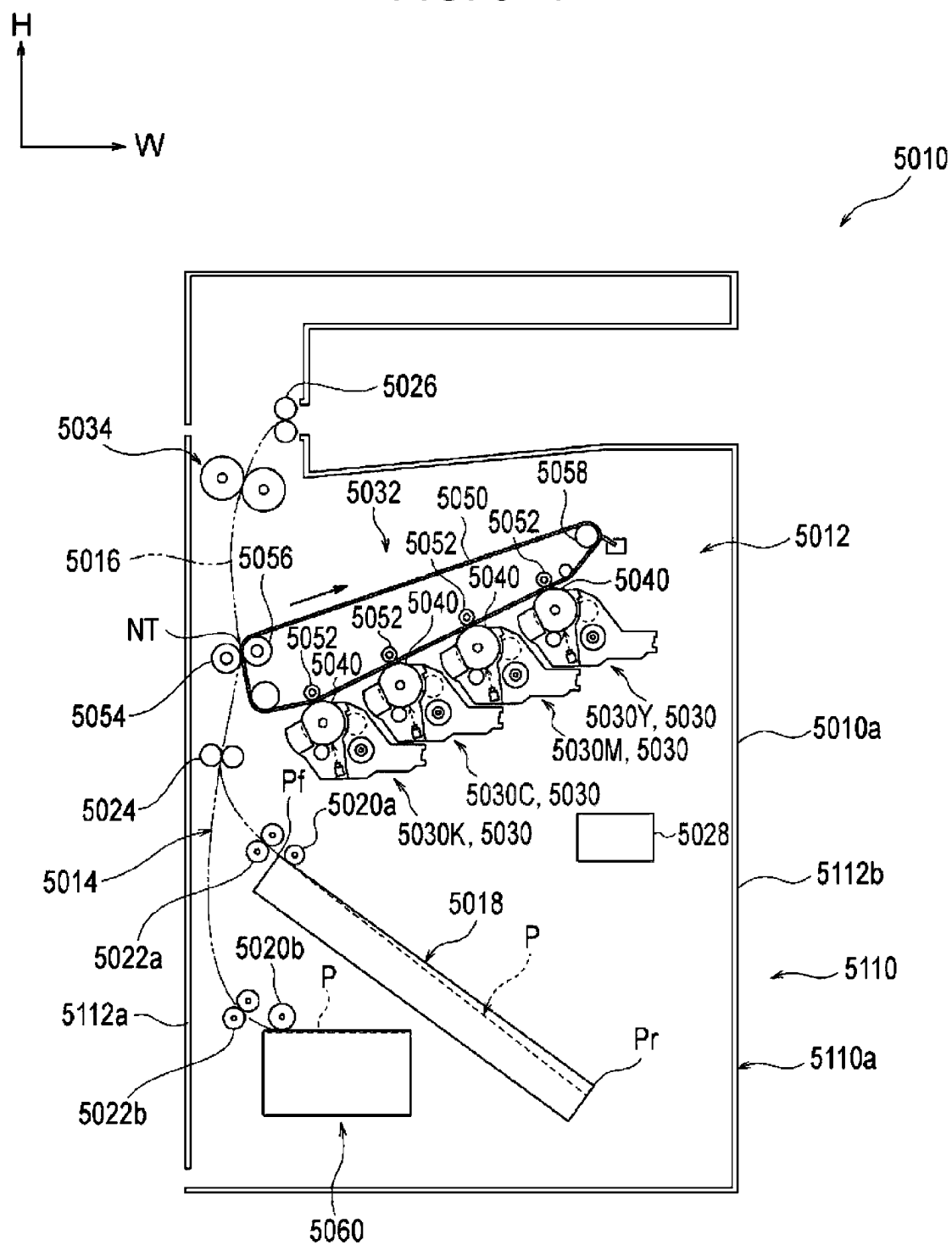


FIG. 5-2

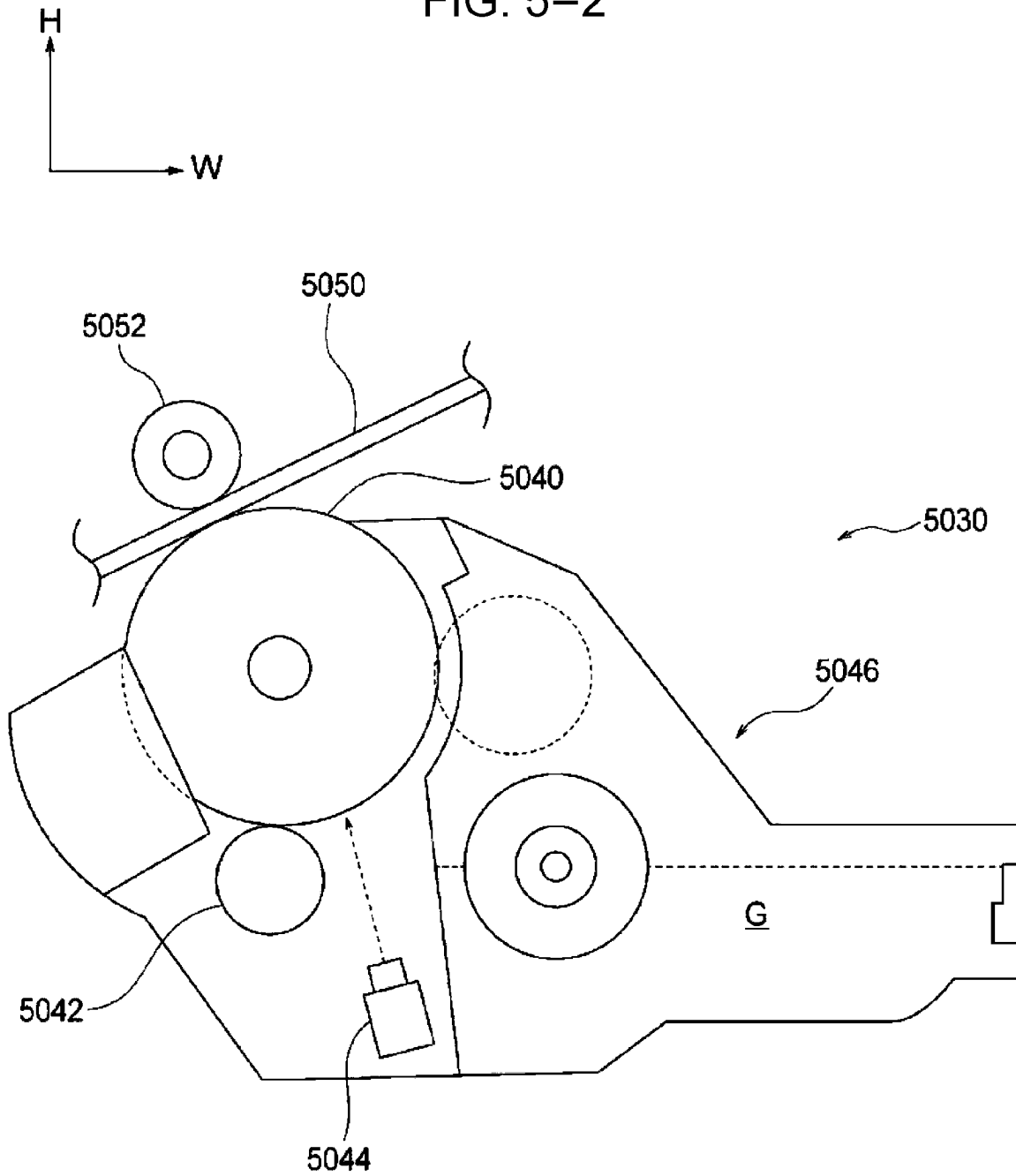


FIG. 5-3A

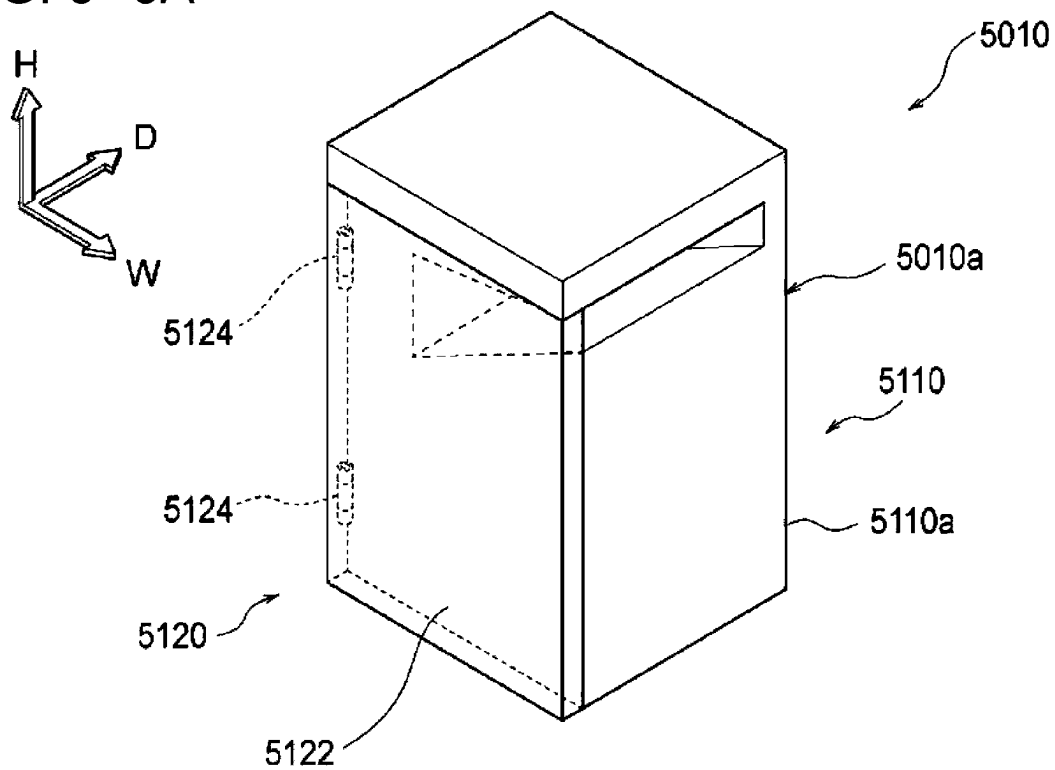


FIG. 5-3B

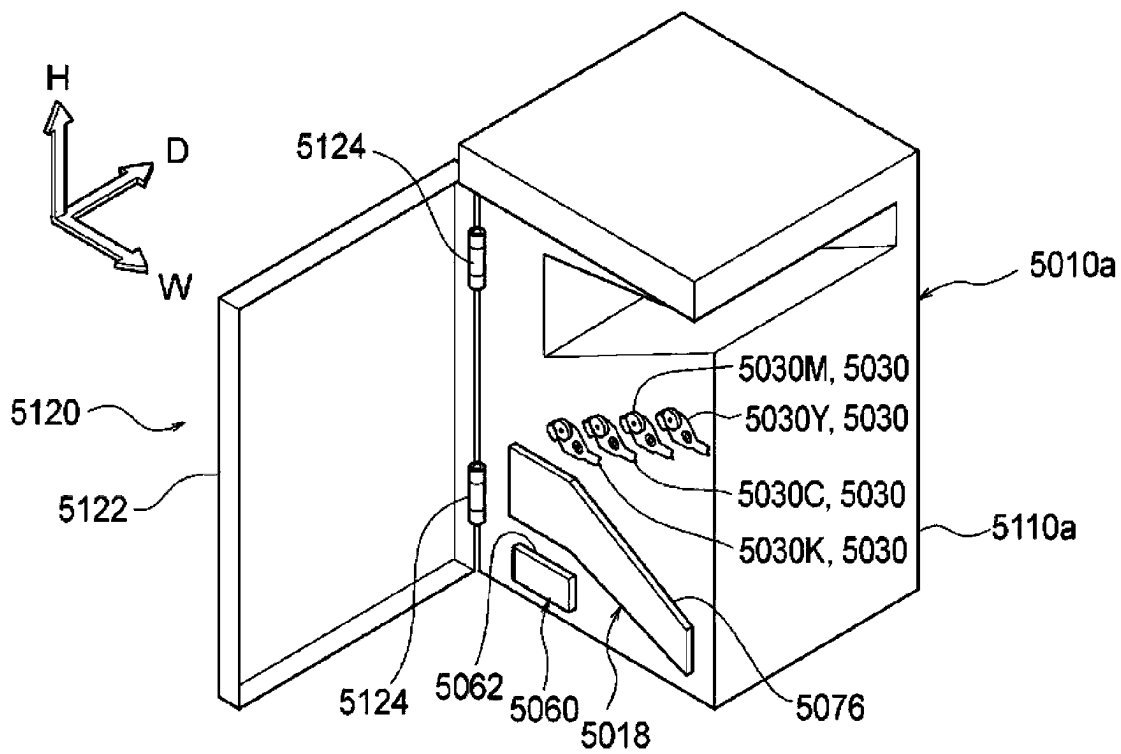


FIG. 5-4A

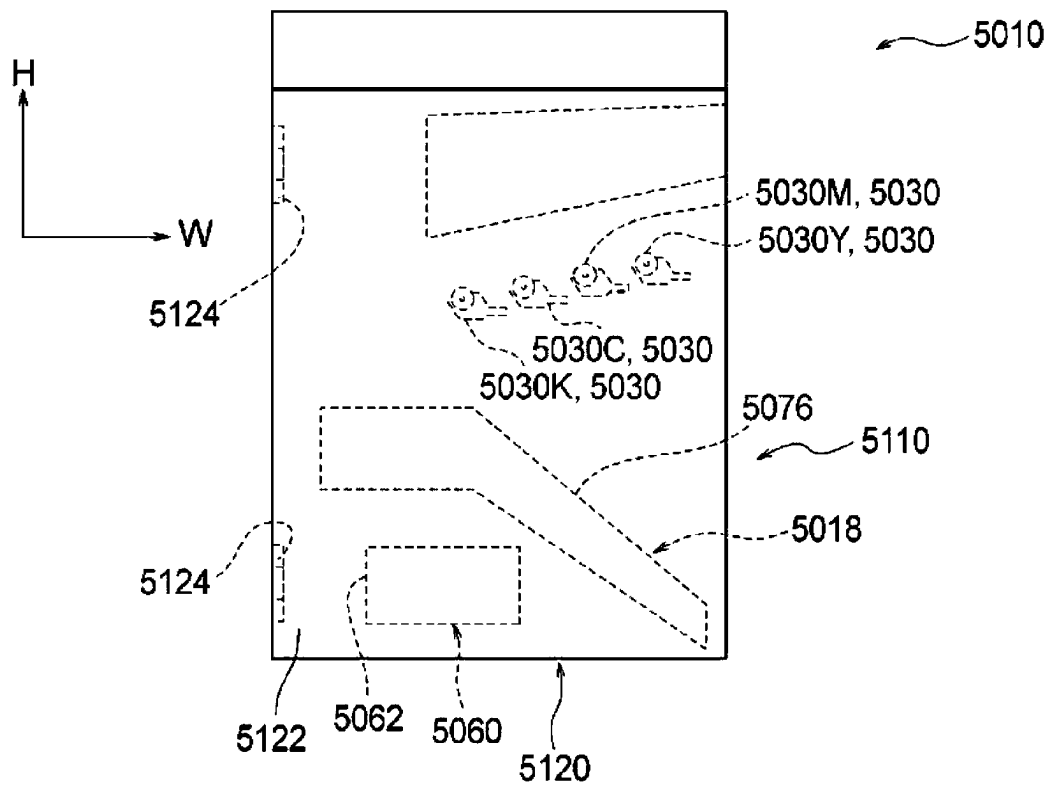


FIG. 5-4B

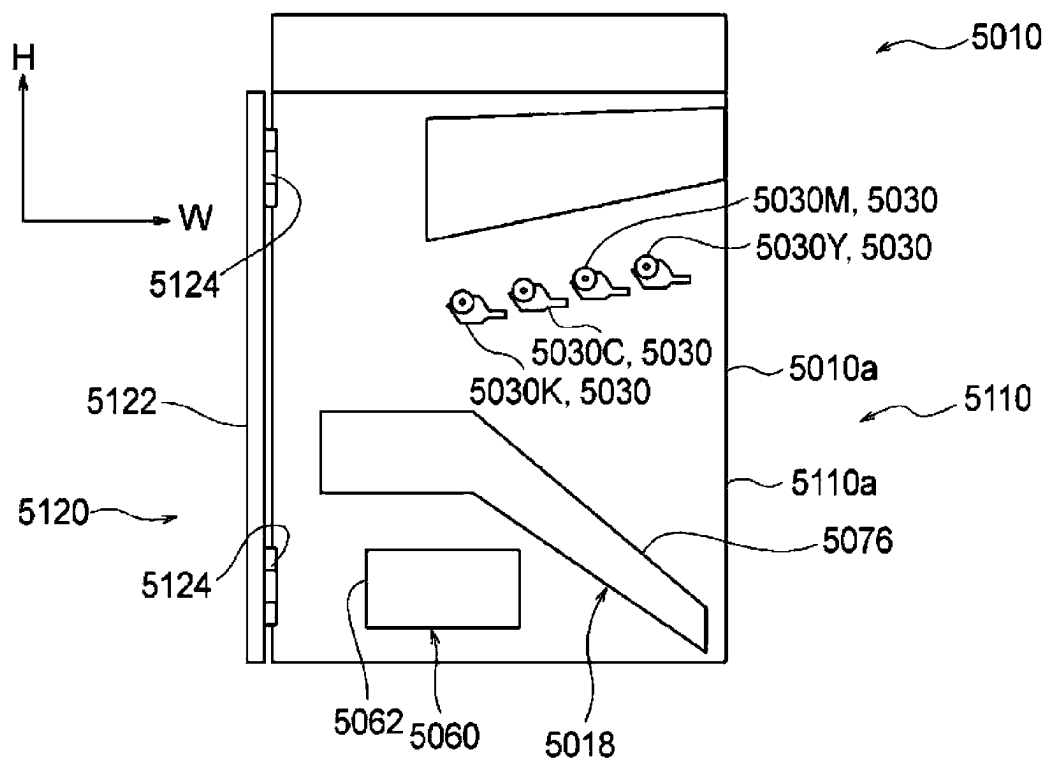


FIG. 5-6A

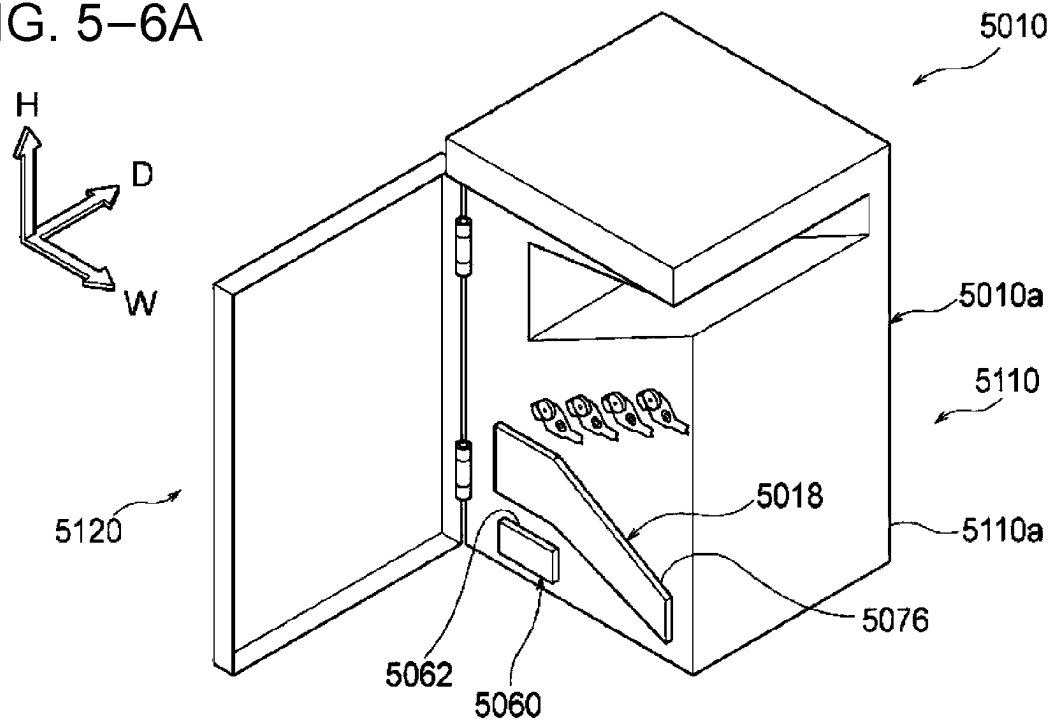


FIG. 5-6B

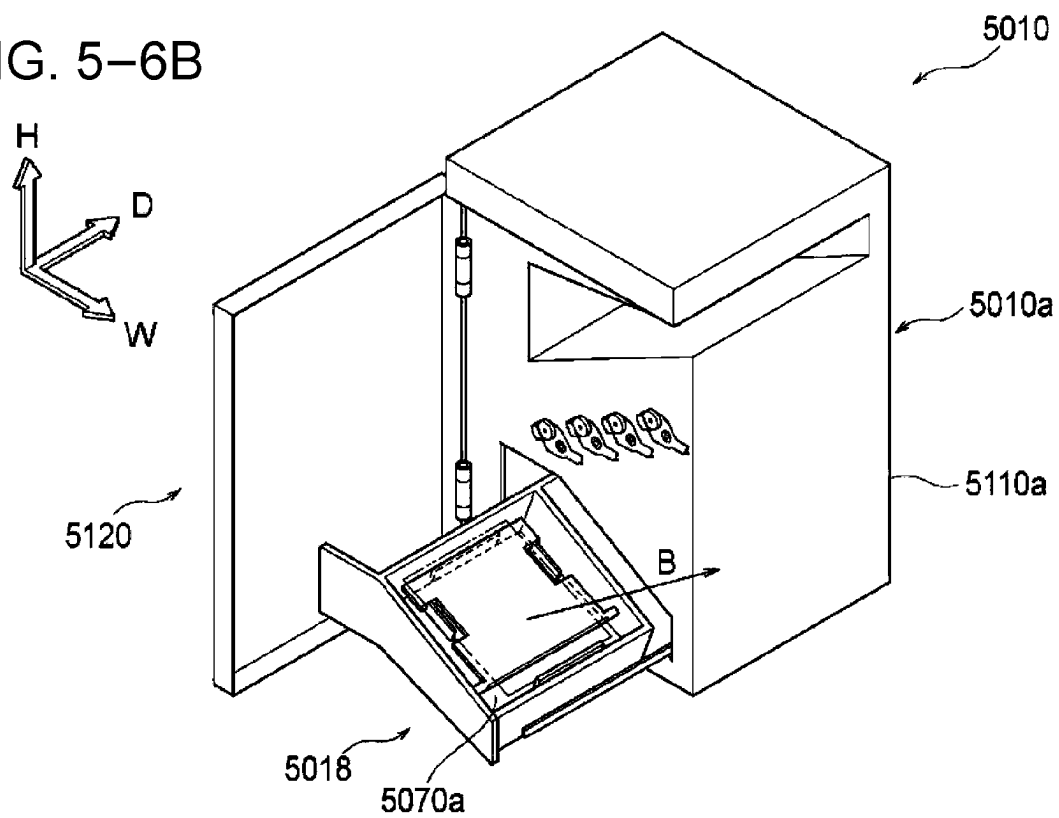


FIG. 5-7A

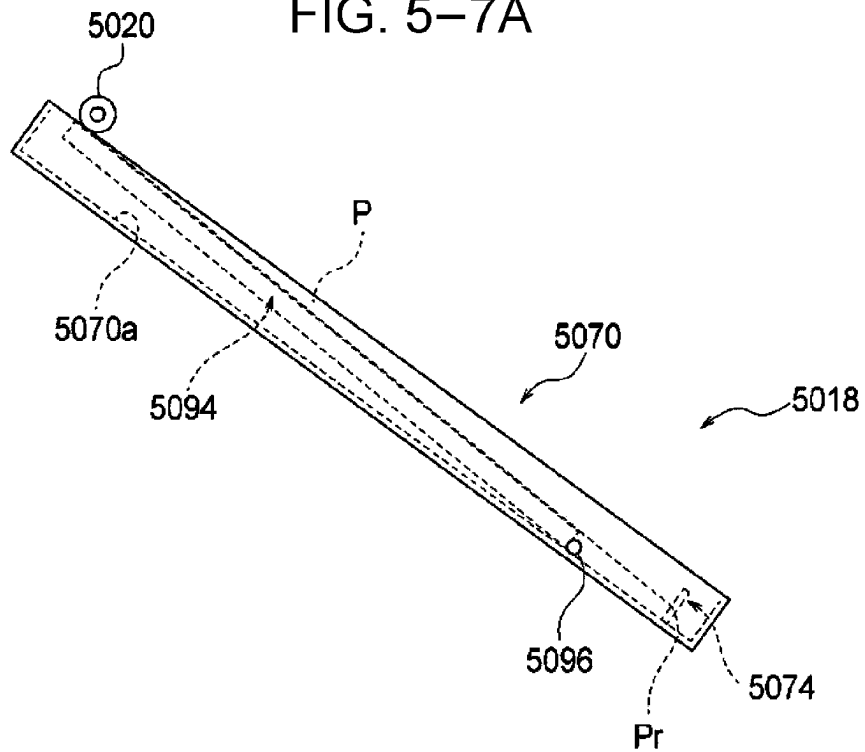


FIG. 5-7B

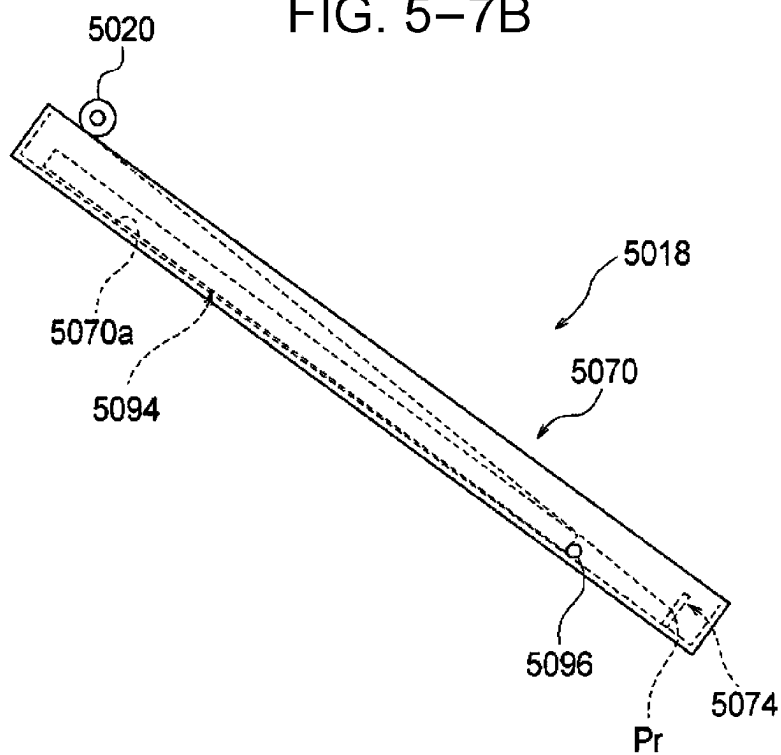


FIG. 5-8A

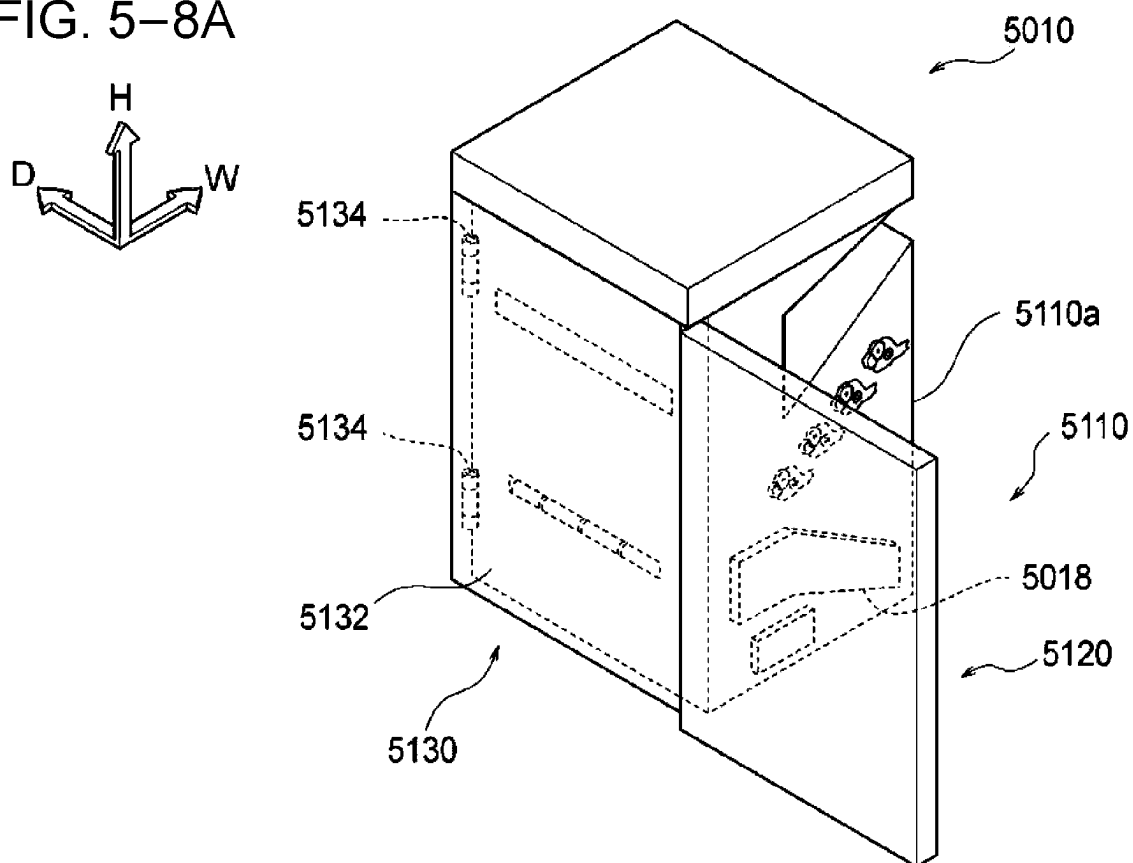


FIG. 5-8B

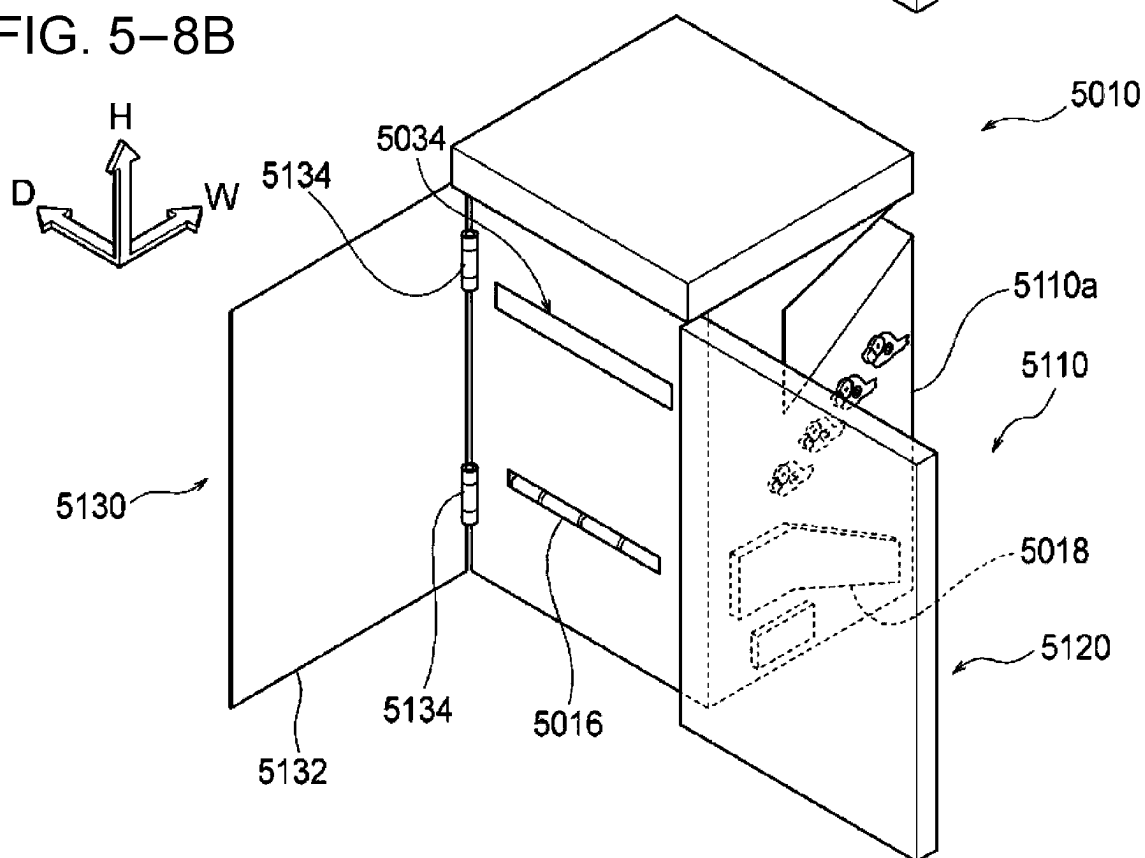


FIG. 5-9

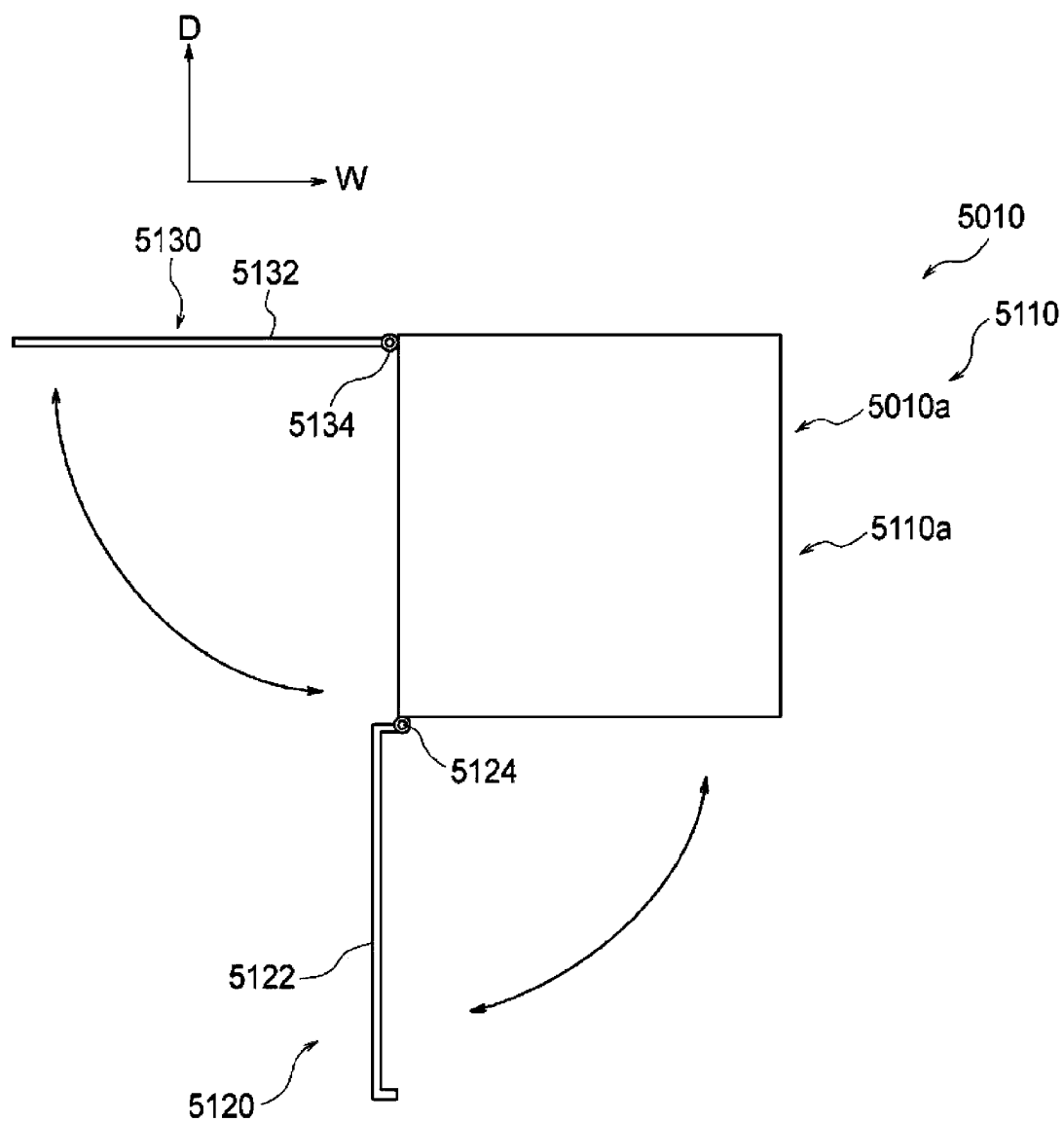


FIG. 6-1A

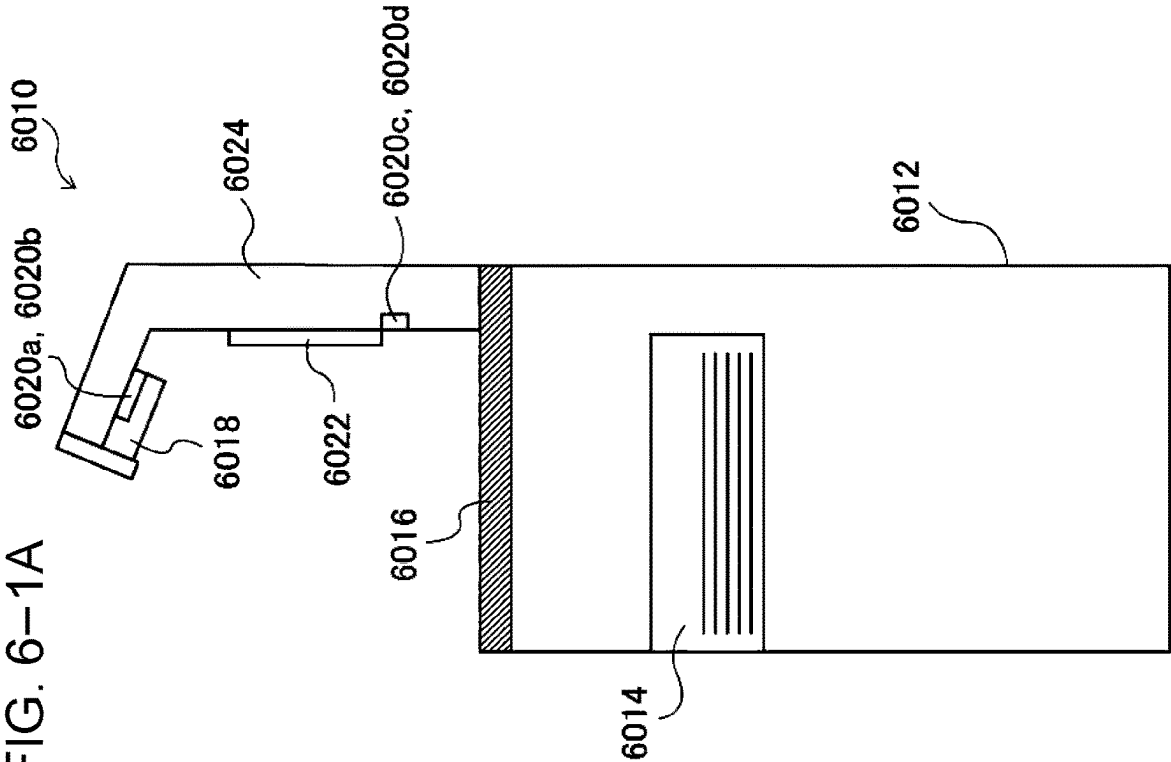


FIG. 6-1B

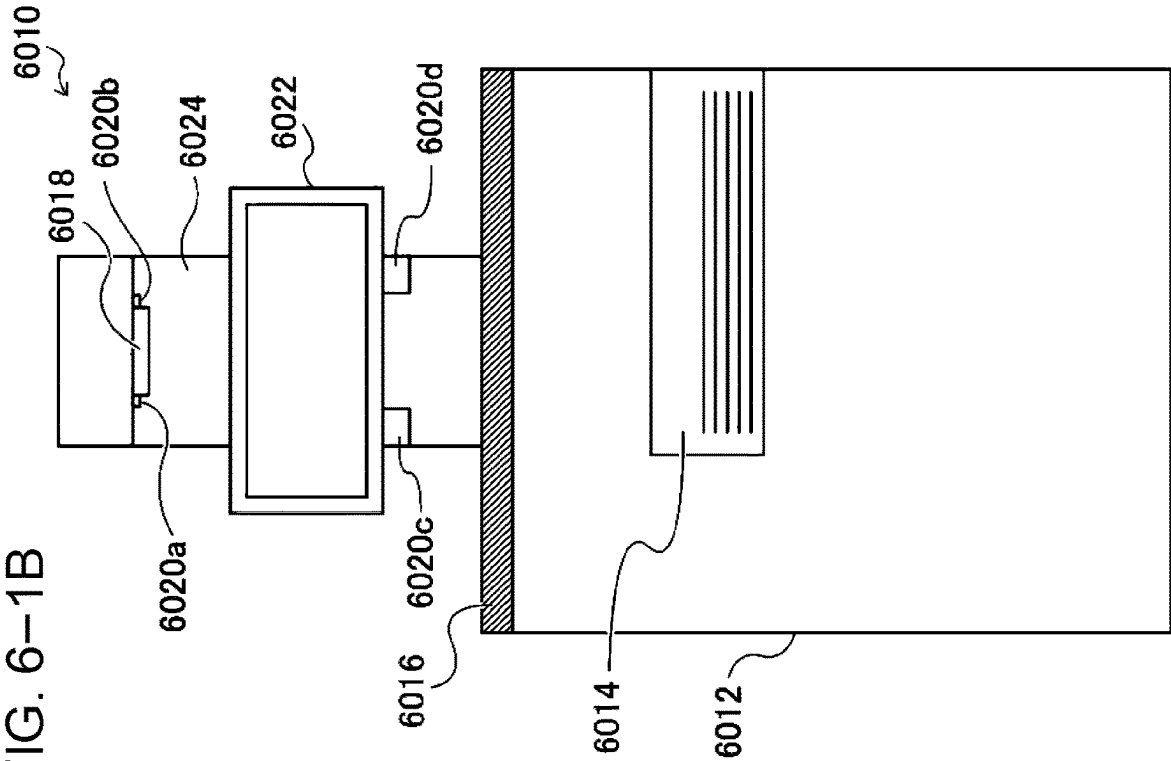


FIG. 6-2

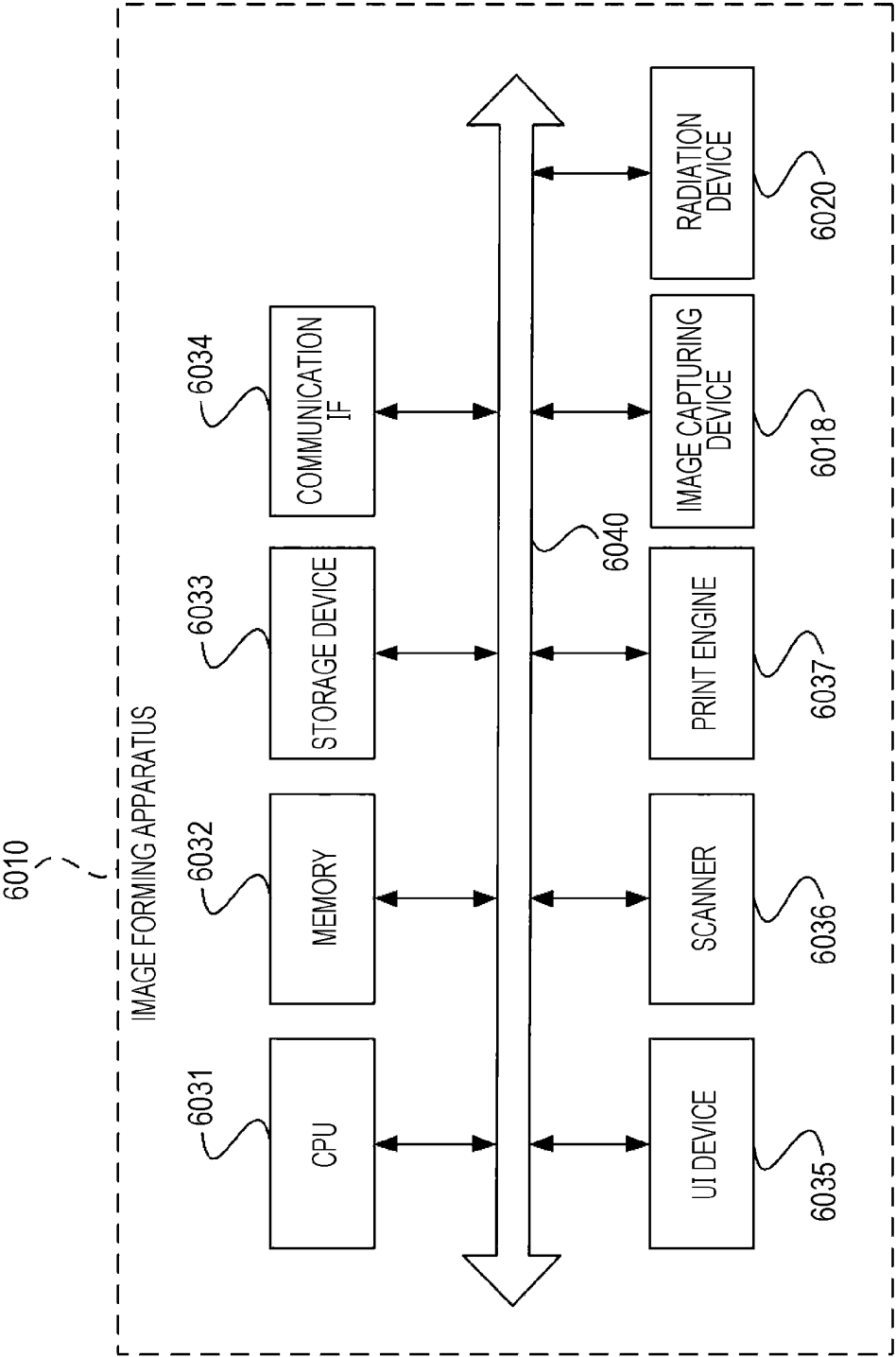


FIG. 6-3

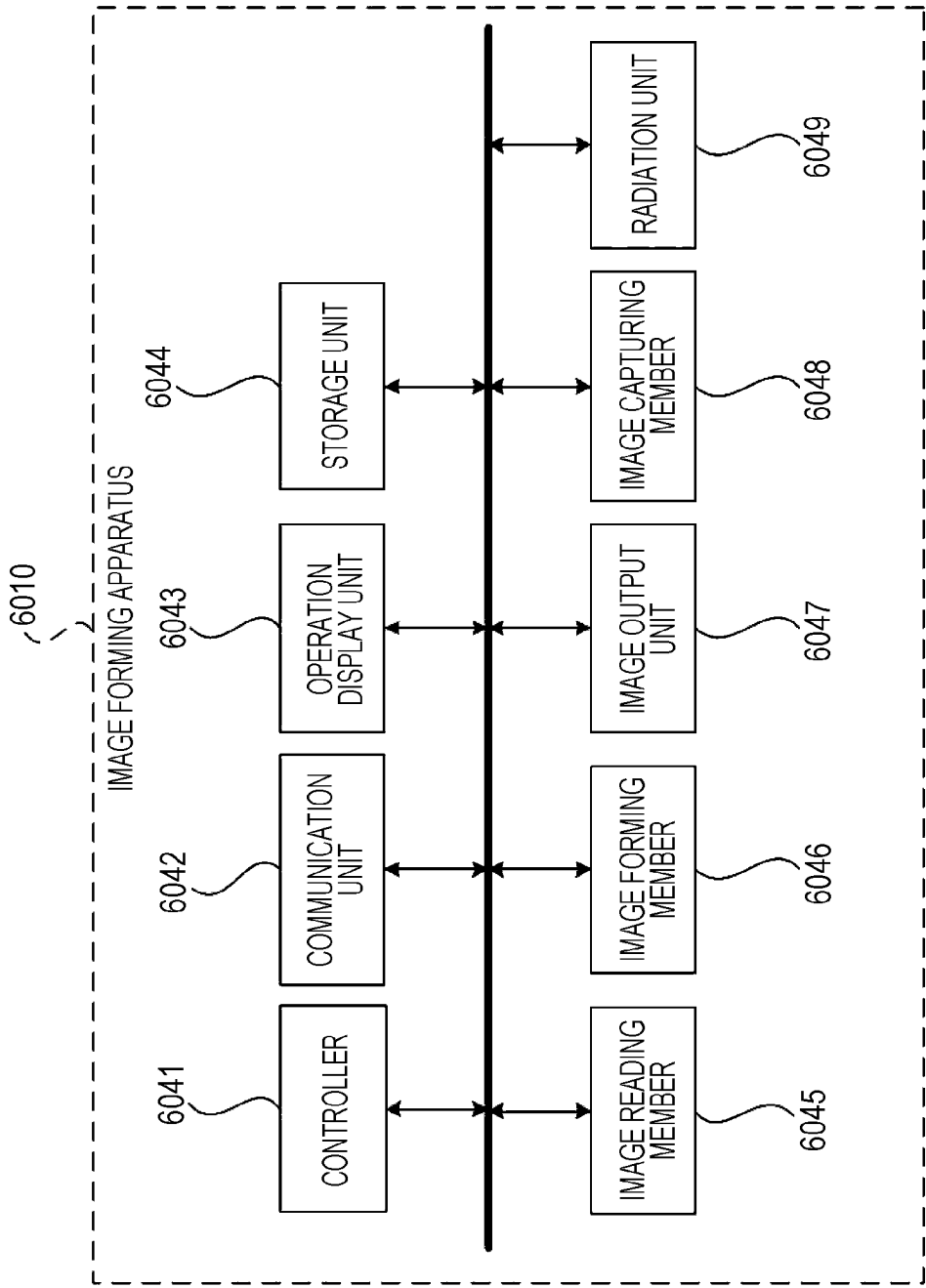


FIG. 6-4

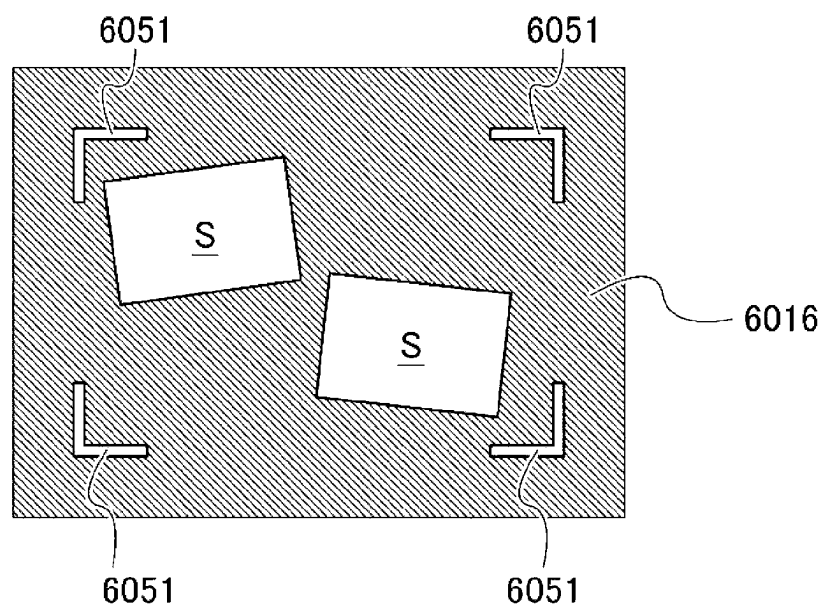


FIG. 6-5

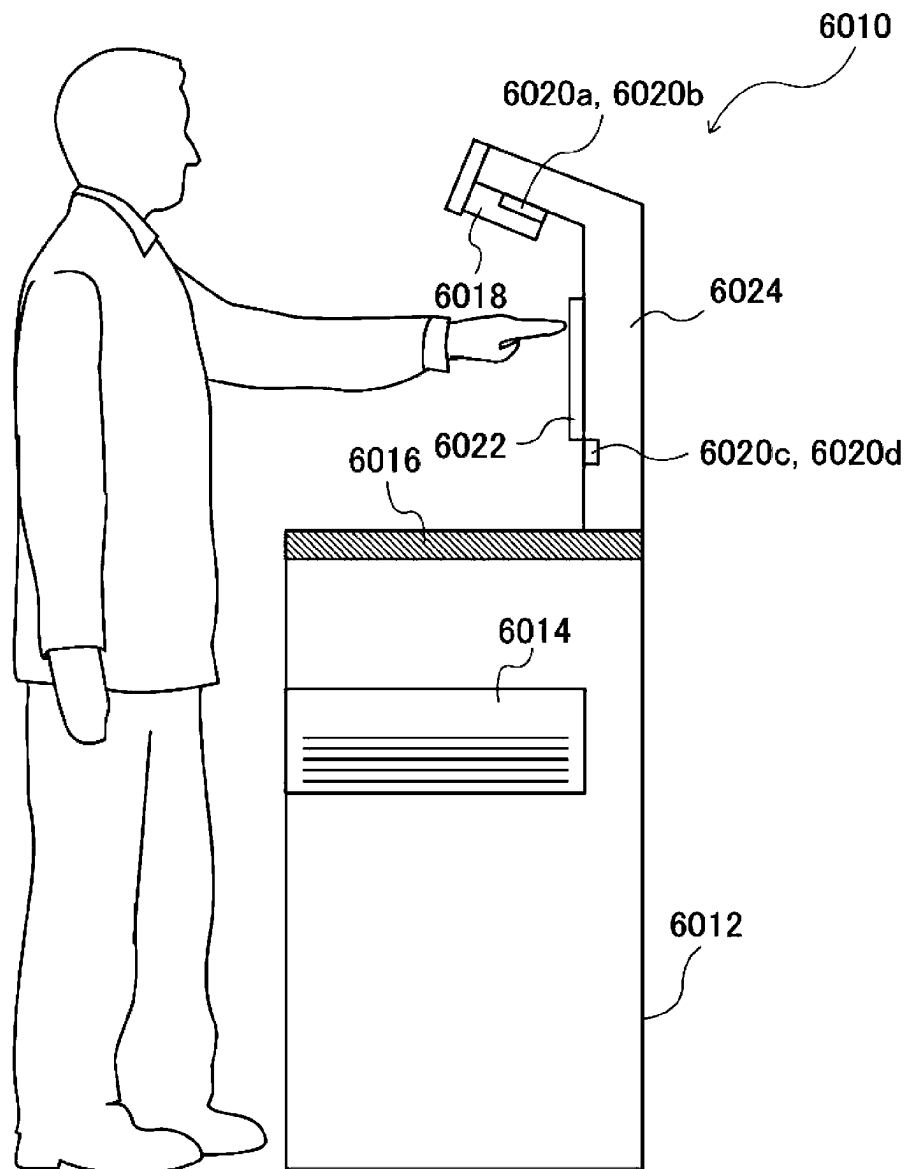


FIG. 6-6B

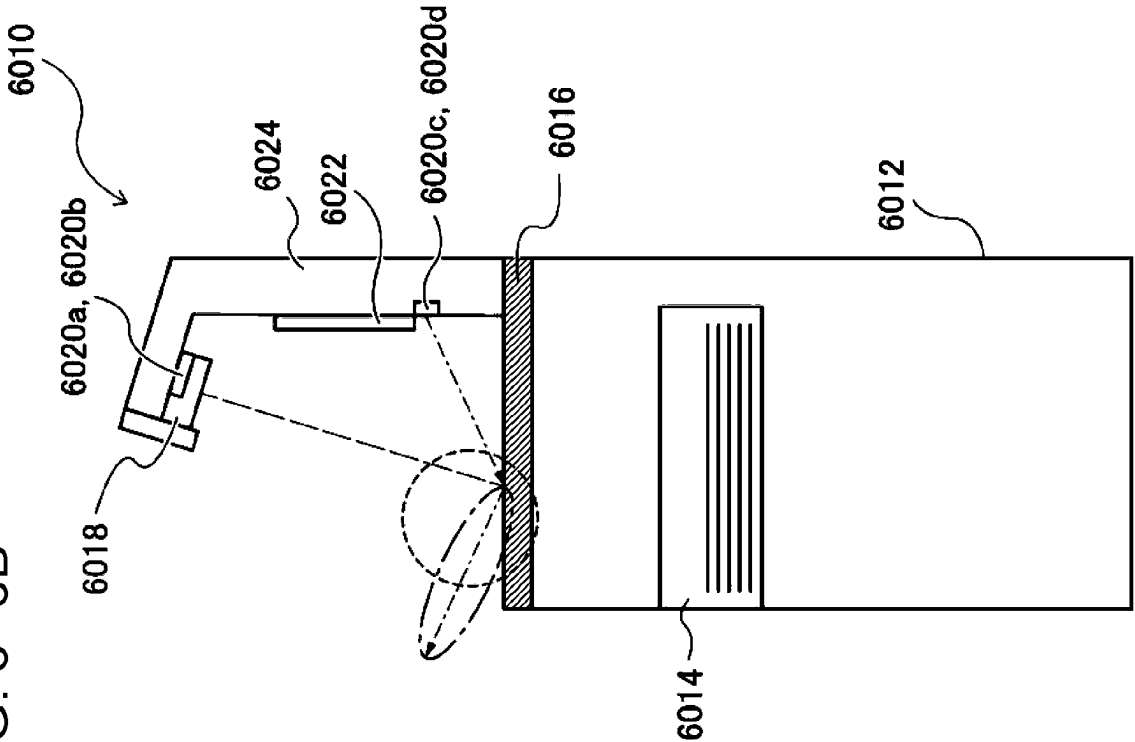


FIG. 6-6A

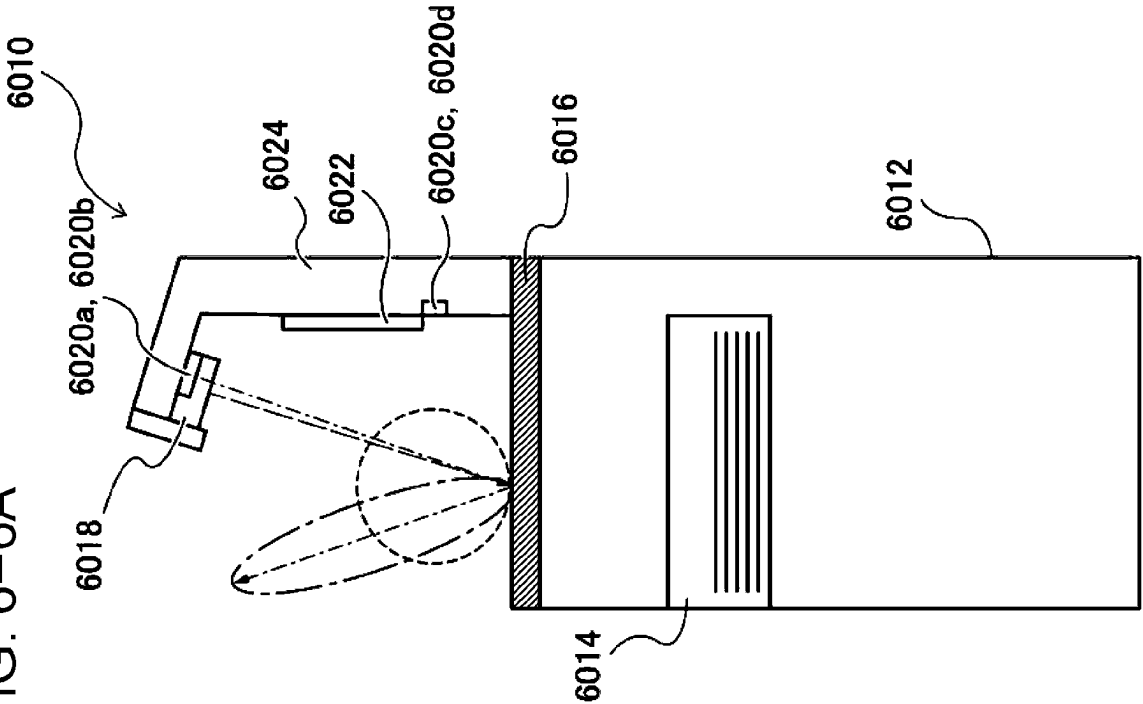


FIG. 6-7

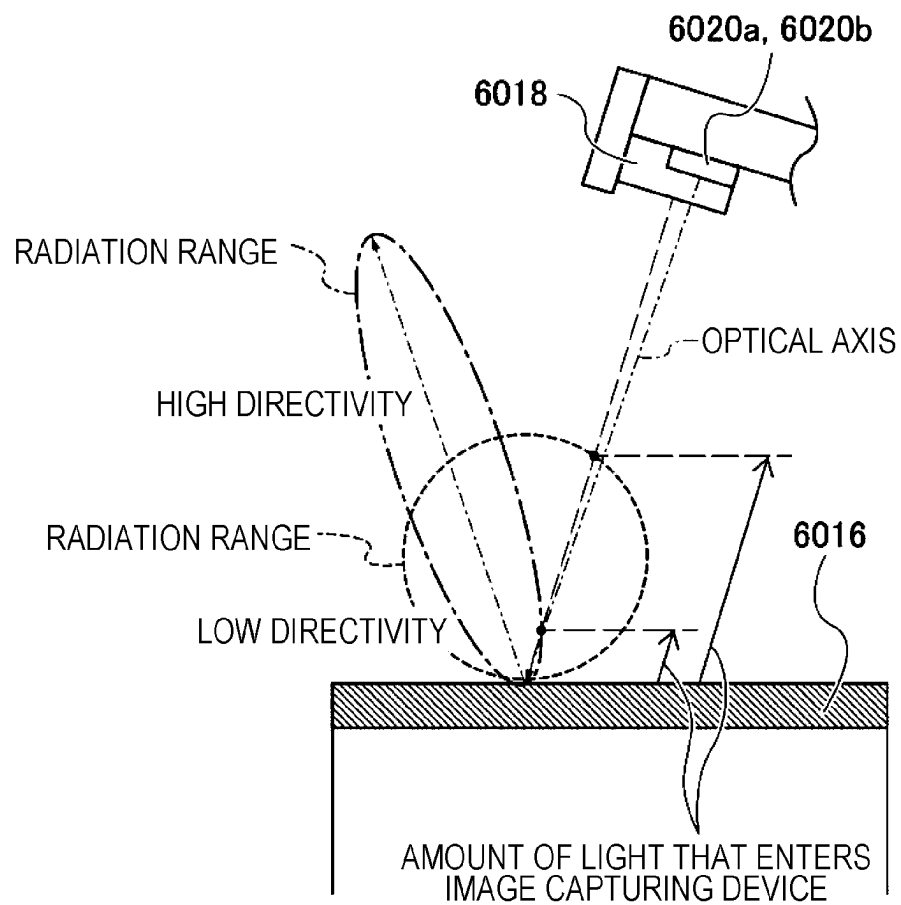


FIG. 6-8A

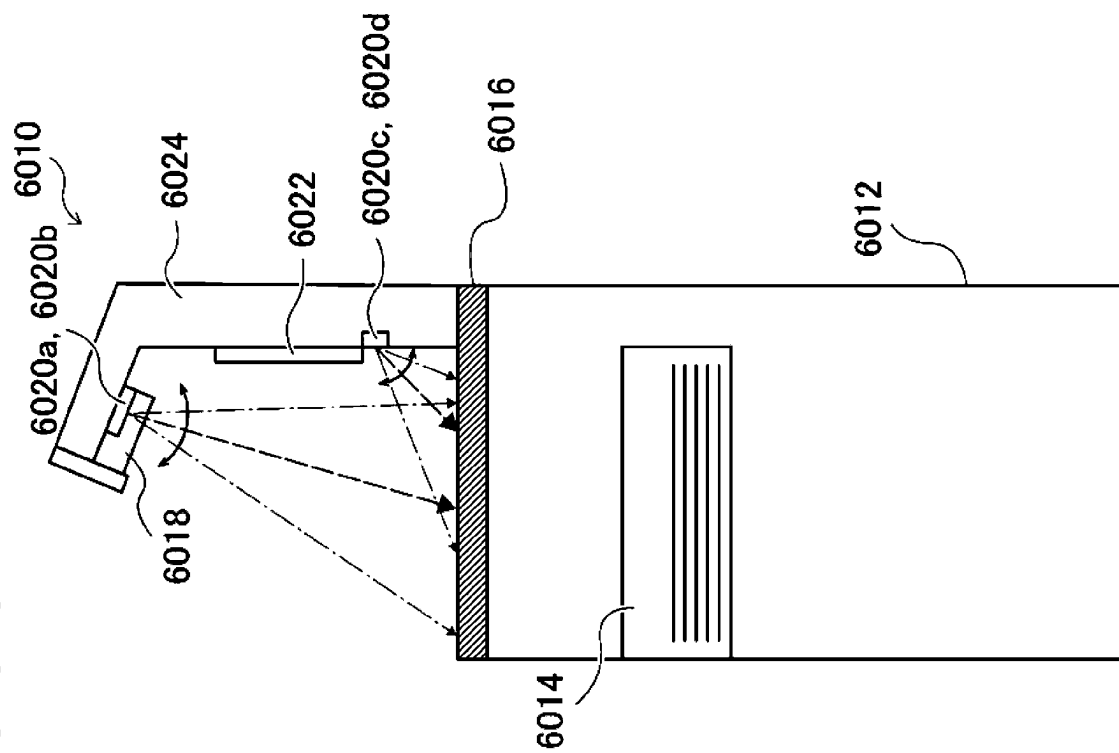


FIG. 6-8B

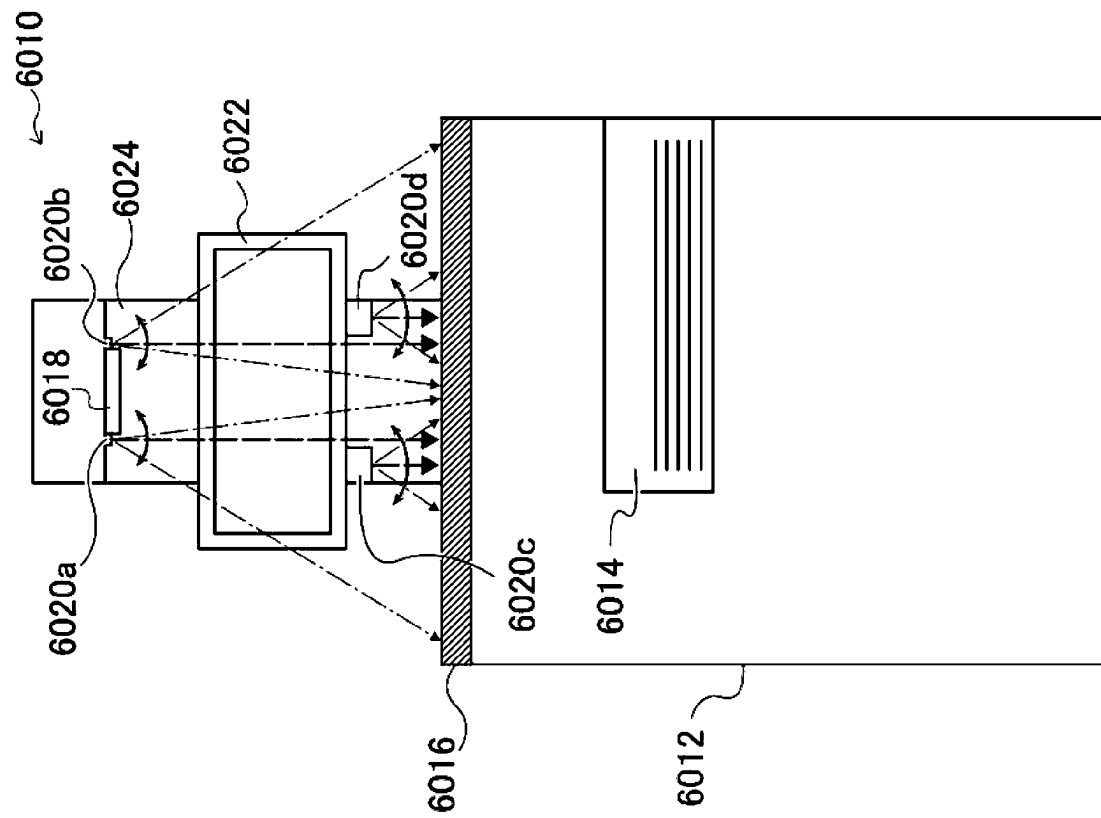


FIG. 6-9

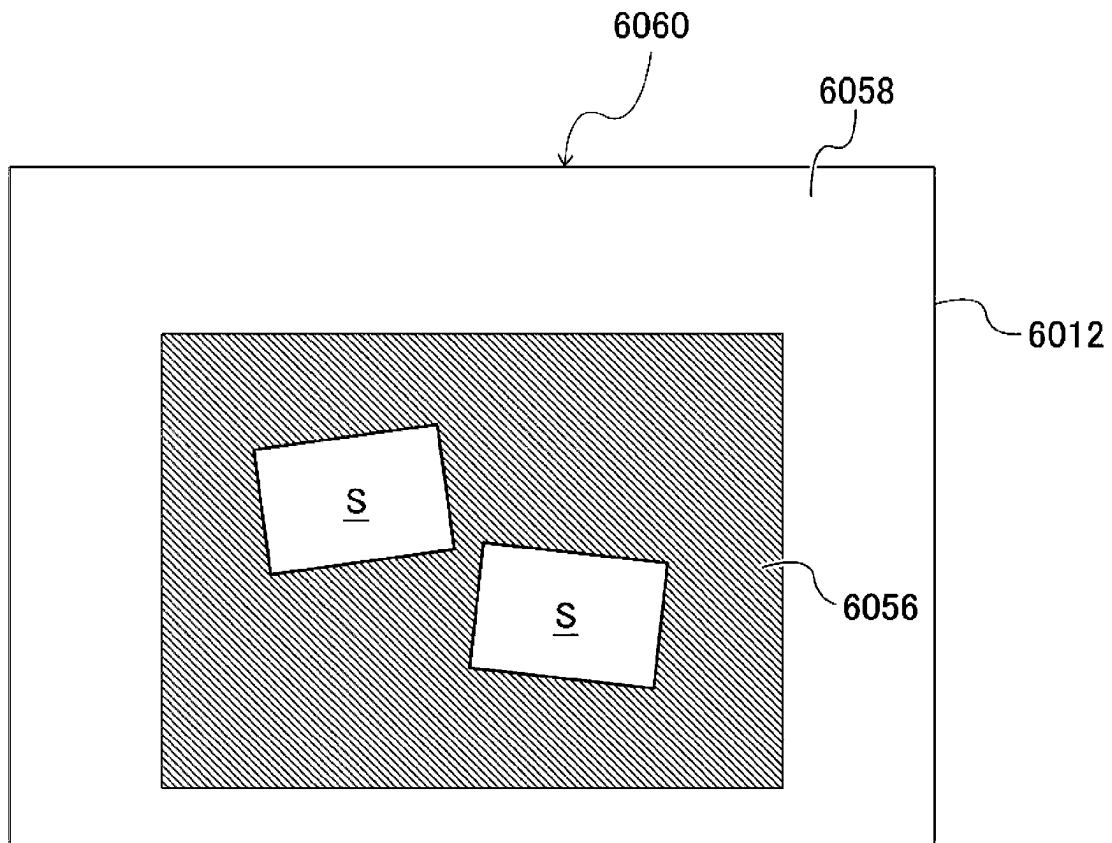


FIG. 6-10

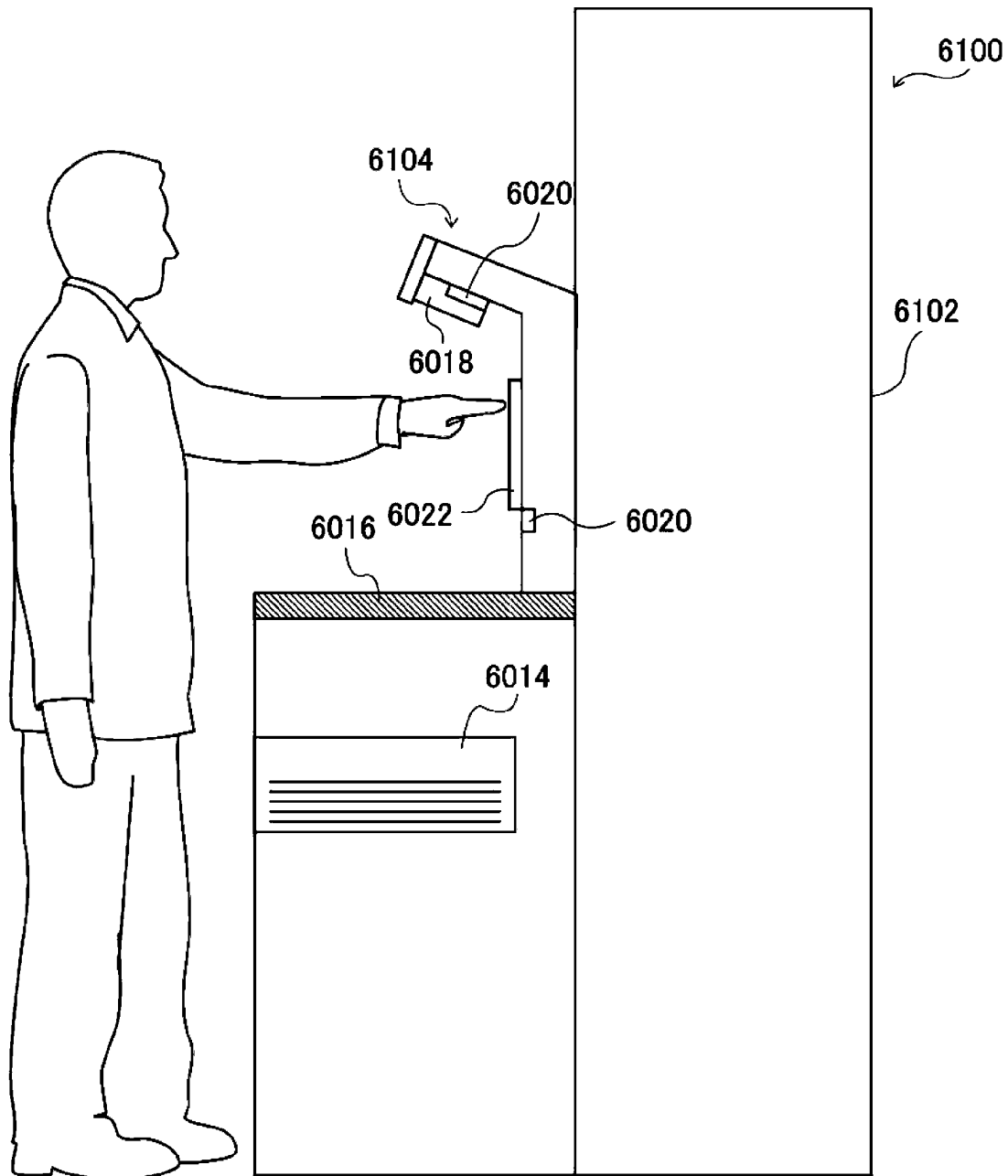


FIG. 7-1

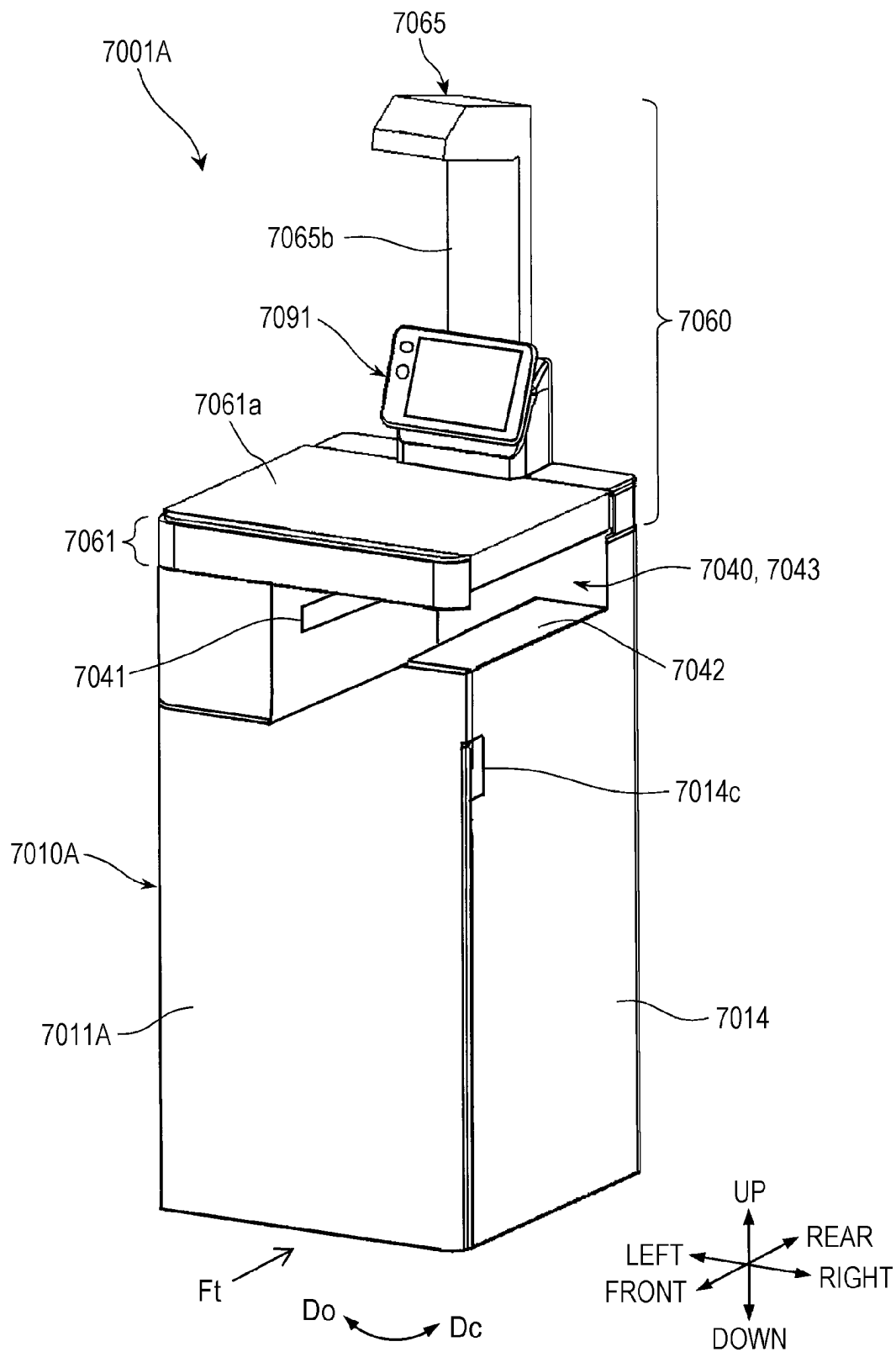


FIG. 7-2

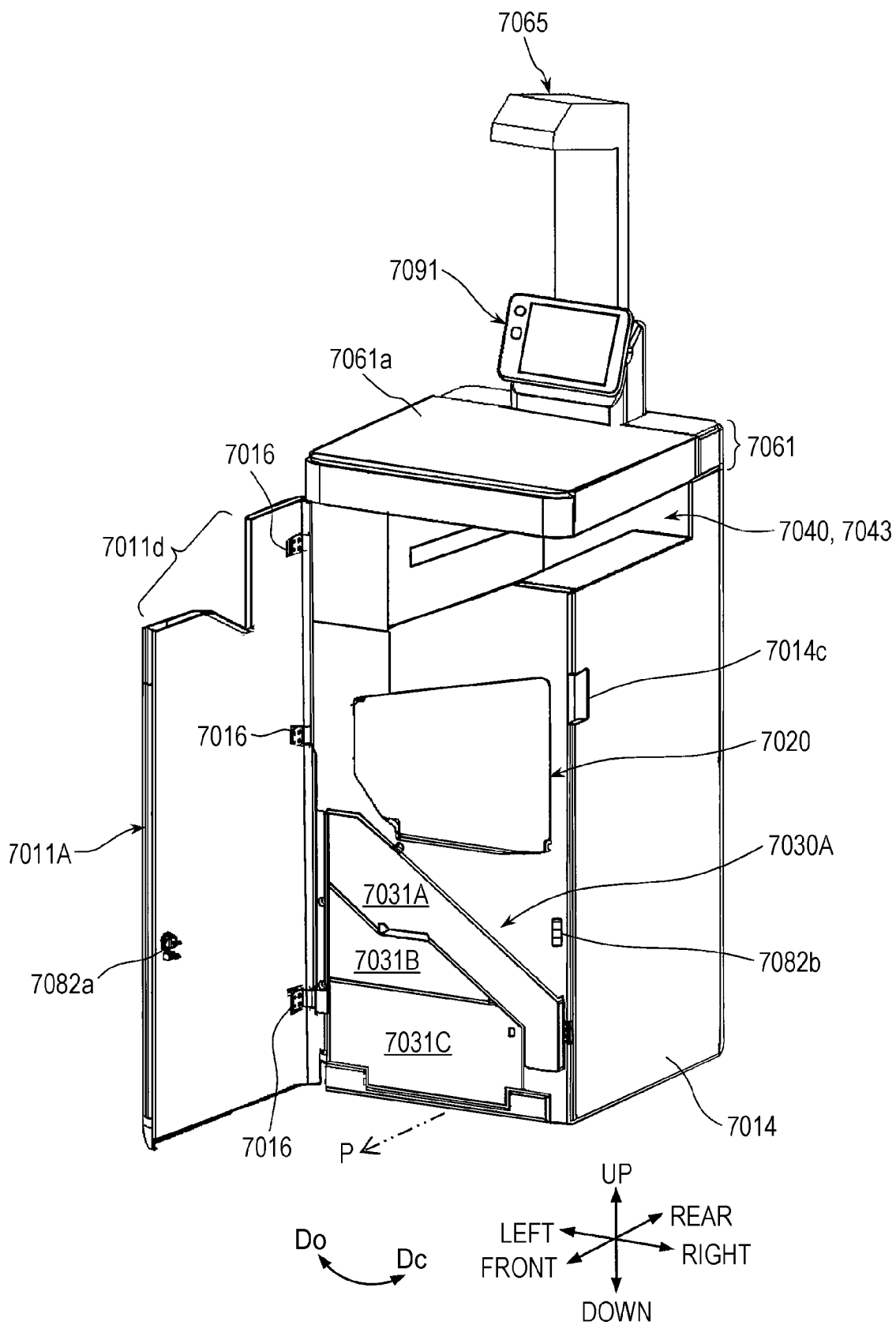


FIG. 7-3

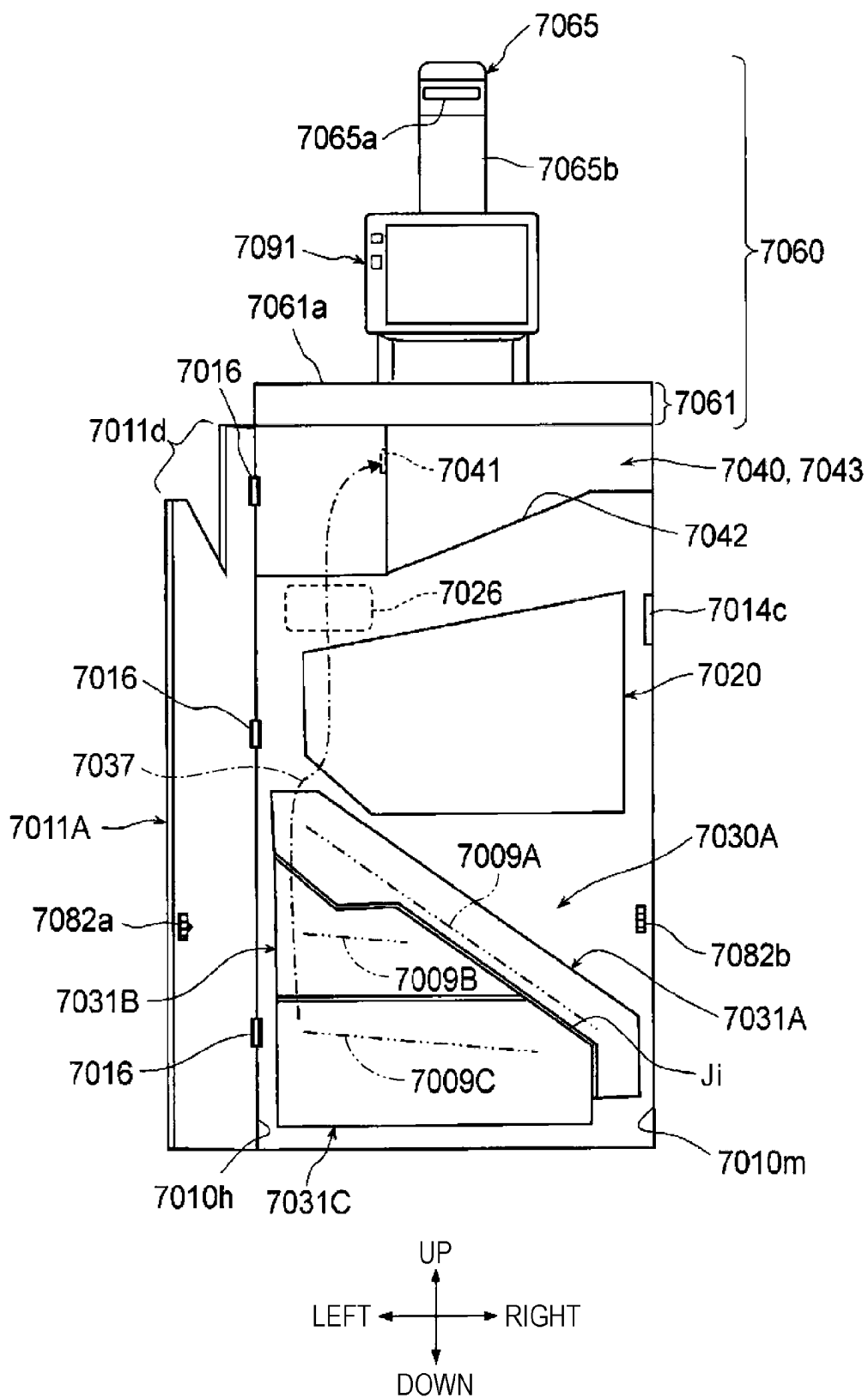


FIG. 7-4

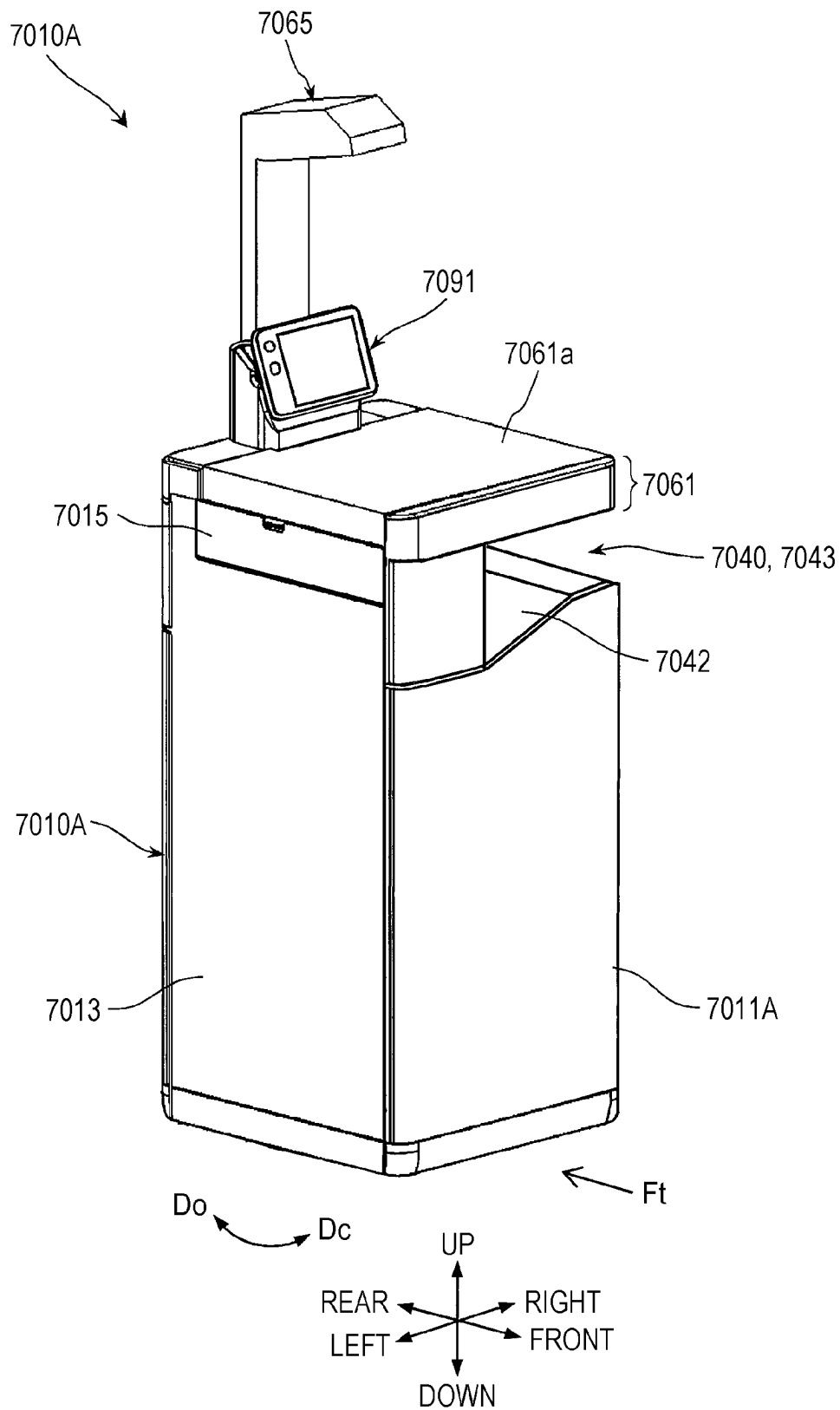


FIG. 7-5

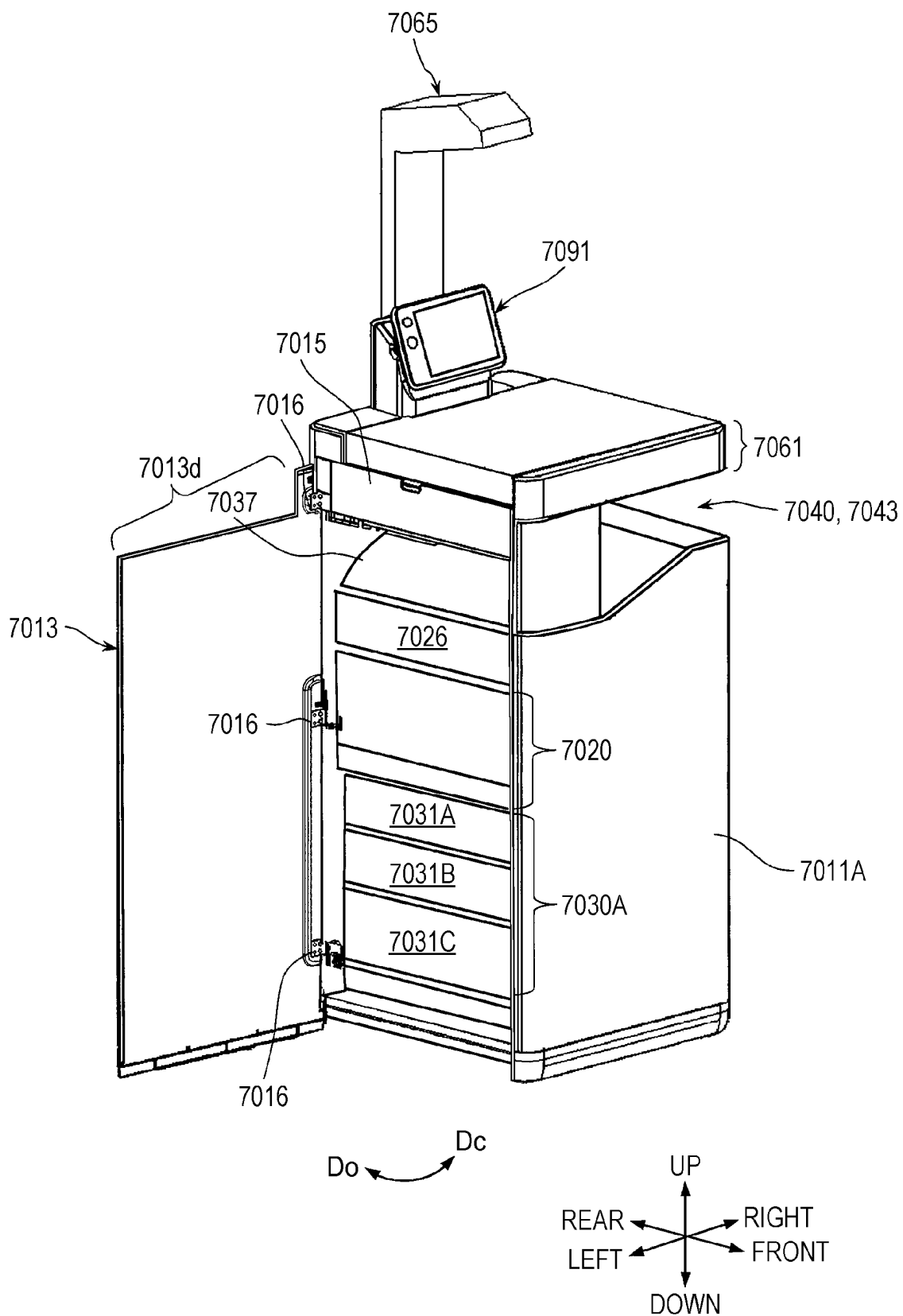


FIG. 7-6

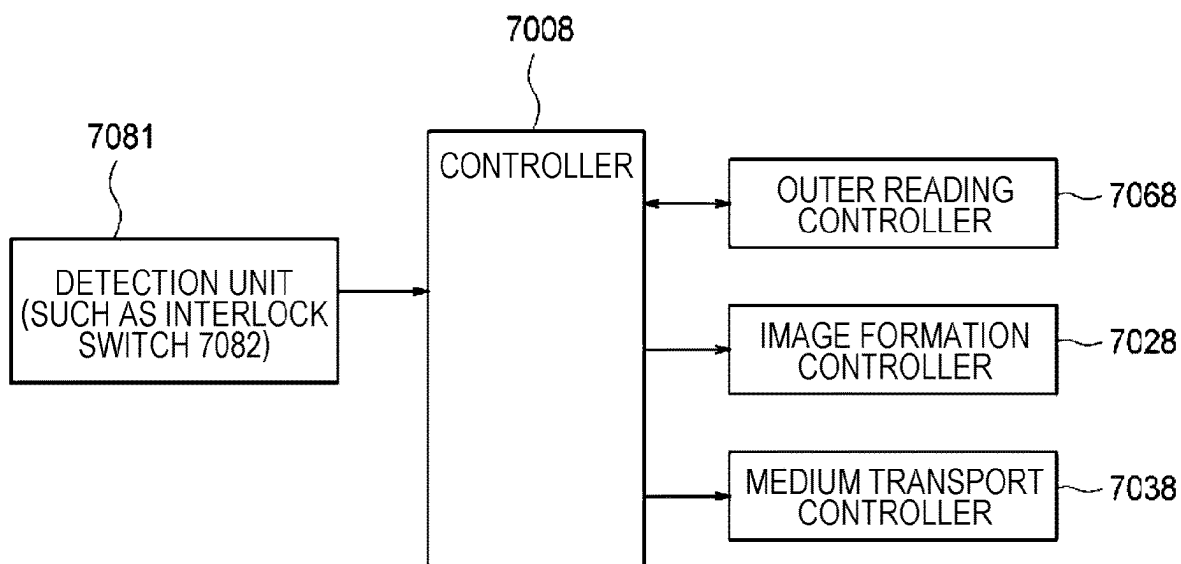


FIG. 7-7

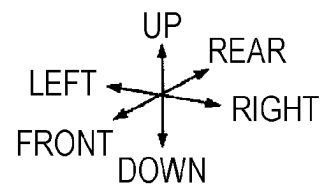
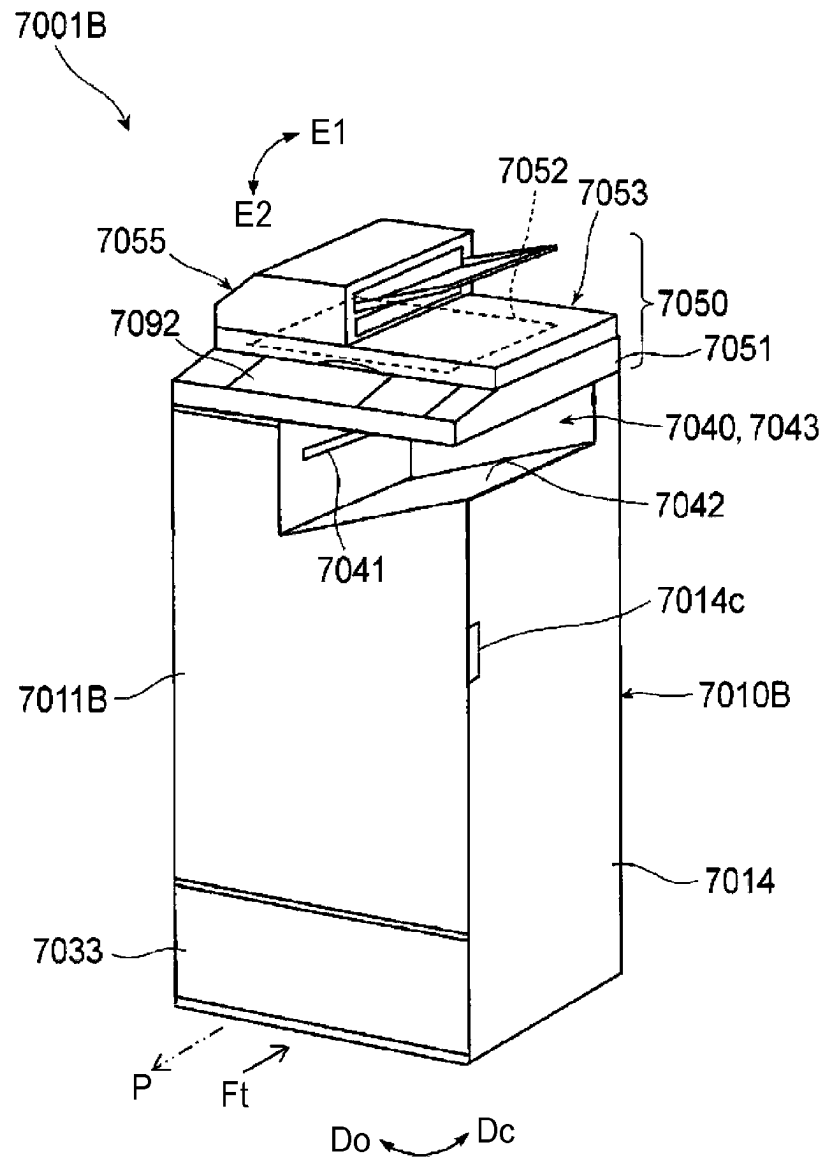


FIG. 7-8

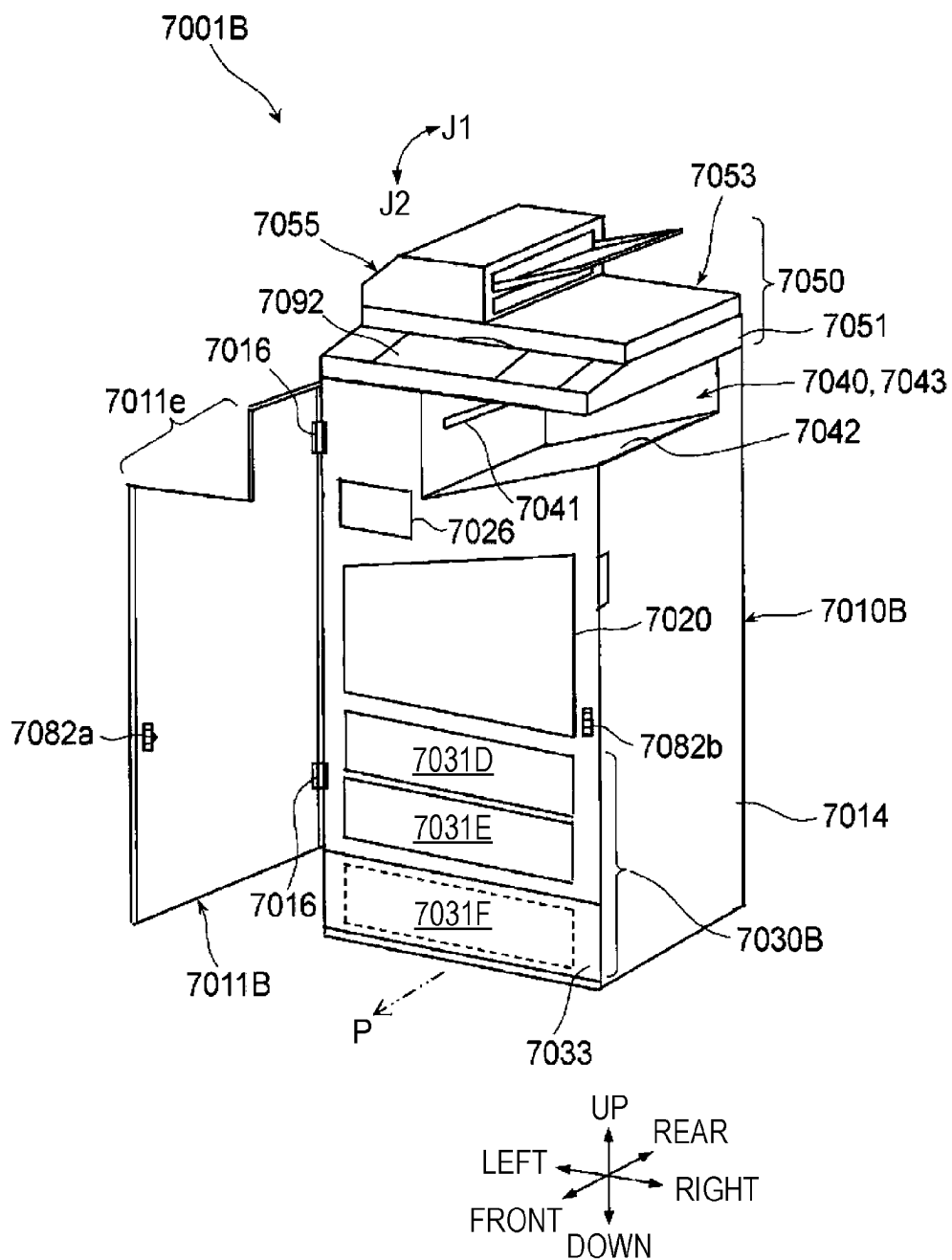


FIG. 7-9

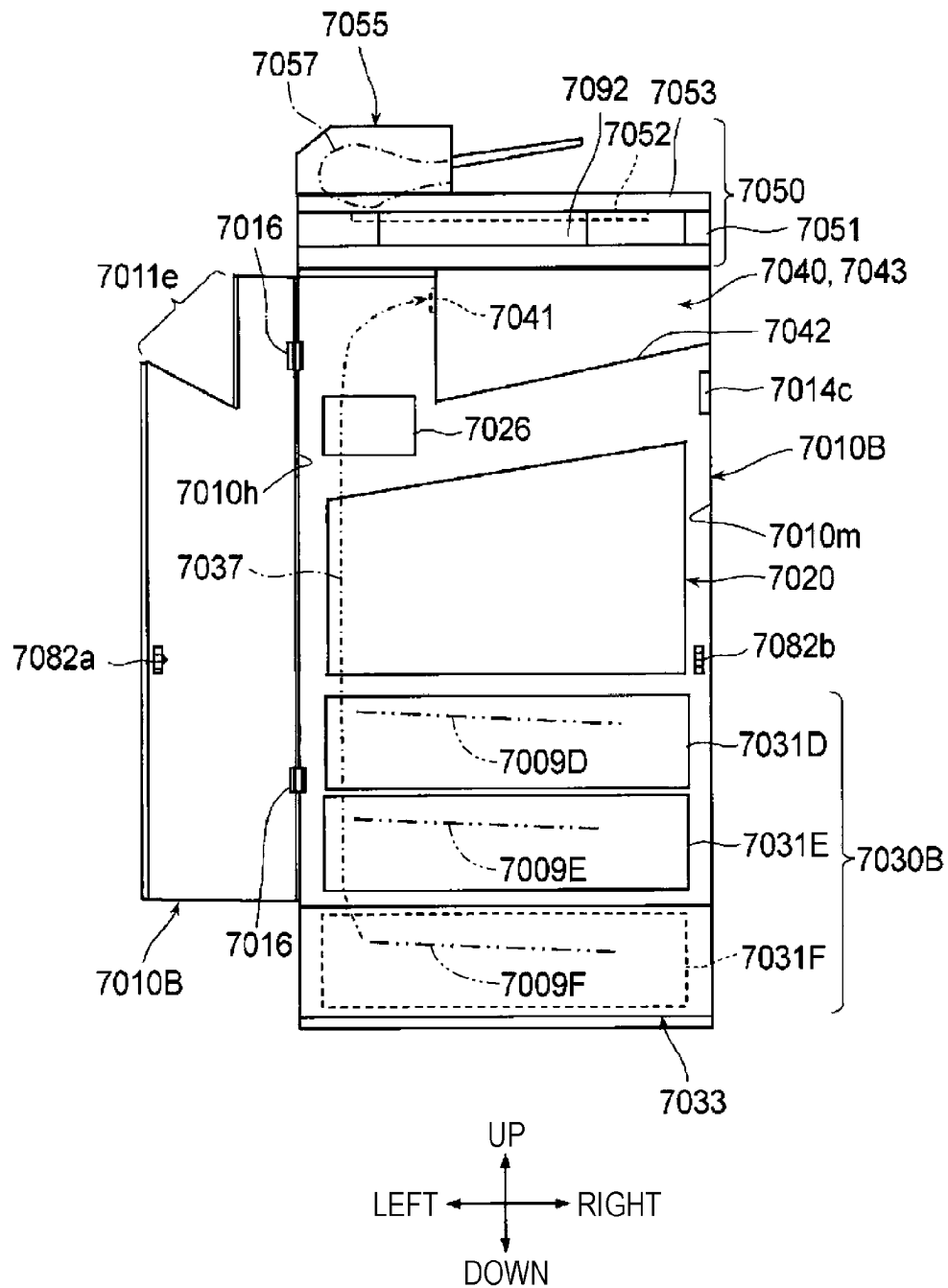


FIG. 7-10

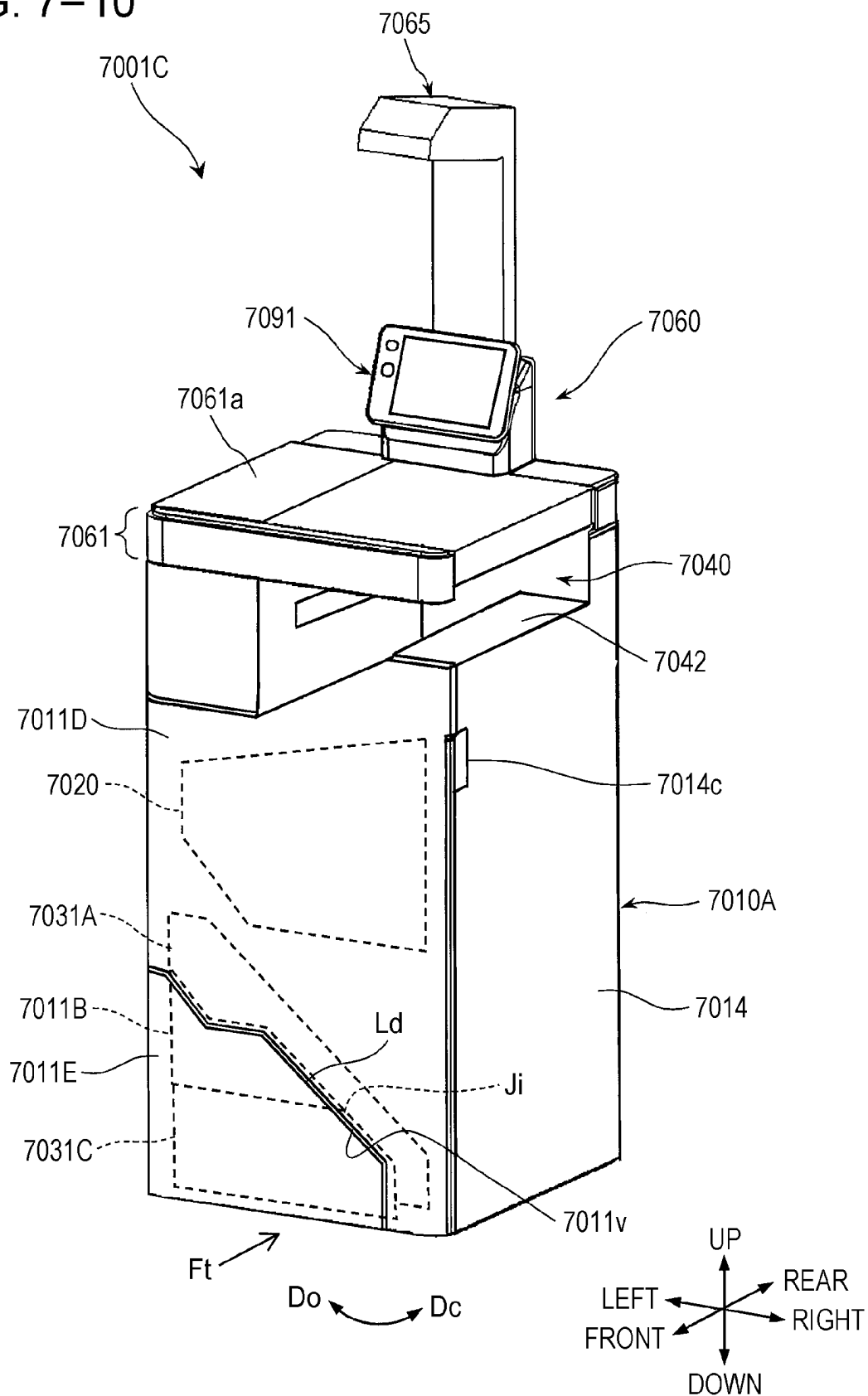


FIG. 7-11

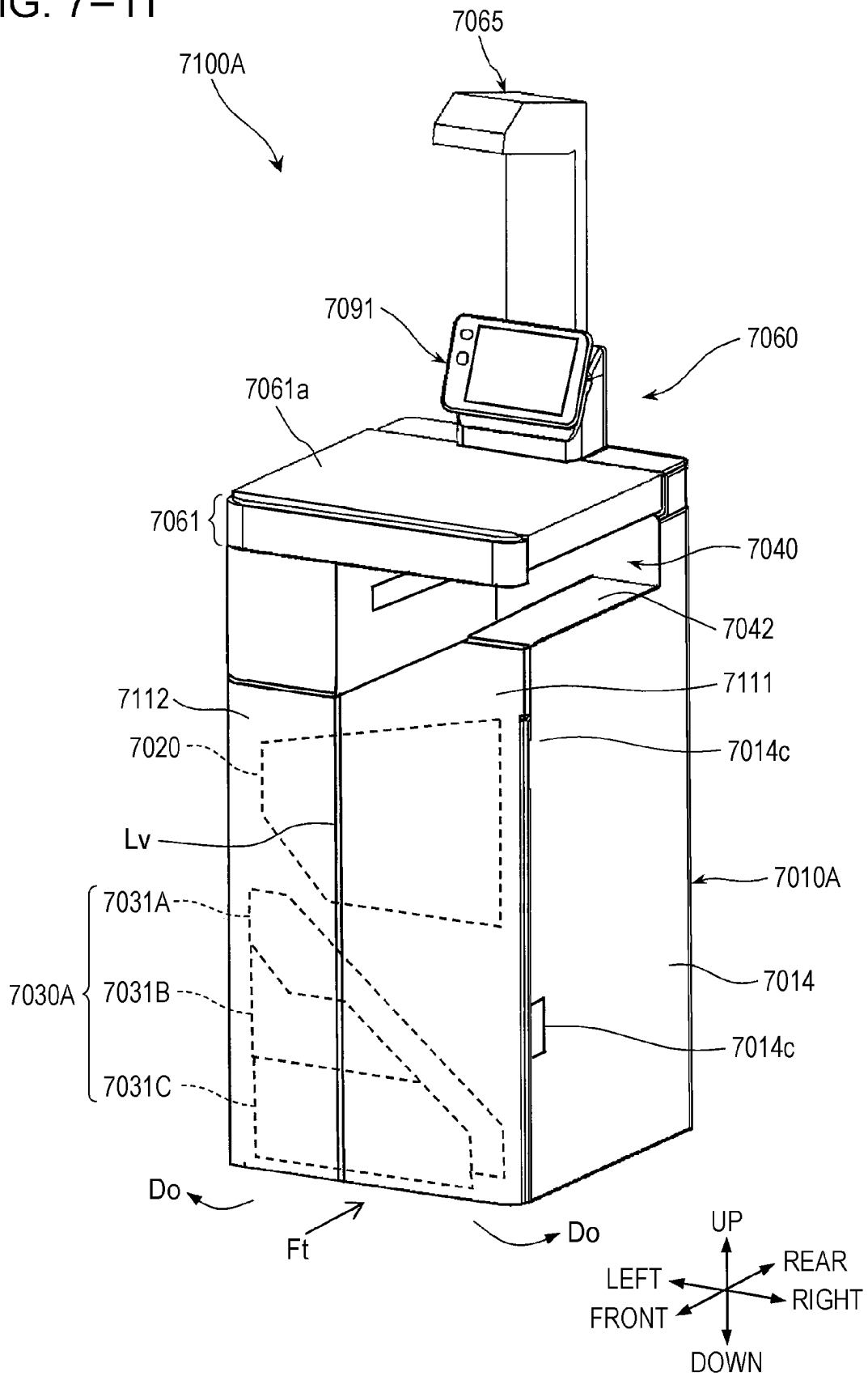


FIG. 7-12

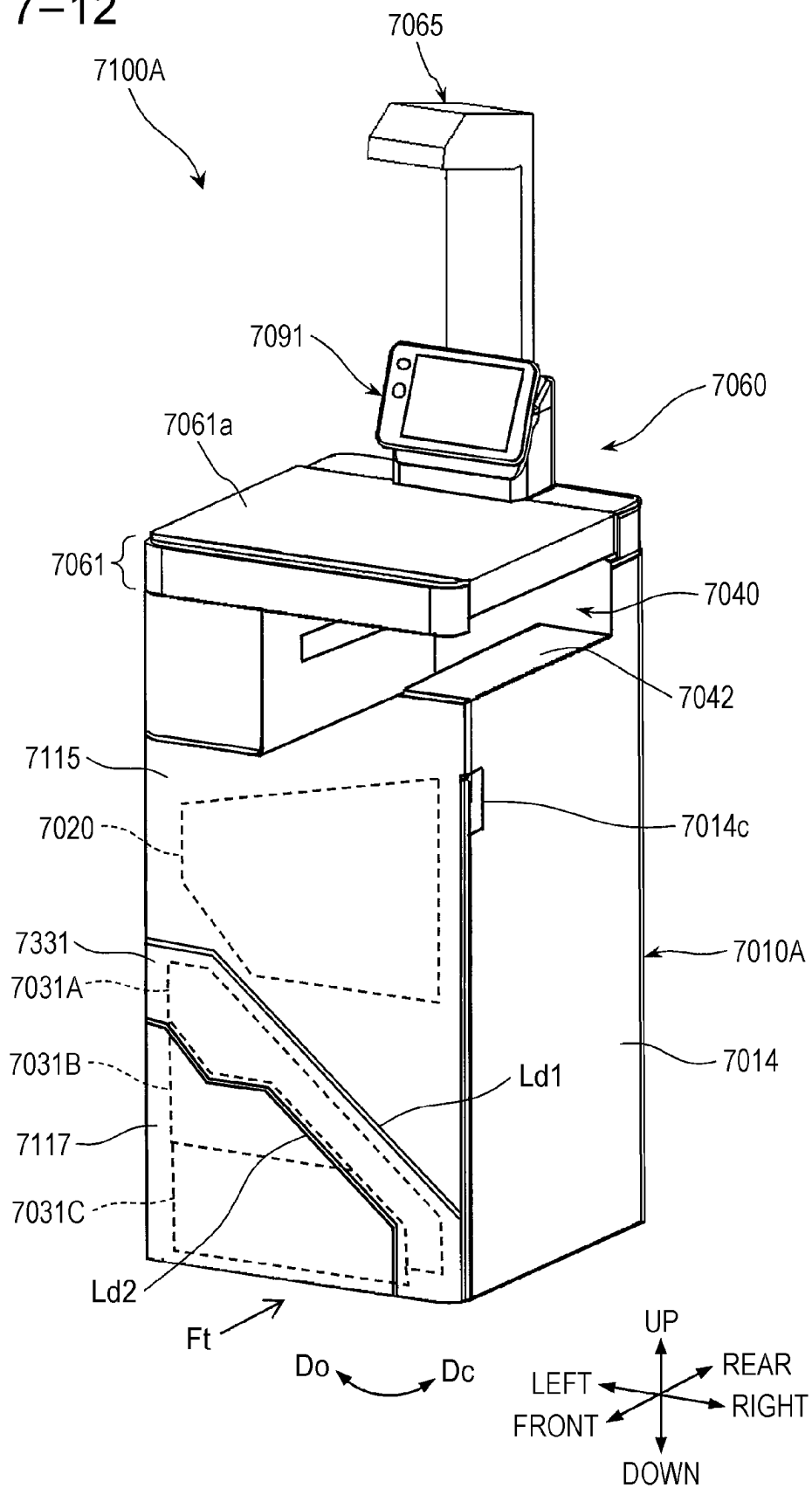


FIG. 8-1

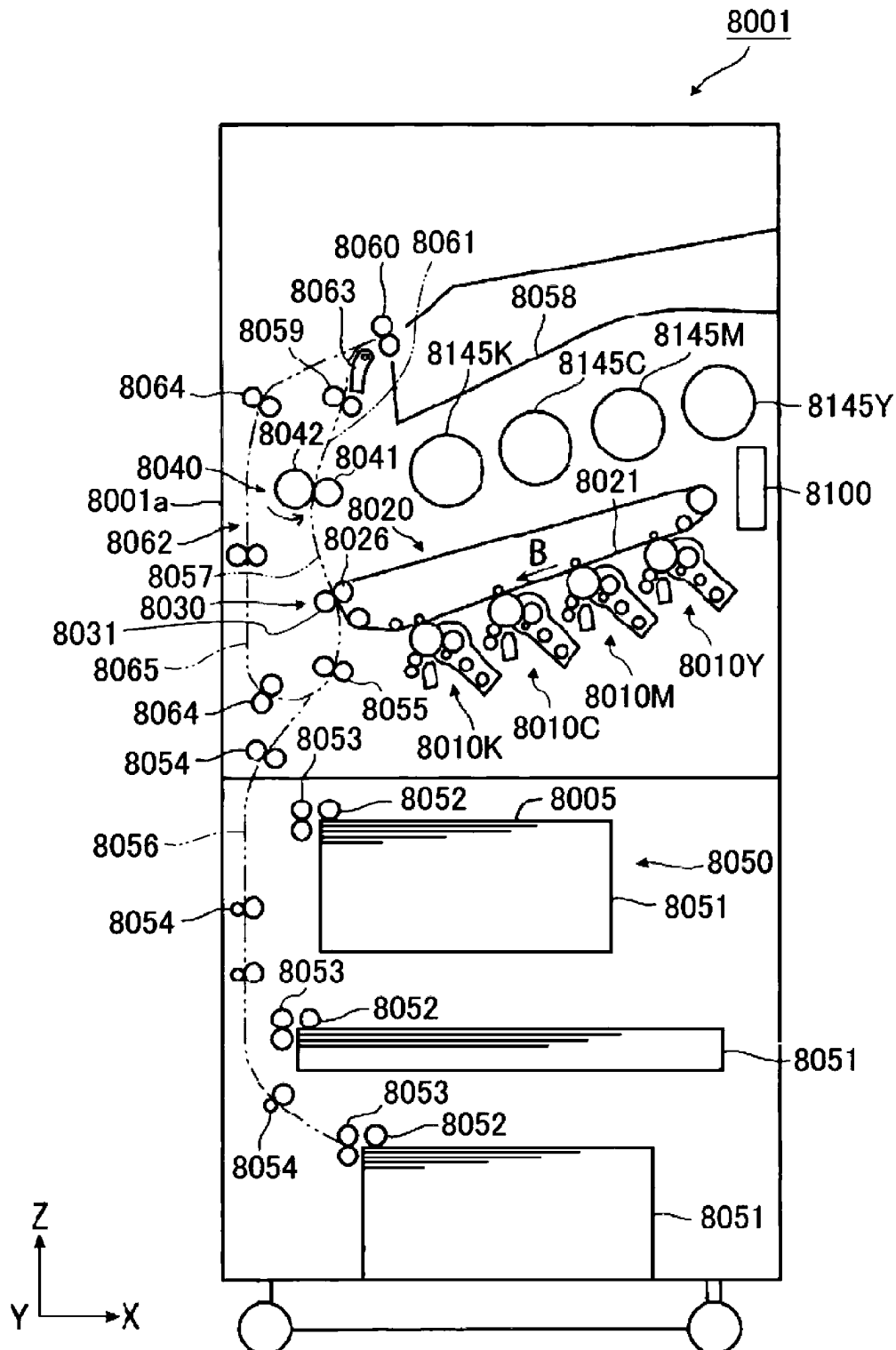


FIG. 8-2

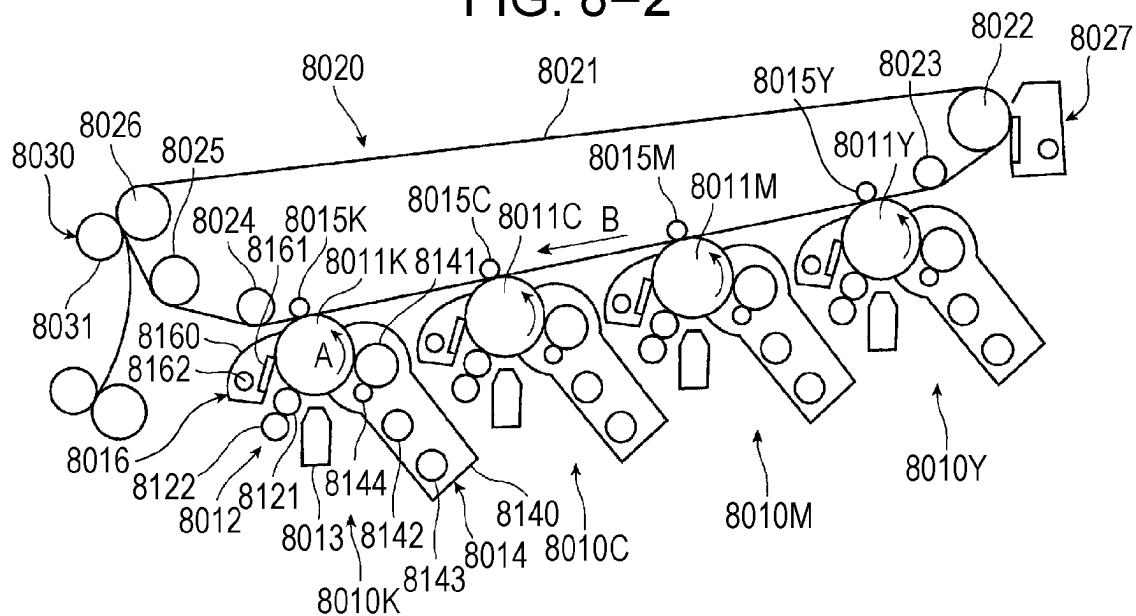


FIG. 8-3

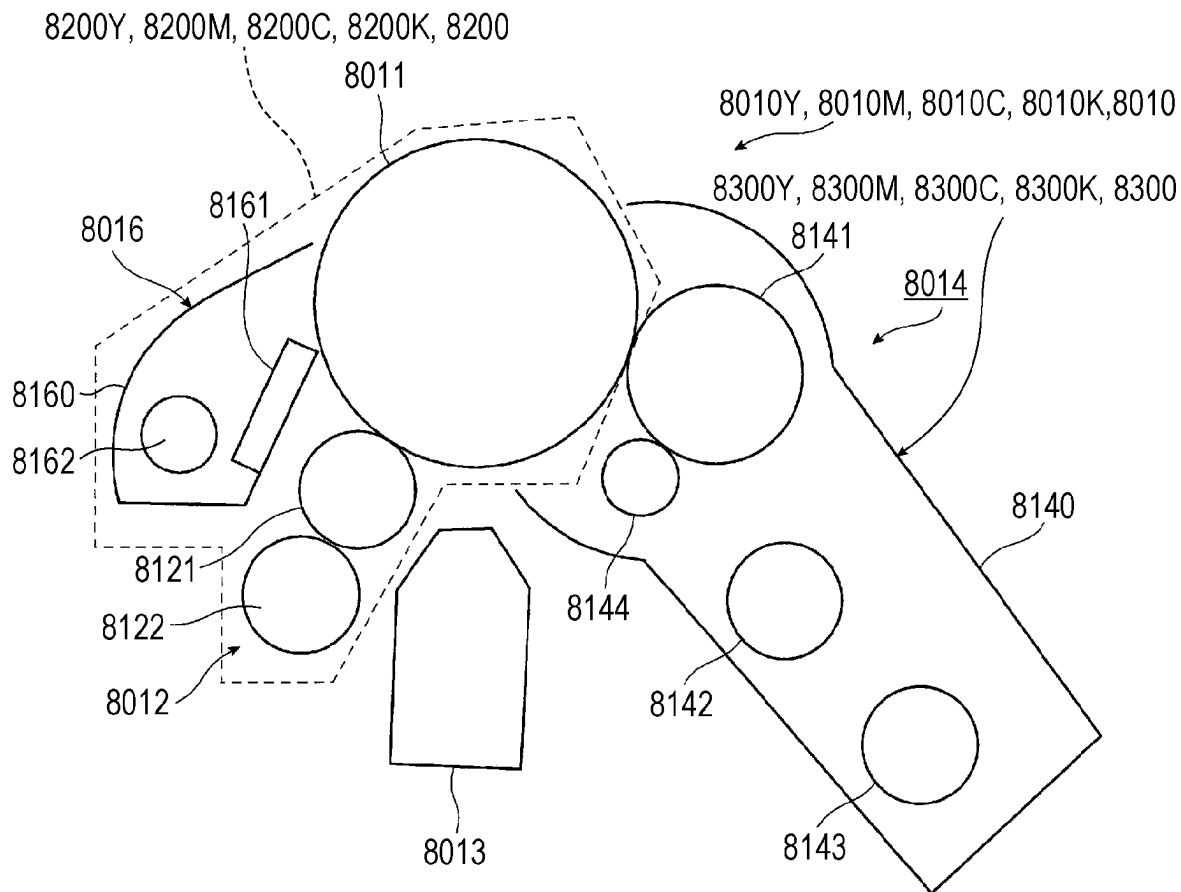


FIG. 8-4

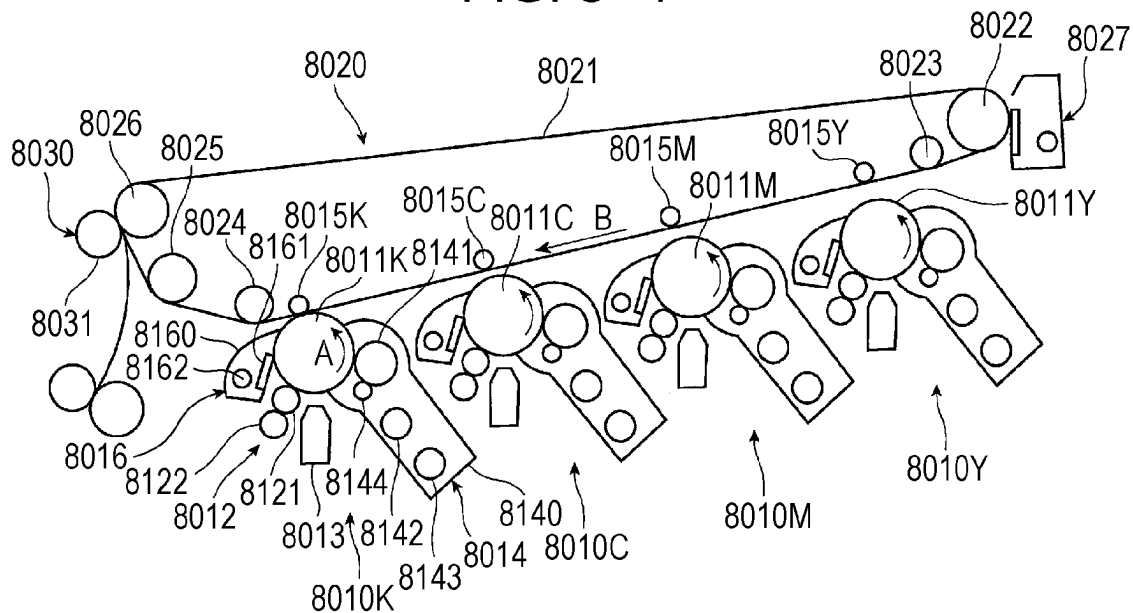


FIG. 8-5

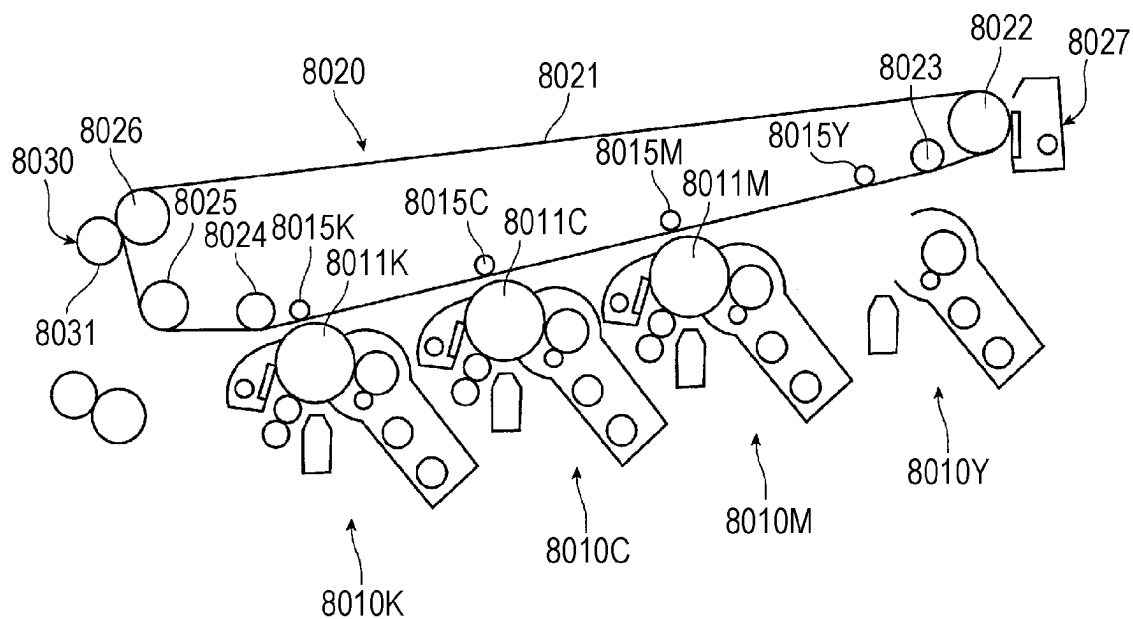


FIG. 8-6

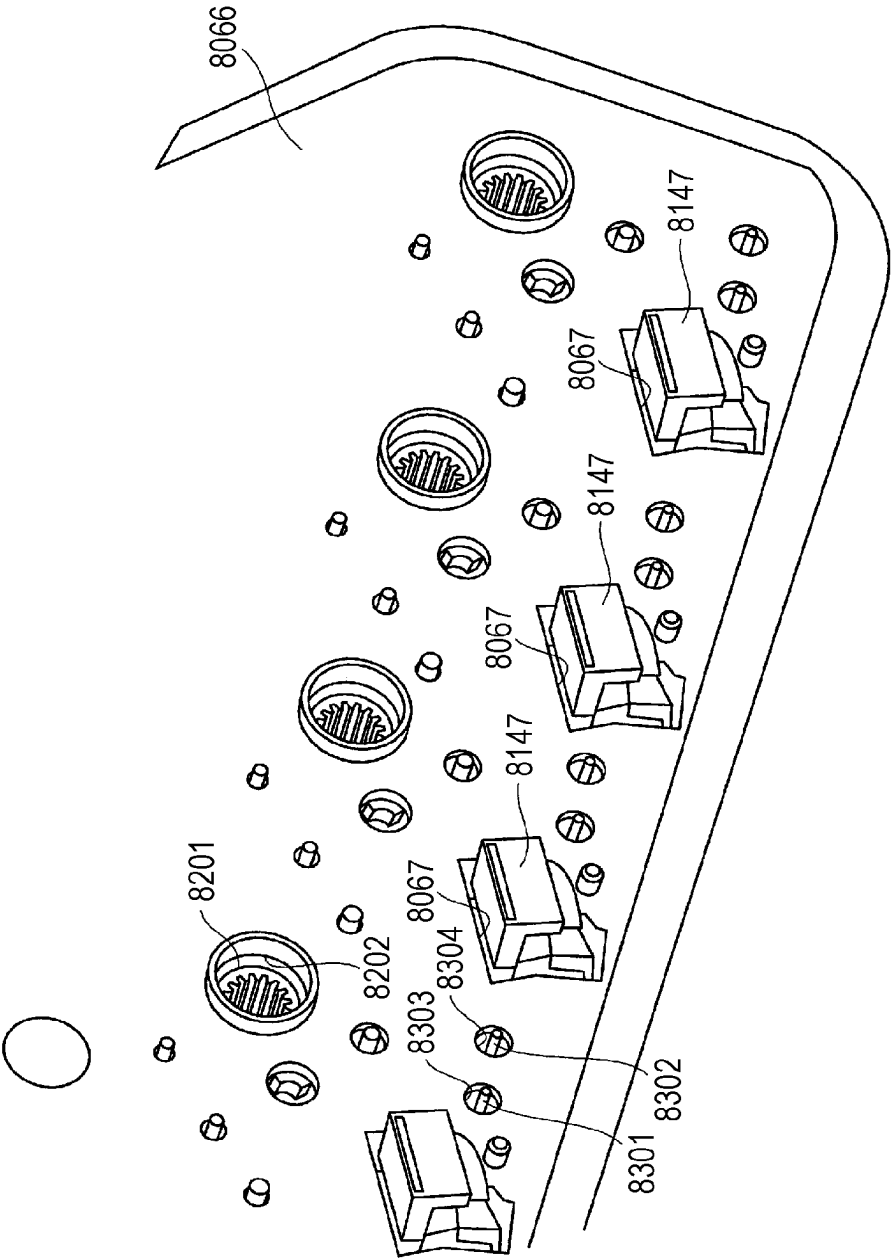


FIG. 8-7

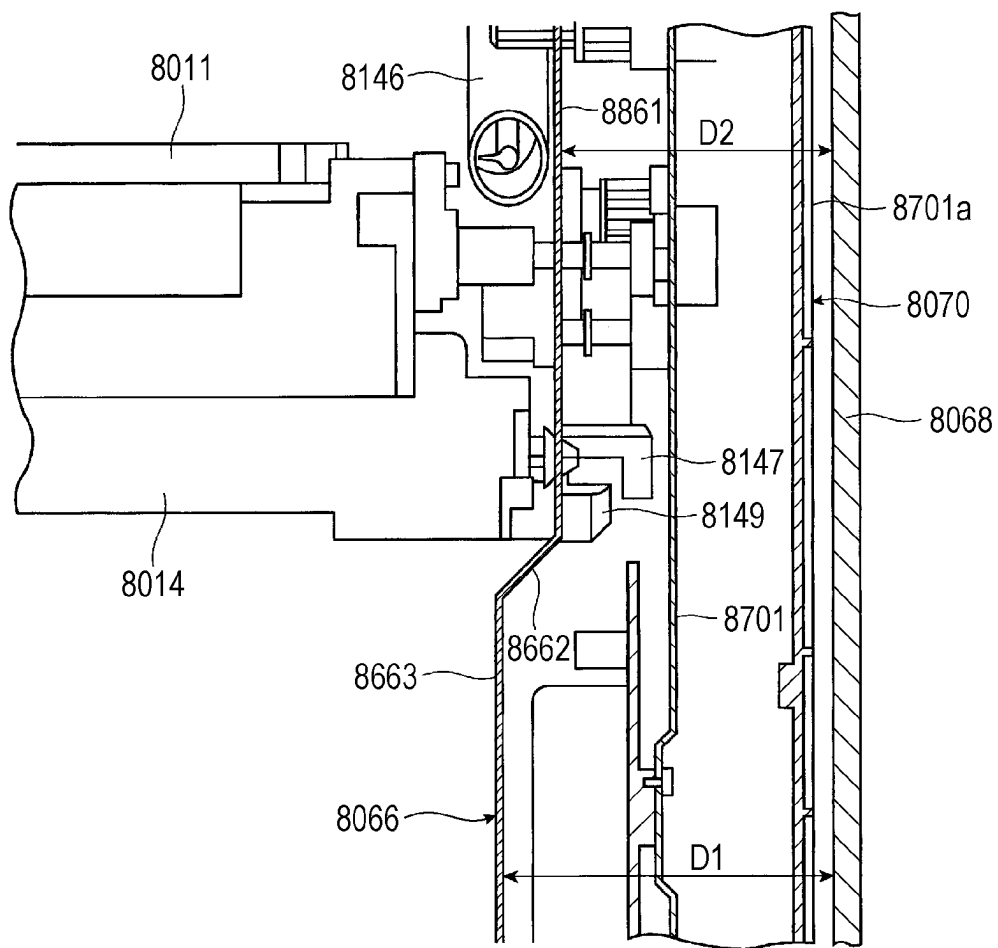


FIG. 8-8

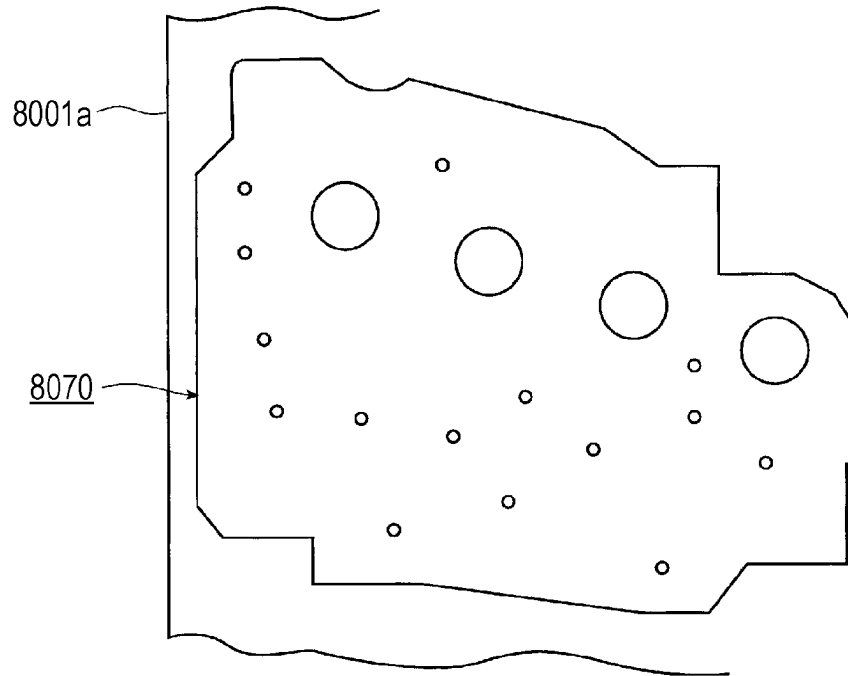


FIG. 8-9

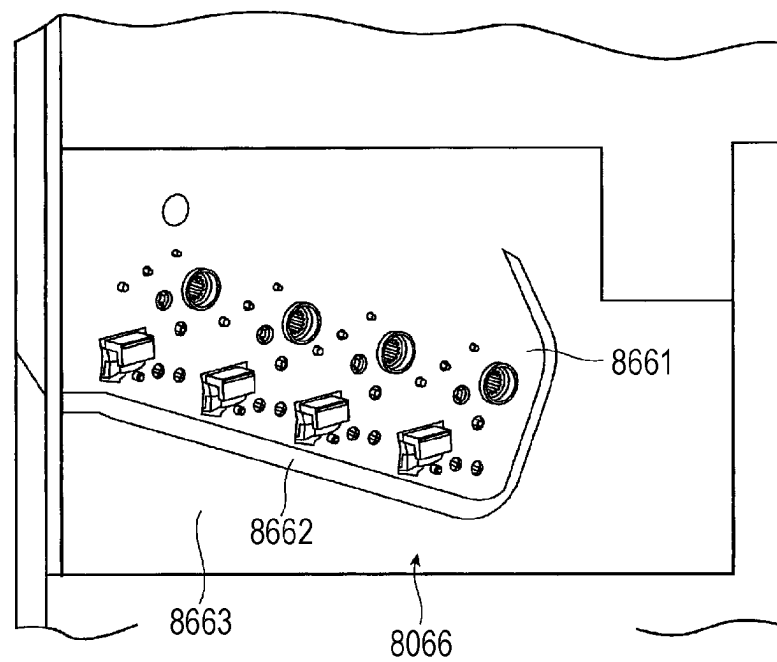


FIG. 8-10

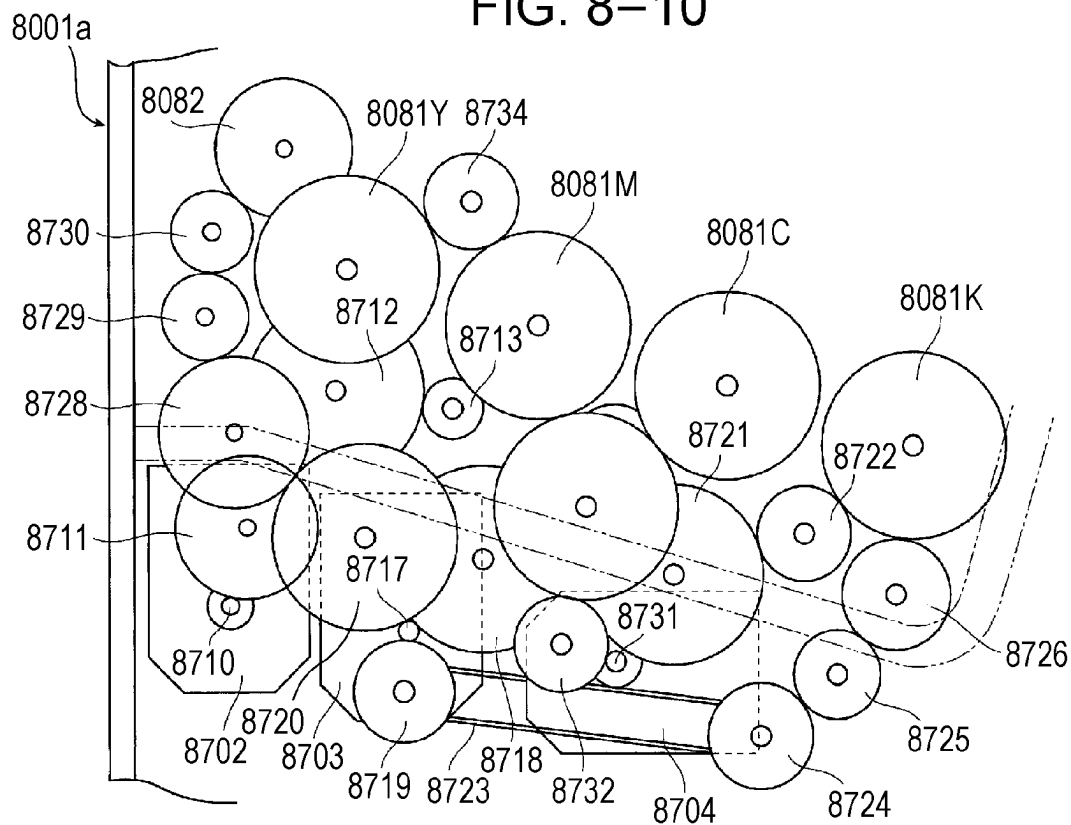


FIG. 8-11

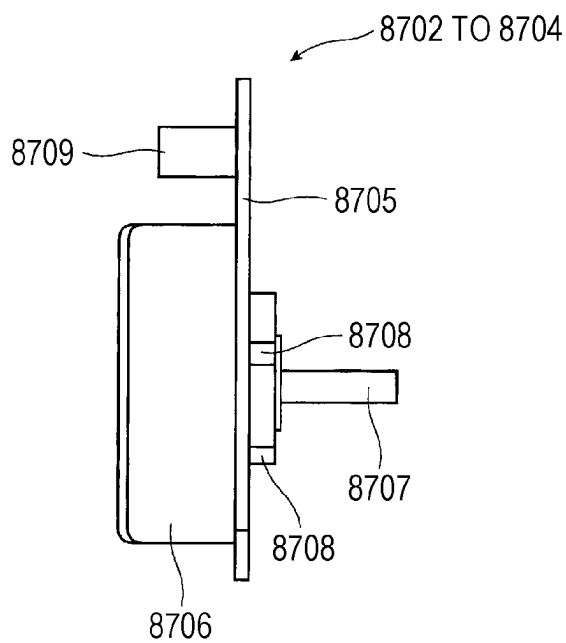


FIG. 8-12

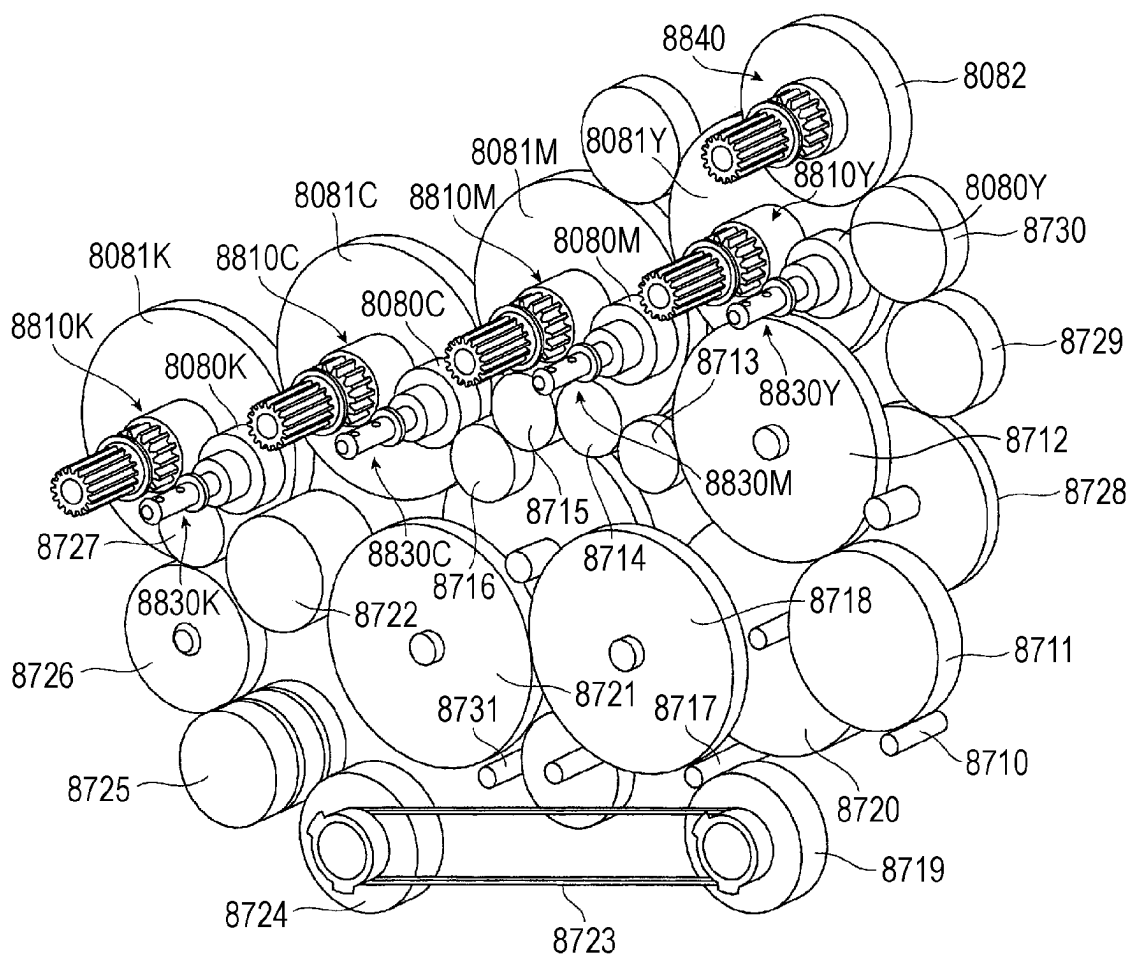


FIG. 8-13

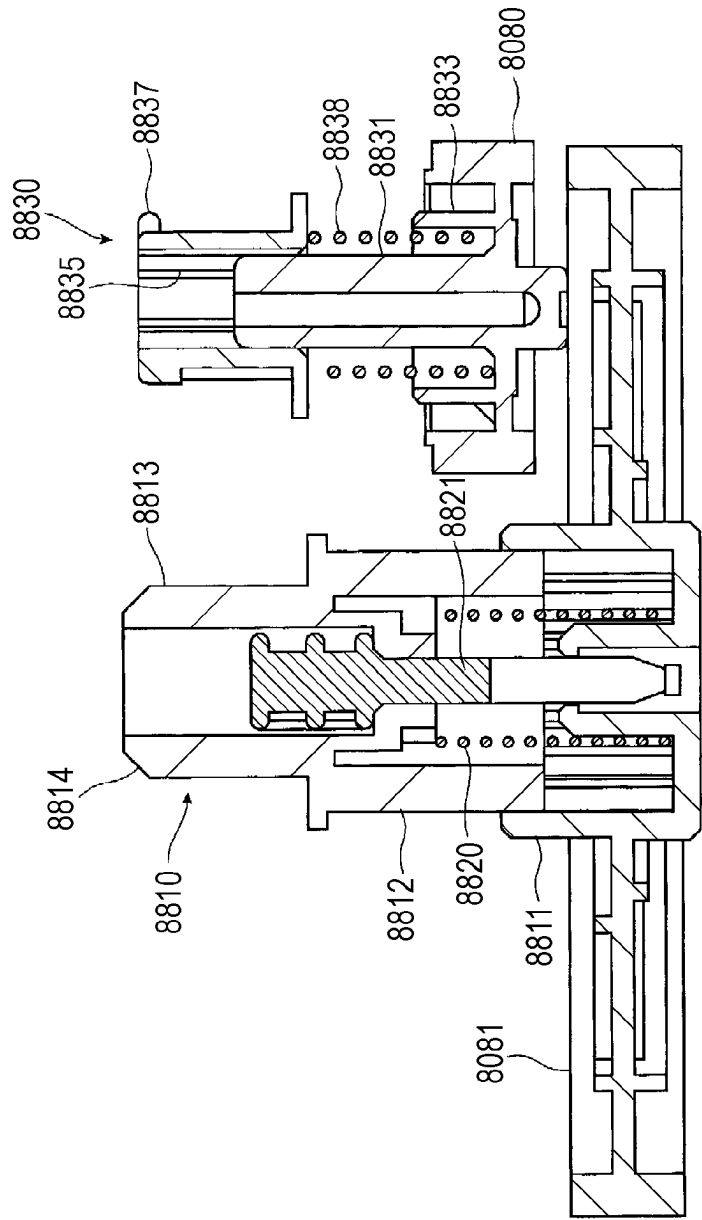


FIG. 8-14

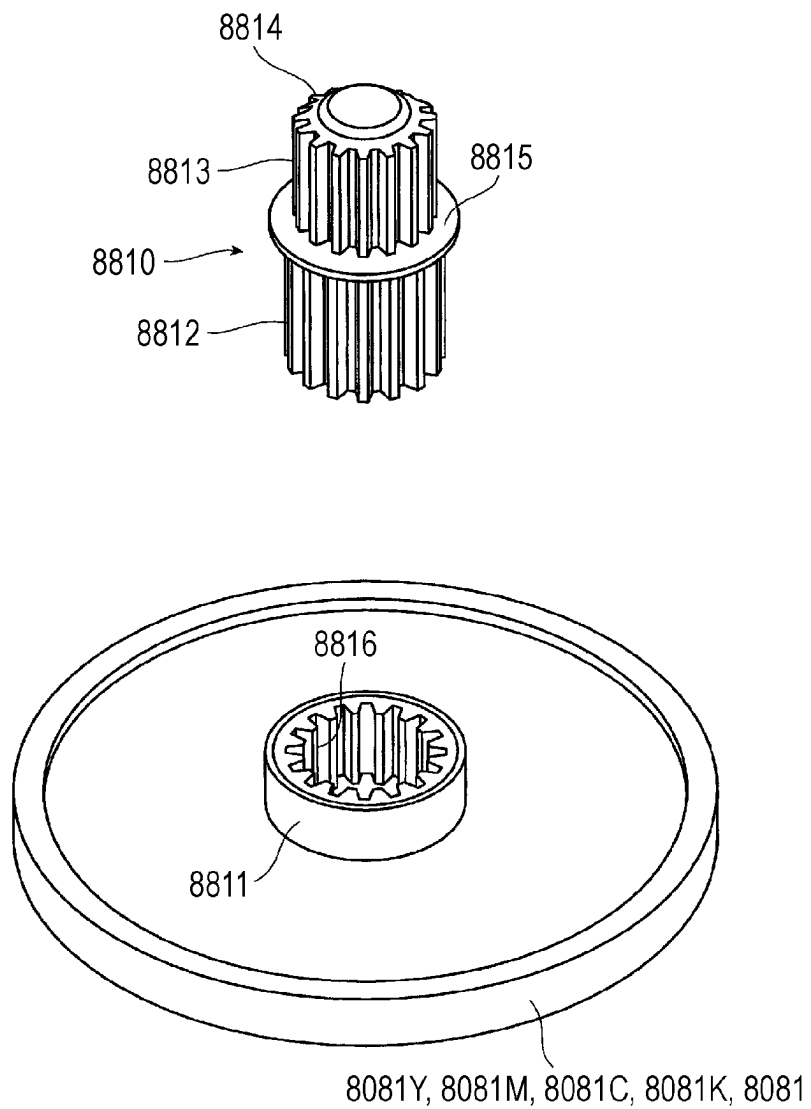


FIG. 8-15

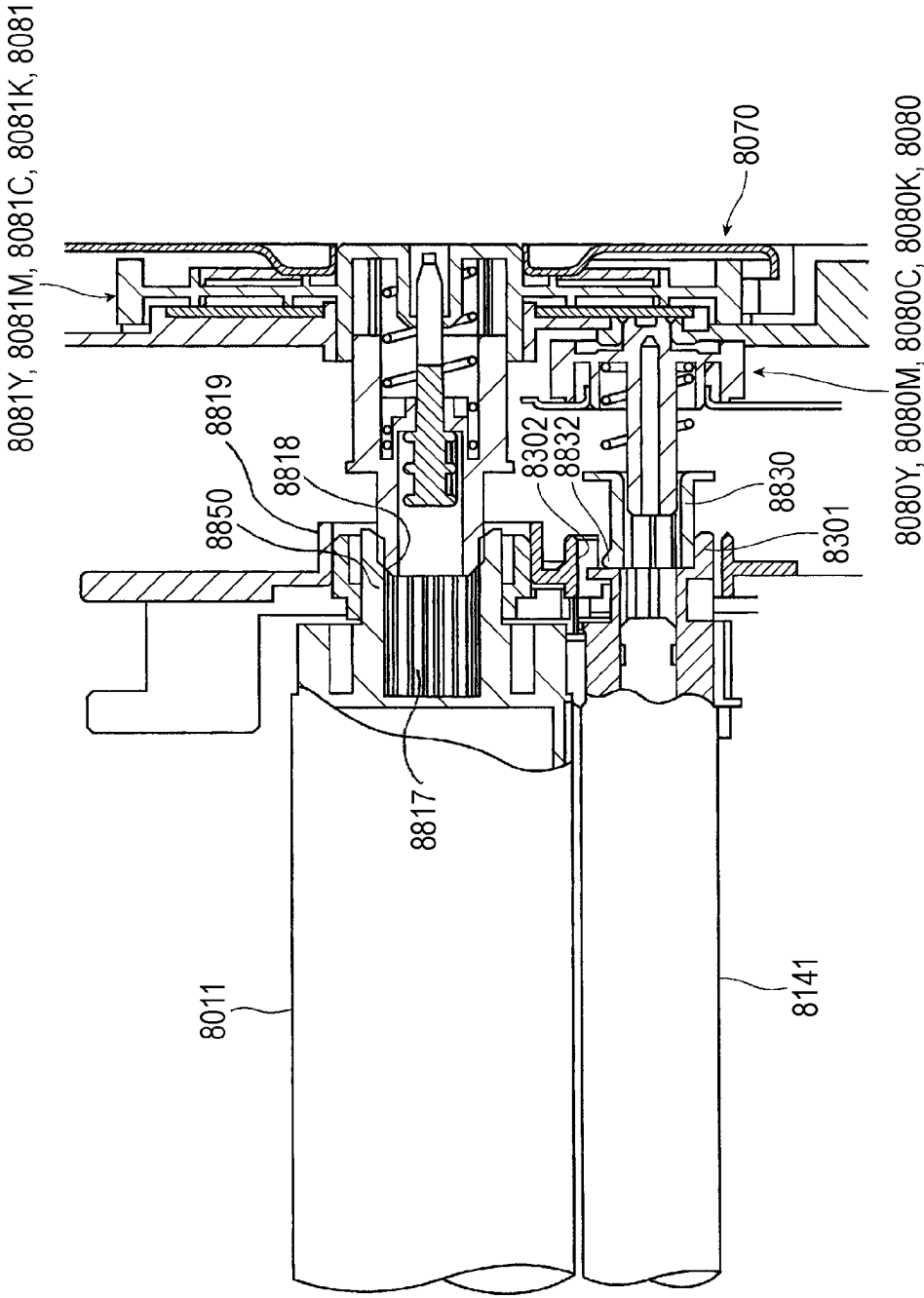
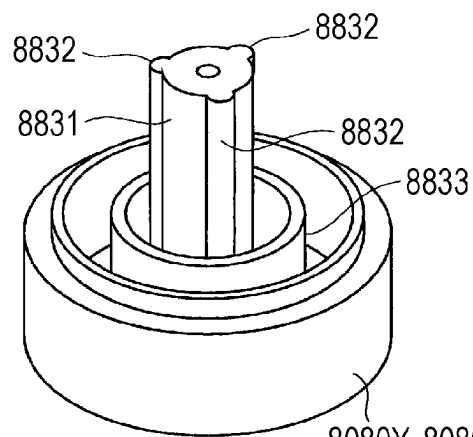
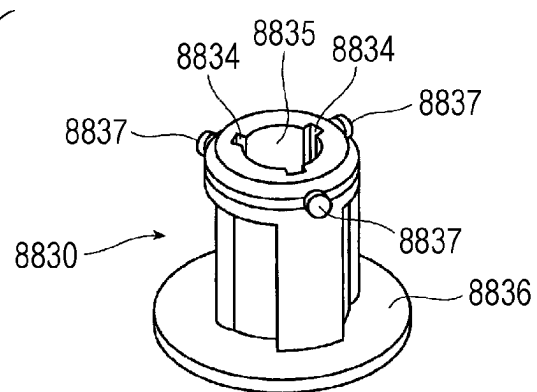
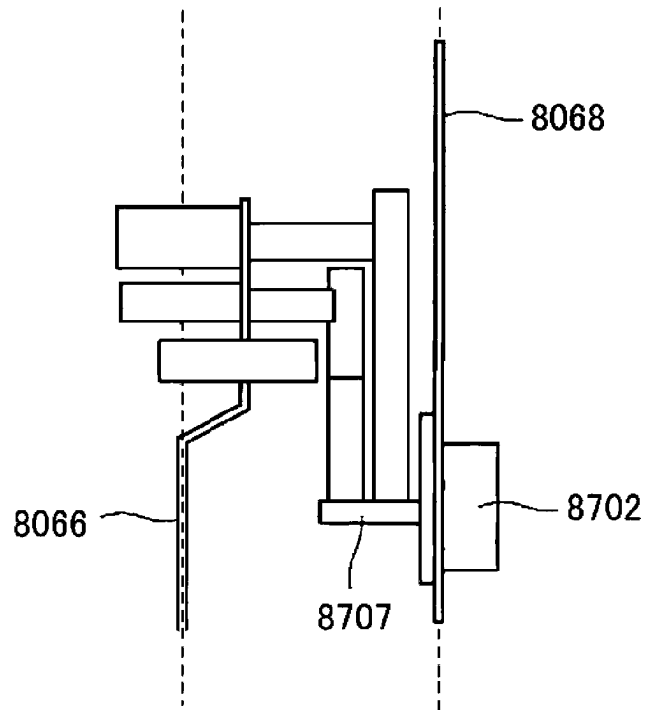


FIG. 8-16



8080Y, 8080M, 8080C, 8080K, 8080

FIG. 8-17



**DOCUMENT READING DEVICE AND
IMAGE FORMING APPARATUS****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2021-135289 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135290 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135424 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135483 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135487 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135489 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135490 filed Aug. 23, 2021, Japanese Patent Application No. 2021-135491 filed Aug. 23, 2021, and Japanese Patent Application No. 2022-003293 filed Jan. 12, 2022.

BACKGROUND**(i) Technical Field**

The present disclosure relates to a document reading device and an image forming apparatus.

(ii) Related Art

Japanese Unexamined Patent Application Publication No. 2021-68980 discloses an image reading device that includes a document placing table that includes an image reader and an auto document feeder that is rotatably supported by the document placing table such that the auto document feeder is rotatable between a closed position at which the auto document feeder covers an upper surface of the document placing table and an opened position at which the auto document feeder uncovers the upper surface of the document placing table. The auto document feeder includes a first tray that is a document tray that supports a document sheet before reading or a document discharge tray that supports the document sheet after reading, a second tray that is disposed below the first tray with a space interposed therebetween and that is the other tray of the document tray or the document discharge tray, a conveyer that conveys the document sheet from the document tray to the document discharge tray, a front wall that is secured to a front side of the first tray and that has a cutout part for opening a front part of the space in a width direction perpendicular to a front-rear direction and an up-down direction, and a hand grip that is located on an upper periphery of the cutout part of the front wall and that is used when the auto document feeder is lifted from the closed position toward the opened position.

SUMMARY

A document reading device that reads a document, an image of which is captured by an image capturing member that captures the image of the document, from above a placement portion on which the document is placed, precision with which the document is read is low if there is a gap line that is a line in a gap between members within the capturing range of the image capturing member and outside the placed document.

Aspects of non-limiting embodiments of the present disclosure relate to a document reading device that reads a document, an image of which is captured by an image

capturing member that captures the image of the document, from above a placement portion on which the document is placed and that improves precision with which the document is read, unlike the case where there is a gap line that is a line in a gap between members within the capturing range of the image capturing member and outside the placed document.

Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

According to an aspect of the present disclosure, there is provided a document reading device including: an image capturing member that captures an image of a document; a placement portion that is disposed below the image capturing member and that has a boundary line between a plurality of surfaces at a position at which the boundary line is covered by the document when the document has a maximum size that the image capturing member is capable of imaging and the document is placed on the placement portion; and a document reading member that reads the document that is placed on the placement portion.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the present disclosure will be described in detail based on the following figures, wherein:

FIG. 1-1 is a perspective view of an image forming apparatus according to a first exemplary embodiment;

FIG. 1-2 schematically illustrates a front view of the configuration of the image forming apparatus according to the first exemplary embodiment;

FIG. 1-3 schematically illustrates a front view of the configuration of a document reading member according to the first exemplary embodiment;

FIG. 1-4 schematically illustrates a front view of a state in which a loading portion of the document reading member according to the first exemplary embodiment is uncovered;

FIG. 1-5 is a front sectional view of a placing portion according to the first exemplary embodiment;

FIG. 1-6 is a plan view of the placing portion according to the first exemplary embodiment viewed from above;

FIG. 1-7 is a block diagram illustrating the hardware configuration of a document reading device according to the first exemplary embodiment;

FIG. 1-8 is a front sectional view of a placing portion according to a first comparative exemplary embodiment relative to the first exemplary embodiment;

FIG. 1-9 is a front sectional view of a placing portion according to a second comparative exemplary embodiment relative to the first exemplary embodiment;

FIG. 1-10 is a front sectional view of a placing portion according to a third comparative exemplary embodiment relative to the first exemplary embodiment;

FIG. 1-11 is a front sectional view of a placing portion according to a modification to the first exemplary embodiment;

FIG. 1-12 is a front sectional view of a placing portion according to a modification to the first exemplary embodiment;

FIG. 2-1 is a perspective view of an image forming apparatus according to a second exemplary embodiment;

FIG. 2-2 schematically illustrates a front view of the configuration of the image forming apparatus according to the second exemplary embodiment;

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FIG. 2-3 schematically illustrates a front view of the configuration of a document reading device according to the second exemplary embodiment;

FIG. 2-4 schematically illustrates a front view of a state in which a loading portion of the document reading device according to the second exemplary embodiment is uncovered;

FIG. 2-5 is a plan view of a placement portion according to the second exemplary embodiment viewed from above;

FIG. 2-6 is a block diagram illustrating the hardware configuration of the document reading device according to the second exemplary embodiment;

FIG. 3-1 schematically illustrates an image reading apparatus according to a third exemplary embodiment;

FIG. 3-2 is a perspective view of the image reading apparatus according to the third exemplary embodiment;

FIG. 3-3 is a perspective view of the configuration illustrated in FIG. 3-2 with a document table moved to an open position;

FIG. 3-4 is a sectional view of an upper portion of the image reading apparatus according to the third exemplary embodiment;

FIG. 3-5 is a sectional view of the configuration illustrated in FIG. 3-4 with the document table moved to the open position;

FIG. 3-6 is a plan view of the image reading apparatus according to the third exemplary embodiment;

FIG. 3-7 is a plan view of the configuration illustrated in FIG. 3-6 with the document table moved to the open position;

FIG. 3-8 is a sectional view of a movement mechanism in the image reading apparatus according to the third exemplary embodiment;

FIG. 3-9 is a sectional view of the configuration illustrated in FIG. 3-8 with the document table moved to the open position;

FIG. 4-1 schematically illustrates the configuration of an image forming apparatus according to a fourth exemplary embodiment;

FIG. 4-2 illustrates the configuration of an image forming member of the image forming apparatus according to the fourth exemplary embodiment;

FIG. 4-3 illustrates the configuration of a container apparatus according to the fourth exemplary embodiment;

FIG. 4-4A and FIG. 4-4B illustrate perspective views of the container apparatus according to the fourth exemplary embodiment with a container unit installed in an apparatus body and with the container unit separated therefrom;

FIG. 4-5A and FIG. 4-5B illustrate perspective views of the image forming apparatus according to the fourth exemplary embodiment with a covering closed and with the covering opened;

FIG. 4-6A and FIG. 4-6B illustrate perspective views of a container apparatus according to a first comparative exemplary embodiment relative to the fourth exemplary embodiment with a container unit installed in an apparatus body and with the container unit separated therefrom;

FIG. 4-7 schematically illustrates the configuration of the image forming apparatus according to a second comparative exemplary embodiment relative to the fourth exemplary embodiment;

FIG. 4-8 illustrates the configuration of an image forming apparatus according to a modification to the fourth exemplary embodiment;

FIG. 5-1 illustrates the configuration of an image forming apparatus according to a fifth exemplary embodiment;

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FIG. 5-2 illustrates the configuration of an image forming member of the image forming apparatus according to the fifth exemplary embodiment;

FIG. 5-3A and FIG. 5-3B illustrate perspective views of a sheet containing apparatus and the image forming apparatus according to the fifth exemplary embodiment with a covering portion opened and with the covering portion closed;

FIG. 5-4A and FIG. 5-4B illustrate front views of the sheet containing apparatus and the image forming apparatus according to the fifth exemplary embodiment with the covering portion opened and with the covering portion closed;

FIG. 5-5 is a perspective view of a container unit of the sheet containing apparatus according to the fifth exemplary embodiment;

FIG. 5-6A and FIG. 5-6B illustrate perspective views of the sheet containing apparatus and the image forming apparatus according to the fifth exemplary embodiment with the container unit installed and with the container unit removed;

FIG. 5-7A and FIG. 5-7B illustrate front views of the sheet containing apparatus according to the fifth exemplary embodiment with the container unit installed and with the container unit removed;

FIG. 5-8A and FIG. 5-8B illustrate perspective views of the sheet containing apparatus and the image forming apparatus according to the fifth exemplary embodiment with another covering portion opened and with the other covering portion closed;

FIG. 5-9 is a plan view of the sheet containing apparatus and the image forming apparatus according to the fifth exemplary embodiment and illustrates trajectories of the covering portion and the other covering portion;

FIG. 6-1A is a side view of an image forming apparatus according to a sixth exemplary embodiment;

FIG. 6-1B is a front view of the image forming apparatus in FIG. 6-1A;

FIG. 6-2 is a block diagram illustrating the hardware configuration of a control device that is used for the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-3 is a block diagram illustrating the functional configuration of the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-4 illustrates a placement portion in the image forming apparatus according to the sixth exemplary embodiment viewed from above;

FIG. 6-5 illustrates a state in which a user operates the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-6A and FIG. 6-6B illustrate diagrams for describing reflected light distribution of a radiation device in the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-7 is a diagram for describing the amount of light that enters an image capturing device in the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-8A and FIG. 6-8B illustrate diagrams for describing the movable ranges of light sources in the image forming apparatus according to the sixth exemplary embodiment;

FIG. 6-9 illustrates a placement portion in an image forming apparatus according to another sixth exemplary embodiment viewed from above;

FIG. 6-10 illustrates the image forming apparatus according to the other sixth exemplary embodiment;

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FIG. 7-1 is a perspective view of an image forming apparatus according to a first aspect of a seventh exemplary embodiment;

FIG. 7-2 is a perspective view of the image forming apparatus in FIG. 7-1 with a front opening-closing covering opened;

FIG. 7-3 is a schematic front view of the image forming apparatus in FIG. 7-2;

FIG. 7-4 is a perspective view of the image forming apparatus in FIG. 7-1 viewed in another direction;

FIG. 7-5 is a perspective view of the image forming apparatus in FIG. 7-4 with a side opening-closing covering opened;

FIG. 7-6 is a block diagram illustrating the configuration of a control system in FIG. 7-1;

FIG. 7-7 is a perspective view of an image forming apparatus according to a second aspect of the seventh exemplary embodiment;

FIG. 7-8 is a perspective view of the image forming apparatus in FIG. 7-7 with a front opening-closing covering opened;

FIG. 7-9 is a schematic front view of the image forming apparatus in FIG. 7-7;

FIG. 7-10 is a perspective view of an image forming apparatus in a reference example;

FIG. 7-11 is a perspective view of an image forming apparatus in a first comparative example;

FIG. 7-12 is a perspective view of an image forming apparatus in a second comparative example;

FIG. 8-1 illustrates the overall configuration of an image forming apparatus that uses a drive device according to an eighth exemplary embodiment of the disclosure;

FIG. 8-2 illustrates the configuration of an imaging device of the image forming apparatus according to the eighth exemplary embodiment of the disclosure;

FIG. 8-3 illustrates the configuration of an image formation unit of the image forming apparatus according to the eighth exemplary embodiment of the disclosure;

FIG. 8-4 illustrates the configuration of the imaging device of the image forming apparatus according to the eighth exemplary embodiment of the disclosure;

FIG. 8-5 illustrates the configuration of the imaging device of the image forming apparatus according to the eighth exemplary embodiment of the disclosure;

FIG. 8-6 is a perspective view of the configuration of an inner frame of the image forming apparatus according to the eighth exemplary embodiment of the disclosure;

FIG. 8-7 is a sectional view of the configuration of the drive device according to the eighth exemplary embodiment of the disclosure;

FIG. 8-8 is a back view of the configuration of the drive device according to the eighth exemplary embodiment of the disclosure;

FIG. 8-9 is a perspective view of an exterior covering of the image forming apparatus according to the eighth exemplary embodiment of the disclosure and a back surface with the drive device removed;

FIG. 8-10 illustrates the configuration of the drive device according to the eighth exemplary embodiment of the disclosure;

FIG. 8-11 is a side view of the configuration of a drive motor;

FIG. 8-12 is a perspective view of the configuration of the drive device according to the eighth exemplary embodiment of the disclosure;

FIG. 8-13 is a sectional view of the configuration of a coupling;

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FIG. 8-14 is an exploded perspective view of the coupling;

FIG. 8-15 is a sectional view of the configuration of the drive device according to the eighth exemplary embodiment of the disclosure;

FIG. 8-16 is an exploded perspective view of a coupling; and

FIG. 8-17 schematically illustrates an existing drive device.

DETAILED DESCRIPTION

First Exemplary Embodiment

An image forming apparatus **100** according to a first exemplary embodiment will be described with reference to the drawings.

In the following description, a vertical direction of the apparatus (a vertical direction), a width direction of the apparatus (a horizontal direction), and a depth direction of the apparatus (a horizontal direction) are respectively described as a H direction, a W direction, and a D direction in a front view of the image forming apparatus **100** from a position at which a user (not illustrated) stands. In the case where it is necessary to distinguish between one direction and the other direction of the vertical direction of the apparatus, the width direction of the apparatus, and the depth direction of the apparatus, an upward direction is described as a +H direction, a downward direction is described as a -H direction, a right-hand direction is described as a +W direction, a left-hand direction is described as a -W direction, a rear direction is described as a -D direction, and a front direction is described as a +D direction in a front view of the image forming apparatus **100**.

Image Forming Apparatus

As illustrated in FIG. 1-1, the image forming apparatus **100** according to the present exemplary embodiment includes a document reading device **10**, a formation device **110**, a panel member **102**, and a controller (not illustrated). The document reading device **10** reads images that are formed on a front surface and a back surface of a document G (not illustrated) and converts the read images into electronic data. The formation device **110** is disposed below the document reading device **10**, forms a copy image of the document G, based on the electronic data of the images that is converted by the document reading device **10**, and records the copy image on a sheet material P as a medium. The panel member **102** is disposed between a camera unit **20** and a document table **30** described later and is a touch screen display that allows the user to perform a touch input operation and that displays information about the image forming apparatus **100** in the +D direction. The panel member **102** is an example of a display unit. The controller (not illustrated) controls the operation of components of the image forming apparatus **100**. The document reading device **10** will be described in detail later.

Formation Device

As illustrated in FIG. 1-2, the formation device **110** includes a container unit **120**, a transport unit **160**, a formation member **130**, and a discharge portion **170**.

The container unit **120** contains sheet materials P having different sizes and supplies the sheet materials P to the formation member **130**. The container unit **120** according to the present exemplary embodiment includes a first container unit **122** and a second container unit **124**. The first container unit **122** is capable of containing up to 500 sheet materials P having an A3 size or smaller. The second container unit

124 is capable of containing up to 1500 sheet materials **P** having an A4 size or smaller. That is, the size of the largest number of sheet materials **P** that are containable in the container unit **120** is the A4 size. The sheet materials **P** are selectively fed from the container unit **120** by using the controller (not illustrated).

The transport unit **160** is configured so as to include multiple roller members **162** and transports the sheet material **P** that is selectively fed by using the controller (not illustrated) toward the formation member **130**.

The formation member **130** has a function of forming an image on the sheet material **P** by using an electrophotographic system. The formation member **130** includes photoconductor member units **132** that form toner images in multiple colors, a transfer member **140** that transfers the toner images that are formed by the photoconductor member units **132** to the sheet material **P**, and a fixing portion **150** that fixes the toner images that are transferred to the sheet material **P** to the sheet material **P**.

The photoconductor member units **132** are provided so as to form the toner images in the respective colors. According to the present exemplary embodiment, the photoconductor member units **132** for four colors of yellow (**Y**), magenta (**M**), cyan (**C**), and black (**K**) are provided. FIG. 1-2 illustrates **Y**, **M**, **C**, and **K** that represent components associated with the colors described above. In the case where the colors (**Y**, **M**, **C**, and **K**) are not particularly distinguished, the end of an alphabet is omitted for description.

The photoconductor member units **132Y** to **132K** (**132Y**, **132M**, **132C**, and **132K**) basically have the same configuration except for toner that is used.

As illustrated in FIG. 1-2, the photoconductor member units **132Y** to **132K** are arranged along an outer circumferential portion of a transfer belt **141** (described in detail later) of the transfer member **140**.

As illustrated in FIG. 1-2, each of the photoconductor member units **132** includes a photoconductor drum **133** that rotates in the direction of an arrow **A** in the figure and a charger **134** that charges the photoconductor drum **133**. Each of the photoconductor member units **132** also includes an exposure device **135** that exposes the photoconductor drum **133** charged by the charger **134** to light and that forms an electrostatic latent image and a developing device **136** that develops the electrostatic latent image by using toner and that forms the toner image.

The transfer member **140** has a function of first-transferring the toner images in the respective colors on the photoconductor drums **133** such that the toner images are superposed on an intermediate transfer body and second-transferring the superposed toner images to the sheet material **P**. Specifically, as illustrated in FIG. 1-2, the transfer member **140** includes the transfer belt **141** that serves as the intermediate transfer body, a drive roller **142**, first transfer rollers **143**, a second transfer roller **144**, and a backup roller **145**.

The transfer belt **141** extends in the width direction of the apparatus, has no ends, and is wound around the drive roller **142** that is disposed at a folded portion in the +**W** direction and the second transfer roller **144** that is disposed in the -**W** direction. The drive roller **142** is connected to a drive unit such as a motor, not illustrated, and has a function of turning the transfer belt **141** by being rotated by the drive unit.

The first transfer rollers **143** are roll members that are disposed opposite the photoconductor drums **133** for the respective colors with the transfer belt **141** interposed therebetween. The first transfer rollers **143** have a function of transferring the toner images that are formed on the photo-

conductor drums **133** to the transfer belt **141** at first transfer positions **T** between the photoconductor drums **133** and the first transfer rollers **143**.

The second transfer roller **144** is a roll member that is disposed inside a folded portion of the transfer belt **141** opposite the drive roller **142**. The backup roller **145** is a roll member that is disposed opposite the second transfer roller **144** with the transfer belt **141** interposed therebetween. The second transfer roller **144** and the backup roller **145** have a function of transferring the toner images that are transferred to the transfer belt **141** to the sheet material **P** at a second transfer position **NT** between the transfer belt **141** and the backup roller **145**.

The fixing portion **150** heats and compresses the sheet material **P** that is transported from the formation member **130** and consequently fixes the images that are transferred to the sheet material **P** by using the formation member **130** to the sheet material **P**. The fixing portion **150** transports the sheet material **P** to which the images are fixed to the discharge portion **170**.

The discharge portion **170** is configured so as to include a discharge table **172** that is located below a discharge portion **18** (described in detail later) of the document reading device **10** and a pair of discharge rollers **174** that is disposed between the discharge table **172** and the fixing portion **150**. The discharge portion **170** discharges the sheet material **P** that is transported from the fixing portion **150** to the discharge table **172** by using the pair of discharge rollers **174**. Document Reading Device

The configuration of the document reading device **10** according to the present exemplary embodiment will now be described. The document reading device **10** is disposed above the formation device **110**. The document reading device **10** includes the camera unit **20** and the document table **30** that is disposed below the camera unit **20**. The document reading device **10** also includes a loading portion **40** that is disposed below the document table **30**, a reading member **50**, and the discharge portion **18**.

The document reading device **10** has a function of reading the image that is formed on an upper surface (the front surface) of the document **G** by imaging the document **G** that is placed on the document table **30** by using the camera unit **20**. The document reading device **10** has a function of reading the document **G** that is placed on the loading portion **40** by using the reading member **50** while transporting the document **G** and discharging the document **G** to the discharge portion **18**. The document **G** has ground color, brightness of which is high. The ground color of the document **G** according to the present exemplary embodiment is white color.

FIG. 1-7 is a block diagram illustrating the hardware configuration of the document reading device **10**. As illustrated in FIG. 1-7, the document reading device **10** includes a central processing unit (CPU) or a processor **11**, a read only memory (ROM) **12**, a random access memory (RAM) **13**, and a storage **14**.

Camera Unit

The camera unit **20** is a so-called document camera and has a function of imaging the document **G** that is placed on the document table **30** described later. The camera unit **20** is an example of an imaging unit.

As illustrated in FIG. 1-6, the imaging range (the angle of view) **AC** of the camera unit **20** overlaps four marks **M** (described in detail later) that are formed on the document table **30** in association with the document **G** having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. According to the present

exemplary embodiment, the document G having the A3 size is a document having the maximum size that the camera unit 20 is capable of imaging.

Document Table

As illustrated in FIG. 1-2, the document table 30 is disposed below the camera unit 20 and is an example of a placing portion on which the document G to be imaged by the camera unit 20 is placed. The document table 30 is constituted by a covering 52 of the reading member 50 that is disposed in the -W direction and that will be described later and a flat plate 32 that is disposed next to the covering 52 in the +W direction and that extends along a D-W plane. The covering 52 is an example of a first placing portion. The flat plate 32 is an example of a second placing portion. As illustrated in FIG. 1-6, the covering 52 and the flat plate 32 have a rectangular shape when viewed in the vertical direction of the apparatus. The document table 30 extends along the D-W plane and has a placing surface 30a on which the document G is placed. The placing surface 30a is formed when an edge of the flat plate 32 in the -W direction and an edge of the covering 52 in the +W direction are adjacent to each other so as to meet with a gap S (described in detail later) interposed therebetween and is surrounded by the four marks M. That is, the placing surface 30a corresponds to a portion that is surrounded by the four marks M in an upper surface 52a of the covering 52 and an upper surface 32a of the flat plate 32 adjacent to the covering 52 such that the flat plate 32 and the covering 52 meet with the gap S interposed therebetween. The placing surface 30a matches the imaging range (the angle of view) AC of the camera unit 20.

The document G may be placed on the document table 30 such that the position of the center of the document G overlaps that of the center of the imaging range (the angle of view) AC of the camera unit 20. In particular, the document G having the A4 size may be placed on the document table 30 such that the position of the center of the document G overlaps that of the center of the imaging range (the angle of view) AC of the camera unit 20.

The document table 30 has the four marks M for adjusting the position of the document G to be placed on the placing surface 30a. The four marks M are associated with the document G having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. The marks M are formed by painting the upper surface 52a of the covering 52 and the upper surface 32a of the flat plate 32.

Each mark M has a L-shape when viewed in the vertical direction of the apparatus, and the four marks are formed at respective associated four corners of the document G having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. Among the four marks M, the marks M in the -W direction are formed on the upper surface 52a of the covering 52 of the reading member 50, and the marks M in the +W direction are formed on the upper surface 32a of the flat plate 32. The four marks M define the placing surface 30a when the covering 52 and the flat plate 32 are adjacent to each other so as to meet with the gap S interposed therebetween. The position of the center of the document G having the A3 size associated with the four marks M in the width direction of the apparatus overlaps that of the center of the capturing range (the angle of view) AC of the camera unit 20.

As illustrated in FIG. 1-3 and FIG. 1-4, the flat plate 32 is a plate member that is disposed so as to be slidable in the width direction of the apparatus with respect to the covering 52 of the reading member 50 by using a guide mechanism, not illustrated. The flat plate 32 constitutes the placing

surface 30a and has the upper surface 32a that faces upward. The upper surface 32a of the flat plate 32 is an example of a second surface. The flat plate 32 has a function of uncovering an upper surface of the loading portion 40 described later (see FIG. 1-4) by sliding in the +W direction from a state in which the flat plate 32 is adjacent to the covering 52 so as to form the placing surface 30a (see FIG. 1-3). When the flat plate 32 is adjacent to the covering 52 so as to form the placing surface 30a, the flat plate 32 blocks the upper surface of the loading portion 40. The length of the flat plate 32 in the width direction of the apparatus is greater than half of the length of the placing surface 30a in the width direction of the apparatus.

Detector

As illustrated in FIG. 1-7, the CPU 11 is a central processing unit, runs various programs, and controls components. That is, the CPU 11 reads a program from the ROM 12 or the storage 14 and runs the program with the RAM 13 used as a work area. The CPU 11 controls the components described above and performs various kinds of arithmetic processing in accordance with the program that is recorded in the ROM 12 or the storage 14. According to the present exemplary embodiment, the ROM 12 or the storage 14 stores a document detection program for detecting an edge of the document G by performing a process of detecting a color difference or a luminance difference on the image that is imaged by the camera unit 20. The ROM 12 or the storage 14 stores a recognition reading program for recognizing the document G, based on the edge of the document G that is detected by the document detection program and reading the image that is formed on the upper surface (the front surface) of the document G. The document detection program causes the CPU 11 to function as a detector 11a that detects the edge of the document G that is placed on the document table 30 from the image that is imaged by the camera unit 20 by using the color difference or the luminance difference between the placing surface 30a and the document G. The recognition reading program causes the CPU 11 to function as a recognition reading unit 11b that recognizes the document G, based on the edge of the document G that is detected by the document detection program and reading the image that is formed on the document G.

The configuration to detect the edge of the document G that is placed on the document table 30 from the image that is imaged by the camera unit 20 is that the luminance difference between the color of the placing surface 30a and the color of the document G is a predetermined value or more, and the color difference therebetween is a predetermined value or more. However, the exemplary embodiment is not a limitation, provided that the edge of the document G is detectable. For example, even when the color difference between the color of the placing surface 30a and the color of the document G is the predetermined value or less, the luminance difference that is the predetermined value or more suffices provided that the edge of the document G is detectable. Specifically, for example, in the case of an 8-bits environment, the luminance difference may be 50 LSB or more, and at least a luminance difference of 20 LSB or more suffices. In contrast, even when the luminance difference between the color of the placing surface 30a and the color of the document G is the predetermined value or less, the color difference that is the predetermined value or more suffices provided that the edge of the document G is detectable.

The ROM 12 stores various programs and various kinds of data. The RAM 13 serves as a work area and temporarily stores a program or data. The storage 14 is constituted by a

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hard disk drive (HDD) or a solid state drive (SSD) and stores various programs including an operating system and various kinds of data.

Loading Portion

As illustrated in FIG. 1-3, the loading portion 40 is disposed below the document table 30 and is a part of the housing of the document reading device 10 that has a bottom surface 42 that is dented from the placing surface 30a and that has a rectangular shape. As for an end portion of the bottom surface 42 in the -W direction, the position in the width direction of the apparatus overlaps that of an end surface 52e of the covering 52 described later. The bottom surface 42 tilts with respect to the horizontal direction such that the end portion in the -W direction is lower than an end portion in the +W direction.

As illustrated in FIG. 1-4, the loading portion 40 has a function of enabling multiple documents G to be loaded on the bottom surface 42 in a state in which an end portion of the flat plate 32 in the -W direction slides to a location away from the loading portion 40 in the +W direction and uncovers the upper surface of the loading portion 40. In this state, the loading portion 40 has a function of enabling the document G that has a length in the width direction of the apparatus greater than that of A4 (a short side) to be placed thereon with the document G extending across the upper surface 32a of the flat plate 32 slid and the bottom surface 42.

Reading Member

The reading member 50 is an auto document feeder (a so-called ADF) and has a function of reading the document G that is placed on the loading portion 40 while transporting the document G. The reading member 50 includes the covering 52, an intake port 54, a transport unit 56, a reading unit 60, and an outlet 58.

The intake port 54 is adjacent to the loading portion 40 in the -W direction. The intake port 54 is an example of an opening. As for the outlet 58, the position in the vertical direction of the apparatus is lower than that of the intake port 54 and is higher than that of the discharge portion 170 of the formation device 110. The outlet 58 is adjacent to the discharge portion 18 in the -W direction.

The transport unit 56 includes a transport path 56a that extends from the intake port 54 to the outlet 58 and that has a substantially U-shape and multiple roller members 56b that are disposed along the transport path 56a. The transport unit 56 takes in the document G that is placed on the loading portion 40 via the intake port 54 by using the roller members 56b and transports the document G toward the outlet 58 along the transport path 56a.

The reading unit 60 is a stationary contact image sensor that is disposed along the transport path 56a and has a function of reading the images that are formed on the front surface and the back surface of the document G that is transported by the transport unit 56. The reading unit 60 includes a front surface reading member 60a that reads the image that is formed on the front surface of the document G and a back surface reading member 60b that reads the image that is formed on the back surface of the document G.

The covering 52 is a panel member that has an L-shaped section when viewed in the depth direction of the apparatus and that covers an upper surface of the reading member 50. The covering 52 has the upper surface 52a that constitutes the placing surface 30a and that faces upwards. The upper surface 52a of the covering 52 is an example of a first surface. The covering 52 has an end surface 52b that faces the end surface of the flat plate 32 in the -W direction and includes a projecting portion 52c that projects from the end

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surface 52b below the flat plate 32 into a rectangular shape. That is, the projecting portion 52c is integrally formed with the covering 52. The projecting portion 52c has an upward surface 52d that faces upward and the end surface 52e that faces in the +W direction.

As illustrated in FIG. 1-3, when the covering 52 and the flat plate 32 form the placing surface 30a, the projecting portion 52c is located below the end portion of the flat plate 32 in the -W direction. That is, when the placing surface 30a is formed, the projecting portion 52c of the covering 52 and the end portion of the flat plate 32 in the -W direction overlaps in the vertical direction of the apparatus. In other words, when the placing surface 30a is formed, the covering 52 and the flat plate 32 form an overlapping portion 70 by using the projecting portion 52c of the covering and a part of the end portion of the flat plate 32 in the -W direction. At this time, the upward surface 52d of the projecting portion 52c faces a lower surface of the flat plate 32. At this time, the projecting portion 52c supports the flat plate 32 from below by using a support unit (not illustrated) such as a rib that is disposed on the upward surface 52d. That is, the projecting portion 52c is capable of supporting the flat plate 32 from below.

According to the present exemplary embodiment, the end surface 52e of the covering 52 extends toward the flat plate 32 (that is, in the +W direction) beyond an end surface 32e of the flat plate 32 in the -W direction. The end surface 52e of the covering 52 is an example of an "end portion of the first placing portion adjacent to the second placing portion", and the end surface 32e of the flat plate 32 in the -W direction is an example of an "end portion of the second placing portion adjacent to the first placing portion".

As illustrated in FIG. 1-5, when the covering 52 and the flat plate 32 form the placing surface 30a, the flat plate 32 is disposed on an imaginary straight line KC that connects a lower edge 52f of the end surface 52e of the covering 52 and the camera unit 20 to each other. At this time, the flat plate 32 is disposed on an imaginary straight line that is located above an upper edge 52g of the end surface 52e of the covering 52 and that connects the upper edge 52g and the camera unit 20 to each other. In other words, when the overlapping portion 70 is formed, the flat plate 32 shields the end surface 52e of the covering 52 from the camera unit 20. When the overlapping portion 70 is formed, the flat plate 32 shields the lower edge 52f of the end surface 52e of the covering 52 from the camera unit 20. The lower edge 52f of the end surface 52e of the covering 52 is an example of a lower edge of the overlapping portion 70.

Gap

When the covering 52 and the flat plate 32 form the placing surface 30a, the gap S is formed between the covering 52 and the flat plate 32. The end surface 52b and the upward surface 52d of the covering 52 and the lower surface and the end surface of the flat plate 32 that faces in the -W direction form the gap S that has an L-shape when viewed in the depth direction of the apparatus. In other words, when the overlapping portion 70 is formed, the overlapping portion 70 has the gap S that nonlinearly extends from the placing surface 30a to the upper edge 52g of the covering 52 that is disposed below the flat plate 32 when viewed in the depth direction of the apparatus. The gap S includes a vertical gap TS that extends in the vertical direction of the apparatus from the placing surface 30a to the lower edge of the end surface of the flat plate 32 in the -W direction and a lateral gap YS that extends in the width direction of the apparatus below the lower edge of the end surface of the flat plate 32 in the -W direction when viewed

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in the depth direction of the apparatus. The lower edge of the end surface of the flat plate **32** in the $-W$ direction corresponds to the lower edge of the vertical gap TS. The vertical gap TS is covered by the projecting portion **52c** from below. In other words, the lower edge of the end surface of the flat plate **32** in the $-W$ direction is covered by the projecting portion **52c** from below. In FIG. 1-2, FIG. 1-3, FIG. 1-5, and FIG. 1-6, the gap S is emphatically illustrated so as to be formed between the covering **52** and the flat plate **32**.

When the covering **52** and the flat plate **32** form the placing surface **30a**, the flat plate **32** is disposed on an imaginary straight line that connects the lower edge of the end surface of the flat plate **32** in the $-W$ direction and the camera unit **20** to each other. In other words, when the overlapping portion **70** is formed, the flat plate **32** shields the lower edge of the end surface of the flat plate **32** in the $-W$ direction from the camera unit **20**.

The gap S forms a boundary line K on the placing surface **30a**. In other words, as illustrated in FIG. 1-5, the boundary line K is formed by the gap S between the upper surface **52a** of the covering **52** and the upper surface **32a** of the flat plate **32**. As illustrated in FIG. 1-6, the single boundary line K linearly extends in the depth direction of the apparatus when viewed in the vertical direction of the apparatus.

The surfaces of the covering **52** and the flat plate **32** are colored in unbright color. According to the present exemplary embodiment, the color of the surfaces of the covering **52** and the flat plate **32** is black color. That is, the color of the surface of the projecting portion **52c** is the black color. For this reason, according to the present exemplary embodiment, the color differences between the overlapping portion **70** and the covering **52** and between the overlapping portion **70** and the flat plate **32** are smaller than the color differences between the covering **52** and the document G and between the flat plate **32** and the document G, where the surfaces of the overlapping portion **70**, the covering **52**, and the flat plate **32** have the black color, and the ground color of the document G is the white color.

The surfaces of the covering **52** and the flat plate **32** may be colored, for example, in bright color (specifically, such as the white color). In this case, the ground color of the document G may be, for example, unbright color (specifically, such as the black color). Various colors may be used as the colors of the document G, the covering **52**, and the flat plate **32**, provided that there is a color difference between the placing surface **30a** and the document G.

The color of the placing surface **30a** is a color that is detected as the color of the placing surface **30a** by using the camera unit **20** (an example of the imaging unit) and is not necessarily the color of the placing surface **30a** itself. The color of the document G is a color that is detected as the color of the document G by using the camera unit **20** and is not necessarily the color of the document G itself.

The covering **52** is configured so as to be capable of uncovering the upper surface of the reading member **50** by using an opening-closing mechanism, not illustrated, and exposing the transport path **56a**. The reading member **50** has a function of uncovering the upper surface of the reading member **50** by using the opening-closing mechanism of the covering **52** and enabling the document G that is jammed on the transport path **56a** to be manually taken out when the document G is jammed on the transport path **56a** due to a transport failure of the transport unit **56**.

As illustrated in FIG. 1-2, the discharge portion **18** is a tray member that is disposed below the outlet **58** and above the discharge portion **170** of the formation device **110**. The discharge portion **18** has a function of receiving the docu-

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ment G that is transported by the transport unit **56** and that is discharged via the outlet **58**.

Action and Effect

Action and effect according to the present exemplary embodiment will now be described. In the case where the same component as that of the image forming apparatus **100** according to the present exemplary embodiment, for example, is used in the description when a comparative exemplary embodiment relative to the present exemplary embodiment is described, the reference character and name of the component, for example, are used as they are for the description.

The document table **30** of the document reading device **10** according to the present exemplary embodiment includes the overlapping portion **70** at which the covering **52** and the flat plate **232** overlap in the vertical direction of the apparatus below the placing surface **30a**. The document reading device **10** according to the present exemplary embodiment is compared with a document reading device **210** according to a first comparative exemplary embodiment described below.

As illustrated in FIG. 1-8, the document reading device **210** according to the first comparative exemplary embodiment includes a covering **252** instead of the covering **52** according to the present exemplary embodiment. The covering **252** does not have a configuration that corresponds to the projecting portion **52c** of the covering **52** according to the present exemplary embodiment. Specifically, an end surface of the covering **252** in the $+W$ direction and an end surface of a flat plate **232** in the $-W$ direction according to the first comparative exemplary embodiment face each other in the width direction of the apparatus and do not overlap in the vertical direction of the apparatus. That is, the document reading device **210** according to the first comparative exemplary embodiment does not have a configuration that corresponds to the overlapping portion **70** according to the present exemplary embodiment. When the placing surface **30a** is formed, a gap S2 is formed between the end surface of the covering **252** in the $+W$ direction and the end surface of the flat plate **232** in the $-W$ direction instead of the gap S according to the present exemplary embodiment. The gap S2 linearly extends in the vertical direction of the apparatus from the placing surface **30a** to the back surfaces (surfaces opposite the placing surface **30a**) of the covering **252** and the flat plate **232** when viewed in the depth direction of the apparatus. Except for the matters described above, the document reading device **210** according to the first comparative exemplary embodiment has the same configuration as that of the document reading device **10** according to the present exemplary embodiment.

The document reading device **210** according to the first comparative exemplary embodiment does not have a configuration that corresponds to the overlapping portion **70** according to the present exemplary embodiment. Therefore, as for the document reading device **210** according to the first comparative exemplary embodiment, light that passes through the gap S2 and that is subsequently reflected from a member that is disposed below the flat plate **232** reaches the camera unit **20**. In this case, there is a possibility that the gap S2 is imaged by the camera unit **20** such that the luminance difference or the color difference between the gap S2 and the document table **30** is large depending on the color of the member that is disposed below the flat plate **232**. For this reason, as for a configuration for detecting the edge of the document G from the luminance difference or the color difference between the document G and the document table

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30 regarding an image that is imaged by the camera unit 20 and that represents the document G that is placed on the document table 30, there is a possibility that the detector 11a mistakenly detects the gap S2 as the edge of the document G.

Since the document table 30 of the document reading device 10 according to the present exemplary embodiment includes the overlapping portion 70, visible light L reaches the upward surface 52d of the projecting portion 52c that is higher than the lower edge 52f of the covering 52 even when the visible light L enters the gap S from the placing surface 30a. For this reason, as for the document reading device 10 according to the present exemplary embodiment, the gap S is imaged by the camera unit 20 such that the color difference between the gap S and the document table 30 is small, and the possibility that the detector 11a mistakenly detects the gap S2 as the edge of the document G is lower than that in the case of the document reading device 210 according to the comparative exemplary embodiment. In this way, the document reading device 10 according to the present exemplary embodiment that is configured such that the edge of the document G that is placed on the document table 30 is detected by using the color difference between the placing surface 30a and the document G may inhibit the detector 11a from failing to detect the edge of the document G, unlike the document reading device 210 according to the first comparative exemplary embodiment.

The document reading device 10 according to the present exemplary embodiment is configured so as to include the flat plate 32 that is disposed on the imaginary straight line KC that connects the lower edge 52f of the projecting portion 52c of the covering 52 and the camera unit 20 to each other. The document reading device 10 according to the present exemplary embodiment is compared with a document reading device 310 according to a second comparative exemplary embodiment described below.

As illustrated in FIG. 1-9, the document reading device 310 according to the second comparative exemplary embodiment includes a covering 352 and a flat plate 332 instead of the covering 52 and the flat plate 32 according to the present exemplary embodiment. An end surface 352b of the covering 352 in the +W direction and an end surface 332b of the flat plate 332 in the -W direction face each other at a position away from the camera unit 20 in the -W direction. The end surface 352b of the covering 352 in the +W direction and the end surface 332b of the flat plate 332 in the -W direction tilt in the -W direction from an upper side to a lower side when viewed in the depth direction of the apparatus. When the covering 352 and the flat plate 332 form the placing surface 30a, the end portion of the covering 352 in the +W direction is disposed above the end portion of the flat plate 332 in the -W direction, and these overlap in the vertical direction of the apparatus. That is, the document reading device 310 according to the second comparative exemplary embodiment includes an overlapping portion. A gap S3 that is formed between the end surface 352b of the covering 352 in the +W direction and the end surface 332b of the flat plate 332 in the -W direction linearly extends so as to tilt in the -W direction from the upper side to the lower side when viewed in the depth direction of the apparatus. As for the document reading device 310 according to the second comparative exemplary embodiment, in the case where the visible light L is incident from the camera unit 20 toward the gap S3, the visible light L linearly reaches the lower edge of the end surface 352b of the covering 352 and the lower edge of the end surface 332b of the flat plate 332. That is, the document reading device 310 according to the second com-

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parative exemplary embodiment is configured so as to enable the visible light L to linearly reach the lower edge of the overlapping portion from the camera unit 20. Except for the matters described above, the document reading device 310 according to the second comparative exemplary embodiment has the same configuration as that of the document reading device 10 according to the present exemplary embodiment.

The document reading device 310 according to the second comparative exemplary embodiment is configured so as to enable the visible light L to linearly reach the lower edge of the overlapping portion from the camera unit 20, and consequently, the visible light L that enters the gap S3 from the camera unit 20 is not shielded but reaches a position below the covering 352 and the flat plate 332. In this case, there is a possibility that the gap S3 is imaged by the camera unit 20 such that the color difference between the gap S3 and the document table 30 is large. For this reason, as for a configuration for detecting only the edge of the document G from the color difference between the document G and the document table 30 regarding the image that is imaged by the camera unit 20 and that represents the document G that is placed on the document table 30, there is a possibility that the detector 11a mistakenly detects the gap S3 as the edge of the document G.

The document reading device 10 according to the present exemplary embodiment is configured such that the flat plate 32 is disposed on the imaginary straight line KC that connects the lower edge 52f of the projecting portion 52c of the covering 52 and the camera unit 20 to each other. For this reason, the visible light L that travels from the camera unit 20 toward the lower edge 52f along the imaginary straight line KC that connects the lower edge 52f of the projecting portion 52c of the covering 52 is shielded by the flat plate 32. The visible light L that enters the gap S from the camera unit 20 reaches the upward surface 52d of the projecting portion 52c as described above. In this way, the document reading device 10 according to the present exemplary embodiment that is configured such that the edge of the document G that is placed on the document table 30 is detected by using the color difference between the placing surface 30a and the document G may inhibit the detector 11a from failing to detect the edge of the document G, unlike the document reading device 310 according to the second comparative exemplary embodiment.

The overlapping portion 70 of the document reading device 10 according to the present exemplary embodiment has the gap S that nonlinearly extends from the placing surface 30a to a position below the document table 30 when viewed in the depth direction of the apparatus. The document reading device 10 according to the present exemplary embodiment is compared with a document reading device 410 according to a third comparative exemplary embodiment described below.

As illustrated in FIG. 1-10, the document reading device 410 according to the third comparative exemplary embodiment includes a covering 452 and a flat plate 432 instead of the covering 52 and the flat plate 32 according to the present exemplary embodiment. An end surface 452b of the covering 452 in the +W direction and an end surface 432b of the flat plate 432 in the -W direction face each other at a position away from the camera unit 20 in the -W direction. The end surface 452b of the covering 452 in the +W direction and the end surface 432b of the flat plate 432 in the -W direction tilt in the +W direction from an upper side to a lower side when viewed in the depth direction of the apparatus. When the covering 452 and the flat plate 432 form

the placing surface 30a, the end portion of the covering 452 in the +W direction is disposed below the end portion of the flat plate 432 in the -W direction, and these overlap in the vertical direction of the apparatus. That is, the document reading device 410 according to the third comparative exemplary embodiment includes an overlapping portion. A gap S4 that is formed between the end surface 452b of the covering 452 in the +W direction and the end surface 432b of the flat plate 432 in the -W direction linearly extends so as to tilt in the +W direction from the upper side to the lower side when viewed in the depth direction of the apparatus. Except for the matters described above, the document reading device 410 according to the third comparative exemplary embodiment has the same configuration as that of the document reading device 10 according to the present exemplary embodiment.

As for the document reading device 410 according to the third comparative exemplary embodiment, the visible light L that travels along an imaginary straight line KC4 that connects the camera unit 20 and the lower edge of the gap S4 to each other toward the lower edge is shielded by the flat plate 432. However, light that enters the gap S4 from the camera unit 20 is diffracted along the gap S4 that linearly extends and reaches a position below the covering 452 and the flat plate 432, and consequently, there is a possibility that the detector 11a mistakenly detects the gap S4 as the edge of the document G.

The document reading device 10 according to the present exemplary embodiment has the gap S that nonlinearly extends from the placing surface 30a to a position below the document table 30 when viewed in the depth direction of the apparatus. In this way, the document reading device 10 according to the present exemplary embodiment that is configured such that the edge of the document G that is placed on the document table 30 is detected by using the color difference between the placing surface 30a and the document G may inhibit the detector 11a from failing to detect the edge of the document G, unlike the document reading device 410 according to the third comparative exemplary embodiment. The third comparative exemplary embodiment described above is included in the technical idea of the exemplary embodiment of the present disclosure as a modification to the present exemplary embodiment.

As for the document reading device 10 according to the present exemplary embodiment, the projecting portion 52c is integrally formed with the covering 52 of the document table 30. Therefore, the document reading device 10 according to the present exemplary embodiment has a smaller number of components than that in a configuration in which the projecting portion 52c is separated from the covering 52.

The projecting portion 52c according to the present exemplary embodiment is integrally formed with the covering 52 and contains the same material as that of the covering 52. That is, the projecting portion 52c according to the present exemplary embodiment has the same surface roughness and the same surface color as those of the covering 52. Therefore, the color difference between the surface of the projecting portion 52c and the upper surface 52a of the covering 52 is smaller than the color difference between a projecting portion that is disposed separately from the covering 52 and the covering 52. For this reason, the color difference between the surface of the projecting portion 52c and the document G is equal to the color difference between the upper surface 52a of the covering 52 and the document G. In this way, the document reading device 10 according to the present exemplary embodiment may inhibit the detector 11a from failing to detect the edge of the document G, unlike a

configuration in which the material of projecting portion 52c differs from that of the covering 52.

As for the document reading device 10 according to the present exemplary embodiment, the color differences between the overlapping portion 70 and the covering 52 and between the overlapping portion 70 and the flat plate 32 are smaller than the color differences between the covering 52 and the document G and between the flat plate 32 and the document G, where the surfaces of the overlapping portion 70, the covering 52, and the flat plate 32 have the black color, and the ground color of the document G is the white color. In this way, the document reading device 10 according to the present exemplary embodiment may inhibit the detector 11a from failing to detect the edge of the document G, unlike a configuration in which the color difference between the overlapping portion 70 and the document table 30 is larger than the color difference between the document table 30 and the document G.

The document reading device 10 according to the present exemplary embodiment also includes the reading member 50. Therefore, the document reading device 10 according to the present exemplary embodiment is capable of using the reading member 50 in addition to the camera unit 20 when the document G is read.

As for the document reading device 10 according to the present exemplary embodiment, the covering 52 of the reading member 50 constitutes the document table 30. In this way, the document reading device 10 according to the present exemplary embodiment may have a compact size, unlike the case where a member of two members that constitute the document table 30 is disposed between the flat plate 32 and the covering 52 of the reading member 50.

The document reading device 10 according to the present exemplary embodiment includes the reading member 50 that transports the document G via the intake port 54 adjacent to the loading portion 40 in the -W direction and the flat plate 32 that is capable of moving in the +W direction with respect to the reading member 50 and uncovering the upper surface of the loading portion 40. In this way, the document reading device 10 according to the present exemplary embodiment that is configured so as to include the reading member 50 may load the document G having an increased size across the loading portion 40 and the flat plate 32 that moves in the +W direction, unlike the case where the flat plate 32 moves in the depth direction of the apparatus.

The document reading device 10 according to the present exemplary embodiment is configured such that the flat plate 32 moves by sliding in the width direction of the apparatus. In this way, the document reading device 10 according to the present exemplary embodiment may facilitate movement of the flat plate 32 to uncover the upper surface of the loading portion 40, unlike the case where the flat plate 32 moves by rotating.

The document reading device 10 according to the present exemplary embodiment includes the projecting portion 52c that enables the covering 52 of the reading member 50 to support the flat plate 32 from below. In this way, the document reading device 10 according to the present exemplary embodiment may increase the load capacity of the flat plate 32 when the upper surface of the loading portion 40 is blocked, unlike a configuration in which the flat plate 32 supports the covering 52 of the reading member 50 from below at the overlapping portion.

Also, in this way, the image forming apparatus 100 according to the present exemplary embodiment may inhibit the camera unit 20 from failing to copy the document G,

unlike a configuration that includes the document reading device **210** according to the first comparative exemplary embodiment.

The image forming apparatus **100** according to the present exemplary embodiment includes the panel member **102** that displays the information about the image forming apparatus **100** in the +D direction. In this way, as for the image forming apparatus **100** according to the present exemplary embodiment that is configured so as to include the panel member **102** that displays the information in the +D direction, the flat plate **32** that has the upper surface **32a** is readily moved while the content of display of the panel member **102** is checked, unlike a configuration in which the flat plate **32** moves in the +D direction.

The first exemplary embodiment is described in detail above, but the present disclosure is not limited to the exemplary embodiment described above, and various modifications, alterations, and improvements may be made within the range of the technical idea of the exemplary embodiment of the present disclosure.

For example, the formation device **110** according to the present exemplary embodiment is an electrophotographic formation device. However, the formation device **110** may be, for example, an ink-jet formation device or an offset print formation device.

The gap S according to the present exemplary embodiment has an L-shape when viewed in the depth direction of the apparatus. According to the exemplary embodiment of the present disclosure, however, the shape of the gap that is formed such that the first placing portion or the second placing portion is disposed on the imaginary straight line that connects the lower edge of the overlapping portion and the imaging unit to each other when viewed in the depth direction of the apparatus is not limited to an L-shape.

For example, the shape of the gap when viewed in the depth direction of the apparatus may be a linear shape, provided that the first placing portion or the second placing portion is disposed on the imaginary straight line that connects the lower edge of the overlapping portion and the imaging unit to each other as in the document reading device **410** according to the third comparative exemplary embodiment described above (see FIG. 1-10).

The shape of the gap when viewed in the depth direction of the apparatus may be the shape of a hat the top of which faces sideways as illustrated in FIG. 1-11. Specifically, the gap may be formed between a covering **552** that has an end portion in the +W direction projecting in the +W direction and a flat plate **532** that has an end portion in the -W direction having a recessed portion that fits to the end portion of the covering **552** in the +W direction when viewed in the depth direction of the apparatus.

As illustrated in FIG. 1-12, the shape of the gap when viewed in the depth direction of the apparatus may be a sideways V-shape. Specifically, the gap may be formed between a covering **652** that has an end portion in the +W direction projecting in the +W direction into a V-shape and a flat plate **632** that has an end portion in the -W direction having a V-shaped recessed portion that fits to the end portion of the covering **652** in the +W direction when viewed in the depth direction of the apparatus. The shape of the gap when viewed in the depth direction of the apparatus may be a U-shape instead of the V-shape.

The projecting portion **52c** that forms the overlapping portion **70** according to the present exemplary embodiment is integrally formed with the covering **52**. However, the overlapping portion according to the exemplary embodiment of the present disclosure may be formed separately

from the first placing portion or the second placing portion. The first placing portion, the second placing portion, a part of the overlapping portion that is exposed at least from the gap according to the exemplary embodiment of the present disclosure may have the same surface roughness and the same surface color.

The placing surface **30a** according to the present exemplary embodiment corresponds to the portion that is surrounded by the four marks M in the upper surface **52a** of the covering **52** and the upper surface **32a** of the flat plate **32** adjacent to the covering **52** such that the flat plate **32** and the covering **52** meet with the gap S interposed therebetween. However, the placing surface **30a** is not particularly limited but may be formed by the entire upper surface **52a** of the covering **52** and the entire upper surface **32a** of the flat plate **32**.

The document table **30** according to the present exemplary embodiment is constituted by the covering **52** of the reading member **50** and the flat plate **32**. However, the document table **30** may be constituted by three or more members including the first surface of the first placing portion and the second surface of the second placing portion. The number of the gap S and the number of the boundary line K may be two or more.

The single boundary line K according to the present exemplary embodiment linearly extends when viewed in the vertical direction of the apparatus. However, the boundary line K may bend or may curve when viewed in the vertical direction of the apparatus.

The flat plate **32** according to the present exemplary embodiment moves by sliding in the width direction of the apparatus. However, the flat plate **32** may be configured so as to slide in the depth direction of the apparatus. The flat plate **32** may be configured so as to move by rotating to uncover the upper surface of the loading portion **40** or may be configured so as to be installable on and removable from the document table **30**.

The length of the flat plate **32** according to the present exemplary embodiment in the width direction of the apparatus is greater than half of the length of the placing surface **30a** in the width direction of the apparatus. However, the length of the flat plate **32** in the width direction of the apparatus is not limited to a length greater than half of the length of the placing surface **30a** in the width direction of the apparatus but may be equal to half of the length of the placing surface **30a** in the width direction of the apparatus or may be less than half of the length of the placing surface **30a** in the width direction of the apparatus.

The document reading device **10** may be configured so as not to include the reading member **50**. For example, the document reading device **10** that is thus configured reads the image of the document G by using only the camera unit **20**. As for the document reading device **10** that is thus configured, the flat plate **32** may be configured so as to slide in the +W direction from the position illustrated in FIG. 1-3 and to consequently uncover the upper surface of the discharge portion **170**. The document reading device **10** that is thus configured has the same action and effect as those of the document reading device **10** that includes the reading member **50** described above.

As for the document reading device **10** according to the present exemplary embodiment, the maximum size of the document G that has an image that the camera unit **20** is capable of capturing is the A3 size. However, the maximum size of the document that has an image that the camera unit **20** is capable of capturing is not limited to the A3 size but

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may be a size larger than the A3 size such as A2 or B3 or may be a size smaller than the A3 size such as A4 or B4.

As for the image forming apparatus **100** according to the present exemplary embodiment, the size of the largest number of sheet materials **P** that are containable in the container unit **120** is A4. However, the size of the largest number of sheet materials **P** that are containable in the container unit **120** is not limited to A4 but may be A3 or may be B5.

Second Exemplary Embodiment

An image forming apparatus **2100** according to a second exemplary embodiment will be described with reference to the drawings.

In the following description, a vertical direction of the apparatus (a vertical direction), a width direction of the apparatus (a horizontal direction), and a depth direction of the apparatus (a horizontal direction) are respectively described as a H direction, a W direction, and a D direction in a front view of the image forming apparatus **2100** from a position at which a user (not illustrated) stands. In the case where it is necessary to distinguish between one direction and the other direction of the vertical direction of the apparatus, the width direction of the apparatus, and the depth direction of the apparatus, an upward direction is described as a +H direction, a downward direction is described as a -H direction, a right-hand direction is described as a +W direction, a left-hand direction is described as a -W direction, a rear direction is described as a -D direction, and a front direction is described as a +D direction in a front view of the image forming apparatus **2100**.

Image Forming Apparatus

As illustrated in FIG. 2-1, the image forming apparatus **2100** according to the present exemplary embodiment includes a document reading device **2010**, a formation device **2110**, a panel member **2102**, and a controller (not illustrated). The document reading device **2010** reads images that are formed on a front surface and a back surface of a document **2000G** and converts the read images into electronic data. The formation device **2110** is disposed below the document reading device **2010**, forms a copy image of the document **2000G**, based on the electronic data of the images that is converted by the document reading device **2010**, and records the copy image on a sheet material **2000P** as a medium. The panel member **2102** is disposed between an image capturing member **2020** and a document table **2030** described later and is a touch screen display that allows the user to perform a touch input operation and that displays information about the image forming apparatus **2100** in the +D direction. The panel member **2102** is an example of the display unit. The controller (not illustrated) controls the operation of components of the image forming apparatus **2100**. The document reading device **2010** will be described in detail later.

Formation Device

As illustrated in FIG. 2-2, the formation device **2110** includes a container unit **2120**, a transport unit **2160**, a formation member **2130**, and a discharge portion **2170**.

The container unit **2120** contains sheet materials **2000P** having different sizes and supplies the sheet materials **2000P** to the formation member **2130**. The container unit **2120** according to the present exemplary embodiment includes a first container unit **2122** and a second container unit **2124**. The first container unit **2122** is capable of containing up to 500 sheet materials **2000P** having the A3 size or smaller. The second container unit **2124** is capable of containing up to

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1500 sheet materials **2000P** having the A4 size or smaller. That is, the size of the largest number of sheet materials **2000P** that are containable in the container unit **2120** is the A4 size. The sheet material **2000P** is a kind of the sheet material **2000P**. The sheet materials **2000P** are selectively fed from the container unit **2120** by using the controller (not illustrated).

The transport unit **2160** is configured so as to include multiple roller members **2162** and transports the sheet material **2000P** that is selectively fed by using the controller (not illustrated) toward the formation member **2130**.

The formation member **2130** has a function of forming an image on the sheet material **2000P** by using an electrophotographic system. The formation member **2130** includes photoconductor member units **2132** that form toner images in multiple colors, a transfer member **2140** that transfers the toner images that are formed by the photoconductor member units **2132** to the sheet material **2000P**, and a fixing portion **2150** that fixes the toner images that are transferred to the sheet material **2000P** on the sheet material **2000P**.

The photoconductor member units **2132** are provided so as to form the toner images in the respective colors. According to the present exemplary embodiment, the photoconductor member units **2132** for the four colors of yellow (Y), magenta (M), cyan (C), and black (K) are provided. FIG. 2-2 illustrates Y, M, C, and K that represent components associated with the colors described above. In the case where the colors (Y, M, C, and K) are not particularly distinguished, the end of an alphabet is omitted for description.

The photoconductor member units **2132Y**, **2132M**, **2132C**, and **2132K** basically have the same configuration except for toner that is used.

As illustrated in FIG. 2-2, the photoconductor member units **2132Y**, **2132M**, **2132C**, and **2132K** are arranged along an outer circumferential portion of a transfer belt **2141** (described in detail later) of the transfer member **2140**.

As illustrated in FIG. 2-2, each of the photoconductor member units **2132** includes a photoconductor drum **2133** that rotates in the direction of an arrow **2000A** in the figure and a charger **2134** that charges the photoconductor drum **2133**. Each of the photoconductor member units **2132** also includes an exposure device **2135** that exposes the photoconductor drum **2133** that is charged by the charger **2134** to light and that forms an electrostatic latent image and a developing device **2136** that develops the electrostatic latent image by using toner and that forms the toner image.

The transfer member **2140** has a function of first-transferring the toner images in the respective colors on the photoconductor drums **2133** such that the toner images are superposed on an intermediate transfer body and second-transferring the superposed toner images to the sheet material **2000P**. Specifically, as illustrated in FIG. 2-2, the transfer member **2140** includes the transfer belt **2141** that serves as the intermediate transfer body, a drive roller **2142**, a first transfer roller **2143**, a second transfer roller **2144**, and a backup roller **2145**.

The transfer belt **2141** extends in the width direction of the apparatus, has no ends, and is wound around the drive roller **2142** that is disposed at a folded portion in the +W direction and the second transfer roller **2144** that is disposed in the -W direction. The drive roller **2142** is connected to a drive unit such as a motor not illustrated and has a function of turning the transfer belt **2141** by being rotated by the drive unit.

The first transfer rollers **2143** are roll members that are disposed opposite the photoconductor drums **2133** for the respective colors with the transfer belt **2141** interposed

therebetween. The first transfer rollers **2143** have a function of transferring the toner images that are formed on the photoconductor drums **2133** to the transfer belt **2141** at first transfer positions **2000T** between the photoconductor drums **2133** and the first transfer roller **2143**.

The second transfer roller **2144** is a roll member that is disposed inside a folded portion of the transfer belt **2141** opposite the drive roller **2142**. The backup roller **2145** is a roll member that is disposed opposite the second transfer roller **2144** with the transfer belt **2141** interposed therebetween. The second transfer roller **2144** and the backup roller **2145** have a function of transferring the toner images that are transferred to the transfer belt **2141** to the sheet material **2000P** at a second transfer position between the transfer belt **2141** and the backup roller **2145**.

The fixing portion **2150** heats and compresses the sheet material **2000P** that is transported from the formation member **2130** and consequently fixes the images that are transferred to the sheet material **2000P** by using the formation member **2130** to the sheet material **2000P**. The fixing portion **2150** transports the sheet material **2000P** to which the images are fixed to the discharge portion **2170**.

The discharge portion **2170** is configured so as to include a discharge table **2172** that is located below a discharge portion **2018** (described in detail later) of the document reading device **2010** and a pair of discharge rollers **2174** that is disposed between the discharge table **2172** and the fixing portion **2150**. The discharge portion **2170** discharges the sheet material **2000P** that is transported from the fixing portion **2150** to the discharge table **2172** by using the pair of discharge rollers **2174**.

Document Reading Device

The configuration of the document reading device **2010** according to the present exemplary embodiment will now be described. The document reading device **2010** is disposed above the formation device **2110**. The document reading device **2010** includes the image capturing member **2020**, the document table **2030** that is disposed below the image capturing member **2020**, and a detection reading member **2011a**.

The document reading device **2010** also includes a loading portion **2040** that is disposed below the document table **2030**, a reading member **2050**, and the discharge portion **2018**. The document reading device **2010** has a function of reading the image that is formed on an upper surface (the front surface) of the document **2000G** by capturing the image of the document **2000G** that is placed on the document table **2030** by using the image capturing member **2020**. The document reading device **2010** has a function of reading the document **2000G** that is placed on the loading portion **2040** by using the reading member **2050** while transporting the document **2000G** and discharging the document **2000G** to the discharge portion **2018**.

FIG. 2-6 is a block diagram illustrating the hardware configuration of the document reading device **2010**. As illustrated in FIG. 2-6, the document reading device **2010** includes a central processing unit (CPU) or a processor **2011**, a read only memory (ROM) **2012**, a random access memory (RAM) **2013**, and a storage **2014**.

Image Capturing Member

The image capturing member **2020** is a so-called document camera and has a function of capturing the image of the document **2000G** that is placed on the document table **2030** described later. The size of the document **2000G** that is placed on the document table **2030** may be any size such as a standard size (for example, a document having the A3 or

A4 size) found in a market or a size that enables the document to be contained in the container unit **2120**.

As illustrated in FIG. 2-5, the capturing range **2000AC** of the image capturing member **2020** overlaps marks **2000M** (described in detail later) that are formed on the document table **2030** in association with the document **2000G** having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. According to the present exemplary embodiment, the document **2000G** having the A3 size is a document having the maximum size that has an image that the image capturing member **2020** is capable of capturing. The document **2000G** having the A3 size is an example of a document having the maximum size that has an image that the image capturing member **2020** is capable of capturing.

Document Table

As illustrated in FIG. 2-2, the document table **2030** is disposed below the image capturing member **2020** and is an example of a placement portion on which the document **2000G** the image of which is captured by the image capturing member **2020** is placed. The document table **2030** is constituted by a covering **2052** of the reading member **2050** that is disposed in the -W direction and that will be described later and a flat plate **2032** that is disposed next to the covering **2052** in the +W direction and that extends along a D-W plane. The covering **2052** and the flat plate **2032** are examples of a member that constitutes the document table **2030**. That is, the document table **2030** is constituted by the covering **2052** and the flat plate **2032** that are two members. The covering **2052** and the flat plate **2032** have a rectangular shape when viewed in the vertical direction of the apparatus. The document table **2030** extends along the D-W plane and has a placement surface **2030a** on which the document **2000G** is placed. The placement surface **2030a** is constituted by an upper surface **2052a** of the covering **2052** and an upper surface **2032a** of the flat plate **2032** that has an edge in the -W direction adjacent to an edge of the covering **2052** in the +W direction such that these edges meet with a gap **2000S** interposed therebetween.

As illustrated in FIG. 2-2, FIG. 2-3, and FIG. 2-5, the gap **2000S** extends in the H direction from the placement surface **2030a** to the back surfaces (surfaces opposite the placement surface **2030a**) of the covering **2052** and the flat plate **2032** when viewed in the depth direction of the apparatus. The gap **2000S** forms a boundary line **2000K** on the placement surface **2030a**. In other words, as illustrated in FIG. 2-5, the boundary line **2000K** is formed by the gap **2000S** between the upper surface **2052a** of the covering **2052** and the upper surface **2032a** of the flat plate **2032**. The single boundary line **2000K** linearly extends in the depth direction of the apparatus. In FIG. 2-2, FIG. 2-3, and FIG. 2-5, the gap **2000S** is emphatically illustrated so as to be formed between the covering **2052** and the flat plate **2032**.

The gap **2000S** may be formed with a part of the covering **2052** and a part of the flat plate **2032** overlapping in a plan view. In this case, the part of the covering **2052** may be disposed above the part of the flat plate **2032**, or the part of the flat plate **2032** may be disposed above the part of the covering **2052**.

The document **2000G** may be placed on the document table **2030** such that the position of the center of the document **2000G** in the width direction of the apparatus overlaps that of the center of the capturing range (the angle of view) of the image capturing member **2020**. In particular, the document **2000G** having the A4 size may be placed on the document table **2030** such that the position of the center of the document **2000G** in the width direction of the appa-

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ratus overlaps that of the center of the capturing range (the angle of view) of the image capturing member 2020. The document 2000G having the A4 size may be placed on the document table 2030 such that the center of the document 2000G overlaps the center of the document 2000G having the A3 size associated with the marks 2000M that are formed on the document table 2030. When the document 2000G having the A4 size is thus placed on the document table 2030, a part of the boundary line 2000K is covered by the document 2000G having the A4 size regardless of whether the document 2000G is vertically or laterally placed.

Accordingly, the boundary line 2000K is positioned between the “center of the capturing range (or the center of the position at which the document 2000G is placed)” and “an edge in the width direction of the apparatus of the document 2000G having the A3 size that is placed at the center of the capturing range (or the center of the position at which the document 2000G is placed)”. According to the present exemplary embodiment, the boundary line 2000K is formed between the center of the capturing range and the marks 2000M.

The document table 2030 has the marks 2000M for adjusting the position of the document 2000G to be placed on the placement surface 2030a. The marks 2000M are associated with the document 2000G having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. The marks 2000M are formed by painting the placement surface 2030a.

Each mark 2000M has a L-shape when viewed in the vertical direction of the apparatus, and the four marks are formed on the placement surface 2030a at respective associated four corners of the document 2000G having the A3 size the longitudinal direction of which coincides with the width direction of the apparatus. Among the four marks 2000M, the marks 2000M in the -W direction are formed on the upper surface 2052a of the covering 2052 of the reading member 2050, and the marks 2000M in the +W direction are formed on the upper surface 2032a of the flat plate 2032. The position of the center of the document 2000G having the A3 size associated with the four marks 2000M in the width direction of the apparatus overlaps that of the center of the capturing range (the angle of view) of the image capturing member 2020.

The marks 2000M are formed on the upper surface 2052a of the covering 2052 or the upper surface 2032a of the flat plate 2032 at positions away from the boundary line 2000K. In other words, the boundary line 2000K is formed at a position away from the marks 2000M along the placement surface 2030a. According to the present exemplary embodiment, the boundary line 2000K is formed at a position at which the boundary line 2000K is covered by the document 2000G with the document 2000G having the A3 size placed on the document table 2030. The boundary line 2000K is not located at a position at which the boundary line 2000K is not covered by the document 2000G having the A3 size within the capturing range of the image capturing member 2020. In other words, the boundary line 2000K is formed only in the range in which the boundary line 2000K is covered by the document 2000G having the A3 size with the document 2000G having the A3 size placed on the placement surface 2030a within the capturing range of the image capturing member 2020. Outside the capturing range, the boundary line 2000K may be located also out of a range in which the boundary line 2000K is covered by the document 2000G having the A3 size.

As illustrated in FIG. 2-3 and FIG. 2-4, the flat plate 2032 is a plate member that is disposed so as to be slidable in the

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width direction of the apparatus with respect to the covering 2052 of the reading member 2050 by using a guide mechanism, not illustrated. The flat plate 2032 has the upper surface 2032a that faces upward. The upper surface 2032a of the flat plate 2032 is an example of a surface that constitutes the placement surface 2030a. The flat plate 2032 has a function of uncovering an upper surface of the loading portion 2040 described later (see FIG. 2-4) by sliding in the +W direction from a state in which the flat plate 2032 is adjacent to the covering 2052 (see FIG. 2-3). The length of the flat plate 2032 in the width direction of the apparatus is greater than half of the length of the placement surface 2030a in the width direction of the apparatus. That is, the boundary line 2000K between the covering 2052 and the flat plate 2032 is away in the -W direction from a bisector that bisects the placement surface 2030a in the width direction of the apparatus.

It may be also said that the document table 2030 has the boundary line 2000K that is formed by the covering 2052 and the flat plate 2032 (examples of multiple components) on the placement surface 2030a (an example of a surface). The boundary line 2000K is located on a central portion when the placement surface 2030a is divided into three portions.

Specifically, the boundary line 2000K is located in a central portion when the placement surface 2030a is divided into three portions in the width direction of the apparatus. Detection Reading Member

As illustrated in FIG. 2-6, the CPU 2011 is a central processing unit, runs various programs, and controls components. That is, the CPU 2011 reads a program from the ROM 2012 or the storage 2014 and runs the program with a RAM 2013 used as a work area. The CPU 2011 controls the components described above and performs various kinds of arithmetic processing in accordance with the program that is recorded in the ROM 2012 or the storage 2014. According to the present exemplary embodiment, the ROM 2012 or the storage 2014 stores a document detection program for detecting an edge of the document 2000G by performing a process of detecting the color difference on the image that is captured by the image capturing member 2020. The ROM 2012 or the storage 2014 stores a captured image reading program for reading the image that is formed on the upper surface (the front surface) of the detected document 2000G the image of which is captured. The CPU 2011 functions as the detection reading member 2011a that uses the programs to measure the color difference between the document 2000G and the placement surface 2030a of the document table 2030 from the image that is captured by the image capturing member 2020, detects the edge of the document 2000G, and reads the image that is formed on the document 2000G. That is, the detection reading member 2011a has a function of detecting only the document 2000G from the image of the document 2000G, placed on the placement surface 2030a, which is captured by the image capturing member 2020, and reading the document 2000G.

The ROM 2012 stores various programs and various kinds of data. The RAM 2013 serves as a work area and temporarily stores a program or data. The storage 2014 is constituted by a hard disk drive (HDD) or a solid state drive (SSD) and stores various programs including an operating system and various kinds of data.

Loading Portion

As illustrated in FIG. 2-2, the loading portion 2040 is disposed below the document table 2030 and is a part of the housing of the document reading device 2010 that has a bottom surface 2042 that is dented from the placement

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surface **2030a** and that has a rectangular shape. As for an end portion of the bottom surface **2042** in the $-W$ direction, the position in the width direction of the apparatus overlaps that of the boundary line **2000K** of the document table **2030**. The bottom surface **2042** tilts with respect to the horizontal direction such that the end portion in the $-W$ direction is lower than an end portion in the $+W$ direction.

As illustrated in FIG. 2-4, the loading portion **2040** has a function of enabling multiple documents **2000G** to be loaded on the bottom surface **2042** in a state in which an end portion of the flat plate **2032** in the $-W$ direction slides to a location away from the loading portion **2040** in the $+W$ direction and uncovers the upper surface of the loading portion **2040**. In this state, the loading portion **2040** has a function of enabling the document **2000G** that has a length in the width direction of the apparatus greater than that of A4 (a short side) to be placed thereon with the document **2000G** extending across the upper surface **2032a** of the flat plate **2032** slid and the bottom surface **2042**.

Reading Member

The reading member **2050** is an auto document feeder (a so-called ADF) and has a function of reading the document **2000G** that is placed on the loading portion **2040** while transporting the document **2000G**. The reading member **2050** includes the covering **2052**, an intake port **2054**, a transport unit **2056**, a reading unit **2060**, and an outlet **2058**.

The intake port **2054** is adjacent to the loading portion **2040** in the $-W$ direction. The intake port **2054** is an example of the opening. As for the outlet **2058**, the position in the vertical direction of the apparatus is lower than that of the intake port **2054** and is higher than that of the discharge portion **2170** of the formation device **2110**. The outlet **2058** is adjacent to the discharge portion **2018** in the $-W$ direction.

The transport unit **2056** includes a transport path **2056a** that extends from the intake port **2054** to the outlet **2058** and that has a substantially U-shape and multiple roller members **2056b** that are disposed along the transport path **2056a**. The transport unit **2056** takes in the document **2000G** that is placed on the loading portion **2040** via the intake port **2054** by using the roller members **2056b** and transports the document **2000G** toward the outlet **2058** along the transport path **2056a**.

The reading unit **2060** is a stationary contact image sensor that is disposed along the transport path **2056a** and has a function of reading the images that are formed on the front surface and the back surface of the document **2000G** that is transported by the transport unit **2056**. The reading unit **2060** includes a front surface reading member **2060a** that reads the image that is formed on the front surface of the document **2000G** and a back surface reading member **2060b** that reads the image that is formed on the back surface of the document **2000G**.

The covering **2052** is a panel member that has an L-shaped section when viewed in the depth direction of the apparatus and that covers an upper surface of the reading member **2050**. The covering **2052** has the upper surface **2052a** that faces upward. The upper surface **2052a** of the covering **2052** is an example of the surface that constitutes the placement surface **2030a** as described above. The upper surface **2052a** of the covering **2052** is substantially flush with the upper surface **2032a** of the flat plate **2032** when the flat plate **2032** is adjacent to the covering **2052** such that the edge of the flat plate **2032** in the $-W$ direction and the edge of the covering **2052** in the $+W$ direction meet with the gap **2000S** interposed therebetween (see FIG. 2-2 and FIG. 2-3).

The covering **2052** is configured so as to be capable of uncovering the upper surface of the reading member **2050**

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by using an opening-closing mechanism, not illustrated, and exposing the transport path **2056a**. The reading member **2050** has a function of uncovering the upper surface of the reading member **2050** by using the opening-closing mechanism of the covering **2052** and enabling the document **2000G** that is jammed on the transport path **2056a** to be manually taken out when the document **2000G** is jammed on the transport path **2056a** due to a transport failure of the transport unit **2056**.

As illustrated in FIG. 2-2, the discharge portion **2018** is a tray member that is disposed below the outlet **2058** and above the discharge portion **2170** of the formation device **2110**. The discharge portion **2018** has a function of receiving the document **2000G** that is transported by the transport unit **2056** and that is discharged via the outlet **2058**.

Action and Effect

Action and effect according to the present exemplary embodiment will now be described. In the case where the same component as that of the image forming apparatus **2100** according to the present exemplary embodiment, for example, is used in the description when a comparative exemplary embodiment relative to the present exemplary embodiment is described, the reference character and name of the component, for example, are used as they are for the description.

As for the document reading device **2010** according to the present exemplary embodiment, the boundary line **2000K** is formed at a position at which the boundary line **2000K** is covered by the document **2000G** with the document **2000G** having the A3 size placed on the document table **2030**.

As for the document reading device **2010** according to the present exemplary embodiment, the boundary line **2000K** is not located at the position at which the boundary line **2000K** is not covered by the document **2000G** having the A3 size within the capturing range of the image capturing member **2020**.

The document reading device **2010** according to the present exemplary embodiment is configured such that the boundary line **2000K** is formed by the gap **2000S** between the covering **2052** and the flat plate **2032** on the placement surface **2030a**.

The document reading device **2010** according to the present exemplary embodiment is configured such that the boundary line **2000K** is formed at the position away from the marks **2000M** for adjusting the position of the document **2000G** along the placement surface **2030a**.

The document reading device **2010** according to the present exemplary embodiment is configured so as to have the single boundary line **2000K**. Therefore, the document reading device **2010** according to the present exemplary embodiment looks good, unlike the case where there are multiple boundary lines **2000K**.

The document reading device **2010** according to the present exemplary embodiment also includes the reading member **2050**. Therefore, the document reading device **2010** according to the present exemplary embodiment is capable of reading the document **2000G** by selecting one from the image capturing member **2020** and the reading member **2050**.

As for the document reading device **2010** according to the present exemplary embodiment, a member of two members that constitute the document table **2030** is the covering **2052** that covers the upper surface of the reading member **2050**. In this way, the document reading device **2010** according to the present exemplary embodiment may have a compact size, unlike the case where the member of the two members

that constitute the document table **2030** is disposed between the flat plate **2032** and the covering **2052** of the reading member **2050**.

The document reading device **2010** according to the present exemplary embodiment includes the reading member **2050** that transports the document **2000G** from the intake port **2054** adjacent to the loading portion **2040** in the $-W$ direction and the flat plate **2032** that is capable of moving in the $+W$ direction with respect to the reading member **2050** and uncovering the upper surface of the loading portion **2040**. In this way, the document reading device **2010** according to the present exemplary embodiment that is configured so as to include the reading member **2050** may load the document **2000G** having an increased size across the loading portion **2040** and the flat plate **2032** that moves in the $+W$ direction, unlike the case where the flat plate **2032** moves in the depth direction of the apparatus.

The document reading device **2010** according to the present exemplary embodiment is configured such that the flat plate **2032** moves by sliding in the width direction of the apparatus. In this way, the document reading device **2010** according to the present exemplary embodiment may facilitate movement of the flat plate **2032** to uncover the upper surface of the loading portion **2040**, unlike the case where the flat plate **2032** moves by rotating.

The document reading device **2010** according to the present exemplary embodiment is configured such that the length of the covering **2052** in the width direction of the apparatus is less than half of the length of the placement surface **2030a** in the width direction of the apparatus. That is, the document reading device **2010** according to the present exemplary embodiment is configured such that the length of the flat plate **2032** in the width direction of the apparatus is greater than half of the length of the placement surface **2030a** in the width direction of the apparatus. In this way, the document reading device **2010** according to the present exemplary embodiment may load the document **2000G** having an increased size across the loading portion **2040** and the flat plate **2032**, unlike the case where the length of the covering **2052** in the width direction of the apparatus is greater than half of the length of the placement surface **2030a** in the width direction of the apparatus.

The image forming apparatus **2100** according to the present exemplary embodiment that is configured such that the image of the document **2000G** is captured by the image capturing member **2020** may be capable of inhibiting the image capturing member **2020** from failing to copy the document **2000G**, unlike the case where the boundary line **2000K** is formed within the capturing range of the image capturing member **2020** and outside the document **2000G**.

In particular, the image forming apparatus **2100** according to the present exemplary embodiment is configured such that a part of the boundary line **2000K** is covered by the document **2000G** when the document **2000G** having the A4 size equal to the sizes of the largest number of the sheet materials **2000P** having the A4 size that are containable in the container unit **2120** is placed on the placement surface **2030a**. In this way, the image forming apparatus **2100** according to the present exemplary embodiment may be capable of inhibiting the image capturing member **2020** from failing to copy the document **2000G**, unlike the case where the boundary line **2000K** is formed within the capturing range of the image capturing member **2020** and outside the document **2000G** having the A4 size.

The image forming apparatus **2100** according to the present exemplary embodiment includes the panel member **2102** that displays the information about the image forming

apparatus **2100** in the $+D$ direction. In this way, as for the image forming apparatus **2100** according to the present exemplary embodiment that is configured so as to include the panel member **2102** that displays the information in the $+D$ direction, the flat plate **2032** that has the upper surface **2032a** is readily moved while the content of display of the panel member **2102** is checked, unlike a configuration in which the flat plate **2032** moves in the $+D$ direction.

The second exemplary embodiment is described in detail above, but the present disclosure is not limited to the exemplary embodiment described above, and various modifications, alterations, and improvements may be made within the range of the technical idea of the exemplary embodiment of the present disclosure.

It is not necessary for the placement surface **2030a** according to the present exemplary embodiment to be completely flat. That is, the placement surface **2030a** may have any shape, provided that the document **2000G** is placed on the placement surface **2030a**, and the image capturing member **2020** is capable of capturing the image. Accordingly, for example, the placement surface **2030a** may have unevenness, provided that the document **2000G** is placed on the placement surface **2030a**, and the image capturing member **2020** is capable of capturing the image.

For example, the formation device **2110** according to the present exemplary embodiment is an electrophotographic formation device. However, the formation device **2110** may be, for example, an ink-jet formation device or an offset print formation device.

The document table **2030** according to the present exemplary embodiment is constituted by two members of the covering **2052** of the reading member **2050** and the flat plate **2032**. However, the document table **2030** may be constituted by a single member an upper surface of which is divided into multiple surfaces by a single or multiple grooves. The single or multiple grooves is examples of a boundary line. The document table **2030** may be constituted by three or more members. That is, the number of the gap **2000S** between members according to the exemplary embodiment of the present disclosure and the number of the boundary line **2000K** that is formed by the gap **2000S** may be two or more.

The single boundary line **2000K** according to the present exemplary embodiment linearly extends. However, the boundary line **2000K** may bend or may curve on the placement surface **2030a**.

The flat plate **2032** according to the present exemplary embodiment moves by sliding in the width direction of the apparatus. However, the flat plate **2032** according to the present exemplary embodiment may be configured so as to slide in the depth direction of the apparatus. The flat plate **2032** according to the present exemplary embodiment may be configured so as to move by rotating to uncover the upper surface of the loading portion **2040** or may be configured so as to be installable on and removable from the document table **2030**.

The length of the flat plate **2032** according to the present exemplary embodiment in the width direction of the apparatus is greater than half of the length of the placement surface **2030a** in the width direction of the apparatus. However, the length of the flat plate **2032** according to the present exemplary embodiment in the width direction of the apparatus is not limited to a length greater than half of the length of the placement surface **2030a** in the width direction of the apparatus but may be equal to half of the length of the placement surface **2030a** in the width direction of the

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apparatus or may be less than half of the length of the placement surface **2030a** in the width direction of the apparatus.

As for the document reading device **2010** according to the present exemplary embodiment, the maximum size of the document **2000G** that has an image that the image capturing member **2020** is capable of capturing is the A3 size. However, the maximum size of the document that has an image that the image capturing member **2020** according to the present exemplary embodiment is capable of capturing is not limited to the A3 size but may be a size larger than the A3 size such as A2 or B3 or may be a size smaller than the A3 size such as A4 or B4.

As for the image forming apparatus **2100** according to the present exemplary embodiment, the size of the largest number of sheet materials **2000P** that are containable in the container unit **2120** is A4. However, the size of the largest number of sheet materials **2000P** that are containable in the container unit **2120** according to the present exemplary embodiment is not limited to A4 but may be A3 or may be B5.

Third Exemplary Embodiment

A third exemplary embodiment will now be described by way of example with reference to the drawings.

Image Reading Apparatus **3010**

An image reading apparatus **3010** according to the third exemplary embodiment will be described. FIG. **3-1** schematically illustrates the image reading apparatus **3010** according to the present exemplary embodiment. An arrow H illustrated in figures represents a vertical direction of the apparatus. An arrow W represents a width direction of the apparatus (specifically, a horizontal direction). An arrow D represents a depth direction of the apparatus (specifically, a horizontal direction). The vertical direction of the apparatus, the width direction of the apparatus, and the depth direction of the apparatus intersect each other (specifically, at right angles).

The image reading apparatus **3010** illustrated in FIG. **3-1** reads an image. Specifically, as illustrated in FIG. **3-1**, the image reading apparatus **3010** includes a transport unit **3014**, an image forming member **3012**, a reading member **3030**, a transport mechanism **3040**, a document table **3050**, a discharge portion **3060**, a document table **3070**, and a camera **3080**.

According to the present exemplary embodiment, the image reading apparatus **3010** has a function of forming an image and may be thought to be an image forming apparatus as described later.

Transport Unit **3014** and Image Forming Member **3012**

The transport unit **3014** (see FIG. **3-1**) transports a recording medium **3000P** such as paper that is contained in a container unit **3016**. Specifically, as illustrated in FIG. **3-1**, the transport unit **3014** includes transport members **3014A** such as multiple transport rollers and transports the recording medium **3000P** by using the transport members **3014A**.

The image forming member **3012** (see FIG. **3-1**) forms an image on the recording medium **3000P** that is transported by the transport unit **3014**. The image forming member **3012** is capable of forming an image that is read by the reading member **3030** or the camera **3080** on the recording medium **3000P**.

Specifically, the image forming member **3012** forms a toner image (an example of an image) on the recording medium **3000P** by using an electrophotographic system. Specifically, as illustrated in FIG. **3-1**, the image forming

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member **3012** includes toner image forming members **3020Y**, **3020M**, **3020C**, and **3020K** (referred to below as **3020Y** to **3020K**), a transfer body **3024**, and fixing portions **3026**.

As for the image forming member **3012**, the toner image forming members **3020Y** to **3020K** perform charging, exposing, developing, and transferring processes and form toner images in respective colors of yellow (Y), magenta (M), cyan (C), and black (K) on the transfer body **3024**. The image forming member **3012** transfers the toner images in the respective colors that are formed on the transfer body **3024** to the recording medium **3000P**. The fixing portions **3026** fix the toner images to the recording medium **3000P**. The image forming member **3012** thus uses an intermediate transfer system that transfers the images to the recording medium **3000P** via the transfer body **3024**.

Reading Member **3030** and Transport Mechanism **3040**

The reading member **3030** illustrated in, for example, FIG. **3-1** is a component that reads the image of a document **3000G** (see FIG. **3-3** and FIG. **3-5**) that is transported. As illustrated in FIG. **3-1**, the reading member **3030** is installed above the image forming member **3012**.

Specifically, the reading member **3030** includes reading sensors **3032** and **3034** and a covering **3033**.

The reading sensor **3032** is a functional unit that has a function of reading an image on one of surfaces of the document **3000G** that is transported. The reading sensor **3034** is a functional unit that has a function of reading an image on the other surface of the document **3000G** that is transported. Examples of the reading sensors **3032** and **3034** include a contact image sensor abbreviated as a CIS.

The covering **3033** is a component that covers the reading sensor **3032**. Specifically, as illustrated in FIG. **3-1**, the covering **3033** includes a top plate **3035** and side plates **3037** and **3039**.

The top plate **3035** is installed above the reading sensor **3032** and covers the reading sensor **3032** from above. As illustrated in FIG. **3-2** and FIG. **3-4**, a part of the document **3000G** the image of which is read by the camera **3080** is placed on the top plate **3035**.

The side plate **3037** is installed at a position away from the reading sensor **3032** in one of the width directions of the apparatus (the left-hand direction in FIG. **3-1**) and covers the reading sensor **3032** from the position away therefrom in one of the width directions of the apparatus. The side plate **3039** is installed at a position away from the reading sensor **3032** in the other width direction of the apparatus (the right-hand direction in FIG. **3-1**) and covers the reading sensor **3032** from the position away therefrom in the other width direction of the apparatus.

The one of the width directions of the apparatus corresponds to the left-hand direction of the image reading apparatus **3010**, and accordingly, the one of the width directions of the apparatus is referred to below as the left-hand direction. The other width direction of the apparatus corresponds to the right-hand direction of the image reading apparatus **3010**, and accordingly, the other width direction of the apparatus is referred to below as the right-hand direction. These directions are defined for convenience of description, and the configuration of the apparatus is not limited by the directions.

The side plate **3039** has an inlet **3039A** via which the document **3000G** that is placed on the document table **3050** enters a location inside the covering **3033**. The inlet **3039A** is configured as an opening that extends in the depth direction of the apparatus.

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An outlet 3039B via which the document 3000G is discharged from the location inside the covering 3033 onto the discharge portion 3060 is formed below the inlet 3039A of the side plate 3039. The outlet 3039B is configured as an opening that extends in the depth direction of the apparatus.

The transport mechanism 3040 transports the document 3000G the image of which is read by the reading member 3030. The transport mechanism 3040 is disposed inside the covering 3033. In other words, the transport mechanism 3040 is covered by the covering 3033.

Specifically, the transport mechanism 3040 includes multiple transport members 3040A such as transport rollers. The transport mechanism 3040 transports the document 3000G that is placed on the document table 3050 from the inlet 3039A to the outlet 3039B (that is, the discharge portion 3060) along a C-shaped transport path by using the transport members 3040A.

As for the image reading apparatus 3010, the transport mechanism 3040 thus transports the document 3000G from the document table 3050 to the discharge portion 3060, and the reading member 3030 reads the image of the document 3000G that is transported by the transport mechanism 3040. Document Table 3050

As illustrated in FIG. 3-3, FIG. 3-5, and FIG. 3-7, the document table 3050 is a table on which the document 3000G the image of which is read by the reading member 3030 (see FIG. 3-1) is placed. That is, the document table 3050 may be referred to as a component on which the document 3000G that is transported by the transport mechanism 3040 (see FIG. 3-1) is placed. The document table 3050 is an example of a second placement portion.

As for the document table 3050, the document 3000G is placed on an upper surface 3050A. Accordingly, the upper surface 3050A of the document table 3050 is referred to as a placement surface on which the document 3000G is placed. According to the present exemplary embodiment, the document 3000G is placed on the document table 3050 from above with the document table 3070 positioned at an open position as described later.

According to the present exemplary embodiment, the meaning of the “document table” represents a component on which the document 3000G is placed. Accordingly, the meaning of the “table” does not include a meaning that specifies a shape.

As illustrated in FIG. 3-1, the document table 3050 is installed adjacent to the reading member 3030 in the right-hand direction. The document table 3050 is installed below the document table 3070 and below a top surface 3035A of the top plate 3035. Accordingly, there is a step between the upper surface 3050A of the document table 3050 and the top surface 3035A of the top plate 3035.

The document table 3050 has a plate shape that extends from the side plate 3039 of the covering 3033 in the right-hand direction. The upper surface 3050A of the document table 3050 extends from the inlet 3039A in the right-hand direction. The upper surface 3050A is an inclined surface that gradually extends upward from the inlet 3039A while extending in the right-hand direction.

As illustrated in FIG. 3-3 and FIG. 3-7, the document table 3050 includes a restriction portion 3058 (a so-called a side guide). The restriction portion 3058 comes into contact with side edges of the document 3000G that is placed on the document table 3050 and restricts movement of the document 3000G in both of the depth directions of the apparatus.

One of the depth directions of the apparatus corresponds to the front direction of the image reading apparatus 3010, and accordingly, the one of the depth directions of the

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apparatus is referred to below as the front direction. The other depth direction of the apparatus corresponds to the rear direction of the image reading apparatus 3010, and accordingly, the other depth direction of the apparatus is referred to below as the rear direction. These directions are defined for convenience of description, and the configuration of the apparatus is not limited by the directions.

Discharge Portion 3060

The discharge portion 3060 illustrated in, for example, FIG. 3-5 is a component onto which the document 3000G the image of which is read by the reading member 3030 is discharged. That is, the discharge portion 3060 may be referred to as a component on which the document 3000G that is transported by the transport mechanism 3040 is placed. As for the discharge portion 3060, the document 3000G is placed on an upper surface 3060A. Accordingly, the upper surface 3060A of the discharge portion 3060 is referred to as a placement surface on which the document 3000G is placed.

The discharge portion 3060 is installed below the top surface 3035A of the top plate 3035 and below the document table 3050. The discharge portion 3060 has a plate shape that extends from the side plate 3039 of the covering 3033 in the right-hand direction.

The upper surface 3060A of the discharge portion 3060 extends from the outlet 3039B in the right-hand direction. The upper surface 3060A is configured as an inclined surface that gradually extends upward from the outlet 3039B while extending in the right-hand direction.

Document Table 3070

As illustrated in FIG. 3-2 and FIG. 3-4, the document table 3070 is a table on which the document 3000G the image of which is read by the camera 3080 is placed. The document table 3070 is an example of a placement portion and an example of a first placement portion.

As for the document table 3070, the document 3000G is placed on an upper surface 3070A. Accordingly, the upper surface 3070A of the document table 3070 is referred to as a placement surface on which the document 3000G is placed. As illustrated in FIG. 3-4, the upper surface 3070A of the document table 3070 is flush with the top surface 3035A of the top plate 3035. That is, the upper surface 3070A of the document table 3070 and the top surface 3035A of the top plate 3035 are on the same plane.

The document table 3070 is installed above the discharge portion 3060 and above the document table 3050. Accordingly, the document table 3070 covers the upper surface 3050A of the document table 3050 from above. The document table 3070 is disposed at a height such that a user who operates the image reading apparatus 3010 is able to perform an operation of placing the document 3000G with the user standing. Specifically, the document table 3070 is disposed, for example, at a height of about 1000 mm away from a floor on which the image reading apparatus 3010 is installed.

Specifically, as illustrated in FIG. 3-3, the document table 3070 includes a top plate 3071 and side plates 3072, 3074, and 3075. The top plate 3071 has a plate shape the thickness direction of which coincides with the vertical direction as illustrated in FIG. 3-4 and has a substantially rectangular shape in a plan view as illustrated in FIG. 3-6. As for the document table 3070, as illustrated in FIG. 3-2, FIG. 3-4, and FIG. 3-6, the document 3000G is placed on the top plate 3071. Accordingly, the upper surface of the top plate 3071 constitutes the upper surface 3070A of the document table 3070.

As illustrated in FIG. 3-4, the top plate 3071 is installed above the document table 3050. The dimensions of the top

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plate **3071** in the width direction of the apparatus and in the depth direction of the apparatus are larger than the dimensions of the document table **3050** in the width direction of the apparatus and in the depth direction of the apparatus.

The side plate **3075** extends downward from a right-hand edge portion of the top plate **3071**. The side plate **3075** has a plate shape the thickness direction of which coincides with the width direction of the apparatus. The side plate **3075** is installed at a position away from the document table **3050** in the right-hand direction and covers the document table **3050** from the position away therefrom in the right-hand direction.

The side plates **3072** and **3074** extend downward from a front edge portion and a rear edge portion of the top plate **3071** (see FIG. 3-3). The side plates **3072** and **3074** have a plate shape the thickness direction of which coincides with the depth direction of the apparatus. The side plates **3072** and **3074** are installed at positions in front of and behind the document table **3050** and cover the document table **3050** from the positions in front of and behind the document table **3050**.

The document table **3070** is movable to a covering position (a position illustrated in FIG. 3-1, FIG. 3-2, FIG. 3-4, and FIG. 3-6) at which the document table **3070** covers the document table **3050** from above and an open position (a position illustrated in FIG. 3-3, FIG. 3-5, and FIG. 3-7) at which the document table **3070** uncovers the upper surface of the document table **3050**. According to the present exemplary embodiment, the document table **3070** is supported by an apparatus body **3011** that serves as a support body so as to be movable to the covering position and the open position.

As illustrated in FIG. 3-8 and FIG. 3-9, the side plates **3072** and **3074** include respective rails **3076** that have long holes that extend in a left-right direction. Two shaft portions **3057** that are disposed on the front surface and rear surface of the document table **3050** are inserted in the rails **3076**. The two shaft portions **3057** guide the document table **3070** to the covering position (a position illustrated in FIG. 3-8) and the open position (a position illustrated in FIG. 3-9) along the rails **3076**. The document table **3070** moves between the covering position (the position illustrated in FIG. 3-8) and the open position (the position illustrated in FIG. 3-9) in the width directions of the apparatus.

As illustrated in FIG. 3-3, FIG. 3-5, and FIG. 3-7, the document table **3070** is located opposite the document table **3050** with respect to the reading member **3030** (that is, away therefrom in the right-hand direction) when being positioned at the open position. A part of the document **3000G** that is placed on the document table **3050** is placed on the document table **3070** that is positioned at the open position.

The image reading apparatus **3010** thus includes a movement mechanism that moves the document table **3070** by using the two shaft portions **3057** and the rails **3076**. The movement mechanism is not limited by the configuration described above and may be constituted by various mechanical factors.

The document table **3070** has a function of a covering that covers the document table **3050** as described above. That is, the document table **3070** doubles as the covering that covers the document table **3050**. The document table **3070** has the function of the covering and may be thought to be the covering (a covering member) of the document table **3050**. Camera **3080**

The camera **3080** is a camera (a so-called a document camera) that is capable of reading the image of the document **3000G** (an example of an object) that is placed on the

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document table **3070**. The camera **3080** is installed above the document table **3070**. The camera **3080** faces downward and is capable of reading the image of the document **3000G** that is placed on the upper surface **3070A** of the document table **3070**. The camera **3080** is an example of an image capturing member that captures the image of the object.

An example of the camera **3080** is a digital camera that includes an optical system such as a lens and an imaging element that converts photosensitivity into an electrical signal. Examples of the imaging element include a charge coupled device (CCD) and a complementary metal oxide semiconductor (CMOS). The optical axis of the camera **3080** coincides with, for example, the vertical direction when viewed from the front.

The camera **3080** is supported by a support portion **3085** that is disposed on the apparatus body **3011**. The support portion **3085** includes a support rod **3087** that extends upward from the apparatus body **3011** at the rear of the document table **3070** and a mounting portion **3089** that extends forward from an upper end portion of the support rod **3087**. The camera **3080** is mounted on the mounting portion **3089**.

As illustrated in FIG. 3-6, the reading range **3000 MH** of the camera **3080** is across the reading member **3030** and the document table **3070**. The top surface **3035A** of the top plate **3035** of the reading member **3030** and the upper surface **3070A** of the document table **3070** are flat at least in the reading range **3000 MH** of the camera **3080**. According to the present exemplary embodiment, the entire top surface **3035A** of the top plate **3035** and the entire upper surface **3070A** of the document table **3070** are flat. The reading range **3000 MH** is an example of a capturing range.

The meaning of "being flat" described herein represents that the surfaces are flat to such an extent that the document **3000G** that is supported has a posture that is suitable for being read by the camera **3080**, and unevenness is permitted provided that the document is supported in the posture that is suitable for being read by the camera **3080**.

The posture of the document **3000G** that is suitable for reading is a smooth posture that enables at least the degree of a warp to be smaller than that in the case where the document **3000G** (specifically, plain paper) is placed across the top surface **3035A** of the top plate **3035** and the upper surface **3050A** of the document table **3050**.

The top surface **3035A** of the top plate **3035** of the reading member **3030** and the upper surface **3070A** of the document table **3070** may be perpendicular to the optical axis of the camera **3080**. As for the image reading apparatus **3010**, at least an angle between the top surface **3035A** and the optical axis of the camera **3080**, that is, the vertical direction and an angle between the upper surface **3070A** and the vertical direction are smaller than an angle between the upper surface **3050A** of the document table **3050** and the vertical direction when viewed from the front.

As for the image reading apparatus **3010**, the camera **3080** reads the image in a manner in which the document **3000G** is placed on the document table **3070** that is positioned at the covering position and the top surface **3035A** of the top plate **3035** of the reading member **3030**, and the document **3000G** is imaged by the camera **3080**. The image is formed on the document **3000G** and includes, for example, a character, a picture, and a photograph.

The top surface **3035A** and the upper surface **3070A** that function as the placement surfaces on which the document **3000G** is placed may not be flat. That is, the top surface **3035A** and the upper surface **3070A** may not be flat, provided that the top surface **3035A** and the upper surface

3070A enable the document **3000G** that is placed on the top surface **3035A** and the upper surface **3070A** to be imaged by the camera **3080**. Accordingly, for example, the top surface **3035A** and the upper surface **3070A** may have unevenness, provided that the document **3000G** is placed on the top surface **3035A** and the upper surface **3070A** and imaged by the camera **3080**.

Action According to Present Exemplary Embodiment

The image reading apparatus **3010** includes the reading member **3030** that reads the image of the document **3000G** that is transported, the discharge portion **3060** onto which the document **3000G** the image of which is read by the reading member **3030** is discharged, the document table **3070** that is installed above the discharge portion **3060**, and the camera **3080** that is installed above the document table **3070** and that is capable of reading the image of the document **3000G** (an example of the object) that is placed on the document table **3070**. In this way, as for the image reading apparatus **3010** that includes the reading member **3030** that reads the image of the document **3000G** transported, both of the reading member **3030** and the camera **3080** that reads the image of the document **3000G** and that is installed above the document table **3070** that is installed above the discharge portion **3060** to which the document **3000G** is discharged may be installable.

As for the image reading apparatus **3010**, the upper surface **3070A** of the document table **3070** is flat at least in the reading range **3000 MH** of the camera **3080**. In this way, the document **3000G** may be placed in a posture that is more suitable for being read by the camera **3080** than the case where the upper surface **3070A** of the document table **3070** has unevenness in the reading range **3000 MH** of the camera **3080** such that the document is difficult to support in the posture that is suitable for being read.

As for the image reading apparatus **3010**, the top surface **3035A** of the top plate **3035** of the reading member **3030** and the upper surface **3070A** of the document table **3070** are flat at least in the reading range **3000 MH** of the camera **3080**. In this way, the document **3000G** may be placed in a posture that is more suitable for being read by the camera **3080** than in the case where the upper surface **3070A** of the document table **3070** and the top surface **3035A** of the top plate **3035** have unevenness in the reading range **3000 MH** of the camera **3080** such that the document is difficult to support in the posture that is suitable for being read.

As for the image reading apparatus **3010**, the document **3000G** is placed on the upper surface **3070A** of the document table **3070** and the top surface **3035A** of the top plate **3035**. In this way, a range in which the document **3000G** is placed may be increased to a range wider than that in the case where the document **3000G** is placed on only the upper surface **3070A** of the document table **3070**.

As for the image reading apparatus **3010**, the document table **3070** is capable of moving to the open position at which the document table **3070** uncovers the upper surface of the document table **3050**. In this way, the document **3000G** may be placed on the document table **3050** from above the document table **3050**.

As for the image reading apparatus **3010**, a part of the document **3000G** that is placed on the document table **3050** is placed on the document table **3070** that is positioned at the open position. In this way, a range in which the document **3000G** is placed may be increased to a range wider than that in the case where the document **3000G** is placed on only the document table **3050**.

As for the image reading apparatus **3010**, the document table **3070** is disposed at the height such that the user who

operates the image reading apparatus **3010** is able to perform the operation of placing the document **3000G** with the user standing. In this way, the user may perform the operation of placing the document **3000G** on the document table **3070** while the user is not sitting.

The image reading apparatus **3010** includes the image forming member **3012** that is capable of forming the image that is read by the reading member **3030** or the camera **3080** on the recording medium **3000P**. In this way, the image of the document **3000G** that is transported or the image of the document **3000G** that is placed on the document table **3070** may be formed on the recording medium **3000P**.

Modification to Image Forming Member **3012**

Examples of the image forming member are not limited to the image forming member **3012** described above. An example of the image forming member may be an image forming member that uses a direct transfer method in which the toner image forming members **3020Y** to **3020K** directly form the toner images on the recording medium **3000P** without using the transfer body **3024**. Another example of the image forming member may be an image forming member that forms an image by spraying ink to the recording medium **3000P**, provided that the image forming member has a function of forming the image on the recording medium **3000P**.

Modification to Reading Member **3030**

The reading member **3030** uses a contact image sensor as a functional unit that has a function of reading an image but is not limited thereto. An example of the functional unit may be an image sensor such as a CCD, provided that the functional unit is capable of reading the image.

According to the present exemplary embodiment, the reading member **3030** includes the reading sensors **3032** and **3034** as functional units that have a function of reading an image but is not limited thereto. The reading member **3030** may be configured so as to include only one of the reading sensors **3032** and **3034**. That is, the reading member **3030** is capable of reading at least the image on one of the surfaces of the document **3000G**.

According to the present exemplary embodiment, the covering **3033** is thought to be a component that constitutes a part of the reading member **3030** that includes the reading sensors **3032** and **3034** but is not limited thereto. For example, the covering **3033** may be thought to be a component that constitutes a part of a transport unit that includes the transport mechanism **3040**.

Modification to Camera **3080**

An example of the image that is read by the camera **3080** is not limited to an image that includes, for example, a character, a picture, and a photograph. An example of the image that is read by the camera **3080** may be a still image or a video image, provided that the camera **3080** is capable of reading the image.

Examples of the object are not limited to the document **3000G**. An example of the object may be a video device (such as a smart phone or a mobile terminal) that captures an image, a book, and a card, provided that the camera **3080** is capable of imaging the object.

The object may be any object, provided that the object is placed on the document table **3070** and the covering **3033**. Accordingly, the camera **3080** may be configured so as to be capable of capturing not only the image but also the still image or video image of the object itself.

Modification to Document Table **3070**

As illustrated in FIG. 3-4, the upper surface **3070A** of the document table **3070** is flush with the top surface **3035A** of the top plate **3035** but is not limited thereto. The upper

surface **3070A** of the document table **3070** and the top surface **3035A** of the top plate **3035** may have a step. For example, the step is formed such that the document is supported in the posture that is suitable for being read by the camera **3080**.

Other Modifications

As for the image reading apparatus **3010**, the top surface **3035A** of the top plate **3035** of the reading member **3030** and the upper surface **3070A** of the document table **3070** may be flat at least in the reading range of the camera **3080** and may have unevenness outside the reading range of the camera **3080** such that the document is difficult to support in the posture that is suitable for being read.

The image reading apparatus **3010** is configured such that the document **3000G** the image of which is read by the camera **3080** is placed on the document table **3070** that is positioned at the covering position and the top surface **3035A** of the top plate **3035** of the reading member **3030** but may be configured such that the document **3000G** is placed on only the document table **3070**. In this case, the reading range **3000 MH** of the camera **3080** may be a range that covers only the document table **3070**.

As for the image reading apparatus **3010**, the document table **3070** is used as an example of the placement portion but is not limited thereto. As for the image reading apparatus **3010**, for example, the document table **3050** may be used as an example of the placement portion. That is, an object such as the document **3000G** the image of which is read by the camera **3080** may be placed on the document table **3050**. In this case, for example, the document table **3070** may be configured so as to be secured at the open position or the document table **3070** may not be provided.

The image reading apparatus **3010** includes the image forming member **3012** but may not include the image forming member **3012**.

The present disclosure is not limited to the exemplary embodiment described above but may be modified, altered, and improved in various ways without departing from the spirit thereof. For example, the modifications described above may be configured by appropriately combining some of these.

Fourth Exemplary Embodiment

An example of an image forming apparatus and a container apparatus according to a fourth exemplary embodiment of the present disclosure will be described with reference to FIG. 4-1 to FIG. 4-7. An arrow H illustrated in figures represents the vertical direction of the apparatus that is vertical. An arrow W represents the width direction of the apparatus that is horizontal. An arrow D represents the depth direction of the apparatus that is horizontal.

Entire Configuration of Image Forming Apparatus

As illustrated in FIG. 4-1, an image forming apparatus **4010** includes an image forming member **4012** that forms a toner image by using an electrophotographic system and a container apparatus **4110** that includes a transport unit **4014** that transports a recording medium P along a transport path **4016** and container units **4060**, **4070**, and **4080** that contain recording media P. The image forming apparatus **4010** also includes a controller **4028** that controls components and a principal power supply **4036** that supplies the power of a commercial principal power supply to components.

As for the image forming apparatus **4010** that has the above configuration, the recording media P are contained in the container units **4060**, **4070**, and **4080**, and the recording medium P that is contained in any one of the container units

4060, **4070**, and **4080** is transported along the transport path **4016** by using the transport unit **4014**. The toner image that is formed by the image forming member **4012** is formed on the transported recording medium P, and the recording medium P on which the toner image is formed is discharged to a location outside an apparatus body **4010a**.

Image Forming Member 4012

As illustrated in FIG. 4-1, the image forming member **4012** includes multiple toner image forming members **4030** that form toner images in respective colors and a transfer member **4032** that transfers the toner images that are formed by the toner image forming members **4030** to the recording medium P. The image forming member **4012** also includes a fixing device **4034** that fixes the toner images that are transferred to the recording medium P by using the transfer member **4032** to the recording medium P.

Toner Image Forming Member 4030

The toner image forming members **4030** are provided so as to form the toner images in the respective colors. According to the present fourth exemplary embodiment, the toner image forming members **4030** for the four colors of yellow (Y), magenta (M), cyan (C), and black (K) are provided. In the following description, Y, M, C, and K in reference characters are omitted when it is not necessary to distinguish among yellow (Y), magenta (M), cyan (C), and black (K).

The toner image forming members **4030** for the respective colors are basically configured in the same manner except for toner that is used. As illustrated in FIG. 4-2, each toner image forming member **4030** includes an image carrier **4040** that is rotatable and that is cylindrical and a charger **4042** that charges the image carrier **4040**. Each toner image forming member **4030** also includes an exposure device **4044** that radiates exposure light to the image carrier **4040** that is charged and that forms an electrostatic latent image and a developing device **4046** that develops the electrostatic latent image into the toner image by using a developer G that contains the toner. Consequently, the toner image forming members **4030** for the respective colors form images in the respective colors by using the toner in the respective colors.

As illustrated in FIG. 4-1, the image carriers **4040** for the respective colors are in contact with a transfer belt **4050** (described in detail later) that turns. The toner image forming members **4030** for yellow (Y), magenta (M), cyan (C), and black (K) are disposed in this order from an upstream position in the direction in which the transfer belt **4050** turns (see an arrow in the figure).

Transfer Member 4032

As illustrated in FIG. 4-1, the transfer member **4032** includes the transfer belt **4050** and first transfer rollers **4052** that are disposed opposite the image carriers **4040** for the respective colors with the transfer belt **4050** interposed therebetween and that transfer the toner images that are formed by the image carriers **4040** for the respective colors to the transfer belt **4050**.

The transfer member **4032** also includes a winding roller **4056** around which the transfer belt **4050** is wound and a drive roller **4058** around which the transfer belt **4050** is wound for transmitting rotational force to the transfer belt **4050**. Consequently, the transfer belt **4050** turns in the direction of the arrow in the figure.

The transfer member **4032** also includes a second transfer roller **4054** that is disposed opposite the winding roller **4056** with the transfer belt **4050** interposed therebetween and that transfers the toner images that are transferred to the transfer belt **4050** to the recording medium P. A transfer nip NT at

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which the toner images are transferred to the recording medium P is formed between the second transfer roller **4054** and the transfer belt **4050**.

With this configuration, the toner images are first-transferred to the transfer belt **4050** in the order of yellow (Y), magenta (M), cyan (C), and black (K) by using the first transfer rollers **4052**. The second transfer roller **4054** transfers the toner images from the transfer belt **4050** to the recording medium P that is transported between the transfer belt **4050** and the second transfer roller **4054**. The recording medium P to which the toner images are transferred is transported toward the fixing device **4034**.

Fixing Device **4034**

As illustrated in FIG. 4-1, the fixing device **4034** is disposed downstream of the transfer nip NT in the direction in which the recording medium P is transported. The fixing device **4034** heats and compresses the toner images that are transferred to the recording medium P and fixes the toner images to the recording medium P.

Container Apparatus **4110**

As illustrated in FIG. 4-1, the container apparatus **4110** is located in a lower portion of the image forming apparatus **4010** and includes the three container units **4060**, **4070**, and **4080** that contain the recording media P, and the transport unit **4014** that transports the recording media P. The container unit **4060** that is located at the highest position tilts with respect to the horizontal direction. The container apparatus **4110** will be described in detail later.

Controller **4028** and Principal Power Supply **4036**

The controller **4028** and the principal power supply **4036** are disposed in a triangular region that is formed between the container unit **4060** that tilts and the image forming member **4012**.

Configuration of Principal Component

The container apparatus **4110** will now be described. As illustrated in FIG. 4-1, the container apparatus **4110** is located in the lower portion of the image forming apparatus **4010**. As illustrated in FIG. 4-3, the container apparatus **4110** includes an apparatus body **4110a**, the container unit **4060** that contains the recording medium P, the container unit **4070** that contains the recording medium P, and the container unit **4080** that contains the recording medium P. The container apparatus **4110** also includes sliding rails **4064** that enable the container unit **4060** to move in the depth direction of the apparatus, sliding rails **4074** that enable the container unit **4070** to move in the depth direction of the apparatus, and sliding rails **4084** that enable the container unit **4080** to move in the depth direction of the apparatus. The container apparatus **4110** also includes the transport unit **4014** that transports the recording media P and a covering **4130** (see FIG. 4-5A and FIG. 4-5B) that covers the apparatus body **4110a** in the width direction of the apparatus.

The container unit **4060** is an example of a first container unit, the container unit **4070** is an example of a second container unit, and the container unit **4080** is an example of a third container unit. The depth direction of the apparatus is an example of a first direction. The width direction of the apparatus is an example of "sideways". The apparatus body **4110a** of the container apparatus **4110** corresponds to a lower portion of the apparatus body **4010a** of the image forming apparatus **4010**.

The container unit **4060**, the container unit **4070**, and the container unit **4080** are arranged in this order downward from above. The maximum size of the recording medium P that is containable in the container unit **4070** is smaller than the maximum size of the recording medium P that is

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containable in the container unit **4060**. The maximum size of the recording medium P that is containable in the container unit **4080** is smaller than the maximum size of the recording medium P that is containable in the container unit **4060** and is larger than the maximum size of the recording medium P that is containable in the container unit **4070**.

The maximum size of the recording medium P that is containable in the container unit **4060** is an example of a first size. The maximum size of the recording medium P that is containable in the container unit **4070** is an example of a second size. The maximum size of the recording medium P that is containable in the container unit **4080** is an example of a third size.

According to the present fourth exemplary embodiment, the container unit **4060** principally contains the recording medium P having the A3 size, and the A3 size is the maximum size of the recording medium P that is containable in the container unit **4060**. The container unit **4070** principally contains the recording medium P that has a postcard size, and the postcard size is the maximum size of the recording medium P that is containable in the container unit **4070**. The container unit **4080** principally contains the recording medium P having the A4 size, and the A4 size is the maximum size of the recording medium P that is containable in the container unit **4080**.

As for the length of each contained recording medium P in a front-rear direction, the recording medium P that is contained in the container unit **4060** is longest, and the recording medium P that is contained in the container unit **4080** is longer than the recording medium P that is contained in the container unit **4070**. The length of each recording medium P in the front-rear direction described herein corresponds to the length of the recording medium P in the direction in which the recording medium P is fed to the transport path **4016**.

The number of the recording media P that are containable in the container unit **4080** is larger than the number of the recording media P that are containable in the container unit **4060** and the number of the recording media that are containable in the container unit **4070**.

According to the present fourth exemplary embodiment, the container unit **4060** is capable of containing 200 recording media P, the container unit **4070** is capable of containing 100 recording media P, and the container unit **4080** is capable of containing 1000 recording media P.

As for the total thickness of the containable recording media P in the thickness direction, the total thickness of the recording media P that are containable in the container unit **4080** is the greatest thickness, and the total thickness of the recording media P that are containable in the container unit **4060** is greater than the total thickness of the recording media P that are containable in the container unit **4070**. That is, the total thickness of the recording media P that are containable in the container unit **4070** is the least thickness.

As for the image forming apparatus **4010**, the consumption of the recording media P having the A4 size is highest. That is, the number of the recording media P that are containable in the container unit **4080** and that have the highest consumption is larger than the number of the recording media P that are containable in the container unit **4060** and the number of the recording media P that are containable in the container unit **4070**.

Transport Unit **4014**

As illustrated in FIG. 4-1, the transport unit **4014** includes a feed roller **4020a** that feeds the recording medium P that is contained in the container unit **4060** to the transport path

4016 and prevention rollers 4022a that prevent multiple recording media P that are fed by the feed roller 4020a from being transported.

The transport unit 4014 also includes a feed roller 4020b that feeds the recording medium P that is contained in the container unit 4070 to the transport path 4016 and prevention rollers 4022b that prevent multiple recording media P that are fed by the feed roller 4020b from being transported.

The transport unit 4014 also includes a feed roller 4020c that feeds the recording medium P that is contained in the container unit 4080 to the transport path 4016 and prevention rollers 4022c that prevent multiple recording media P that are fed by the feed roller 4020c from being transported.

The transport unit 4014 also includes adjustment rollers 4024 that are disposed downstream of the prevention rollers 4022a, 4022b, and 4022c in the direction in which the recording medium P is transported and that adjust a timing with which the recording medium P is transported to the transfer nip NT. The transport unit 4014 also includes discharge rollers 4026 that discharge the recording medium P to which the toner images are fixed by the fixing device 4034 to the location outside the apparatus body 4010a.

Container Unit 4060 and Sliding Rail 4064

As illustrated in FIG. 4-3, the container unit 4060 has a box shape that opens upward. The two sliding rails 4064 are mounted to the respective edges of the container unit 4060 in the width direction of the apparatus. Each sliding rail 4064 includes an outer member, an intermediate member, and an inner member. The outer member is mounted on the apparatus body 4110a. The inner member is mounted on the container unit 4060. These enable the container unit 4060 to move in the depth direction of the apparatus with respect to the apparatus body 4110a.

As illustrated in FIG. 4-3 and FIG. 4-4A, the container unit 4060 tilts with respect to the horizontal direction with the container unit 4060 installed in the apparatus body 4110a such that a first end (a left-hand end portion in the figures) and a second end in the width direction of the apparatus are located at different heights when viewed in the depth direction of the apparatus. Specifically, the container unit 4060 tilts with respect to the horizontal direction such that the first end in the width direction of the apparatus is higher than the second end when viewed in the depth direction of the apparatus. The first end and the second end described herein correspond to portions of the container unit 4060 that are farthest from each other in the width direction of the apparatus. In other words, the first end of the container unit 4060 is a downstream end portion of the container unit 4060, and the second end of the container unit 4060 is an upstream end portion of the container unit 4060. The upstream end portion or the downstream end portion of the container unit 4060 means an intersection point between a plane on which the recording medium P that is contained in the container unit 4060 is placed and the container unit 4060.

The posture of the recording medium P that is contained in the container unit 4060 is that a surface of the recording medium P is along a bottom plate of the container unit 4060.

According to the present fourth exemplary embodiment, the container unit 4060 tilts with respect to the horizontal direction such that a leading edge Pf of the recording medium P that is contained in the container unit 4060 is higher than a trailing edge Pr. The leading edge Pf of the recording medium P described herein is an edge in the direction in which the recording medium P is fed to the transport path 4016 (see FIG. 4-1), and the trailing edge Pr of the recording medium P is an edge opposite the leading edge Pf.

The recording medium P that is contained in the container unit 4060 is transportable by using the transport unit 4014 with the container unit 4060 installed in the apparatus body 4110a. In other words, the container unit 4060 that is installed in the apparatus body 4110a is positioned at a first transport position at which the contained recording medium P is transportable.

The user pulls the container unit 4060 that is installed in the apparatus body 4110a forward in the depth direction of the apparatus, and consequently, the container unit 4060 is guided by the sliding rails 4064, comes into contact with a stopper not illustrated, stops, and is separated from the apparatus body 4110a as illustrated in FIG. 4-4B. The user pushes the container unit 4060 that is separated from the apparatus body 4110a backward in the depth direction of the apparatus, and consequently, the container unit 4060 is guided by the sliding rails 4064 and is installed in the apparatus body 4110a. In a state of separation, the recording medium P is containable in the container unit 4060. According to the present fourth exemplary embodiment, the container unit 4060 is not removed from the apparatus body 4110a but is supported by the apparatus body 4110a, and the recording medium P is containable in the container unit 4060 with the container unit 4060 separated from the apparatus body 4110a.

The container unit 4060 tilts with respect to the horizontal direction such that the first end and the second end in the width direction of the apparatus are located at different heights with the container unit 4060 separated from the apparatus body 4110a when viewed in the depth direction of the apparatus. Specifically, the container unit 4060 that is separated from the apparatus body 4110a overlaps the container unit 4060 that is installed in the apparatus body 4110a when viewed in the depth direction of the apparatus. In other words, the container unit 4060 that is separated from the apparatus body is located within the width of the apparatus body 4110a when viewed in the depth direction of the apparatus.

The container unit 4060 opens upward and enables the recording medium P to be supplied to the container unit 4060 with the container unit 4060 separated from the apparatus body 4110a. In other words, the container unit 4060 that is separated from the apparatus body 4110a is positioned at a first supply position at which the container unit 4060 enables the recording medium P to be supplied to the container unit 4060. The first supply position is an example of a second position.

Container Unit 4070 and Sliding Rail 4074

As illustrated in FIG. 4-3, the container unit 4070 has a box shape that opens upward. The two sliding rails 4074 are mounted to the respective edges of the container unit 4070 in the width direction of the apparatus.

Each sliding rail 4074 includes an outer member, an intermediate member, and an inner member. The outer member is mounted on the apparatus body 4110a. The inner member is mounted on the container unit 4070.

The user pulls the container unit 4070 that is installed in the apparatus body 4110a forward in the depth direction of the apparatus, and consequently, the container unit 4070 is guided by the sliding rails 4074 and is separated from the apparatus body 4110a. The user pushes the container unit 4070 that is separated from the apparatus body 4110a backward in the depth direction of the apparatus, and consequently, the container unit 4070 is guided by the sliding rails 4074 and is installed in the apparatus body 4110a.

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As illustrated in FIG. 4-3, the container unit 4070 is horizontally disposed with the container unit 4070 installed in the apparatus body 4110a, and the container unit 4070 is horizontally disposed with the container unit 4070 separated from the apparatus body 4110a when viewed in the depth direction of the apparatus. According to the present fourth exemplary embodiment, an example of the state in which the container unit 4070 is horizontally disposed includes the state in which the recording medium P that is contained in the container unit 4070 does not move due to the tilt although the container unit 4070 is permitted to slightly tilt, provided that the container unit 4070 extends in the horizontal direction.

The container unit 4070 that is installed in the apparatus body 4110a is positioned at a second transport position at which the contained recording medium P is transportable. The container unit 4070 that is separated from the apparatus body 4110a is positioned at a second supply position at which the container unit 4070 enables the recording medium P to be supplied to the container unit 4070.

The container unit 4070 that is installed in the apparatus body 4110a is pulled forward from the apparatus body 4110a in the depth direction of the apparatus regardless of the position of the container unit 4060.

The recording medium P having the maximum size that is contained in the container unit 4070 is located within the range (R1 in the figure) in which the recording medium P having the maximum size is contained in the container unit 4060 in the width direction of the apparatus when viewed in the depth direction of the apparatus. The recording medium P having the maximum size that is contained in the container unit 4070 is located within the range (R2 in the figure) in which the recording medium P having the maximum size is contained in the container unit 4060 in the vertical direction of the apparatus when viewed in the depth direction of the apparatus.

Container Unit 4080 and Sliding Rail 4084

As illustrated in FIG. 4-3, the container unit 4080 has a box shape that opens upward. The two sliding rails 4084 are mounted to the respective edges of the container unit 4080 in the width direction of the apparatus.

Each sliding rail 4084 includes an outer member, an intermediate member, and an inner member. The outer member is mounted on the apparatus body 4110a. The inner member is mounted on the container unit 4080.

The user pulls the container unit 4080 that is installed in the apparatus body 4110a forward in the depth direction of the apparatus, and consequently, the container unit 4080 is guided by the sliding rails 4084 and is separated from the apparatus body 4110a. The user pushes the container unit 4080 that is separated from the apparatus body 4110a backward in the depth direction of the apparatus, and consequently, the container unit 4080 is guided by the sliding rails 4084 and is installed in the apparatus body 4110a.

As illustrated in FIG. 4-3, the container unit 4080 is horizontally disposed with the container unit 4080 installed in the apparatus body 4110a, and the container unit 4080 is horizontally disposed with the container unit 4080 separated from the apparatus body 4110a when viewed in the depth direction of the apparatus. According to the present fourth exemplary embodiment, an example of the state in which the container unit 4080 is horizontally disposed includes the state in which the recording medium P that is contained in the container unit 4080 does not move due to the tilt

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although the container unit 4080 is permitted to slightly tilt, provided that the container unit 4080 extends in the horizontal direction.

The container unit 4080 that is installed in the apparatus body 4110a is positioned at a third transport position at which the contained recording medium P is transportable. The container unit 4080 that is separated from the apparatus body 4110a is positioned at a third supply position at which the container unit 4080 enables the recording medium P to be supplied to the container unit 4080.

The container unit 4080 that is installed in the apparatus body 4110a is pulled forward from the apparatus body 4110a in the depth direction of the apparatus regardless of the position of the container unit 4060.

The recording medium P having the maximum size that is contained in the container unit 4080 is located within the range R1 in which the recording medium P having the maximum size is contained in the container unit 4060 in the width direction of the apparatus when viewed in the depth direction of the apparatus.

Covering 4130

As illustrated in FIG. 4-5A and FIG. 4-5B, the covering 4130 is mounted on the apparatus body 4110a and moves to a covering position at which the covering 4130 covers a first side surface of the apparatus body 4110a in the width direction of the apparatus and an open position at which the covering 4130 uncovers the first side surface of the apparatus body 4110a in the width direction of the apparatus.

The covering 4130 that is positioned at the covering position covers at least a part of the transport path 4016, along which the recording medium P is transported by the transport unit 4014, in the width direction of the apparatus. For this reason, the covering 4130 that is positioned at the open position uncovers at least a part of the transport path 4016 (see FIG. 4-1), along which the recording medium P is transported by the transport unit 4014, in the width direction of the apparatus.

According to the present fourth exemplary embodiment, the covering 4130 has a function of covering the first side surface of the apparatus body 4010a of the image forming apparatus 4010. That is, the covering 4130 moves to the covering position at which the covering 4130 covers the fixing device 4034 that is installed in the apparatus body 4010a in the width direction of the apparatus and the open position at which the covering 4130 uncovers the fixing device 4034 in the width direction of the apparatus.

As illustrated in FIG. 4-5A, the covering 4130 that is positioned at the covering position includes a body 4132 that has a rectangular shape and that extends in the vertical direction of the apparatus when viewed in the width direction of the apparatus and support portions 4134 that rotatably support the body 4132.

As illustrated in FIG. 4-5A and FIG. 4-5B, the two support portions 4134 are mounted on a rear portion of the apparatus body 4110a in the depth direction of the apparatus and are separated from each other in the vertical direction of the apparatus.

With this structure, the covering 4130 moves to the covering position (see FIG. 4-5A) at which the covering 4130 covers the apparatus body 4110a of the container apparatus 4110 in the width direction of the apparatus and the open position (see FIG. 4-5B) at which the covering 4130 uncovers the apparatus body 4110a in the width direction of the apparatus. Specifically, the covering 4130 comes into contact with a stopper not illustrated and stops at the covering position. The covering 4130 that is positioned

at the covering position swings, consequently comes into contact with a stopper not illustrated, and stops at the open position.

Conclusion

The action of the container apparatus **4110** is compared with those of a container apparatus **4310** of an image forming apparatus **4210** according to a first comparative exemplary embodiment and a container apparatus **4510** of an image forming apparatus **4410** according to a second comparative exemplary embodiment, and this will now be described. Differences between the configuration of the container apparatus **4310** and the configuration of the container apparatus **4110** and between the configuration of the container apparatus **4510** and the configuration of the container apparatus **4110** will be principally described.

Configuration of Container Apparatus **4310**

As illustrated in FIG. **4-6A** and FIG. **4-6B**, the container apparatus **4310** includes three container units **4260**, **4070**, and **4080** that contain the recording media **P**.

As illustrated in FIG. **4-6A**, the container unit **4260** tilts with respect to the horizontal direction with the container unit **4260** installed in an apparatus body **4310a** as in the container unit **4060**.

As illustrated in FIG. **4-6B**, the container unit **4260** is guided by a sliding and link mechanism not illustrated with the container unit **4260** separated from the apparatus body **4310a** and is horizontally disposed. The container unit **4260** projects from the apparatus body **4310a** in the width direction of the apparatus when viewed in the depth direction of the apparatus.

Configuration of Container Apparatus **4510**

As illustrated in FIG. **4-7**, the container apparatus **4510** includes a container unit **4560**, a container unit **4570**, and a container unit **4580**. The container unit **4560**, the container unit **4570**, and the container unit **4580** are arranged in this order downward from above in the vertical direction of the apparatus.

The maximum size of the recording medium **P** that is containable in the container unit **4570** is smaller than the maximum size of the recording medium **P** that is containable in the container unit **4560**. The maximum size of the recording medium **P** that is containable in the container unit **4580** is smaller than the maximum size of the recording medium that is containable in the container unit **4570**.

The number of the recording media **P** that are containable in the container unit **4560**, the number of the recording media **P** that are containable in the container unit **4570**, and the number of the recording media **P** that are containable in the container unit **4580** are equal to each other. Consequently, the total thickness of the containable recording media **P** in the thickness direction is equal among the container unit **4560**, the container unit **4570**, and the container unit **4580**.

The container unit **4560** tilts with respect to the horizontal direction such that a first end in the width direction of the apparatus is higher than a second end when viewed in the depth direction of the apparatus. The container unit **4570** tilts at the same angle as the container unit **4560**. The container unit **4580** tilts at the same angle as the container unit **4560**.

The recording medium **P** having the maximum size that is contained in the container unit **4570** and the recording medium **P** having the maximum size that is contained in the container unit **4580** project from the range (**R4** in the figure) in which the recording medium **P** having the maximum size is contained in the container unit **4560** in the width direction of the apparatus. In other words, the recording media **P** stick

out from the range **R4**. As for the container apparatus **4510** according to the second comparative exemplary embodiment, a part of the recording medium **P** having the maximum size that is contained in the container unit **4570** and a part of the recording medium **P** having the maximum size that is contained in the container unit **4580** stick out from the range **R4** in the width direction of the apparatus, but the recording media **P** may entirely stick out from the range **R4**.

The recording medium **P** having the maximum size that is contained in the container unit **4570** projects from the range (**R5** in the figure) in which the recording medium **P** having the maximum size is contained in the container unit **4560** in the vertical direction of the apparatus. In other words, the recording medium **P** sticks out from the range **R5**. As for the container apparatus **4510** according to the second comparative exemplary embodiment, a part of the recording medium **P** having the maximum size that is contained in the container unit **4570** sticks out from the range **R5** in the vertical direction of the apparatus, but the recording medium **P** may entirely stick out from the range **R5**.

Action of Container Apparatus **4110**

As for the container apparatus **4110**, the container unit **4060** that is separated from the apparatus body **4110a** tilts with respect to the horizontal direction such that the first end and the second end in the width direction of the apparatus are located at different heights when viewed in the depth direction of the apparatus as described above. For this reason, when the recording medium **P** is supplied, the range in which the container unit **4060** is disposed in the width direction of the apparatus is smaller than that of the container apparatus **4310** according to the first comparative exemplary embodiment.

As for the container apparatus **4110**, the container unit **4060** that is separated from the apparatus body **4110a** is located within the range of the apparatus body **4110a** in the width direction of the apparatus when viewed in the depth direction of the apparatus. For this reason, when the recording medium **P** is supplied, the range in which the container unit **4060** is disposed in the width direction of the apparatus is smaller than that of the container apparatus **4310** according to the first comparative exemplary embodiment.

As for the container apparatus **4110**, the recording medium **P** having the maximum size that is contained in the container unit **4070** that is installed in the apparatus body **4110a** is located within the range **R1** in which the recording medium **P** having the maximum size is contained in the container unit **4060** that is installed in the apparatus body **4110a** in the width direction of the apparatus when viewed in the depth direction of the apparatus. In this way, the range in which the recording medium **P** is contained in the width direction of the apparatus may be decreased to a range narrower than that of the container apparatus **4510** according to the second comparative exemplary embodiment when it is assumed that the container unit **4060** and the container unit **4560** that tilt have the same shape.

As for the container apparatus **4110**, the covering **4130** that is positioned at the open position uncovers at least a part of the transport path **4016** for the recording medium **P**. In this way, it may make easy to remove the recording medium **P** that is transported from the container unit **4060**, the container unit **4070**, or the container unit **4080** and that is jammed on the transport path **4016**, unlike the case where the covering **4130** is not included.

As for the container apparatus **4110**, the recording medium **P** having the maximum size that is contained in the container unit **4070** is located within the ranges in which the recording medium **P** having the maximum size is contained

in the container unit **4060** in the vertical direction of the apparatus and in the width direction of the apparatus. In this way, the ranges in which the recording medium P is contained in the width direction of the apparatus and in the vertical direction of the apparatus may be decreased to ranges narrower than those of the container apparatus **4510** according to the second comparative exemplary embodiment when it is assumed that the container unit **4060** and the container unit **4560** that tilt have the same shape.

As for the container apparatus **4110**, the recording medium P having the maximum size that is contained in the container unit **4080** is located within the range in which the recording medium P having the maximum size is contained in the container unit **4060** in the width direction of the apparatus. In this way, the range in which the recording medium P is contained in the width direction of the apparatus may be decreased to a range narrower than that of the container apparatus **4510** according to the second comparative exemplary embodiment when it is assumed that the container unit **4060** and the container unit **4560** that tilt have the same shape.

As for the container apparatus **4110**, the degree of tilt of the container unit **4070** and the degree of tilt of the container unit **4080** are lower than the degree of tilt of the container unit **4060**. In this way, the recording media P that are contained in the container unit **4070** and in the container unit **4080** may be inhibited from moving in a tilt direction, unlike the case where the degree of tilt of the container unit **4070** and the degree of tilt of the container unit **4080** are equal to the degree of tilt of the container unit **4060**.

As for the container apparatus **4110**, the container unit **4070** and the container unit **4080** are horizontally disposed. In this way, the recording media P that are contained in the container unit **4070** and in the container unit **4080** may be inhibited from moving in a tilt direction, unlike the case where the container unit **4070** and the container unit **4080** tilt with respect to the horizontal direction.

As for the container apparatus **4110**, the number of the recording media P that are containable in the container unit **4080** is larger than the number of the recording media P that are containable in the container unit **4060** and the number of the recording media P that are containable in the container unit **4070**. Consequently, the position of the center of gravity of the container apparatus **4110** with the recording media P contained is a lower position than that in the case where the number of the recording media P that are containable in the container unit **4080** is smaller than the number of the recording media P that are containable in the container unit **4060** and the number of the recording media P that are containable in the container unit **4070**.

As for the container apparatus **4110**, the container unit **4070** and the container unit **4080** that are installed in the apparatus body **4110a** are capable of being pulled from the apparatus body **4110a** in the depth direction of the apparatus regardless of the position of the container unit **4060**. In this way, it may make easy to supply the recording medium P to the container unit **4070** or the container unit **4080** with the container unit **4060** separated from the apparatus body **4110a**, unlike a configuration in which neither the container unit **4070** nor the container unit **4080** is capable of being pulled.

As for the image forming apparatus **4010**, a range that is needed when the recording medium P is supplied to the image forming apparatus **4010** is smaller than that in the case where the container apparatus **4110** is not provided. The specific fourth exemplary embodiment of the present disclosure is described in detail. The present disclosure is not

limited to the fourth exemplary embodiment. It is clear for a person skilled in the art that the present disclosure includes different fourth exemplary embodiments within the scope of the present disclosure. For example, according to the fourth exemplary embodiment described above, the container apparatus **4110** is used for the electrophotographic image forming apparatus **4010**. However, the container apparatus **4110** may be used for, for example, an ink-jet image forming apparatus.

According to the fourth exemplary embodiment described above, the container apparatus **4110** includes the container unit **4060**, the container unit **4070**, and the container unit **4080** but may include neither the container unit **4070** nor the container unit **4080**. However, this does not take the action that is carried out by including the container unit **4070** and the container unit **4080**.

According to the fourth exemplary embodiment described above, the recording medium P having the maximum size that is contained in the container unit **4070** is located within the range R1 in the width direction of the apparatus when viewed in the depth direction of the apparatus but may project to a location outside the range R1. However, this does not take the action that is carried out by locating the recording medium P within the range R1.

According to the fourth exemplary embodiment described above, the container unit **4070** and the container unit **4080** are horizontally disposed but may not be horizontally disposed, provided that the angles at which the container unit **4070** and the container unit **4080** tilt are smaller than the angle at which the container unit **4060** tilts. In this way, the recording media P that are contained in the container unit **4070** and in the container unit **4080** may be inhibited from moving in a tilt direction, unlike the case where the angles at which the container unit **4070** and the container unit **4080** tilt are equal to the angle at which the container unit **4060** tilts or the case where the angles at which the container unit **4070** and the container unit **4080** tilt are larger than the angle at which the container unit **4060** tilts.

According to the fourth exemplary embodiment described above, the container unit **4070** and the container unit **4080** that are installed in the apparatus body **4110a** are capable of being pulled from the apparatus body **4110a** in the depth direction of the apparatus regardless of the position of the container unit **4060**. However, at least the container unit **4070** or the container unit **4080** may be capable of being pulled. In this way, it may make easy to supply the recording medium P, unlike the case where neither the container unit **4070** nor the container unit **4080** is capable of being pulled.

According to the fourth exemplary embodiment described above, the number of the recording media P that are containable in the container unit **4080** is larger than the number of the recording media P that are containable in the container unit **4060** and the number of the recording media P that are containable in the container unit **4070**. However, the relationship in the number of the recording media P that are contained in each container unit may differ. In this case, the action that is carried out by increasing the number of the recording media P that are containable in the container unit **4080** to the largest number is not taken.

According to the fourth exemplary embodiment described above, the container unit **4060** that is installed in the apparatus body **4110a** is movable to the supply position regardless of the position of the container unit **4070** and the position of the container unit **4080**, although this is not particularly described. In this way, it may make easy to supply the recording medium P to the container unit **4060**, unlike the case where movement of the container unit **4060**

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is restricted by the position of the container unit **4070** and the position of the container unit **4080**.

According to the fourth exemplary embodiment described above, as illustrated in FIG. 4-8, a guide member **4200** such as a rib that guides the recording medium P that is transported from the container unit **4070** may be formed at an upper edge portion of the container unit **4060**, and a part of the transport path **4016** that extends from the container unit **4070** to the guide member **4200** may be located within the range in which the container unit **4060** is disposed in the width direction of the apparatus, although this is not particularly described. In this way, the width of the apparatus body **4110a** may be inhibited from increasing due to the part of the transport path **4016** that extends from the container unit **4070** to the guide member **4200**, unlike the case where the part of the transport path **4016** that extends from the container unit **4070** to the guide member **4200** is located out of the range in which the container unit **4060** is disposed in the width direction of the apparatus.

According to the fourth exemplary embodiment described above, the image forming member **4012** may be located within the range in which the container unit **4060** is disposed in the width direction of the apparatus, although this is not particularly described. In this way, at least the width of the apparatus body **4010a** that includes the image forming member **4012** may be inhibited from increasing, unlike the case where the image forming member **4012** is located out of the range in which the container unit **4060** is disposed.

According to the fourth exemplary embodiment described above, a reverse path along which the recording medium P is reversed upside down may be located within the range in which the container unit **4060** is disposed in the width direction of the apparatus, although this is not particularly described. In this way, the width of the apparatus body **4010a** may be inhibited from increasing, unlike the case where the reverse path is located out of the range in which the container unit **4060** is disposed in the width direction of the apparatus.

According to the fourth exemplary embodiment described above, the transfer nip NT at which the image forming member **4012** transfers the image to the recording medium P may be disposed opposite a portion of the container unit **4060** from which the recording medium P is fed in the width direction of the apparatus, although this is not particularly described. The reverse path along which the recording medium P is reversed upside down may be located in the range in which the container unit **4060** is disposed in the width direction of the apparatus. In this way, the width of the apparatus body **4010a** may be inhibited from increasing, unlike the case where the reverse path is located out of the range in which the container unit **4060** is disposed in the width direction of the apparatus.

According to the fourth exemplary embodiment described above, the container apparatus **4110** does not include the transport unit **4014** in some cases, although this is not particularly described.

According to the fourth exemplary embodiment described above, the container unit **4070** may tilt with respect to the horizontal direction, although this is not particularly described, provided that the recording medium P having the maximum size that is contained in the container unit **4070** is located within the range R1 when viewed in the depth direction of the apparatus.

According to the fourth exemplary embodiment described above, the container unit **4080** may tilt with respect to the horizontal direction, although this is not particularly described, provided that the recording medium P having the

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maximum size that is contained in the container unit **4080** is located within the range R1 when viewed in the depth direction of the apparatus.

In the description according to the fourth exemplary embodiment described above, the container unit tilts with respect to the horizontal direction. As for the container unit, however, the position of the first end and the position of the second end may differ from each other. In this way, a range in which the container unit is disposed in the width direction may be smaller than that in the case where the container unit horizontally extends, for example, even when the container unit is disposed in the vertical direction.

Fifth Exemplary Embodiment

An example of an image forming apparatus and a sheet containing apparatus according to a fifth exemplary embodiment of the present disclosure will be described with reference to FIG. 5-1 to FIG. 5-9. An arrow H illustrated in figures represents the height direction of the apparatus that is vertical. An arrow W represents the width direction of the apparatus that is horizontal. An arrow D represents the depth direction of the apparatus that is horizontal.

Entire Configuration of Image Forming Apparatus

As illustrated in FIG. 5-1, an image forming apparatus **5010** includes an image forming member **5012** that forms a toner image by using an electrophotographic system, a sheet containing apparatus **5110** that includes container units **5018** and **5060** that contain sheets P and a transport unit **5014** that transports the sheets P along a transport path **5016**, and a controller **5028** that controls components.

As for the image forming apparatus **5010** that has the configuration described above, each of the sheets P that are contained in the container units **5018** and **5020** is transported along the transport path **5016** by using the transport unit **5014**. The toner image that is formed by the image forming member **5012** is formed on the transported sheet P, and the sheet P on which the toner image is formed is discharged to a location outside an apparatus body **5010a**.

Image Forming Member 5012

As illustrated in FIG. 5-1, the image forming member **5012** includes multiple toner image forming members **5030** that form toner images in respective colors and a transfer member **5032** that transfers the toner images that are formed by the toner image forming members **5030** to the sheet P. The image forming member **5012** also includes a fixing device **5034** that fixes the toner images that are transferred to the sheet P by using the transfer member **5032** to the sheet P. The toner image forming members are examples of an image formation unit, and the fixing device **5034** is an example of a fixing portion.

Toner Image Forming Member 5030

The toner image forming members **5030** are provided so as to form the toner images in the respective colors. According to the present fifth exemplary embodiment, the toner image forming members **5030** for the four colors of yellow (Y), magenta (M), cyan (C), and black (K) are provided. In the following description, Y, M, C, and K in reference characters are omitted when it is not necessary to distinguish among yellow (Y), magenta (M), cyan (C), and black (K).

The toner image forming members **5030** for the respective colors are basically configured in the same manner except for toner that is used. As illustrated in FIG. 5-2, each toner image forming member **5030** includes an image carrier **5040** that is rotatable and that is cylindrical and a charger **5042** that charges the image carrier **5040**. Each toner image forming member **5030** also includes an exposure device

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5044 that radiates exposure light to the image carrier **5040** that is charged and that forms an electrostatic latent image and a developing device **5046** that develops the electrostatic latent image into the toner image by using a developer **G** that contains the toner.

Consequently, the toner image forming members **5030** for the respective colors form images in the respective colors by using the toner in the respective colors.

As illustrated in FIG. 5-1, the image carriers **5040** for the respective colors are in contact with a transfer belt **5050** (described in detail later) that turns. The toner image forming members **5030** for yellow (Y), magenta (M), cyan (C), and black (K) are disposed in this order from an upstream position in the direction in which the transfer belt **5050** turns (see an arrow in the figure).

Transfer Member **5032**

As illustrated in FIG. 5-1, the transfer member **5032** includes the transfer belt **5050** and first transfer rollers **5052** that are disposed opposite the image carriers **5040** for the respective colors with the transfer belt **5050** interposed therebetween and that transfer the toner images that are formed by the image carriers **5040** for the respective colors to the transfer belt **5050**.

The transfer member **5032** also includes a winding roller **5056** around which the transfer belt **5050** is wound and a drive roller **5058** around which the transfer belt **5050** is wound for transmitting rotational force to the transfer belt **5050**. Consequently, the transfer belt **5050** turns in the direction of the arrow in the figure.

The transfer member **5032** also includes a second transfer roller **5054** that is disposed opposite the winding roller **5056** with the transfer belt **5050** interposed therebetween and that transfers the toner images that are transferred to the transfer belt **5050** to the sheet **P**. A transfer nip **NT** at which the toner images are transferred to the sheet **P** is formed between the second transfer roller **5054** and the transfer belt **5050**.

With this configuration, the toner images are first-transferred to the transfer belt **5050** in the order of yellow (Y), magenta (M), cyan (C), and black (K) by using the first transfer rollers **5052**. The second transfer roller **5054** transfers the toner images from the transfer belt **5050** to the sheet **P** that is transported between the transfer belt **5050** and the second transfer roller **5054**. The sheet **P** to which the toner images are transferred is transported toward the fixing device **5034**.

Fixing Device **5034**

As illustrated in FIG. 5-1, the fixing device **5034** is disposed downstream of the transfer nip **NT** in the direction in which the sheet **P** is transported. The fixing device **5034** heats and compresses the toner images that are transferred to the sheet **P** and fixes the toner images to the sheet **P**.

Sheet Containing Apparatus **5110**

The sheet containing apparatus **5110** is located in a lower portion of the image forming apparatus **5010**. The sheet containing apparatus **5110** will be described in detail later. Configuration of Principal Component

The sheet containing apparatus **5110** will now be described.

As illustrated in FIG. 5-1, the sheet containing apparatus **5110** is located in the lower portion of the image forming apparatus **5010**. The sheet containing apparatus **5110** includes an apparatus body **5110a**, the container unit **5018** that contains the sheet **P**, and the container unit **5060** that contains the sheet **P**. The sheet containing apparatus **5110** also includes the transport unit **5014** that transports the sheets that are contained in the container unit **5018** and the container unit **5060**. As illustrated in FIG. 5-8A and FIG.

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5-8B, the sheet containing apparatus **5110** also includes a covering portion **5120** that covers the apparatus body **5110a** in the depth direction of the apparatus and a covering portion **5130** that covers the apparatus body **5110a** in the width direction of the apparatus. As illustrated in FIG. 5-5, the sheet containing apparatus **5110** also includes rail members **5084** that are capable of moving the container unit **5018** in the depth direction of the apparatus.

The container unit **5018** is an example of the first container unit, and the container unit **5060** is an example of the second container unit. The depth direction of the apparatus is an example of the first direction, and the width direction of the apparatus is an example of "sideways".

The apparatus body **5110a** of the sheet containing apparatus **5110** corresponds to a lower portion of the apparatus body **5010a** of the image forming apparatus **5010**.

Container Unit **5018**

As illustrated in FIG. 5-1, the container unit **5018** contains a sheet **P** having the A3 size and is disposed below the toner image forming members **5030**. The A3 size is an example of the first size.

The container unit **5018** tilts with respect to the horizontal direction such that a first end and a second end in the width direction of the apparatus are located at different heights when viewed in the depth direction of the apparatus. Specifically, the container unit **5018** tilts such that an end portion of the container unit **5018** in one direction (the left-hand direction in the figure) of the width direction of the apparatus is higher than an end portion of the container unit **5018** in the other direction. The first end and the second end described herein correspond to portions of the container unit **5018** that are farthest from each other in the width direction of the apparatus. The apparatus body **5110a** has a side wall **5112a** that faces an upper edge of the container unit **5018** in one direction of the width direction of the apparatus and a side wall **5112b** that faces a lower edge of the container unit **5018** in the other direction of the width direction of the apparatus. The lower edge of the container unit **5018** is nearer than the side wall **5112a** that faces the upper edge of the container unit **5018** to the side wall **5112b** that faces the lower edge of the container unit **5018**.

As illustrated in FIG. 5-6A and FIG. 5-6B, the container unit **5018** is installable in and removable from the apparatus body **5110a** in the depth direction of the apparatus with the covering portion **5120** that is described later positioned at an open position.

As illustrated in FIG. 5-5, the container unit **5018** includes a container body **5070** and a trailing edge restriction portion **5074** that comes into contact with the trailing edge **Pr** of the contained sheet **P** and that restricts the position of the trailing edge **Pr** of the sheet **P**. The container unit **5018** also includes two side edge restriction portions **5082** that come into contact with respective side edges **Ps** of the sheet **P** and that restrict the positions of the side edges **Ps** of the sheet **P** and a support portion **5094** that supports the contained sheet **P** from below.

The container unit **5018** tilts such that the leading edge **Pf** of the sheet **P** that is contained in the container unit **5018** is higher than the trailing edge **Pr**. The sheet **P** is fed to the transport path **5016** with the leading edge **Pf** of the sheet **P** facing forward.

Container Body **5070**

As illustrated in FIG. 5-5, the container body **5070** has a box shape that opens upward. The container body **5070** includes a bottom plate **5070a** that faces a back surface of the contained sheet **P** in the thickness direction of the sheet **P** (referred to below as a "medium thickness direction") and

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a rear wall **5070b** that faces the trailing edge **Pr** of the sheet **P** in the front-rear direction of the sheet **P** (referred to below as a “medium front-rear direction”). The container unit **5018** also includes a front wall **5070c** that faces the leading edge **Pf** of the contained sheet **P** in the medium front-rear direction and two side walls **5072** (a and b) that face the respective side edges **Ps** of the sheet **P** in the width direction of the sheet **P** (referred to below as a “medium width direction”).

The two side walls **5072** are a side wall **5072a** at the rear in the depth direction of the apparatus and a side wall **5072b** at the front in the depth direction of the apparatus. The side wall **5072a** has a rectangular shape that extends from the rear wall **5070b** to the front wall **5070c** when viewed in the depth direction of the apparatus. The side wall **5072b** includes a foundation portion **5073a** that has a rectangular shape that extends from the rear wall **5070b** to the front wall **5070c** and an extension portion **5073b** that extends from the foundation portion **5073a** in one direction of the width direction of the apparatus when viewed in the depth direction of the apparatus.

As illustrated in FIG. 5-3B and FIG. 5-4B, a division line **5076** that divides the container unit **5018** and the apparatus body **5110a** and that tilts is present between the side wall **5072b** and the apparatus body **5110a** with the container unit **5018** installed in the apparatus body **5110a**. With the container unit **5018** installed in the apparatus body **5110a**, the container unit **5018** is nearer than the covering portion **5120** that covers the apparatus body **5110a** from the front in the depth direction of the apparatus to the apparatus body **5110a**. Trailing Edge Restriction Portion **5074**

As illustrated in FIG. 5-5, the trailing edge restriction portion **5074** is disposed in the container body **5070** and is mounted on the bottom plate **5070a** of the container body **5070**. The trailing edge restriction portion **5074** is guided by a pair of slits (not illustrated) that is formed in the bottom plate **5070a** and that extends in the medium front-rear direction and moves in a predetermined range in the medium front-rear direction. Movement of the trailing edge restriction portion **5074** that moves is restricted by a lock mechanism not illustrated.

With this configuration, the trailing edge restriction portion **5074** comes into contact with the trailing edge **Pr** of the sheet **P** that is contained in the container unit **5018** in the medium front-rear direction and consequently restricts the position of the trailing edge **Pr** of the sheet **P**. Side Edge Restriction Portion **5082**

As illustrated in FIG. 5-5, the side edge restriction portions **5082** are disposed in the container body **5070** and are mounted on the bottom plate **5070a** of the container body **5070**. The side edge restriction portions **5082** are disposed on both sides of the sheet **P** in the medium width direction. The two side edge restriction portions **5082** are symmetrical in the medium width direction.

The side edge restriction portions **5082** are guided by a pair of slits (not illustrated) that is formed in the bottom plate **5070a** and that extends in the medium width direction and moves in a predetermined range in the medium width direction. The side edge restriction portions **5082** that move stop there due to frictional force that acts between the side edge restriction portions **5082** and the bottom plate **5070a**.

The container unit **5018** uses a center registration method. The container unit **5018** include a mechanism not illustrated, and when one of the side edge restriction portions **5082** is moved in the medium width direction, the mechanism moves the other side edge restriction portion **5082** the same distance in the medium width direction.

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With this configuration, the two side edge restriction portions **5082** come into contact with the side edges **Ps** of the sheet **P** that is contained in the container unit **5018** in the medium width direction and consequently restrict the positions of the side edges **Ps** of the sheet **P**.

Support Portion **5094**

As illustrated in FIG. 5-5, the support portion **5094** is disposed in the container body **5070**.

The support portion **5094** is formed by using a metal plate, is symmetrical in the medium width direction when viewed in the medium thickness direction and has a shape obtained by notching a rectangle at two portions.

Specifically, notches **5094a** that have a U-shape and that are used to avoid interference with the side edge restriction portions **5082** that move are formed in the support portion **5094**. The position of a rear edge **5094b** of the support portion **5094** is determined so as to avoid interference with the trailing edge restriction portion **5074** that moves.

Shaft portions **5096** that project in the medium width direction are formed on respective end portions of the rear edge **5094b** of the support portion **5094**. The shaft portions **5096** are inserted in through-holes **5078** that are formed in the respective side walls **5072** of the container unit **5018**. Consequently, the support portion **5094** is rotatable about the shaft portions **5096** such that a front edge portion of the support portion **5094** moves up and down.

As illustrated in FIG. 5-7A, the front edge portion of the support portion **5094** is urged upward by an urging member that is disposed on the apparatus body **5110a** and that is not illustrated with the container unit **5018** installed in the apparatus body **5110a** (see FIG. 5-1), and the support portion **5094** turns about the shaft portions **5096**. The portion of the leading edge of the sheet **P** that is supported by the support portion **5094** comes into contact with the feed roller **5020**.

As illustrated in FIG. 5-7B, the urging force of the urging member is released, and the front edge portion of the support portion **5094** comes into contact with the bottom plate **5070a** of the container body **5070** with the container unit **5018** separated from the apparatus body **5110a** (see FIG. 5-1). As in the case where the container unit **5018** is installed in the apparatus body **5110a**, the container unit **5018** tilts with respect to the horizontal direction even with the container unit **5018** separated from the apparatus body **5110a**. A state of being separated is a state in which the container unit is pulled from the apparatus body, and the sheet **P** is to be supplied to the container unit. Specifically, the container unit **5018** is not removed from the apparatus body **5110a** but is supported by the apparatus body **5110a**, and the sheet **P** is containable in the container unit **5018** with the container unit **5018** separated from the apparatus body **5110a**.

Container Unit **5060**

The container unit **5060** contains the sheet **P** having the A4 size and is disposed below an upper portion of the container unit **5018** as illustrated in FIG. 5-1. The container unit **5060** includes a container body, a trailing edge restriction portion, two side edge restriction portions, and a support portion that supports the sheet **P** from below as in the container unit **5018** although a detailed description is omitted. The A4 size is an example of the second size. According to the present fifth exemplary embodiment, the first size and the second size differs from each other but may be the same size.

The container unit **5060** is disposed in the horizontal direction when viewed in the depth direction of the apparatus. In other words, the container unit **5060** is disposed

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such that movement force does not act on the sheet P that is contained in the container unit **5060**.

As illustrated in FIG. 5-3B and FIG. 5-4B, a division line **5062** that divides the apparatus body **5110a** and the container unit **5060** is present between the container unit **5060** and the apparatus body **5110a** with the container unit **5060** installed in the apparatus body **5110a**. The division line according to the present fifth exemplary embodiment is not limited to a division line between a container unit and the apparatus body **5110a**, but a division line between a container unit and another member such as another container unit, for example, is acceptable.

The container unit **5060** is installable in and removable from the apparatus body **5110a** in the depth direction of the apparatus with the covering portion **5120** that is described later positioned at the open position.

Rail Member **5084**

As illustrated in FIG. 5-5, the two rail members **5084** are mounted on a lower edge portion of the rear wall **5070b** of the container body **5070** and a lower edge portion of the front wall **5070c** of the container body **5070**.

Each rail member **5084** is a so-called slide rail, includes an outer member, an intermediate member, and an inner member, the outer member is mounted on the apparatus body **5110a**, and the inner member is mounted on the container body **5070**.

With this configuration, the rail members **5084** guide the container unit **5018** in the depth direction of the apparatus. Specifically, the rail members **5084** guide the container unit **5018** to an installation position (see FIG. 5-6A) at which the container unit **5018** is installed in the apparatus body **5110a** and a separation position (see FIG. 5-6B) at which the container unit **5018** is separated from the apparatus body **5110a**.

Also, the container unit **5060** includes rail members for installing the container unit **5060** in and removing the container unit **5060** from the apparatus body **5110a** although a detailed description is omitted.

Transport Unit **5014**

As illustrated in FIG. 5-1, the transport unit **5014** includes a feed roller **5020a** that feeds the sheet P that is contained in the container unit **5018** to the transport path **5016** and prevention rollers **5022a** that prevent multiple sheets P that are fed by the feed roller **5020a** from being transported. The transport unit **5014** also includes a feed roller **5020b** that feeds the sheet P that is contained in the container unit **5060** to the transport path **5016** and prevention rollers **5022b** that prevent multiple sheets P that are fed by the feed roller **5020b** from being transported.

The transport unit **5014** also includes adjustment rollers **5024** that adjust a timing with which the sheet P is transported to the transfer nip NT and discharge rollers **5026** that discharge the sheet P to which the toner images are fixed by the fixing device **5034** to the location outside the apparatus body **5010a**.

Covering Portion **5120**

The covering portion **5120** is a so-called front covering, is mounted on the apparatus body **5110a** as illustrated in FIG. 5-3A and FIG. 5-3B, moves to the covering position at which the covering portion **5120** covers the container unit **5018** that is installed in the apparatus body **5110a** in the depth direction of the apparatus and the open position at which the covering portion **5120** uncovers the container unit **5018** in the depth direction of the apparatus.

According to the present fifth exemplary embodiment, the covering portion **5120** has a function of covering the apparatus body **5010a** of the image forming apparatus **5010** in the

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depth direction of the apparatus. Consequently, the covering portion **5120** moves to the covering position at which the covering portion **5120** covers the toner image forming members **5030** in the respective colors that are installed in the apparatus body **5010a** in the depth direction of the apparatus and the open position at which the covering portion **5120** uncovers the toner image forming members **5030** in the depth direction of the apparatus.

As illustrated in FIG. 5-3A, the covering portion **5120** that is positioned at the covering position includes a body **5122** that has a rectangular shape extending in the height direction of the apparatus and that includes a flange on an outer circumferential edge when viewed in the depth direction of the apparatus and support portions **5124** that rotatably support the body **5122**.

As illustrated in FIG. 5-3A and FIG. 5-3B, the two support portions **5124** are mounted on a portion of the apparatus body **5110a** in one direction of the width direction of the apparatus and are separated from each other in the height direction of the apparatus.

With this structure, the covering portion **5120** moves to the covering position (see FIG. 5-3A and FIG. 5-4A) at which the covering portion **5120** covers the container unit **5018** in the depth direction of the apparatus and the open position (see FIG. 5-3B and FIG. 5-4B) at which the covering portion **5120** uncovers the container unit **5018** in the depth direction of the apparatus. Specifically, the covering portion **5120** comes into contact with a stopper not illustrated and stops at the covering position. The covering portion **5120** that is positioned at the covering position swings, consequently comes into contact with a stopper not illustrated, and stops at the open position.

As illustrated in FIG. 5-4A, the covering portion **5120** covers the entire container unit **5018** and the entire container unit **5060** that are installed in the apparatus body **5110a** when viewed in the depth direction of the apparatus with the covering portion **5120** positioned at the covering position. In other words, the covering portion **5120** covers the entire division line **5076** of the container unit **5018** and the entire division line **5062** of the container unit **5060** when viewed in the depth direction of the apparatus.

As illustrated in FIG. 5-4B, the covering portion **5120** is nearer than the lower edge of the container unit **5018** to the upper edge with the covering portion **5120** positioned at the open position when viewed in the depth direction of the apparatus. The entire container unit **5018** and the entire container unit **5060** are uncovered in the depth direction of the apparatus with the covering portion **5120** positioned at the open position when viewed in the depth direction of the apparatus. In other words, the entire division line **5076** of the container unit **5018** and the entire division line **5062** of the container unit **5060** are directly visible with the covering portion **5120** positioned at the open position when viewed in the depth direction of the apparatus.

The covering portion **5120** that is positioned at the covering position limits work on separation of the container unit **5018** and the container unit **5060** that are installed in the apparatus body **5110a**, but the covering portion **5120** that is positioned at the open position does not limit work on separation of the container unit **5018** and the container unit **5060** that are installed in the apparatus body **5110a** as above. In other words, work on separation of the container unit **5018** that is installed in the apparatus body **5110a** is permitted.

When the container unit **5018** that is installed in the apparatus body **5110a** is pulled in the depth direction of the apparatus and is separated from the apparatus body **5110a**

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with the covering portion 5120 positioned at the open position, the bottom plate 5070a of the separated container unit 5018 is uncovered in the thickness direction of the bottom plate 5070a. Specifically, as illustrated in FIG. 5-6B, the covering portion 5120 is not present in the thickness direction (an arrow B illustrated in FIG. 5-6B) of the bottom plate 5070a, and the bottom plate 5070a of the separated container unit 5018 is uncovered.

Covering Portion 5130

The covering portion 5130 is a so-called side covering, is mounted on the apparatus body 5110a as illustrated in FIG. 5-8A and FIG. 5-8B, moves to the covering position at which the covering portion 5130 covers the apparatus body 5110a in the width direction of the apparatus and the open position at which the covering portion 5130 uncovers the apparatus body 5110a in the width direction of the apparatus. The covering portion 5130 is another example of the covering portion. According to the present fifth exemplary embodiment, the covering portion 5130 is constituted by the single covering portion.

The covering portion 5130 that is positioned at the covering position covers the apparatus body 5110a in a direction in which the sheet P is transported by the transport unit 5014. For this reason, the covering portion 5130 that is positioned at the open position uncovers at least a part of the transport path 5016 (see FIG. 5-1), along which the sheet P is transported by the transport unit 5014, in the width direction of the apparatus.

According to the present fifth exemplary embodiment, the covering portion 5130 also has a function of covering the apparatus body 5010a of the image forming apparatus 5010 in the width direction of the apparatus. Consequently, the covering portion 5130 moves to the covering position at which the covering portion 5130 covers the fixing device 5034 that is installed in the apparatus body 5010a in the width direction of the apparatus and the open position at which the covering portion 5130 uncovers the fixing device 5034 in the width direction of the apparatus.

As illustrated in FIG. 5-8A, the covering portion 5130 that is positioned at the covering position includes a body 5132 that has a rectangular shape and that extends in the height direction of the apparatus when viewed in the width direction of the apparatus and support portions 5134 that rotatably support the body 5132.

As illustrated in FIG. 5-8A and FIG. 5-8B, the two support portions 5134 are mounted on a rear portion of the apparatus body 5110a in the depth direction of the apparatus and are separated from each other in the height direction of the apparatus.

With this structure, the covering portion 5130 moves to the covering position (see FIG. 5-8A) at which the covering portion 5130 covers at least a part of the transport path 5016 in the width direction of the apparatus and the open position (see FIG. 5-8B) at which the covering portion 5130 uncovers at least a part of the transport path 5016 in the width direction of the apparatus. Specifically, the covering portion 5130 comes into contact with a stopper not illustrated and stops at the covering position. The covering portion 5130 that is positioned at the covering position swings, consequently comes into contact with a stopper not illustrated, and stops at the open position.

Conclusion

As for the sheet containing apparatus 5110, as illustrated in FIG. 5-4A, the covering portion 5120 covers the entire container unit 5018 that is installed in the apparatus body

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5110a when viewed in the depth direction of the apparatus with the covering portion 5120 positioned at the covering position as described above. In this way, a sense of beauty may be inhibited from being reduced due to the division line 5076 that tilts, unlike a configuration that does not include the covering portion 5120 that covers the container unit 5018.

As for the sheet containing apparatus 5110, as illustrated in FIG. 5-4A, the covering portion 5120 covers the entire container unit 5060 that is installed in the apparatus body 5110a when viewed in the depth direction of the apparatus with the covering portion 5120 positioned at the covering position. In this way, the sense of beauty may be inhibited from being reduced due to the division line 5062, unlike a configuration that does not include the covering portion 5120 that covers the container unit 5060.

As for the sheet containing apparatus 5110, as illustrated in FIG. 5-4B, the covering portion 5120 is nearer than the lower edge of the container unit 5018 to the upper edge with the covering portion 5120 positioned at the open position when viewed in the depth direction of the apparatus. In this way, workability when the sheet P is contained in the container unit 5018 may be increased to a level higher than that in the case where the covering portion 5120 that is positioned at the open position is nearer than the upper edge of the container unit 5018 to the lower edge when viewed in the depth direction of the apparatus.

As for the sheet containing apparatus 5110, when the container unit 5018 is separated from the apparatus body 5110a with the covering portion 5120 positioned at the open position, the bottom plate 5070a of the separated container unit 5018 is uncovered in the thickness direction of the bottom plate 5070a. In this way, the workability when the sheet P is contained in the container unit 5018 may be increased to a level higher than that in the case where the bottom plate 5070a of the container unit 5018 that is separated from the apparatus body 5110a is uncovered in the thickness direction of the bottom plate 5070a.

As for the sheet containing apparatus 5110, the covering portion 5120 that is positioned at the covering position limits work on separation of the container unit 5018 that is installed in the apparatus body 5110a, but the covering portion 5120 that is positioned at the open position does not limit work on separation of the container unit 5018 that is installed in the apparatus body 5110a. In other words, work on separation of the container unit 5018 that is installed in the apparatus body 5110a is permitted. Consequently, separation of the container unit 5018 from the apparatus body 5110a is limited with the covering portion 5120 positioned at the covering position.

As for the sheet containing apparatus 5110, the lower edge of the container unit 5018 is nearer than the side wall 5112a that faces the upper edge of the container unit 5018 to the side wall 5112b that faces the lower edge when viewed in the depth direction of the apparatus. In this way, the workability when the sheet P is contained in the container unit 5018 may be increased to a level higher than that in the case where the lower edge of the container unit is nearer than the side wall that faces the lower edge of the container unit to the side wall that faces the upper edge. As for the sheet containing apparatus 5110, the trajectory of the covering portion 5120 is separated from the trajectory of the covering portion 5130. Consequently, the covering portion 5130 moves regardless of the position to which the covering portion 5120 moves. The two trajectories do not three-dimensionally overlap with the trajectory of the covering portion 5120 separated from the trajectory of the covering portion 5130.

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As for the sheet containing apparatus **5110**, a part of the transport path **5016** for the sheet P is uncovered with the covering portion **5130** positioned at the open position, and the sheet P that is jammed on the transport path **5016** is removable.

As for the image forming apparatus **5010**, the sense of beauty may be inhibited from being reduced due to the division line **5076** that tilts, unlike the configuration that does not include the covering portion **5120** that covers the container unit **5018**.

As for the image forming apparatus **5010**, the toner image forming members **5030** are maintainable with the covering portion **5120** positioned at the open position.

As for the image forming apparatus **5010**, the fixing device **5034** is maintainable with the covering portion **5130** positioned at the open position.

The specific fifth exemplary embodiment of the present disclosure is described in detail. The present disclosure is not limited to the fifth exemplary embodiment. It is clear for a person skilled in the art that the present disclosure includes different fifth exemplary embodiments within the scope of the present disclosure. For example, according to the fifth exemplary embodiment described above, the sheet containing apparatus **5110** is used for the electrophotographic image forming apparatus **5010**. However, the sheet containing apparatus **5110** may be used for, for example, an ink-jet image forming apparatus.

According to the fifth exemplary embodiment described above, the apparatus body **5110a** is opened and closed by swinging the covering portion **5120** and the covering portion **5130**, but the apparatus body **5110a** may be opened and closed by sliding.

According to the fifth exemplary embodiment described above, the covering portion **5120** at the covering position covers the entire container unit **5018** that is installed in the apparatus body **5110a** when viewed in the depth direction of the apparatus but may cover only a part of the container unit **5018**. Consequently, the number of the division line **5076** that tilts and that is exposed in the depth direction of the apparatus decreases. The meaning of being exposed corresponds to the meaning of not being covered by another member and exposed from the surface. The meaning that the number decreases is that the length of an exposed portion along a straight portion of the division line decreases. According to the fifth exemplary embodiment described above, the covering portion **5120** at the covering position covers the entire container unit **5060** that is installed in the apparatus body **5110a** when viewed in the depth direction of the apparatus but may cover only a part of the container unit **5060**. Consequently, the number of the division line **5062** that is exposed in the depth direction of the apparatus decreases.

According to the fifth exemplary embodiment described above, the covering portion **5120** that is positioned at the open position is nearer than the lower edge of the container unit **5018** to the upper edge when viewed in the depth direction of the apparatus. However, the covering portion may be nearer than the upper edge of the container unit **5018** to the lower edge. In this way, the maintenance of a paper feeding unit such as the feed roller of the container unit **5018** may be facilitated more than the case where the covering portion **5120** that is positioned at the open position is nearer than the lower edge of the container unit **5018** to the upper edge.

According to the fifth exemplary embodiment described above, the lower edge of the container unit **5018** is nearer than the side wall **5112a** that faces the upper edge of the

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container unit **5018** to the side wall **5112b** that faces the lower edge when viewed in the depth direction of the apparatus. However, the lower edge of the container unit **5018** may be nearer than the side wall **5112b** that faces the lower edge of the container unit **5018** to the side wall **5112a** that faces the upper edge. In this case, the action that is carried out with the lower edge of the container unit **5018** being nearer than the side wall **5112a** that faces the upper edge of the container unit **5018** to the side wall **5112b** that faces the lower edge is not taken.

According to the fifth exemplary embodiment described above, the mass of the covering portion **5120** is smaller than the mass of the container unit **5018**, and the covering portion **5120** that is positioned at the open position may be prevented from moving to the covering position with the container unit **5018** separated from the apparatus body **5110a** although this is not particularly described. Consequently, when the covering portion **5120** at the open position moves to the covering position with the container unit **5018** separated from the apparatus body **5110a**, the covering portion **5120** comes into contact with the container unit **5018** and stops moving. In this way, the covering portion **5120** may be inhibited from being damaged, unlike a configuration in which the covering portion **5120** at the open position is forcefully movable to the covering position with the container unit **5018** separated from the apparatus body **5110a**. According to the fifth exemplary embodiment described above, the body **5122** of the covering portion **5120** opens and closes about the support portions **5124** that are disposed in the -W direction of the container unit **5018** and moves to the open position, and the container unit **5018** is uncovered in the front direction of the depth direction of the apparatus although this is not particular described. In this way, containing the sheet P in the container unit **5018** by the user may be facilitated because the body **5122** is not disposed in the +W direction of the container unit **5018** with the body **5122** of the covering portion **5120** moved to the open position.

According to the fifth exemplary embodiment described above, the support portions **5124** are disposed in the -W direction of the container unit **5018**. However, the support portions **5124** may be disposed in the +W direction of the container unit **5018**, and the body **5122** may open and close about the support portions. In this case, the body **5122** is not disposed in the -W direction of the container unit **5018** with the body **5122** moved to the open position. In this way, a measure against a jam that happens, for example, at a pickup portion near the feed roller **5020a** may be facilitated.

According to the fifth exemplary embodiment described above, the container unit **5018** and the container unit **5060** are covered in the depth direction of the apparatus with the covering portion **5120** that is constituted by the single covering portion moved to the covering position although this is not particular described. Consequently, the number of the division line due to the covering portion is smaller than that in the case of two covering portions.

Sixth Exemplary Embodiment

FIG. 6-1A and FIG. 6-1B illustrate an image forming apparatus **6010** that is an example of an image processing apparatus according to a sixth exemplary embodiment. The image forming apparatus **6010** is an apparatus that is installed in an office such as a multifunction peripheral. In FIG. 6-1A and FIG. 6-1B, the image forming apparatus **6010** is constituted by, for example, an apparatus body **6012**, a discharge tray **6014** that serves as a medium discharge

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portion, a placement portion **6016**, an image capturing device **6018** that serves as an image capturing member, a radiation device **6020**, and an operation panel **6022**.

Paper that is an example of a medium is discharged onto the discharge tray **6014**. The discharge tray **6014** is disposed at a position at which a user takes out the medium with the user standing.

The placement portion **6016** corresponds to a top surface of the apparatus body **6012** and is disposed on an upper portion of the apparatus body **6012** although this will be described in detail later. The placement portion **6016** is installed above the discharge tray **6014** in the vertical direction.

The placement portion **6016** is configured such that an object is placed thereon. According to the sixth exemplary embodiment, examples of the object include a document such as paper, flat objects such as a card and a receipt, and a three-dimensional object such as a book.

The image capturing device **6018** is installed above the placement portion **6016** and is configured such that the image of the object that is placed on the placement portion **6016** is captured from above. The image capturing device **6018** faces downward so as to face the placement portion **6016**. In the case where the upper surface of the placement portion **6016** is referred to as a reference surface, the phrase “above the placement portion **6016**” described herein is not limited to a position upward from the reference surface in the vertical direction but means a position that is an upper position in the vertical direction and that is higher than that of the reference surface. The image capturing device **6018** is configured such that the object is set on the placement portion **6016** with an image captured surface facing upward for capturing the image.

A support **6024** is disposed on the top surface of the apparatus body **6012** at the rear of the upper surface of the placement portion **6016**. The support **6024** extends upward from the upper surface of the placement portion **6016** in the substantially vertical direction, and an end portion thereof bends toward a location above the placement portion **6016**. The image capturing device **6018** is installed on the end portion of the support **6024**.

An image forming member that forms the image of the object that is placed on the placement portion **6016** is disposed between the discharge tray **6014** and the placement portion **6016**. That is, the image forming member that forms the image on the paper that is discharged onto the discharge tray **6014** is disposed above the discharge tray **6014**.

The radiation device **6020** is constituted by one or more light sources and is constituted by, for example, light sources **6020a**, **6020b**, **6020c**, and **6020d**. The light sources **6020a**, **6020b**, **6020c**, and **6020d** are disposed on the support **6024**. That is, the light sources **6020a**, **6020b**, **6020c**, and **6020d** are disposed above the placement portion **6016**.

Specifically, the light sources **6020a** and **6020b** are disposed on the support **6024** and on both sides of the image capturing device **6018** near the image capturing device **6018** so as to face downward. The light sources **6020c** and **6020d** are disposed on a lower edge of the operation panel **6022** and on both sides of the support **6024** near the operation panel **6022** so as to face in an obliquely downward direction. That is, the light sources **6020a** and **6020b** and the light sources **6020c** and **6020d** are disposed at different height positions from the placement portion **6016**. That is, the light sources **6020a**, **6020b**, **6020c**, and **6020d** are disposed at the positions such that the object is irradiated with light at different angles and are configured so as to radiate the light to the

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object in different directions. In this way, the ability of identification between the object and the placement portion **6016** may be improved.

The operation panel **6022** includes a display screen and is configured so as to be capable of setting the image forming apparatus **6010**. The operation panel **6022** is disposed above the placement portion **6016**. The operation panel **6022** is disposed at the rear of the apparatus body **6012**. The operation panel **6022** is installed on the support **6024** between the placement portion **6016** and the image capturing device **6018**.

A document feeder that has a function of reading both surfaces simultaneously is disposed in the apparatus body **6012** below the top plate of the placement portion **6016**. That is, the document feeder is configured so as to be usable by sliding the top plate of the placement portion **6016** in a plane direction.

The hardware configuration of the image forming apparatus **6010** according to the sixth exemplary embodiment will now be described with reference to FIG. 6-2.

As illustrated in FIG. 6-2, the image forming apparatus **6010** includes a CPU **6031**, a memory **6032**, a storage device **6033** such as a hard disk drive, a communication interface (abbreviated as IF) **6034** for data transmission to and data reception from, for example, an external device via a network, a user interface (abbreviated as UI) device **6035** that includes a touch screen or a liquid-crystal display and a keyboard, a scanner **6036**, a print engine **6037**, the image capturing device **6018**, and the radiation device **6020**. These components are connected to each other with a control bus **6040** interposed therebetween.

The print engine **6037** prints an image on paper through processes such as charging, exposure, development, transfer, and fixing.

The CPU **6031** is a processor that performs a predetermined process, based on the control program that is stored in the memory **6032** or the storage device **6033** and controls the operation of the image forming apparatus **6010**. In the description according to the sixth exemplary embodiment, the CPU **6031** reads and runs the control program that is stored in the memory **6032** or the storage device **6033**, but the program may be stored in a storage medium such as a CD-ROM and provided to the CPU **6031**.

FIG. 6-3 is a block diagram illustrating the functional configuration of the image forming apparatus **6010** that is acquired by running the control program described above.

As illustrated in FIG. 6-3, the image forming apparatus **6010** according to the sixth exemplary embodiment includes a controller **6041**, a communication unit **6042**, an operation display unit **6043**, a storage unit **6044**, an image reading member **6045**, an image forming member **6046**, an image output unit **6047**, an image capturing member **6048**, and a radiation unit **6049**.

The communication unit **6042** receives a print job (an example of a print instruction) that is transmitted from a terminal device.

The controller **6041** implements control to generate image data that is used as print data, based on the print job that is received by the communication unit **6042** and to output the generated image data from the image output unit **6047**.

The storage unit **6044** stores various kinds of data such as the image data that is generated by the controller **6041**.

The image output unit **6047** outputs the image on the paper, based on control of the controller **6041**.

The operation display unit **6043** is controlled by the controller **6041** and displays various kinds of information on the operation panel **6022** or a display screen such as the

terminal device. The operation display unit **6043** inputs various kinds of information about operations that are performed by the user.

The image reading member **6045** performs a scanning operation for reading the image of the object such as a document, based on the control of the controller **6041**.

The image forming member **6046** forms the image of the object that is placed on the placement portion **6016**, based on the control of the controller **6041**.

The image capturing member **6048** is controlled by the controller **6041** and captures the image of the object. The controller **6041** functions as an image acquiring member that acquires the image data of the object the image of which is captured by the image capturing member **6048**.

The radiation unit **6049** is controlled by the controller **6041** and radiates light to the object that is placed on the placement portion **6016**.

The controller **6041** controls the radiation unit **6049** such that the multiple light sources **6020a**, **6020b**, **6020c**, and **6020d** are switched on or switched off, and the brightness or optical axis of the light to be radiated is adjusted.

The controller **6041** implements control such that the operation panel **6022** is switched off when the image capturing member **6048** captures the image of the object.

FIG. 6-4 illustrates the placement portion **6016** viewed from above and illustrates a state in which an object S is placed on the placement portion **6016**.

The placement portion **6016** according to the sixth exemplary embodiment is configured such that the luminance difference between a painted portion or material portion of the whole or a part of an upper portion of the placement portion **6016** and the object S that is placed is a predetermined luminance difference or more, and the color difference between the paint color or material color of the whole or a part of the upper portion of the placement portion **6016** and the color of the object S that is placed is a predetermined color difference or more and such that the object S and the placement portion **6016** are distinguishable from each other. However, the present exemplary embodiment is not a limitation, provided that the object S and the placement portion **6016** are distinguishable. For example, even when the paint color or material color of the whole or a part of the upper portion of the placement portion **6016** and the color of the object S that is placed have the predetermined color difference or less, a luminance difference equal to or more than the predetermined luminance difference suffices provided that the object S and the placement portion **6016** are distinguishable. Specifically, for example, in the case of an 8-bits environment, the luminance difference may be 50 LSB or more, and at least a luminance difference of 20 LSB or more suffices. In contrast, even when the luminance difference between the painted portion or material portion of the whole or a part of the upper portion of the placement portion **6016** and the object S that is placed is the predetermined luminance difference or less, a color difference equal to or more than the predetermined color difference suffices provided that the object S and the placement portion **6016** are distinguishable.

The object S that has color that is represented by a Y-value closer than a value that represents the black color to a value that represents the white color is placed on the placement portion **6016**, for example, where the color is represented by using a Y-value that represents luminance in a YCbCr method. For example, in the case of the 8-bits environment, the object S the Y-value of which is represented by a value closer than 0 to 255 is placed. The color of the upper portion of the placement portion **6016** is represented by the Y-value

closer than the value that represents the white color to the value that represents the black color where the color is represented by using the Y-value that represents the luminance in the YCbCr method such that the object S and the placement portion **6016** are distinguishable in the image that is captured by the image capturing member **6048**. For example, in the case of the 8-bits environment, the color is color the Y-value of which is represented by a value closer than 255 to 0. The color of the whole or a part of the upper portion of the placement portion **6016** is represented by the Y-value closer than the value that represents the white color to the value that represents the black color where the color is represented by using the Y-value that represents the luminance in the YCbCr method.

As for the placement portion **6016** according to the sixth exemplary embodiment, the paint color or material color of the whole or a part of the upper portion is the black color or color having a predetermined optical concentration or more. The light reflectance of the paint color or material color of the whole or a part of the upper surface of the placement portion **6016** is 10% or less. According to the sixth exemplary embodiment, the black color means color chromaticity of which differs from that at an achromatic color point ($x=0.333$, $y=-0.333$, $Y=0$) by a color difference ΔE of 3 or less, color an optical concentration OD of which is 1.0 or more, or color the Y-value representing the luminance in the YCbCr method of which is 50 or less. However, the definition of the black color described above is used for the placement portion **6016** according to the sixth exemplary embodiment and may be changed depending on the actual environment of the vicinity of the apparatus or the performance of the image capturing member.

In addition, the color of the upper portion of the placement portion **6016** is not limited to the paint color or material color of the placement portion **6016** itself but may be color that is reflected on the placement portion **6016** by using the radiation unit **6049** or color that is acquired by superposing the color that is reflected on the placement portion **6016** and shadow.

The object S may be readily identified in a manner in which the color difference or the luminance difference between the color of the placement portion **6016** and the color of the object S that is placed is thus made clear. In this way, the precision of extraction or rotation correction of the image data of the object may be increased.

Marks **6051** are provided at four corners of the upper surface of the placement portion **6016**.

The marks **6051** represent a range in which the image capturing device **6018** is capable of capturing the image of the object S on the placement portion **6016**, and the object the image data of which is acquirable by the controller **6041** is placed on the placement portion **6016**, that is, a position at which the object is placed. As illustrated in FIG. 6-4, the marks **6051** have, for example, a bracket shape. The color of the marks **6051** is represented by the Y-value closer than the value that represents the black color to the value that represents the white color where the color is represented by using the Y-value that represents the luminance in the YCbCr method, and an example thereof is the white color. In other words, the light reflectance of the color of the marks **6051** is a predetermined value or more and is, for example, 80% or more. The object that is placed on the placement portion **6016** may be readily identified in a manner in which the luminance difference between the marks **6051** and the placement portion **6016** is thus made clear. In this way, the precision of extraction or rotation correction of the image data of the object may be increased.

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The marks **6051** are not limited by the bracket shape but may be, for example, guidelines. According to the sixth exemplary embodiment, the marks **6051** are used to distinguish between a region in which the object **S** is placed and a region in which the object **S** is not placed. Specifically, the marks **6051** may be marks that represent the range in which the object is placed by using, for example, a bracket as described above, the color of the placement portion **6016** that is in the range in which the object is placed and that differs from that of a range out of the range, the material of the placement portion **6016** that is in the range in which the object is placed and that differs from the material of a range out of the range, or marks that are represented by irradiating the position at which the object is placed by using the radiation device **6020**.

FIG. **6-5** illustrates a state in which the user operates the image forming apparatus **6010**.

The image forming apparatus **6010** is configured such that the placement portion **6016**, the operation panel **6022**, and the discharge tray **6014**, for example, are disposed at positions so as to be operable by a person with the person standing and are readily operated with the user standing.

As illustrated in FIG. **6-1A**, FIG. **6-1B**, and FIG. **6-5**, the paint color or material color of the placement portion **6016** is configured also on a portion of a surface that is accessed by the user and is configured on the surface that is accessed by the user between the top surface of the apparatus body **6012** and the discharge tray **6014**.

The paint color or material color of the placement portion **6016** and another color are configured between the top surface of the apparatus body **6012** and the discharge tray **6014**. Specifically, the paint color or material color of the placement portion **6016** is configured on an upper portion, and the other color is configured on a lower portion between the top surface of the apparatus body **6012** and the discharge tray **6014**.

In this way, the user is more likely to recognize the position of the placement portion **6016** than in the case where the surface that is accessed by the user has the white color.

A relationship between reflected light distribution due to the light sources **6020a**, **6020b**, **6020c**, and **6020d** and the amount of the light that enters the image capturing device **6018** will now be described with reference to FIG. **6-6A**, FIG. **6-6B**, and FIG. **6-7**.

FIG. **6-6A** illustrates the reflected light distribution in the case where the directivity of the light sources is high and the case where the directivity of the light sources is low when the object is irradiated downward from above by using the light sources **6020a** and **6020b** near the image capturing device **6018** for capturing the image. FIG. **6-6B** illustrates the reflected light distribution in the case where the directivity of the light sources is high and the case where the directivity of the light sources is low when the object is irradiated obliquely downward from above by using the light sources **6020c** and **6020d** near the operation panel **6022** for capturing the image.

When the directivity of the light sources is high, the light that enters the object is reflected with the light having the directivity, the light does not diffuse, and the reflected light distribution does not expand. That is, when the directivity of the light sources is high, the amount of the light that enters the image capturing device **6018** decreases, and it becomes dark. When the directivity of the light sources is low, the light that enters the object is reflected with the light having

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no directivity, the light diffuses, and the reflected light distribution expands, unlike the case where the directivity is high.

That is, the amount of the light that enters the image capturing device **6018** may be increased by setting the directivity of the light sources to a predetermined value or less. For example, in the case where the light is radiated to the object at 45 degrees by using a light source, the directivity may be such that the ratio of the amount of the reflected light to the amount of the radiated light is 50% or less.

In other words, the light sources **6020a**, **6020b**, **6020c**, and **6020d** may be such that the directivity of the light to be radiated is higher than a predetermined range.

The color temperature of the light that is radiated from the light sources **6020a**, **6020b**, **6020c**, and **6020d** has a predetermined value or more and has, for example, 5000 K (kelvin) or more. Examples of the light sources **6020a**, **6020b**, **6020c**, and **6020d** include a LED that is a light source for the white color.

The light sources **6020a**, **6020b**, **6020c**, and **6020d** include respective dimming units that adjust the brightness of the light to be radiated and are configured so as to be capable of dimming the light in consideration of the object and natural light.

The light sources **6020a**, **6020b**, **6020c**, and **6020d** include respective motors that are examples of a changing unit that changes the optical axis of the light to be radiated.

The change operation of the light sources **6020a**, **6020b**, **6020c**, and **6020d** will now be described with reference to FIG. **6-8A** and FIG. **6-8B**.

The controller **6041** causes the motors to change the optical axes of the light sources **6020a**, **6020b**, **6020c**, and **6020d** such that radiation angles for the object change.

The controller **6041** implements control such that the light sources **6020a**, **6020b**, **6020c**, and **6020d** are switched on or switched off.

The controller **6041** adjusts the amount of the light of the light sources **6020a**, **6020b**, **6020c**, and **6020d**, that is, the brightness of the light. That is, the light sources **6020a**, **6020b**, **6020c**, and **6020d** have a dimming function and are capable of adjusting the brightness of the light that is radiated to the object under the control of the controller **6041**.

In this way, the reflected light from the object may be adjusted so as to be constant regardless of the influence of the object and the natural light.

The ratio of the light sources **6020a**, **6020b**, **6020c**, and **6020d** that are switched on is changeable. In this way, the ability of identification may be improved, and the optimal image data of the object may be acquired.

Specifically, the brightness of the light that is radiated to the object may be adjusted, for example, in a manner in which the light sources **6020a** and **6020b** are switched on in a high intensity of 100%, and the light sources **6020c** and **6020d** are switched off, in a manner in which the light sources **6020a**, **6020b**, **6020c**, and **6020d** are switched on in a middle intensity of 60%, or in a manner in which the light sources **6020a** and **6020b** are switched on in a high intensity of 100%, and the light sources **6020c** and **6020d** are switched on in a low intensity of 30%.

A modification to the placement portion **6016** will now be described. FIG. **6-9** illustrates a placement portion **6060** in an image forming apparatus according to another sixth exemplary embodiment viewed from above.

As for the placement portion **6060** according to the present modification, a range **6056** in which the object is

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placed has the black color or color having a predetermined optical concentration or more, and there is a guideline 6058 around the range 6056.

The center of the capturing range of the image capturing device 6018 is nearer than the center of the top surface of the apparatus body 6012 to the surface that is accessed by the user. In other words, the center of the capturing range of the image capturing device 6018 is nearer than the center of the top surface of the apparatus body 6012 to the front of the apparatus body 6012.

The center of the range 6056 that corresponds to the center in the guideline 6058 is nearer than the center of the top surface of the apparatus body 6012 to the surface that is accessed by the user and to the front of the apparatus body 6012.

An image forming apparatus 6100 according to the other sixth exemplary embodiment will now be described with reference to FIG. 6-10.

As for the image forming apparatus 6100, as illustrated in FIG. 6-10, the placement portion 6016 and the image capturing device 6018 are not disposed on the upper portion of the image forming apparatus 6100 but on an intermediate portion of the image forming apparatus 6100. The image forming apparatus 6100 is constituted by an apparatus body 6102 that includes, for example, an image forming member and an image processing unit 6104 that includes, for example, the placement portion 6016 and the discharge tray 6014 described above.

The placement portion 6016 and the discharge tray 6014 are disposed at positions so as to be operable and be capable of being taken out by the user with the user standing. The placement portion 6016 is configured so as to face the user.

The image capturing device 6018 is installed above the placement portion 6016 and is configured such that the image of the object that is placed on the placement portion 6016 is captured from above. The image capturing device 6018 faces downward so as to face the placement portion 6016. The center of the placement portion 6016 is nearer than the center of the top surface of the apparatus body 6102 to the surface that is accessed by the user. In other words, the center of the placement portion 6016 is nearer than the center of the top surface of the apparatus body 6102 to the front of the apparatus body.

The center of the capturing range of the image capturing device 6018 is nearer than the center of the top surface of the apparatus body 6102 to the surface that is accessed by the user. In other words, the center of the capturing range of the image capturing device 6018 is nearer than the center of the top surface of the apparatus body 6102 to the front of the apparatus body 6102.

According to the sixth exemplary embodiment described above, the use of the present disclosure for the image capturing device 6018 and the placement portion 6016 that are included in the image forming apparatus 6010 is described. The present disclosure, however, is not limited thereto. The present disclosure may be used also for an image processing apparatus that includes the image capturing device 6018 and the placement portion 6016 and that is used with the image processing apparatus connected to an image forming apparatus that includes the image forming member 6046 and the image output unit 6047. The present disclosure may be used also for an apparatus that includes a post-processing device.

According to the sixth exemplary embodiment described above, the use of the present disclosure for the image capturing device 6018 and the placement portion 6016 that are included in the image forming apparatus 6010 is

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described. The present disclosure, however, is not limited thereto. The present disclosure may be used also for various image processing apparatuses such as an apparatus in which the image capturing device 6018 and the placement portion 6016 are separated devices.

In the description according to the sixth exemplary embodiment described above, the paint color or material color of the whole or a part of the upper portion of the placement portion 6016 is the black color or color having a predetermined optical concentration or more, but this is not a limitation. The paint color or material color near a part of the upper portion of the placement portion 6016 that is included in a portion at which the object is not placed and that is in contact with at least an edge of the object may be the black color or color having a predetermined optical concentration or more.

According to the sixth exemplary embodiment described above, the color of the marks 6051 is the white color but is not limited thereto and may include another color or may be the black color. That is, color that is distinguishable from the color of the placement portion 6016 suffices.

First Aspect of Seventh Exemplary Embodiment

A first aspect of a seventh exemplary embodiment will now be described with reference to the drawings.

FIG. 7-1 is a perspective view of an image forming apparatus 7001A according to the first aspect of the seventh exemplary embodiment of the present disclosure viewed in an oblique direction in front of the image forming apparatus 7001A. FIG. 7-2 is a perspective view of the image forming apparatus 7001A with a front opening-closing covering opened. FIG. 7-3 is a schematic front view of the image forming apparatus 7001A with the front opening-closing covering opened. Upward, downward, left-hand, right-hand, front, and rear directions illustrated by arrows in the figures such as FIG. 7-1 are directions based on the front Ft of the image forming apparatus 7001A that is supposed to face a user when the image forming apparatus 7001A is used.

As illustrated in FIG. 7-1 to FIG. 7-3, the image forming apparatus 7001A according to the first aspect of the seventh exemplary embodiment includes an image forming member 7020, a medium container unit 7030A, a housing 7010A that includes a medium discharge portion 7040 and an outer reading member 7060, and a front opening-closing covering 7011A that opens or closes such that the front opening-closing covering 7011A covers the front of the housing 7010A.

The housing 7010A is an example of an apparatus body of the image forming apparatus 7001A. The housing 7010A includes an inner frame portion and an outer exterior member (an exterior portion) and has a structural body that has a required shape and structure that includes components such as multiple frames and an exterior covering. As illustrated in FIG. 7-1, the housing 7010A according to the first aspect of the seventh exemplary embodiment has the structural body that has a rectangular cuboid shape that extends in the vertical direction.

For example, the exterior member of the housing 7010A includes exterior coverings such as the front opening-closing covering 7011A, a back covering, a left-hand side opening-closing covering 7013, and a right-hand side covering 7014. The exterior coverings such as the back covering and the right-hand side covering 7014 are non-opening-closing coverings that neither opens nor closes and that are secured. The exterior coverings are configured as untransparent coverings (coverings that are not see-through from the outside) except for the case where there is no particular description. The

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front opening-closing covering **7011A** and the left-hand side opening-closing covering **7013** will be described in detail later.

The image forming member **7020** includes a device for forming an image on a recording medium **7009** (A, B, or C). The image forming member **7020** according to the first aspect of the seventh exemplary embodiment is constituted by an electrophotographic imaging device that finally forms an image that is formed by a developer on the recording medium **7009**.

The image forming member **7020** that includes the electrophotographic imaging device includes an image carrier such as a photoconductor member, devices that are disposed around the image carrier such as a charging device, an exposure device, a developing device, and a transfer device, and a fixing device that is disposed at a position away from the image carrier, although these are not illustrated. The image forming member **7020** also includes devices such as a developer supplying device, an image processing apparatus, and a control device, not illustrated. Among these, the transfer device is a direct transfer device that directly transfers the image that is formed by the developer on the image carrier to the recording medium **7009** or an intermediate transfer device that transfers the image from the image carrier to the recording medium **7009** via an intermediate transfer body. A reference character **7026** that designates a dashed line in FIG. 7-3 represents a fixing portion at which the fixing device that constitutes a part of the image forming member **7020** is disposed.

As illustrated by an arrow P in FIG. 7-2, the image forming member **7020** is mounted so as to be able to be pulled from a location inside the housing **7010A** to a location outside the housing **7010A** (toward the front of the housing **7010A**) with the front opening-closing covering **7011A** opened (see FIG. 7-2 and FIG. 7-3), for example, when being inspected or repaired.

The image forming member **7020** has a function of forming an image that corresponds to image information that is inputted from an external device such as an information terminal that is connected to the image forming apparatus **7001A** and a function of forming an image that corresponds to image information that is reading information about an object that is read by the outer reading member **7060**. For this reason, the image forming member **7020** performs a charging operation by chiefly using the image carrier of the imaging device described above, an exposure operation in accordance with the image information, a developing operation, and a transfer operation in this order. Consequently, the image forming member **7020** creates the image that is formed by the developer on the image carrier, and the image is subsequently transferred from the image carrier to the recording medium **7009**. The image forming member **7020** performs a fixing operation on the recording medium **7009** to which the image is transferred, and the image is consequently fixed to the recording medium **7009**. The recording medium **7009** to which the formed image is fixed is finally discharged onto the medium discharge portion **7040**.

The medium container unit **7030A** includes a device that contains the recording media **7009** that are supplied to the image forming member **7020**. The medium container unit **7030A** is disposed below the image forming member **7020**. The medium container unit **7030A** includes three container units **7031A**, **7031B**, and **7031C**. Examples of the recording media **7009** (**7009A**, **7009B**, and **7009C**) include media such as plain paper sheets, coated paper sheets, and thick paper sheets that have predetermined sizes.

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The container unit **7031A** is configured as an oblique container unit that is obliquely disposed such that a left-hand portion is higher than a right-hand portion when viewed from the front of the housing **7010A** and that is disposed at the highest stage. The oblique container unit **7031A** at the highest stage is capable of containing and supplying the recording medium **7009A** that is sized such that a length in a transport direction is longer than the width of the housing **7010A**. The oblique container unit **7031A** at the highest stage is located nearest to the image forming member **7020** among the container units.

The container unit **7031C** is configured as a typical container unit that is substantially horizontally disposed in a left-right direction when viewed from the front of the housing **7010A** and that is disposed at the lowest stage. The container unit **7031C** at the lowest stage is capable of containing and supplying the recording medium **7009C** that is sized such that a length in the transport direction is slightly smaller than the width of the housing **7010A**. The container unit **7031B** is configured as a small container unit that is disposed in a space between the oblique container unit **7031A** at the highest stage and the container unit **7031C** at the lowest stage and that is disposed at a middle stage.

Each of the container units **7031A**, **7031B**, and **7031C** is constituted by, for example, a container body such as a tray that contains the recording medium **7009** and a feed device that feeds the recording medium **7009** from the container body although these are not illustrated. The container bodies of the container units **7031A**, **7031B**, and **7031C** include respective inner coverings that have contours following the front shapes of the container units **7031A**, **7031B**, and **7031C** at the front. The container body of the oblique container unit **7031A** is obliquely disposed such that a left-hand portion is higher than a right-hand portion. The container bodies of the other container units **7031B** and **7031C** are substantially horizontally disposed.

As illustrated by the arrow P in FIG. 7-2, the container bodies of the container units **7031A**, **7031B**, and **7031C** are mounted so as to be able to be pulled from locations inside the housing **7010A** to locations outside the housing **7010A** (outside the front of the housing **7010A**) with the front opening-closing covering **7011A** opened (see FIG. 7-2 and FIG. 7-3), for example, when the recording media **7009** are put therein.

In the medium container unit **7030A**, the recording medium **7009** that is required is fed from the container unit **7031A**, **7031B**, or **7031C** that contains the recording medium **7009** toward the image forming member **7020** in conjunction with the transfer operation of the image forming member **7020**. A reference character **7037** that designates a one-dot chain line in FIG. 7-3 represents a principal medium transport path on which the recording media **7009** are transported from the medium container unit **7030A** (the container units **7031A**, **7031B**, and **7031C**) to an outlet **7041** in the medium discharge portion **7040** via a part of the image forming member **7020**. The part of the image forming member **7020** corresponds to a portion that transfers the image to the recording medium **7009**. The medium transport path **7037** is constituted by, for example, pairs of transfer drive rollers and a transfer guide member, not illustrated.

The outer reading member **7060** reads the object to be read outside (above) the housing **7010A**. As illustrated in, for example, FIG. 7-1, the outer reading member **7060** is constituted by, for example, a table **7061** on which the object to be read is placed and a reading device unit **7065** that reads the object that is placed on the table **7061**.

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The table **7061** is constituted by a plate member that includes an upper surface portion **7061a** that is flat, and the object to be read may be placed thereon at rest. The table **7061** is mounted on the upper end of the housing **7010A** such that the upper surface portion **7061a** serves as the uppermost surface of the housing **7010A**. The object to be read may be an object that is placed on the table **7061** and read by the reading device unit **7065** from a location above the table **7061**, and examples thereof include three-dimensional objects such as a sheet document on which image information is recorded, a book, a magazine, a food, and a plant.

The reading device unit **7065** is constituted by components such as a reading device **7065a** (see FIG. 7-3) that is capable of optically reading the object that is placed on the table **7061** and a support portion **7065b** that supports and mounts the reading device **7065a**. The reading device unit **7065** according to the first aspect of the seventh exemplary embodiment is mounted such that the reading device **7065a** is supported by the support portion **7065b** with the support portion **7065b** extending upward a required length from the rear edge of the table **7061** and is capable of reading a portion on the table **7061**. Examples of the reading device **7065a** include an imaging element such as a charge coupled device (CCD) and a camera that is constituted by a combination of optical elements such as a lens. The camera that serves as the reading device **7065a** is also referred to as a document camera. The reading device unit **7065** may include an illumination device that illuminates the object that is placed on the table **7061** when the object is read.

As illustrated in, for example, FIG. 7-1, the reading device unit **7065** includes an operation panel **7091** for operation of the image forming apparatus **7001A** at a position near the bottom of the support portion that supports the reading device **7065a** and away forward therefrom. From the perspective of, for example, improvement in operability, the operation panel **7091** may be disposed away upward from the table **7061**. The operation panel **7091** includes a display unit such as a liquid crystal touch screen that is capable of displaying and operating an operation screen but may include a mechanical operation member such as a button.

The medium discharge portion **7040** is a portion onto which the recording media **7009** on which the images are formed by the image forming member **7020** are discharged. The medium discharge portion **7040** is disposed above the image forming member **7020**. The medium discharge portion **7040** has a container space **7043** that is located at the upper right of the front **Ft** of the housing **7010A**. The container space **7043** opens into the outside via a front opening in the housing **7010A** and a right-hand opening continuous thereto. The medium discharge portion **7040** has the outlet **7041** for the recording media **7009** in a left-hand inner wall surface adjacent to the container space **7043** and a placement surface **7042** on which the recording media **7009** that are discharged via the outlet **7041** are to be placed and contained along the bottom of the container space **7043**. For example, the placement surface **7042** is a surface that obliquely extends such that the height thereof increases as a position from the outlet **7041** in the right-hand direction increases, or a slope rising to the right.

As illustrated in, for example, FIG. 7-1 to FIG. 7-3, the front opening-closing covering **7011A** of the image forming apparatus **7001A** is configured as an opening-closing covering that is capable of entirely covering the image forming member **7020** and the medium container unit **7030A** that correspond to a portion at the front of the housing **7010A** except for the medium discharge portion **7040** and the table

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7061 of the outer reading member **7060**. The front opening-closing covering **7011A** opens or closes so as to cover the front of the housing **7010A** from an edge (a left-hand edge **7010h**) to an edge (a right-hand edge **7010m**) in the left-right direction. The meaning of to be covered from the edge to the edge includes not only the meaning of to be completely covered from a boundary portion between the front of the apparatus body and a surface in the left-hand direction to a boundary portion between the front of the apparatus body and a surface in the right-hand direction but also the meaning that the medium container unit is covered at least in the left-right direction.

As illustrated in, for example, FIG. 7-2, the overall shape of the front opening-closing covering **7011A** is a shape obtained by removing an upper right half as a notch **7011d** that has a shape that corresponds to the shape of the front opening of the medium discharge portion **7040** from a rectangle that extends in the vertical direction and that corresponds to a lower portion of the front of the housing **7010A** except for the table **7061**.

As illustrated in, for example, FIG. 7-2, the front opening-closing covering **7011A** is mounted so as to open and close in directions illustrated by arrows **Do** and **Dc** on opening-closing support members **7016** such as multiple hinges (three hinges in this example) on the left-hand edge **7010h** at the front of the housing **7010A**. Consequently, the whole of the front opening-closing covering **7011A** swings sideways in the left-hand direction (the direction illustrated by the arrow **Do**).

As for the image forming apparatus **7001A**, as illustrated in FIG. 7-1 and FIG. 7-2, the right-hand side covering **7014** has a handle guide depression **7014c** on a front edge portion. The handle guide depression **7014c** has a depressed shape for guiding a finger of the user to a handle of the front opening-closing covering **7011A** such that the finger is easily caught on the handle when the front opening-closing covering **7011A** is opened.

As illustrated in FIG. 7-4 or FIG. 7-5, the left-hand side opening-closing covering **7013** of the image forming apparatus **7001A** is configured as an opening-closing covering that is capable of entirely covering the image forming member **7020** and the medium container unit **7030A** that correspond to a portion at the left-hand side of the housing **7010A** except for the table **7061** of the outer reading member **7060** and a part of an upper edge portion (an upper covering portion **7015**). The left-hand side opening-closing covering **7013** is heavier than the front opening-closing covering **7011A**, for example, because there are a part of the medium transport path **7037** and a member such as a thermal insulation member inside the left-hand side opening-closing covering **7013**.

As illustrated in, for example, FIG. 7-5, the overall shape of the left-hand side opening-closing covering **7013** is a shape obtained by removing a part as a notch **7013d** from a rectangle that extends in the vertical direction and that corresponds to a lower portion of the left-hand side of the housing **7010A** except for the table **7061** and the upper covering portion **7015**. The notch **7013d** has a shape that corresponds to a rectangular shape of the upper covering portion **7015** that extends in a front-rear direction and is formed in an upper front portion of the rectangle described above. As illustrated in FIG. 7-5, the left-hand side opening-closing covering **7013** opens or closes so as to cover a fixing portion **7026** and a part of the medium transport path **7037** that corresponds to a part of the image forming member **7020**.

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As illustrated in, for example, FIG. 7-5, the left-hand side opening-closing covering 7013 is mounted so as to open and close in directions illustrated by the arrows Do and Dc on the opening-closing support members 7016 such as multiple hinges (three hinges in this example) on the rear edge of the left-hand side of the housing 7010A. The opening-closing support members 7016 are disposed on the rear edge of both edges in the left-right direction of the left-hand side opening-closing covering 7013 (the front-rear direction of the housing 7010A). Consequently, the whole of the left-hand side opening-closing covering 7013 swings sideways in the rear direction (the direction illustrated by the arrow Do).

As illustrated in FIG. 7-6, the image forming apparatus 7001A includes a control device 7008 that controls the operation of the entire apparatus such as the image forming member 7020, the outer reading member 7060, and the medium transport path 7037. In FIG. 7-6, the control device 7008 is illustrated as a controller.

The control device 7008 is constituted by, for example, an arithmetic processing unit, a storage element, input and output portions, a storage device, and a control device. The control device 7008 transmits a control instruction that is needed for an object to be controlled depending on, for example, a control program, reference data, or detection data that is stored in the storage element or the storage device. For this reason, the control device 7008 is connected to, for example, an image formation controller 7028 that exclusively controls the operation of the image forming member 7020, an outer reading controller 7068 that exclusively controls the operation of the outer reading member 7060, and a medium transport controller 7038 that exclusively controls the operation of the medium transport path 7037. The control device 7008 is connected to a detection unit 7081 that includes various sensors that detect information that is needed for control.

As for the image forming apparatus 7001A, an interlock switch 7082 that is one of safety mechanisms regarding the operation of the image forming apparatus 7001A is used as an example of the detection unit 7081. The interlock switch 7082 is constituted by, for example, a movable connection member 7082a that is disposed on the front opening-closing covering 7011A and a switch member 7082b that is disposed in the housing 7010A.

When the interlock switch 7082 is in a connection state, the control device 7008 implements control such that the operation of the image forming member 7020, the outer reading member 7060, and the medium transport path 7037, for example, is performable. The connection state is a state in which the movable connection member 7082a is connected to the switch member 7082b. When the interlock switch 7082 is in a non-connection state, the control device 7008 implements control such that the operation of the image forming member 7020, the outer reading member 7060, and the medium transport path 7037, for example, is not performable. The non-connection state is a state in which the movable connection member 7082a is separated from the switch member 7082b and is not connected thereto. As for the control in the non-connection state, the operation is not started (no operation instruction is accepted) when the operation is inactive, and the operation is forcefully stopped when the operation is active.

As for the image forming apparatus 7001A that has the above configuration, as illustrated in FIG. 7-1, the image forming member 7020 (including the fixing portion 7026) and the medium container unit 7030A are entirely covered by the single front opening-closing covering 7011A at the

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front exterior of the housing 7010A with the front opening-closing covering 7011A closed.

Accordingly, as for the image forming apparatus 7001A, a vertical division line Lv that extends in the vertical direction across the image forming member 7020 and the medium container unit 7030A as illustrated in FIG. 7-11 by way of example is not present on the front exterior (the front opening-closing covering 7011A) of the housing 7010A between the image forming member 7020 and the medium container unit 7030A, unlike the case where the housing 7010A is covered from the left-hand edge to the right-hand edge by using multiple opening-closing coverings, and in this way, the quality of appearance may be improved.

Regarding this point, an image forming apparatus 7100A in a first comparative example illustrated in FIG. 7-11 is configured such that two opening-closing coverings 7111 and 7112 open or close so as to entirely cover the image forming member 7020 and the medium container unit 7030A. As for the image forming apparatus 7100A in the first comparative example, the vertical division line Lv that corresponds to the boundary line between the opening-closing covering 7111 and the opening-closing covering 7112 is present on the front exterior of the housing 7010A. For this reason, in the case of the image forming apparatus 7100A in the first comparative example, the quality of the appearance of the front of the housing 7010A is degraded due to the vertical division line Lv on the front exterior of the housing 7010A.

The image forming apparatus 7001A is configured such that the front opening-closing covering 7011A covers the three (all) container units 7031A, 7031B, and 7031C of the medium container unit 7030A.

For this reason, as for the image forming apparatus 7001A, no division line is present on the front exterior (the front opening-closing covering 7011A) of the housing 7010A among the multiple container units 7031A, 7031B, and 7031C of the medium container unit 7030A, and in this way, the quality of appearance may be improved. In addition, as for the image forming apparatus 7001A, the front opening-closing covering 7011A may entirely cover and conceal the medium container unit 7030A at once. For this reason, a container unit that has a unique form such as the oblique container unit 7031A may be used as the medium container unit 7030A. In the case where the oblique container unit 7031A is included in the medium container unit 7030A as in the image forming apparatus 7001A, the oblique container unit 7031A may be concealed, and in this way, the quality of appearance may be improved.

An image forming apparatus 7001C that includes a front opening-closing covering 7011D illustrated in FIG. 7-10 instead of the front opening-closing covering 7011A of the image forming apparatus 7001A may be provided. The front opening-closing covering 7011D opens or closes so as to cover the entire image forming member 7020 and the oblique container unit 7031A that corresponds to a part of the medium container unit 7030A at the front of the housing 7010A. A lower edge 7011v thereof serves as an opening-closing covering that has a shape substantially following an oblique contour line Ji of the lower side of the oblique container unit 7031A. Also, the front opening-closing covering 7011D covers the housing 7010A from the edge to the edge in the left-right direction.

However, as illustrated in FIG. 7-10, the image forming apparatus 7001C in a reference example that includes the front opening-closing covering 7011D needs a second front opening-closing covering 7011E that covers the other container units 7031B and 7031C of the medium container unit

7030A. In the case of the image forming apparatus **7001C**, a division line **Ld** that obliquely extends is present between the front opening-closing covering **7011D** and the second front opening-closing covering **7011E**, and it is difficult to improve the quality of appearance due to the division line **Ld** accordingly.

The image forming apparatus **7001C** that includes the front opening-closing covering **7011D** is compared with the image forming apparatus **7100A** in a second comparative example illustrated in FIG. 7-12 as follows.

Firstly, the image forming apparatus **7100A** in the second comparative example includes a front opening-closing covering **7115** that entirely covers the image forming member **7020** but does not cover the medium container unit **7030A** at all. Secondly, the image forming apparatus **7100A** in the second comparative example includes an exterior covering **7331** that covers the oblique container unit **7031A** of the medium container unit **7030A** and a second front opening-closing covering **7117** that covers the two container units **7031B** and **7031C** of the medium container unit **7030A**.

That is, in comparison between the image forming apparatus **7001C** and the image forming apparatus **7100A** in the second comparative example, as for the image forming apparatus **7100A** in the second comparative example, the oblique container unit **7031A** is not covered by the front opening-closing covering **7115**, and two oblique division lines **Ld1** and **Ld2** are present. Regarding this point for the image forming apparatus **7001C** that includes the front opening-closing covering **7011D**, the single oblique division line **Ld** is present, the number of the division line **Ld** is decreased, and in this way, the quality of appearance at the front exterior of the housing **7010A** may be improved.

As for the image forming apparatus **7001A** according to the first aspect of the seventh exemplary embodiment, the lower edge **7011v** of the front opening-closing covering **7011A** is not along the oblique contour line **Ji** of the oblique container unit **7031A** but substantially horizontally extends in the left-right direction. For this reason, the image forming apparatus **7001A** may be capable of covering and concealing the oblique container unit **7031A** such that the existence thereof is not recognized, and in this way, the quality of appearance may be maintained. The image forming apparatus **7001A** even includes the container units **7031B** and **7031C** in addition to the oblique container unit **7031A** as in the medium container unit **7030A** but may not include the second front opening-closing covering **7117** for covering the container units **7031B** and **7031C**, unlike the image forming apparatus **7100A** in the second comparative example.

The image forming apparatus **7001A** is configured such that the left-hand side opening-closing covering **7013** that is heavier than the front opening-closing covering **7011A** opens and closes by using the opening-closing support members **7016** on the rear edge in the left-right direction. For this reason, as for the image forming apparatus **7001A**, a vibration is unlikely to be produced when the left-hand side opening-closing covering **7013** opens and closes, a vibration is inhibited from being transmitted to the reading device **7065a** in the outer reading member **7060** when the left-hand side opening-closing covering **7013** opens and closes, and the reading device **7065a** is inhibited from being vibrated, unlike the case where the left-hand side opening-closing covering **7013** opens and closes on the opening-closing support members **7016** that are disposed on an edge thereof in the vertical direction. This is particularly effective when it is necessary to open or close the left-hand side opening-closing covering **7013** during the reading operation of the outer reading member **7060**.

In addition, the image forming apparatus **7001A** includes the left-hand side opening-closing covering **7013** that opens or closes such that the left-hand side opening-closing covering **7013** covers a portion at the left-hand side of the housing **7010A** across the image forming member **7020** and the medium container unit **7030A**. For this reason, as for the image forming apparatus **7001A**, the vertical division line that extends across the image forming member **7020** and the medium container unit **7030A** is not present on the left-hand side exterior of the housing **7010A**, and in this way, the quality of appearance at the left-hand exterior thereof may be improved.

As for the image forming apparatus **7001A**, when the front opening-closing covering **7011A** is opened, the movable connection member **7082a** of the interlock switch **7082** on the front opening-closing covering **7011A** is separated from the switch member **7082b** of the interlock switch **7082** in the housing **7010A** and the state thereof changes into the non-connection state. In the case where the front opening-closing covering **7011A** is opened while the image forming apparatus **7001A** forms the image, the control device **7008** implements control such that the operation of the image formation controller **7028**, the outer reading controller **7068**, and the medium transport controller **7038** is forcefully stopped in accordance with detection information that represents the state of the interlock switch **7082** changes into the non-contact state.

For this reason, as for the image forming apparatus **7001A**, when the front opening-closing covering **7011A** is opened, the operation (image formation) of the image forming member **7020** and the medium transport path **7037** that are exposed at the front of the housing **7010A** is stopped, and safety is ensured. In the case where there is the recording medium **7009** on the medium transport path **7037** for forming the image, the control device **7008** may implement control such that the recording medium **7009** that is being transported is transported to the medium discharge portion **7040** and discharged when the front opening-closing covering **7011A** is opened and such that the operation of the medium transport path **7037** is subsequently stopped.

Second Aspect of Seventh Exemplary Embodiment

FIG. 7-7 is a perspective view of an image forming apparatus **7001B** according to a second aspect of the seventh exemplary embodiment of the present disclosure viewed in an oblique direction in front of the image forming apparatus **7001B**. FIG. 7-8 is a perspective view of the image forming apparatus **7001B** with a front opening-closing covering opened. FIG. 7-9 is a schematic front view of the image forming apparatus **7001B** with the front opening-closing covering opened.

The image forming apparatus **7001B** according to the second aspect of the seventh exemplary embodiment has the same configuration as that of the image forming apparatus **7001A** according to the first aspect of the seventh exemplary embodiment except that the following points are modified. As illustrated in FIG. 7-7 to FIG. 7-9, one of the changed points is that a document reading member **7050** is used instead of the outer reading member **7060**, another is that a medium container unit **7030B** is used instead of the medium container unit **7030A**, and the other is that a front opening-closing covering **7011B** is used instead of the front opening-closing covering **7011A**. A housing **7010B** that is an example of the apparatus body differs from the housing **7010A** according to the first aspect of the seventh exemplary embodiment in including the medium container unit **7030B**. For this reason, in the following description, common components are designated by reference characters like to those

according to the first aspect of the seventh exemplary embodiment, and the description thereof is omitted unless the description is needed.

The document reading member 7050 includes a device that reads a sheet document. As illustrated in FIG. 7-7 to FIG. 7-9, the document reading member 7050 is constituted by devices or members such as an image reading member 7051, a platen covering portion 7053, and an auto document feeder 7055.

The image reading member 7051 includes, for example, a platen glass 7052 that is an example of a document table on which the document is placed and a reading device that reads the document that is placed on the platen glass 7052. The platen covering portion 7053 opens or closes so as to cover the platen glass 7052. The auto document feeder 7055 is disposed on the platen covering portion 7053 and enables multiple documents to be automatically transported and continuously read.

The auto document feeder 7055 uses a method of reading a document by using a reading device such as an image sensor that is disposed on a document transport path or a method of reading a document that is placed on the platen glass 7052 by using a reading device in the image reading member 7051. A reference character 7057 that designates a one-dot chain line in FIG. 7-9 represents a principal document transport path on which the document is transported in the auto document feeder 7055. The platen covering portion 7053 is coupled with the image reading member 7051 behind the platen glass 7052 by using opening-closing support members such as hinges, not illustrated. Consequently, the platen covering portion 7053 is mounted so as to be swingable in open and close directions illustrated by arrows E1 and E2 on the hinges that are disposed on a rear edge portion thereof and that are not illustrated.

The document reading member 7050 reads the document on the platen glass 7052 or continuously reads the documents that are being transported by the auto document feeder 7055 and transmits image information about each read document from the image reading member 7051 to, for example, the image forming member 7020. Consequently, the image forming apparatus 7001B is configured such that the image forming member 7020 forms the image of each document that is read by the document reading member 7050 on a recording medium 7009 (D, E, or F) for output. As for the document reading member 7050, an operation panel 7092 for the operation of the image forming apparatus 7001B is disposed on a front edge portion of the image reading member 7051. The operation panel 7092 includes mechanical operation members such as a button and a switch, a screen display unit, and a display unit such as a liquid crystal touch screen that is capable of displaying and operating an operation screen.

As illustrated in FIG. 7-8 and FIG. 7-9, the medium container unit 7030B includes three container units 7031D, 7031E, and 7031F that are arranged in the vertical direction.

Each of the three container units 7031D, 7031E, and 7031F is constituted by, for example, a container body such as a tray that contains the recording medium 7009 and a feed device that feeds the recording medium 7009 from the container body although these are not illustrated. Each of the three container units 7031D, 7031E, and 7031F is configured as a typical container unit in which the container body is substantially horizontally disposed in the left-right direction. The recording media 7009D, 7009E, and 7009F that are contained in the respective container units 7031D, 7031E, and 7031F may have the same size or different sizes, or at least one of the sizes may differ from the others. The

container unit 7031D at the highest stage is located nearest to the image forming member 7020 among the container units.

As illustrated in, for example, FIG. 7-7 to FIG. 7-9, the front opening-closing covering 7011B is configured as an opening-closing covering that is capable of entirely covering the image forming member 7020 and that is capable of covering a part (the container units 7031D and 7031E) of the medium container unit 7030B in the housing 7010B. The front opening-closing covering 7011B opens or closes so as to cover the front of the housing 7010B from an edge (the left-hand edge 7010h) to an edge (the right-hand edge 7010m) in the left-right direction.

The overall shape of the front opening-closing covering 7011B is a shape obtained by removing an upper right half as a notch 7011e from a rectangle that extends in the vertical direction and that corresponds to a lower portion of the front of the housing 7010B except for the image reading member 7051, and the shape covers a portion of the medium container unit 7030B except for a lower portion including the container unit 7031F at the lowest stage. The notch 7011e has a shape following the shape of the front opening of the medium discharge portion 7040. As illustrated in, for example, FIG. 7-8, the front opening-closing covering 7011B is mounted so as to open and close in directions illustrated by the arrows Do and Dc on the opening-closing support members 7016 such as multiple hinges (two hinges in this example) on the left-hand edge at the front of the housing 7010B. Consequently, the whole of the front opening-closing covering 7011B swings sideways in the left-hand direction (the direction illustrated by the arrow Do).

As for the image forming apparatus 7001B, the front opening-closing covering 7011B does not cover the container unit 7031F at the lowest stage in the medium container unit 7030B. For this reason, the container unit 7031F at the lowest stage is mounted such that a front outer covering 7033 is secured at the front of the container body instead of the front opening-closing covering 7011B. The container bodies of the container units 7031D and 7031E that are covered by the front opening-closing covering 7011B include respective inner coverings that have contours following the front shapes (rectangles that extend sideways) of the container units 7031D and 7031E at the front.

As illustrated by an arrow P in FIG. 7-8, the container bodies of the container units 7031D and 7031E are mounted so as to be able to be pulled from locations inside the housing 7010B to locations outside the housing 7010B (toward the front of the housing 7010B) with the front opening-closing covering 7011B opened (see FIG. 7-8 and FIG. 7-9), for example, when the recording media 7009 are put therein. The container unit 7031F at the lowest stage is mounted so as to be able to be pulled from a location inside the housing 7010B to a location outside the housing 7010B anytime, for example, when the recording medium 7009 is put therein regardless of whether the front opening-closing covering 7011B opens or closes.

As for the image forming apparatus 7001B that has the above configuration, as illustrated in FIG. 7-7, the entire image forming member 7020 (including the fixing portion 7026) and the container units 7031D and 7031E that correspond to a part of the medium container unit 7030B are covered by the single front opening-closing covering 7011B at the front exterior of the housing 7010B with the front opening-closing covering 7011B closed. Accordingly, as for the image forming apparatus 7001B, the division line Ld (see FIG. 7-10) is not present on the front exterior (the front opening-closing covering 7011B) of the housing 7010B

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between the image forming member **7020** and the medium container unit **7030B**, unlike the case where the image forming member **7020** and the medium container unit **7030B** are covered by respective opening-closing coverings, and in this way, the quality of appearance may be improved.

As for the image forming apparatus **7001B**, the front opening-closing covering **7011B** is configured so as to cover the two container units **7031D** and **7031E** of the medium container unit **7030B**. For this reason, as for the image forming apparatus **7001B**, no division line (Ld) is present on the front exterior (the front opening-closing covering **7011B**) of the housing **7010B** between the container units **7031D** and **7031E** of the medium container unit **7030B**, and in this way, the quality of appearance may be improved.

As for the image forming apparatus **7001B**, the single container unit **7031F** of the medium container unit **7030B** that is not covered by the front opening-closing covering **7011B** is pulled from a location inside the housing **7010B** to a location outside the housing **7010B** anytime, for example, when the recording medium **7009** is put therein regardless of whether the front opening-closing covering **7011B** opens or closes. For this reason, an example of the container unit **7031F** that is not covered by the front opening-closing covering **7011B** may be a container unit that frequently needs supply of the recording medium **7009** because of high consumption of the recording medium **7009**. When the container unit **7031F** thus is not covered by the front opening-closing covering **7011B**, it is not necessary for the front opening-closing covering **7011B** to open and close when the recording medium **7009** is supplied to the container unit **7031F** that is frequently used, and in this way, workability may be improved.

As for the image forming apparatus **7001B**, when the front opening-closing covering **7011B** is opened, the state of the interlock switch **7082** changes into the non-contact state. Specifically, as illustrated in, for example, FIG. 7-8, the movable connection member **7082a** of the interlock switch **7082** on the front opening-closing covering **7011B** is separated from the switch member **7082b** of the interlock switch **7082** in the housing **7010B**. In the case where the front opening-closing covering **7011B** is opened while the image forming apparatus **7001B** forms the image, it is detected that the state of the interlock switch **7082** changes into the non-connection state, and the control device (see FIG. 7-6) receives information about the detection and implements control such that the operation of the image formation controller **7028** and the medium transport controller **7038** is forcefully stopped.

For this reason, as for the image forming apparatus **7001B**, when the front opening-closing covering **7011B** is opened, the operation (image formation) of the image forming member **7020** and the medium transport path **7037** that are exposed at the front of the housing **7010B** is stopped, and safety is ensured. As for the image forming apparatus **7001B**, however, the medium transport path **7037** does not stop a transport operation toward the container unit **7031F** at the lowest stage even when the front opening-closing covering **7011B** is opened while the image is formed. Modification

The present disclosure is not limited by the contents described according to the first aspect of the seventh exemplary embodiment and the second aspect of the seventh exemplary embodiment by way of example but includes, for example, a modification described below.

As for the image forming apparatus **7001A** according to the first aspect of the seventh exemplary embodiment, the left-hand side opening-closing covering **7013** may be con-

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figured so as to open and close on the opening-closing support members **7016** that are disposed on the right-hand edge in the left-right direction of the left-hand side opening-closing covering **7013** (the front edge of the housing **7010A** in the front-rear direction). As for the image forming apparatus **7001A**, the right-hand side covering **7014** may be configured as an opening-closing covering that opens or closes so as to cover the image forming member **7020** and the medium container unit **7030A** at the right-hand side of the housing **7010A** instead of or in addition to the left-hand side opening-closing covering **7013**. In this case, when the right-hand side covering **7014** is heavier than the front opening-closing covering **7011A**, the right-hand side covering **7014** may open and close on the opening-closing support members **7016** that are disposed on one of both edges of the right-hand side covering **7014** in the left-right direction (the front-rear direction of the housing **7010A**). The exterior coverings of the image forming apparatus **7001A** may be a left-hand side covering and a right-hand side covering that neither open nor close.

As for the image forming apparatus **7001B** according to the second aspect of the seventh exemplary embodiment, the front opening-closing covering **7011B** may be configured so as to cover all of the three container units **7031D**, **7031E**, and **7031F** of the medium container unit **7030B** as in the front opening-closing covering **7011A** according to the first aspect of the seventh exemplary embodiment. The front opening-closing covering **7011B**, however, may cover the image forming member **7020** and the container unit **7031D** of the medium container unit **7030B** at the highest stage.

The housings **7010A** and **7010B** are not limited by the structural body that has a rectangular cuboid shape that extends in the vertical direction as described according to the first aspect of the seventh exemplary embodiment and the second aspect of the seventh exemplary embodiment by way of example but may have, for example, a curved side surface portion.

An image forming apparatus may have a configuration obtained by omitting the outer reading member **7060** from the configuration of the image forming apparatus **7001A** according to the first aspect of the seventh exemplary embodiment (including the image forming apparatus **7001C**). An image forming apparatus may have a configuration obtained by omitting the document reading member **7050** from the configuration of the image forming apparatus **7001B** according to the second aspect of the seventh exemplary embodiment. In these cases, an upper end covering, for example, may be provided as an exterior covering on the upper end of the housing **7010A** or **7010B** on which the outer reading member **7060** or the document reading member **7050** is originally disposed. As for another image forming apparatus, the image forming member **7020** is not limited to an electrophotographic image forming member that uses a developer, but the image forming member **7020** may use another image formation method (for example, an ink-jet method).

Eighth Exemplary Embodiment

FIG. 8-1 illustrates an image forming apparatus that uses a drive device according to an eighth exemplary embodiment. In FIG. 8-1, an arrow X represents the horizontal direction (the width direction) of an image forming apparatus **8001**, an arrow Y represents the depth direction of the image forming apparatus **8001**, and an arrow Z represents the vertical direction of the image forming apparatus **8001** (the vertical direction).

Entire Configuration of Image Forming Apparatus

The image forming apparatus **8001** according to the eighth exemplary embodiment is configured as, for example, a color printer. The image forming apparatus **8001** is an example of a container apparatus. As illustrated in FIG. 8-1, the image forming apparatus **8001** includes, for example, multiple imaging devices **8010** that form toner images that are developed by using toner that constitutes a developer, an intermediate transfer device **8020** that holds and finally transports the toner images that are formed by the imaging devices **8010** to a second transfer position at which the toner images are second-transferred to recording paper **8005** that is an example of a recording medium, a paper feeding device **8050** that contains and transports the recording paper **8005** that is required to be supplied to the second transfer position of the intermediate transfer device **8020**, and a fixing device **8040** that fixes the toner images that are second-transferred by the intermediate transfer device **8020** to the recording paper **8005**. In the figure, the apparatus body of the image forming apparatus **8001** is designated as **8001a**. The apparatus body **8001a** is constituted by, for example, a support structure member or an exterior covering. A two-dot chain line in the figure represents a principal transport path on which the recording paper **8005** is transported in the apparatus body **8001a**.

The imaging devices **8010** are constituted by four imaging devices **8010Y**, **8010M**, **8010C**, and **8010K** that exclusively form toner images in the four colors of yellow (Y), magenta (M), cyan (C), and black (K). The four imaging devices **8010** (Y, M, C, and K) are aligned in a column with the imaging device **8010Y** for yellow (Y) tilting at a higher position in a Z-direction and with the imaging device **8010K** for black (K) tilting at a lower position in the Z-direction in an interior space of the apparatus body **8001a**.

The four imaging devices **8010** are constituted by the imaging devices **8010** (Y, M, and C) for colors of yellow (Y), magenta (M), and cyan (C) and the imaging device **8010K** for black (K). The imaging device **8010K** for black is disposed at a most downstream position in a direction B in which an intermediate transfer belt **8021** of the intermediate transfer device **8020** moves. The image formation mode of the image forming apparatus **8001** includes a full color mode in which the imaging devices **8010** (Y, M, and C) for the colors and the imaging device **8010K** for black (K) operate to form a full color image and a monochrome mode in which only the imaging device **8010K** for black (K) operates to form a monochrome image.

As illustrated in FIG. 8-2, the imaging devices **8010** (Y, M, C, and K) include respective photoconductor drums **8011** that rotate and that are examples of an image forming unit (an image carrier). Around the photoconductor drums **8011**, for example, there are charging devices **8012** that charge circumferential surfaces (image carrying surfaces) of the photoconductor drums **8011** on which images are formable at a required potential, exposure devices **8013** that radiate light based on information (signal) about the images to the charged circumferential surfaces of the photoconductor drums **8011** and that form electrostatic latent images (for the respective colors) that have a potential difference, developing devices **8014** (Y, M, C, and K) that are examples of an image forming unit (a developing unit) that develops the electrostatic latent images by using toner of the developer in the associated colors (Y, M, C, and K) into the toner images, first transfer devices **8015** (Y, M, C, and K) that transfer the respective toner images to the intermediate transfer device **8020**, and drum cleaning devices **8016** (Y, M, C, and K) that remove and clean attached substances such as attached

residual toner on the image carrying surfaces of the photoconductor drums **8011** after the first transfer.

In each photoconductor drum **8011**, the image carrying surface that has a photoconductive layer (a photosensitive layer) composed of a photoconductive material is formed on the circumferential surface of a cylindrical or column base material that is grounded. Each photoconductor drum **8011** is supported so as to rotate in a direction illustrated by an arrow A when driving force is transmitted from the drive unit according to the eighth exemplary embodiment as described later.

Each charging device **8012** is constituted by a charging roller **8121** that is disposed such that the charging roller **8121** is in contact with the photoconductor drum **8011**. A cleaning roller **8122** that cleans the surface of the charging roller **8121** is disposed so as to be in contact with a back surface of the charging roller **8121**. Charge voltage is applied to the charging device **8012**. In the case where the developing device **8014** is used for reversal development, voltage, or current, having the same polarity as charge polarity of the toner that is supplied from the developing device **8014** is applied as the charge voltage. The charging roller **8121** and the cleaning roller **8122** are rotated when driving force is transmitted from the photoconductor drum **8011**.

Each of the exposure devices **8013** includes a LED print head that radiates light depending on the image information to the photoconductor drum **8011** by using light emitting diodes (LEDs) that are multiple light emitting elements that are arranged in the axial direction of the photoconductor drum **8011** and forms the electrostatic latent image. The exposure device **8013** may be used for deflection scanning of laser light that is configured depending on the image information in the axial direction of the photoconductor drum **8011**.

Each of the developing devices **8014** (Y, M, C, and K) is configured such that a development roller **8141** that carries the developer and that transports the developer to a development region that faces the photoconductor drum **8011**, a mixing supply member **8142** such as a screw auger that mixes and supplies the developer such that the developer passes through the development roller **8141**, a mixing transport member **8143** such as a screw auger that mixes and transports the developer to the mixing supply member **8142**, and a layer thickness restriction member **8144** that restricts the amount (layer thickness) of the developer that is carried by the development roller **8141** are disposed in a housing **8140** that has an opening portion and a container chamber for the developer. Development voltage is applied from a power supply device, not illustrated, to the developing device **8014** between the development roller **8141** and the photoconductor drum **8011**. The development roller **8141** rotates in a required direction when driving force is transmitted from the drive unit according to the eighth exemplary embodiment as described later. The mixing supply member **8142** and the mixing transport member **8143** are rotated when driving force is transmitted from the development roller **8141**. Two-component developer that contains non-magnetic toner and a magnetic carrier is used as the developer for the four colors.

Each of the first transfer devices **8015** (Y, M, C, and K) is a contact transfer device that includes a first transfer roller that comes into contact with the intermediate transfer belt **8021** along the circumference of the photoconductor drum **8011** and rotates, and a first transfer voltage is applied thereto. As for the first transfer voltage, direct current

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voltage that has polarity opposite the charge polarity of the toner is applied from the power supply device, not illustrated.

Each of the drum cleaning devices **8016** is constituted by, for example, a body **8160** that has a container shape that partly opens, a cleaning plate **8161** that comes into contact with the circumferential surface of the photoconductor drum **8011** at required pressure after the first transfer and that removes and cleans attached substances such as residual toner, and a feed member **8162** such as a screw auger that collects the attached substances such as the toner removed by the cleaning plate **8161** and that feeds and transports the attached substances to a collection system, not illustrated. A plate member (for example, a blade) composed of, for example, a rubber material is used as the cleaning plate **8161**. The feed member **8162** of the drum cleaning device **8016** is rotated when driving force is transmitted from the photoconductor drum **8011**.

As illustrated in FIG. 8-1, the intermediate transfer device **8020** is disposed so as to be located above the imaging devices **8010** (Y, M, C, and K) in the Z-direction. As illustrated in FIG. 8-2, the intermediate transfer device **8020** is constituted by, for example, the intermediate transfer belt **8021** that turns in a direction illustrated by an arrow B while passing through the first transfer position between the photoconductor drums **8011** and the first transfer devices **8015** (the first transfer rollers), multiple belt support rollers **8022** to **8026** that hold and rotatably support the intermediate transfer belt **8021** from an inner surface thereof in a desired state, a second transfer device **8030** that is an example of a second transfer unit that is disposed on the outer circumferential surface (the image carrying surface) of the intermediate transfer belt **8021** supported by the belt support roller **8026** and that second-transfers the toner images on the intermediate transfer belt **8021** to the recording paper **8005**, and a belt cleaning device **8027** that removes and cleans attached substances such as the toner and paper powder that remain and that are attached to the outer circumferential surface of the intermediate transfer belt **8021** after passing through the second transfer device **8030**.

An example of the intermediate transfer belt **8021** is a belt that is composed of a material acquired by diffusing a resistance regulator such as carbon black in synthetic resin such as polyimide resin or polyamide resin and that has no ends. The belt support roller **8022** is configured as a drive roller that doubles as a facing roller of the belt cleaning device **8027** and that is rotated by a drive device **8070** according to the eighth exemplary embodiment described later. Belt support rollers **8023** and **8024** are configured as surfacing rollers that form an image formation surface of the intermediate transfer belt **8021**. A belt support roller **8025** is configured as a tension applying roller that applies tension to the intermediate transfer belt **8021**. The belt support roller **8026** is configured as a facing roller that faces the second transfer device **8030**.

The intermediate transfer device **8020** is configured such that the belt support rollers **8023** and **8024** are movable toward the inside and outside of the intermediate transfer belt **8021** together with the first transfer devices **8015** (Y, M, C, and K). In the full color mode, as illustrated in FIG. 8-2, the belt support rollers **8023** and **8024** are disposed at positions at which the first transfer devices **8015** (Y, M, C, and K) come into contact with the intermediate transfer belt **8021** on the surfaces of the photoconductor drums **8011** (Y, M, C, and K). In the monochrome mode, as illustrated in FIG. 8-4, only the first transfer device **8015K** for black (K) comes into contact with the intermediate transfer belt **8021**

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on the surface of the photoconductor drum **8011K** for black (K), and the belt support roller **8023** moves such that the first transfer devices **8015** (Y, M, and C) for the colors are away from the surface of the photoconductor drums **8011** (Y, M, and C) together with the intermediate transfer belt **8021**.

During maintenance, as illustrated in FIG. 8-5, the belt support rollers **8023** and **8024** move such that the first transfer devices **8015** (Y, M, and C) for the colors and the first transfer device **8015K** for black (K) are away from the surfaces of the photoconductor drums **8011** (Y, M, C, and K) together with the intermediate transfer belt **8021**. FIG. 8-5 illustrates a state in which a photoconductor drum unit **8200Y** for yellow (Y) is removed for convenience.

As illustrated in FIG. 8-1, the second transfer device **8030** is a contact transfer device including a second transfer roller **8031** that comes into contact with the circumferential surface of the intermediate transfer belt **8021** and that rotates, and second transfer voltage is applied thereto at the second transfer position that is a portion of the outer circumferential surface of the intermediate transfer belt **8021** that is supported by the belt support roller **8026** of the intermediate transfer device **8020**. Direct current voltage that has polarity opposite the charge polarity of the toner or the same polarity as the charge polarity of the toner is applied as the second transfer voltage from the power supply device, not illustrated, to the second transfer roller **8031** or the belt support roller **8026** of the intermediate transfer device **8020**.

The fixing device **8040** is configured such that a heat rotating body **8041** in the form of a roller or a belt that rotates in a direction illustrated by an arrow and that is heated by a heating unit such that surface temperature is kept at a predetermined temperature, and a compression rotating body **8042** in the form of a belt or a roller that comes into contact therewith substantially in the axial direction of the heat rotating body **8041** at predetermined pressure and that is rotated are disposed in a housing that has an inlet and an outlet for the recording paper **8005** and that is not illustrated. The fixing device **8040** corresponds to a portion for a fixing process in which a contact portion at which the heat rotating body **8041** and the compression rotating body **8042** are in contact with each other performs the required fixing process (heating and compressing).

The paper feeding device **8050** is disposed below the imaging devices **8010** (Y, M, C, and K). The paper feeding device **8050** is constituted by, for example, a paper container body **8051** (or paper container bodies **8051**) that contains a desired kind of the recording paper **8005** having a desired size with the recording paper **8005** loaded thereon and feed devices **8052** and **8053** that feed the recording paper **8005** from the paper container body **8051** one by one. For example, the paper container body **8051** is mounted so as to be able to be pulled to a location in front of the apparatus body **8001a** (a side surface that the user faces during operation).

Examples of the recording paper **8005** include plain paper for use in an electrophotographic copying machine or printer, thin paper such as tracing paper, and OHP paper. The surface of the recording paper **8005** may be smooth to improve smoothness of the surface of the fixed image. For example, so-called thick paper having a relatively great basis weight such as coated paper acquired by coating the surface of the plain paper with, for example, resin or print art paper may be used.

A single or multiple pairs of paper transport rollers **8054** and single or multiple pairs of paper transport rollers **8055** that transport the recording paper **8005** that is fed from the paper feeding device **8050** to the second transfer position,

and a paper feed transport path **8056** that is constituted by a transport guide, not illustrated, are provided between the paper feeding device **8050** and the second transfer device **8030**. The pair of paper transport rollers **8055** that is disposed just upstream of the second transfer position on the paper feed transport path **8056** is configured, for example, as rollers (resist rollers) that adjust a timing with which the recording paper **8005** is transported. A sheet transport path **8057** on which the recording paper **8005** that is fed from the second transfer device **8030** after the second transfer is transported to the fixing device **8040** is provided between the second transfer device **8030** and the fixing device **8040**. A discharge transport path **8061** is provided near the outlet that is formed in the apparatus body **8001a** of the image forming apparatus **8001** for the paper, and a pair of paper discharge rollers **8059** and a pair of paper discharge rollers **8060** for discharging the recording paper **8005** that is fed from the fixing device **8040** after fixing to a paper discharge portion **8058** that is an upper portion of apparatus body **8001a** are disposed on the discharge transport path **8061**.

The image forming apparatus **8001** also includes a double-sided unit **8062** for forming images on both surfaces of the recording paper **8005**. The double-sided unit **8062** introduces the recording paper **8005** via a switch gate **8063** by rotating the pair of paper discharge rollers **8060** in an opposite direction while the pair of paper discharge rollers **8060** holds the trailing edge of the recording paper **8005** when the recording paper **8005** after the image is formed on one surface is transported to the paper discharge portion **8058** by using the pairs of paper discharge rollers **8059** and **8060**. The double-sided unit **8062** has a double-sided transport path **8065** that is constituted by multiple transport rollers **8064** that transport the introduced recording paper **8005** with the recording paper **8005** turned upside down and a transport guide, not illustrated.

In FIG. 8-1, reference characters **8145** (Y, M, C, and K) designate respective toner cartridges that are examples of a developer container that contains the developer containing at least the toner that is supplied to the respective developing devices **8014**. According to the eighth exemplary embodiment, only the toner is contained in the toner cartridges **8145** (Y, M, C, and K).

In FIG. 8-1, reference characters **8100** designate a control device that collectively controls the operation of the image forming apparatus **8001**. The control device **8100** includes a central processing unit (CPU), a read only memory (ROM), a random access memory (RAM), a bus that connects, for example, the CPU and the ROM, and a communication interface, not illustrated.

Operation of Image Forming Apparatus

A basic image formation operation of the image forming apparatus **8001** will now be described.

Operation in the full color mode in which the full-color image that is constituted by a combination of the toner images in the four colors (Y, M, C, and K) with the four imaging devices **8010** (Y, M, C, and K) described above is formed will now be described.

When the image forming apparatus **8001** receives image information and instruction information about a request for full color image formation operation (print) from, for example, a personal computer or an image reading apparatus, not illustrated, the control device **8100** starts up, for example, the four imaging devices **8010** (Y, M, C, and K), the intermediate transfer device **8020**, the second transfer device **8030**, and the fixing device **8040**.

In the imaging devices **8010** (Y, M, C, and K), as illustrated in FIG. 8-1 and FIG. 8-2, the photoconductor

drums **8011** rotate in a direction illustrated by the arrow A, and the charging devices **8012** charge the surfaces of the photoconductor drums **8011** such that the surfaces have a required polarity (minus polarity according to the eighth exemplary embodiment) Subsequently, the exposure devices **8013** radiate light to the charged surfaces of the photoconductor drums **8011**, based on an image signal acquired by conversion into color components (Y, M, C, and K), and electrostatic latent images having the respective color components that are constituted by a required potential difference are formed on the surfaces.

Subsequently, the imaging devices **8010** (Y, M, C, and K) supply the toner in the respective colors (Y, M, C, and K) that is charged and that has the required polarity (the minus polarity) to the electrostatic latent images having the respective color components that are formed on the photoconductor drums **8011** from the development rollers **8141** and electrostatically stick the toner for development. As a result of the development, the toner images in the four colors (Y, M, C, and K) that are developed by using the toner in the respective colors are acquired from the electrostatic latent images having the respective color components that are formed on the respective photoconductor drums **8011**.

Subsequently, the toner images in the respective colors that are formed on the photoconductor drums **8011** of the imaging devices **8010** (Y, M, C, and K) are transported to the first-transfer position, and the first-transfer devices **8015** (Y, M, C, and K) then first-transfer the toner images such that the toner images in the respective colors are superposed in order on the intermediate transfer belt **8021** of the intermediate transfer device **8020** that turns in the direction illustrated by the arrow B.

As for the imaging devices **8010** (Y, M, C, and K) after the first transfer ends, the drum cleaning devices **8016** scrape and remove the attached substances and clean the surfaces of the photoconductor drums **8011**. Consequently, the imaging devices **8010** (Y, M, C, and K) become capable of performing imaging operation described below.

Subsequently, the intermediate transfer device **8020** carries the toner images that are first-transferred by the intermediate transfer belt **8021** that turns and transports the toner images to the second transfer position. The paper feeding device **8050** feeds the recording paper **8005** that is required to the paper feed transport path **8056** in conjunction with the imaging operation. As for the paper feed transport path **8056**, the pair of paper transport rollers **8055** that serves as the resist rollers feeds and supplies the recording paper **8005** to the second transfer position in conjunction with the timing of the transfer.

At the second transfer position, the second transfer device **8030** collectively second-transfers the toner images on the intermediate transfer belt **8021** to the recording paper **8005**. As for the intermediate transfer device **8020** after the second transfer ends, the belt cleaning device **8027** removes and cleans the attached substances such as the residual toner on the surface of the intermediate transfer belt **8021** after the second transfer.

Subsequently, the recording paper **8005** on which the toner images are second-transferred is separated from the intermediate transfer belt **8021** and is subsequently transported to the fixing device **8040** via the sheet transport path **8057**. The fixing device **8040** introduces the recording paper **8005** after the second transfer ends such that the recording paper **8005** passes through the contact portion between the heat rotating body **8041** that rotates and the compression rotating body **8042** for the needed fixing process (heating and compressing) and fixes the unfixed toner images to the

recording paper **8005**. The recording paper **8005** after fixing is finally discharged onto, for example, the paper discharge portion **8058** that is installed on the upper portion of the apparatus body **8001a** by using the pair of paper discharge rollers **8060**.

As a result of the operation described above, the recording paper **8005** on which the full-color image that is constituted by the combination of the toner images in the four colors is formed is outputted.

When the image forming apparatus **8001** receives the image information and the instruction information about a request for monochrome image formation operation (print) from, for example, a personal computer or an image reading apparatus, not illustrated, and the control device **8100** starts up, for example, the imaging device **8010K** for black (K) among the four imaging devices **8010** (Y, M, C, and K), the intermediate transfer device **8020**, the second transfer device **8030**, and the fixing device **8040**.

In the monochrome mode, as illustrated in FIG. 8-4, only the first-transfer device **8015K** for black (K) comes into contact with the intermediate transfer belt **8021** on the surface of the photoconductor drum **8011K** for black (K), the belt support roller **8023** moves such that the first-transfer devices **8015** (Y, M, and C) for the colors are away from the surfaces of the photoconductor drums **8011** (Y, M, and C) together with the intermediate transfer belt **8021**.

The toner image in black (K) is formed by the imaging device **8010K** for black (K), and the monochrome image is formed on the recording paper **8005**.

Image Formation Unit

As for the image forming apparatus **8001**, the photoconductor layers of the photoconductor drums **8011**, for example, wear due to the image formation operation, and it is necessary for the photoconductor drums **8011** to be replaced with new ones, for example, in the case where the photoconductor drums **8011** reach the lifetime thereof. As for the image forming apparatus **8001**, the developing devices **8014** are replaced with new ones in the case where the developer in the developing devices **8014** deteriorates, and the developing devices **8014** reach the lifetime thereof.

For this reason, as for the image forming apparatus **8001**, the imaging devices **8010** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K) are manufactured as units. As illustrated in FIG. 8-3, each of the imaging devices **8010** (Y, M, C, and K) includes a photoconductor drum unit **8200** that is an example of a customer replacement unit (CRU) that includes the photoconductor drum **8011**, the charging device **8012**, and the drum cleaning device **8016** that are integrally formed, and a development unit **8300** that is constituted by only the developing device **8014**. The exposure devices **8013** are installed in the apparatus body **8001a** of the image forming apparatus **8001**.

The photoconductor drum units **8200** (Y, M, C, and K) and the development units **8300** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K) are configured so as to be separately installable in and removable from the apparatus body **8001a**.

The photoconductor drum units **8200** (Y, M, C, and K) and the development units **8300** (Y, M, C, and K) are positioned and secured at predetermined operation positions by being pushed in a Y-direction from the front (a front surface) of the apparatus body **8001a** toward the rear (a back surface) by using guide rails that are examples of a guide unit that is disposed in the apparatus body **8001a** and that is not illustrated. As illustrated in FIG. 8-6, positioning members **8201** and multiple positioning pins **8301** and **8302** are disposed at positions that face back surfaces of the devel-

opment units **8300** and the photoconductor drum units **8200** so as to project toward the back surfaces. An inner frame **8066** that faces the photoconductor drum units **8200** and the development units **8300** has positioning holes **8202**, **8303**, and **8304** into which the positioning members **8201** and the positioning pins **8301** and **8302** of the photoconductor drum units **8200** and the development units **8300** are inserted. The photoconductor drum units **8200** and the development units **8300** are secured at the operation positions by using a securing unit, not illustrated, after being positioned at the operation positions.

As illustrated in FIG. 8-1, the toner in the respective colors is supplied to the developing devices **8014** (Y, M, C, and K) of the development units **8300** (Y, M, C, and K) from the toner cartridges **8145** (Y, M, C, and K) that are disposed on the upper portion of the intermediate transfer device **8020** with a required timing. The toner in the respective colors that is contained in the toner cartridges **8145** (Y, M, C, and K) is transported in the Y-direction toward the back surface of the apparatus body **8001a**. As illustrated in FIG. 8-7, supply pipes **8146** that extend downward in the Z-direction are connected to end portions of the toner cartridges **8145** (Y, M, C, and K) along the back surface. As illustrated in FIG. 8-6 and FIG. 8-7, coupling portions **8147** that are examples of projecting portions that are coupled with the developing devices **8014** (Y, M, C, and K) and that supply the toner are disposed on lower end portions of the supply pipes **8146**. First shutter members, not illustrated, are mounted on the coupling portions **8147** and normally keep blocking supply holes that are formed in the lower end portions of the supply pipes **8146** and that are not illustrated. As illustrated in FIG. 8-7, the coupling portions **8147** are disposed so as to project from opening portions **8067** (see FIG. 8-6) that are formed in the inner frame **8066** of the apparatus body **8001a** toward the back surface of the apparatus body **8001a**.

As for each of the developing devices **8014** (Y, M, C, and K), a supply member **8149** that is an example of a projecting portion that is connected to the coupling portion **8147** and that supplies the toner to a location inside the housing **8140** is disposed at an end portion of the mixing transport member **8143** that faces the back surface with the supply member **8149** extending (projecting) toward the back surface of the apparatus body **8001a**. The supply member **8149** of each of the developing devices **8014** (Y, M, C, and K) has a supply port, not illustrated, on an upper end portion thereof. A second shutter member that is normally urged in a close direction and that is not illustrated is installed in the supply port of the supply member **8149** so as to be openable and closable.

When each of the developing devices **8014** (Y, M, C, and K) is installed in the apparatus body **8001a**, the end portion of the housing **8140** that faces the back surface pushes the first shutter member and opens the supply hole, and an end portion of the coupling portion **8147** that faces an inner surface of the apparatus body **8001a** pushes the second shutter member and opens the supply port. In this way, the toner may be supplied from the supply pipe **8146** into the developing device **8014**.

Configuration of Drive Device

FIG. 8-8 illustrates the configuration of the drive device of the image forming apparatus according to the eighth exemplary embodiment. FIG. 8-8 is a rear view with an exterior covering **8068** that is an example of an outer wall that is located on a back surface of the image forming apparatus **8001** removed. The exterior covering is nearer

than the inner frame and the drive motor to a location outside the apparatus body **8001a** and covers at least the inner frame and the drive motor.

As illustrated in FIG. 8-8, the drive device **8070** that drives the imaging devices **8010** (Y, M, C, and K) and the intermediate transfer device **8020** for yellow (Y), magenta (M), cyan (C), and black (K) is installed on the back surface of the apparatus body **8001a** so as to face back surfaces of the imaging devices **8010** (Y, M, C, and K) and the intermediate transfer device **8020**.

As illustrated in FIG. 8-9, the apparatus body **8001a** includes the inner frame **8066** inside the back surface. The inner frame **8066** includes a first portion **8661** that has a flat plate shape and that is close to and parallel with the exterior covering **8068** (see FIG. 8-7) that is an example of the outer wall of the apparatus body **8001a** and a second portion **8663** that has a flat plate shape and that is disposed in a portion recessed toward the inside of the apparatus body **8001a** with a drawing process portion **8662** that tilts with respect to the first portion **8661** interposed therebetween, which are formed by performing, for example, press working (a drawing process) on a metal plate. The first portion **8661** of the inner frame **8066** is disposed so as to face the back surfaces of the imaging devices **8010** (Y, M, C, and K) and the intermediate transfer device **8020** for yellow (Y), magenta (M), cyan (C), and black (K). The second portion **8663** of the inner frame **8066** is disposed below the imaging devices **8010** (Y, M, C, and K) and the intermediate transfer device **8020** for yellow (Y), magenta (M), cyan (C), and black (K) so as to face a side surface thereof. The first portion **8661** and the second portion **8663** of the inner frame **8066** are parallel with the exterior covering **8068**. A distance D1 between the second portion **8663** and the exterior covering **8068** of the inner frame **8066** is longer than a distance D2 between the first portion **8661** and the exterior covering **8068** (D1>D2). The lower end portion of the drawing process portion **8662** of the inner frame **8066** tilts at the same angle as the imaging devices **8010** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K).

The first portion **8661** and the second portion **8663** of the inner frame **8066** are not limited to those having a flat plate shape, and examples thereof widely range from a portion having unevenness to a portion including a tilting part. The first portion **8661** and the second portion **8663** of the inner frame **8066** are not necessarily parallel with the exterior covering **8068** but may tilt with respect to the exterior covering **8068**.

As for the inner frame **8066**, the first portion **8661** and the second portion **8663** are not necessarily integrally formed. The first portion **8661** and the second portion **8663** may be composed of, for example, different metal plates and coupled with each other into a single piece or may be configured as separated bodies.

As illustrated in FIG. 8-8, the drive device **8070** has a flat box shape such that a length (thickness) in a depth Y direction of a substantially rectangular shape in a plan view is relatively short (thin). As for the drive device **8070**, as illustrated in FIG. 8-7, a device substrate **8701** is disposed on a side surface of the apparatus body **8001a** with the device substrate **8701** extending in the Z-direction, and an end surface of the apparatus body **8001a** that faces the back surface is covered by a covering **8701a**.

As illustrated in FIG. 8-10, the drive device **8070** includes a first drive motor **8702** that is an example of a drive unit (a drive source) that drives the developing devices **8014** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C), a second drive motor **8703** that is an example of a drive

unit (a drive source) that drives the intermediate transfer belt **8021**, the photoconductor drum **8011K** for black (K), and the developing device **8014K** for black (K), and a third drive motor **8704** that is an example of a drive unit (a drive source) that drives the photoconductor drums **8011** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C). According to the eighth exemplary embodiment, the first to third drive motors **8702** to **8704** are mounted on the device substrate **8701**.

The first to third drive motors **8702** to **8704** are configured in the same manner. As illustrated in FIG. 8-11, each of the first to third drive motors **8702** to **8704** includes a substrate **8705** that has a substantially rectangular shape in a plane view. A motor body **8706** that is an example of a drive member body that includes a motor with built-in reduction mechanism is disposed on a back surface of the substrate **8705** so as to project into a short cylindrical shape. A drive shaft **8707** is disposed on a front surface of the substrate **8705** so as to project from the motor body **8706** in a direction. Mounting internal thread portions **8708** for mounting the first to third drive motors **8702** to **8704** are disposed on the outer circumference of the drive shaft **8707** at three positions along a circumferential direction. In FIG. 8-11, reference characters **8709** designate a column boss for positioning each of the first to third drive motors **8702** to **8704**.

As for each of the first to third drive motors **8702** to **8704**, the rotating shaft of a built-in motor, not illustrated, is rotated, the number of rotation of the rotating shaft is decreased by the built-in reduction mechanism, and the drive shaft **8707** is rotated at a predetermined number of rotation. The first to third drive motors **8702** to **8704** may not include the built-in reduction mechanism.

As illustrated in FIG. 8-9 and FIG. 8-10, the first to third drive motors **8702** to **8704** are disposed on the second portion **8663** of the inner frame **8066**. The first drive motor **8702** is disposed so as to face downward on an upper left part of the second portion **8663** of the inner frame **8066** when viewed from the back surface. The second drive motor **8703** is disposed so as to face downward and so as to be adjacent to the first drive motor **8702** slightly below the first drive motor **8702**. The third drive motor **8704** is horizontally disposed so as to face the second drive motor **8703** at a position slightly away from the second drive motor **8703** to a central portion below the second drive motor **8703**.

As illustrated in FIG. 8-7, the first to third drive motors **8702** to **8704** are disposed with the drive shaft **8707** projecting from the motor body **8706** toward the exterior covering **8068**. The first to third drive motors **8702** to **8704** are configured such that the length of a mountable region in the axial direction of the drive shaft **8707** is longer than the distance D2 between the first portion **8661** of the inner frame **8066** and the exterior covering **8068**.

As illustrated in FIG. 8-10, the drive shaft **8707** of the first drive motor **8702** is configured as a drive gear **8710** that includes, for example, a helical gear or a spur gear. A first transmission gear **8711** that transmits rotation driving force to the developing devices **8014** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C) is engaged with the drive gear **8710** of the first drive motor **8702**. As illustrated in FIG. 8-12, a second transmission gear **8712** that is disposed inside the apparatus body **8001a** and that has a relatively large outer diameter is engaged with the first transmission gear **8711**. A developing device drive gear **8080Y** that faces an end portion of the developing device **8014Y** for yellow (Y) in the axial direction is engaged with the second transmission gear **8712**. A developing device

drive gear **8080M** that faces an end portion of the developing device **8014M** for magenta (M) in the axial direction is engaged with the second transmission gear **8712** with third and fourth transmission gears **8713** and **8714** that have a relatively small outer diameter interposed therebetween. A developing device drive gear **8080C** that faces an end portion of the developing device **8014C** for cyan (C) in the axial direction is engaged with the fourth transmission gear **8714** with fifth and sixth transmission gears **8715** and **8716** that have a relatively small outer diameter interposed therebetween.

Accordingly, the drive device **8070** rotates the first drive motor **8702** and consequently rotates the developing device drive gears **8080** (Y, M, and C) that face the respective end portions of the developing devices **8014** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C) in the axial direction.

As illustrated in FIG. 8-10, the drive shaft **8707** of the second drive motor **8703** is configured as a drive gear **8717** that includes, for example, a helical gear or a spur gear. A seventh transmission gear **8718** that transmits rotation driving force to the photoconductor drum unit **8200K** for black (K), a first transmission pulley **8719** that transmits rotation driving force to the developing device **8014K** for black (K), and an eighth transmission gear **8720** that transmits rotation driving force to the belt support roller **8022** of the intermediate transfer device **8020** are engaged with the drive gear **8717** of the second drive motor **8703**.

A ninth transmission gear **8721** that has a relatively large outer diameter and a tenth transmission gear **8722** that has a relatively small outer diameter are engaged with the seventh transmission gear **8718**. A photoconductor drive gear **8081K** that faces an end portion of the photoconductor drums **8011K** for black (K) in the axial direction is engaged with the tenth transmission gear **8722**. The rotation driving force of the first transmission pulley **8719** is transmitted to a second transmission pulley **8724** via a timing belt **8723**. The first and second transmission pulleys **8719** and **8724** are engaged with the timing belt **8723** and include gears. A developing device drive gear **8080K** that faces an end portion of the developing device **8014K** for black (K) in the axial direction is engaged with the second transmission pulley **8724** with eleventh to thirteenth transmission gears **8725** to **8727** interposed therebetween. The eleventh transmission gear **8725** includes multiple transmission gears that are integrally formed in the axial direction and is configured so as to be capable of transmitting driving force also in the axial direction.

An intermediate transfer body drive gear **8082** that faces an end portion of the belt support roller **8022** of the intermediate transfer device **8020** in the axial direction is engaged with the eighth transmission gear **8720** with a fourteenth transmission gear **8728** that has a relatively small outer diameter and fifteenth and sixteenth transmission gears **8729** and **8730** that have an outer diameter smaller than that interposed therebetween.

Accordingly, the drive device **8070** rotates the second drive motor **8703** and consequently rotates the developing device drive gear **8080K** that faces the end portion of the developing device **8014K** for black (K) in the axial direction, the photoconductor drive gear **8081K** that faces an end portion of the photoconductor drum **8011K** for black (K) in the axial direction, and the intermediate transfer body drive gear **8082** that faces the end portion of the belt support roller **8022** of the intermediate transfer device **8020** in the axial direction.

As illustrated in FIG. 8-10, the drive shaft **8707** of the third drive motor **8704** is configured as a drive gear **8731** that includes, for example, a helical gear or a spur gear. A seventeenth transmission gear **8732** that transmits rotation driving force to the photoconductor drums **8011** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C) is engaged with the drive gear **8731** of the third drive motor **8704**. Photoconductor drive gears **8081** (M and C) that face end portions of the photoconductor drums **8011** (M and C) for magenta (M) and cyan (C) in the axial direction are engaged with the seventeenth transmission gear **8732**. A photoconductor drive gear **8081Y** that faces an end portion of the photoconductor drum **8011Y** for yellow (Y) in the axial direction is engaged with the photoconductor drive gear **8081M** of the photoconductor drums **8011M** for magenta (M) with an eighteenth transmission gear **8734** that has a relatively small outer diameter interposed therebetween.

Accordingly, the drive device **8070** rotates the third drive motor **8704** and consequently rotates the developing device drive gears **8080** (Y, M, and C) that face the end portions of the developing devices **8014** (Y, M, and C) for the colors of yellow (Y), magenta (M), and cyan (C) in the axial direction.

As illustrated in FIG. 8-12, the drive device **8070** includes photoconductor member principal couplings **8810** (Y, M, C, and K) and developing device principal couplings **8830** (Y, M, C, and K) that are examples of first and second connection transmission units that are movable in the axial direction of the photoconductor drive gears **8081** (Y, M, C, and K) and the developing device drive gears **8080** (Y, M, C, and K). The drive device **8070** includes an intermediate transfer device principal coupling **8840** that is an example of a third connection transmission unit that is movable in the axial direction of the belt support roller **8022** of the intermediate transfer device **8020**.

As illustrated in FIG. 8-13, each of the photoconductor drive gears **8081** (Y, M, C, and K) includes a shaft core **8811** that projects toward a side surface, that has a cylindrical shape, and that is integrally formed at the center. Each of the photoconductor member principal couplings **8810** is installed on the shaft core **8811** so as to be urged in a projection direction parallel to the rotation axis direction of the side surface of the photoconductor drive gear **8081**. The photoconductor member principal coupling **8810** includes a first gear portion **8812** that has a spur gear formed on the outer circumference by using an involute gear and that has a cylindrical shape and a second gear portion **8813** that has a spur gear formed on the outer circumference by using an involute gear, that has an outer diameter smaller than that of the first gear portion **8812**, that has a cylindrical shape, and that is at an end of the first gear portion **8812**, and these are integrally formed. A tapered portion **8814** that has a tapered shape is disposed at an end of the second gear portion **8813**. The photoconductor member principal coupling **8810** includes a first contact portion **8815** that projects into an annular shape outward in a radial direction between the first gear portion **8812** and the second gear portion **8813**.

The shaft core **8811** of each photoconductor drive gear **8081** includes an internal gear **8816** that is formed by using a spur gear that is an involute gear that is engaged with the first gear portion **8812** of the photoconductor member principal coupling **8810**. The photoconductor member principal coupling **8810** is configured so as to be movable in the axial direction with the first gear portion **8812** engaged with the internal gear **8816** of the photoconductor drive gear **8081** and with the rotation driving force transmitted. As illustrated in FIG. 8-15, the second gear portion **8813** of the photo-

conductor member principal coupling **8810** is configured so as to be capable of being engaged (connected) with and separated from a photoconductor member sub-coupling **8850** that is an example of the second connection transmission unit that is disposed at an end portion of the photoconductor drum **8011** of the photoconductor drum unit **8200** in the axial direction. The photoconductor member sub-coupling **8850** includes an internal gear **8817** that is formed on the inner circumferential surface thereof by using a spur gear that is an involute gear that is engaged with the second gear portion **8813** of the photoconductor member principal coupling **8810** and has a cylindrical shape. A tapered portion **8818** the diameter of which increases in the direction toward the end portion is disposed at an opening edge of the internal gear **8817**. The photoconductor member sub-coupling **8850** is installed so as to be secured to the end portion of the photoconductor drums **8011** in the axial direction. A protection member **8819** that protects the photoconductor member sub-coupling **8850** and that has a cylindrical shape is disposed on the outer circumference of the photoconductor member sub-coupling **8850** and on a side surface of the photoconductor drum unit **8200** so as to project sideways.

The photoconductor member principal coupling **8810** is not limited to a coupling that has the first and second gear portions **8812** and **8813** that are formed by the involute gears. The photoconductor drums **8011** that are rotated by using the photoconductor member principal coupling **8810** directly affects image quality and may accordingly have a small variation in speed and high rotation precision. The photoconductor member principal coupling **8810** that includes the first and second gear portions **8812** and **8813** that are formed by the involute gears is capable of transmitting rotation driving force to the photoconductor drums **8011** with relatively high rotation precision and may be accordingly used.

As illustrated in FIG. 8-13, the photoconductor member principal coupling **8810** is urged in the projection direction by using a first coil spring **8820** that is an example of a first urging unit that is interposed between an inner end surface of the shaft core **8811** of each photoconductor drive gear **8081** and an inner end surface of the first gear portion **8812**. As for the photoconductor member principal coupling **8810**, the amount of projection in the axial direction of the photoconductor drive gear **8081** is restricted by a stationary shaft **8821** that is installed on the shaft core **8811** of the photoconductor drive gear **8081**.

As illustrated in FIG. 8-16, a developing device principal coupling **8830** is installed on each of the developing device drive gears **8080** (Y, M, C, and K) so as to be movable with the amount of projection restricted in the axial direction of the rotating shaft. The developing device drive gear **8080** includes a drive transmission shaft **8831** that is disposed on a side surface thereof so as to project in the axial direction, that has a substantially column or cylindrical shape, and that is integrally formed. Multiple (three projecting portions in an illustrated example) first projecting portions **8832** that have a substantially semicircular section are integrally formed on the outer circumference of the drive transmission shaft **8831** in the circumferential direction. The three first projecting portions **8832** are disposed on the outer circumference of the drive transmission shaft **8831** and form an angle of 120 degrees therebetween. The first projecting portions **8832** extend over the entire drive transmission shaft **8831** in the axial direction. A first shaft support **8833** that is formed around the outer circumference of the drive transmission shaft **8831** and that has a cylindrical shape is disposed on the developing device drive gear **8080**.

The developing device principal coupling **8830** has a substantially cylindrical shape. The developing device principal coupling **8830** has an installation hole **8835**, along the inner circumferential surface thereof, including multiple first recessed portions **8834** with which the multiple first projecting portions **8832** of the drive transmission shaft **8831** are engaged. A base end portion of the developing device principal coupling **8830** includes a second contact portion **8836** that projects into an annular shape outward in the radial direction and that is integrally formed. Multiple (three projecting portions in an illustrated example) second projecting portions **8837** that have a substantially semicircular shape are integrally formed at an end of the developing device principal coupling **8830** in the circumferential direction so as to project sideways. The second projecting portions **8837** are disposed at the same positions in the circumferential direction of the developing device principal coupling **8830** as those of the first projecting portions **8832** of the developing device drive gears **8080**.

As illustrated in FIG. 8-13, the multiple second projecting portions **8837** of each developing device principal coupling **8830** are configured so as to be capable of being engaged (connected) with and separated from the developing device sub-coupling **8301** that is an example of a second drive transmission unit that is disposed at an end portion of the development roller **8141** of the development unit **8300** in the axial direction. The developing device sub-coupling **8301** includes the multiple second recessed portions **8302** that are engaged with the multiple first projecting portions **8832** of the developing device principal coupling **8830** and that are formed on the inner circumferential surface thereof. As illustrated in FIG. 8-16, the developing device principal coupling **8830** is urged in the projecting direction by using a second coil spring **8838** that is an example of a second urging unit that is interposed between an end surface of each developing device drive gear **8080** and a lower end surface of the second contact portion **8836**. As illustrated in FIG. 8-15, the developing device principal coupling **8830** is in contact with the developing device sub-coupling **8301** of the development roller **8141** and is mounted so as to be rotatable with the position of the projection restricted. The position of the projection of the developing device principal coupling **8830** is restricted by a restriction member, not illustrated.

As illustrated in FIG. 8-12, the intermediate transfer device principal coupling **8840** that is an example of a first connection transmission unit is installed on the intermediate transfer body drive gear **8082** so as to be movable in the axial direction. The intermediate transfer device principal coupling **8840** is configured in the same manner as the photoconductor member principal coupling **8810**.

Operation of Drive Device

As for the drive device **8070** that is used for the image forming apparatus **8001** according to the eighth exemplary embodiment, in which the drive unit is disposed between the outer wall and the inner frame that supports the photoconductor drums **8011** and the developing devices **8014** that are examples of a unit to be driven, a distance between the inner frame and the outer wall may be smaller than that in the case where the drive unit is disposed between a portion of the inner frame that is nearest to the outer wall and the outer wall, which is achieved in the manner described below.

That is, as illustrated in FIG. 8-7, the drive device **8070** according to the eighth exemplary embodiment is disposed at an end portion that faces the back surface inside the apparatus body **8001a**. There is the inner frame **8066** at the end portion that faces the back surface of the apparatus body **8001a**.

The supply members **8149** of the developing devices **8014** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K) are disposed on the first portion **8661** of the inner frame **8066** so as to project from the first portion **8661** toward the back surface.

For this reason, as for the drive device **8070** that drives the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K), it is necessary for the transmission gears that transmit the driving force of the first to third drive motors **8702** to **8704** to the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) to be disposed so as to avoid the supply members **8149** of the developing devices **8014** (Y, M, C, and K) in the past as illustrated in FIG. 8-17.

Consequently, as for an existing drive device, it is necessary for the first to third drive motors **8702** to **8704** to further approach the back surface of the apparatus body **8001a**, a distance between the first portion **8661** of the inner frame **8066** and the exterior covering **8068** increases, and the size of the apparatus body **8001a** increases.

As for the drive device **8070** according to the eighth exemplary embodiment, however, as illustrated in FIG. 8-7, the first to third drive motors **8702** to **8704** are configured so as not to be disposed between the first portion **8661** and the exterior covering **8068** such that the distance from the exterior covering **8068** is shortest in the inner frame **8066**, but the first to third drive motors **8702** to **8704** are disposed on the second portion **8663** such that the distance D1 is longer than that in the case of the first portion **8661** that has the shortest distance from the exterior covering **8068** in the inner frame **8066**.

For this reason, as for the drive device **8070** that drives the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K), as illustrated in FIG. 8-7, the first to third drive motors **8702** to **8704** may be disposed on the second portion **8663** such that the distance D1 is larger than that in the case of the first portion **8661** that has the shortest distance from the exterior covering **8068** in the inner frame **8066** with the drive shaft **8707** projecting from the motor body **8706** toward the exterior covering **8068**.

Therefore, as for the drive device **8070** that drives the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) for yellow (Y), magenta (M), cyan (C), and black (K), only the transmission gears that transmit the driving force of the first to third drive motors **8702** to **8704** to the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) may be disposed so as to avoid the supply members **8149** of the developing devices **8014** (Y, M, C, and K), and the first to third drive motors **8702** to **8704** do not project from the transmission gears toward the back surface.

In this way, the drive device **8070** that is used for the image forming apparatus **8001** according to the eighth exemplary embodiment, in which the first to third drive motors **8702** to **8704** are disposed between the inner frame **8066** that supports the developing devices **8014** (Y, M, C, and K) and the photoconductor drums **8011** (Y, M, C, and K) and the exterior covering **8068**, the distance between the inner frame **8066** and the exterior covering **8068** may be smaller than that in the case where the first to third drive motors **8702** to **8704** are disposed between the first portion **8661** that has the shortest distance from the outer wall in the inner frame **8066** and the exterior covering **8068**.

The form of the drive device according to the exemplary embodiment of the present disclosure includes not only a form in the case where the drive device is configured to be separated from the other components of the image forming apparatus **8001** but also a form in the case where the drive device is configured as a part of the image forming apparatus **8001** as described above. The form of the frame according to the exemplary embodiment of the present disclosure includes not only a form in the case where the frame of the drive device is configured to be separated from the inner frame **8066** of the image forming apparatus **8001** but also a form in the case where the frame of the inner frame **8066** of the image forming apparatus **8001** is common to the frame of the drive device **8070**.

Application to a full color image forming apparatus is described according to the above eighth exemplary embodiment. Needless to say, application to a monochrome image forming apparatus is also acceptable.

Additional Matter of All Exemplary Embodiments

Although the specific exemplary embodiments of the present disclosure are described, it is clear for a person in the art that the present disclosure is not limited to the exemplary embodiments, and various other exemplary embodiments may be achieved within the range of the present disclosure. Modifications, deletions, additions, and combinations may be made without departing from the technical idea of the present exemplary embodiments. For example, a part or all of multiple exemplary embodiments may be appropriately combined for a configuration, or some components described in different exemplary embodiments may be combined for a configuration. For example, an apparatus that includes the boundary line **2000K** that is a feature that is described according to the second exemplary embodiment but is not described according to the fourth exemplary embodiment and the container unit **4060** tilting with respect to the horizontal direction that is a component that is described according to the fourth exemplary embodiment but is not described according to the second exemplary embodiment may be configured.

(1) A component is described by using different wording in the exemplary embodiments even through the wording has the same meaning, and (2) a component is described by using the same wording in the exemplary embodiments even through the wording has different meanings although this is not particularly described according to the above exemplary embodiments. In the cases of (1) and (2), the interpretation of a component recited in claims is based on the technical significance and/or definition of the component described according to each exemplary embodiment. In the case of (2), interpretation is basically made based on the technical significance of the component such that any meaning is included. However, the case where the related art is included as a result of the interpretation including any meaning is not excluded. The meaning of the “same” described in this paragraph includes not only the meaning of “completely the same” but also the meaning of “substantially the same”.

The foregoing description of the exemplary embodiments of the present disclosure has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with the various modifications as are suited to the particular use

contemplated. It is intended that the scope of the disclosure be defined by the following claims and their equivalents.

What is claimed is:

1. A document reading device comprising:
 - an image capturing member that captures an image of a document;
 - a placement portion that is disposed below the image capturing member and that has a boundary line between a plurality of surfaces at a position at which the boundary line is covered by the document when the document has a maximum size that the image capturing member is capable of imaging and the document is placed on the placement portion; and
 - a document reading member that reads the document that is placed on the placement portion, wherein the plurality of surfaces comprises a cover of the document reading member and a plate of the placement portion.
2. The document reading device according to claim 1, wherein the boundary line is not located at a position at which the boundary line is not covered by the document having the maximum size at least within a capturing range of the image capturing member.
3. The document reading device according to claim 1, wherein the placement portion is constituted by a plurality of members, and wherein the boundary line is formed by a gap between the plurality of members.
4. The document reading device according to claim 1, wherein the placement portion has a mark for adjusting a position of the document, and wherein the boundary line is formed at a position away from the mark along a placement surface of the placement portion on which the document is placed.
5. The document reading device according to claim 2, wherein the placement portion has a mark for adjusting a position of the document, and wherein the boundary line is formed at a position away from the mark along a placement surface of the placement portion on which the document is placed.
6. The document reading device according to claim 3, wherein the placement portion has a mark for adjusting a position of the document, and wherein the boundary line is formed at a position away from the mark along a placement surface of the placement portion on which the document is placed.
7. The document reading device according to claim 1, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.
8. The document reading device according to claim 2, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.
9. The document reading device according to claim 3, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.
10. The document reading device according to claim 4, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.

11. The document reading device according to claim 5, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.
12. The document reading device according to claim 6, wherein the placement portion is constituted by two members, and wherein the placement portion includes only one of the boundary line.
13. The document reading device according to claim 1, further comprising:
 - a loading portion that is disposed below a placement surface of the placement portion on which the document is placed and that enables a plurality of documents to be loaded thereon; and
 - a reading member that reads the document that is placed on the loading portion while transporting the document.
14. The document reading device according to claim 13, wherein the placement portion is constituted by two members, wherein the reading member transports the document that is placed on the loading portion via an opening that is adjacent to the loading portion in a direction that intersects a vertical direction, and wherein a member of the two members is a covering that covers an upper surface of the reading member.
15. The document reading device according to claim 14, wherein the other member of the two members is capable of moving in the direction with respect to the reading member and uncovering an upper surface of the loading portion.
16. The document reading device according to claim 15, wherein the other member moves with respect to the reading member by sliding in the direction.
17. The document reading device according to claim 15, wherein a length of the covering of the reading member in the direction is less than half of a length of a placement surface of the placement portion on which the document is placed in the direction.
18. An image forming apparatus comprising:
 - the document reading device according to claim 1;
 - a formation member that forms a copy image of a document that is read by the document reading device; and
 - a container unit that supplies media having different sizes that are contained to the formation member, wherein the boundary line is formed at a position at which the boundary line is covered by the document when a document having a size equal to a size of a largest number of media that are containable in the container unit among the media having the different sizes is placed on the placement portion.
19. The image forming apparatus according to claim 18, further comprising:
 - a display unit that is disposed between the image capturing member and the placement portion and that displays information about the apparatus in an intersecting direction that intersects a vertical direction and a direction that intersects the vertical direction, wherein a part of the plurality of surfaces is movable in the direction.
20. A document reading device comprising:
 - an image capturing member that captures an image of a document;
 - a placement portion on which the document is placed; and
 - a document reading member that reads the document that is placed on the placement portion,

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wherein the placement portion has a boundary line that is formed by a plurality of components on a surface, and the boundary line is located on a central portion when the surface is divided into three portions, wherein the plurality of components of the surface comprises a 5 cover of the document reading member and a plate of the placement portion.

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